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Effects of climate change on Austrian water management





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Media owner and publisher

Federal Ministry of Agriculture, Forestry, Environment and Water Management, Stubenring 1, 1012 Vienna Austrian Water and Waste Management Association (ÖWAV), Marc-Aurel-Straße 5, 1010 Vienna

Editorial team

Dr Reinhard Böhm, MR DI Reinhold Godina, o.Univ.-Prof. DI Dr Hans Peter Nachtnebel, DI Dr Otto Pirker

Editing, layout and typesetting

Fritz Randl (ÖWAV)

Cover photos

(from top left to bottom right): Floods on the Inn near Innsbruck (photo: Josef Pfister, HD Tirol), Mooserboden reservoir (photo: Reinhold Godina, BMLFUW), Grüner See, Styria (photo: Reinhold Godina, BMLFUW), dry cracks (photo: Josef Eitzinger, BOKU)

Pressure

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Climate change: effects on Austrian water management

The latest research reports show what many people have been realising for years. Greenhouse gases released by humans are very likely the cause of the rise in average global temperatures observed since the middle of the 20th century. It is a generally recognised fact that global warming is influenced by humans and has become a challenge for the earth's inhabitants in the 21st century. The worldwide change in the climate system has an impact on nature, society, the economy and, in particular, on water management.

Although Austria is in a very favourable position compared to other regions in Europe with regard to the availability of water, it is important to monitor the possible changes caused by climate change and to take precautions for sustainable water management in current and future measures.

The description of the complex interrelationships of the climate shows how important it is to carry out long-term measurements and basic research in order to be able to precisely record and evaluate climate-related changes and their effects on water management.

The member states of the European Union have included the issue of climate change in the regulations that are important for water management - the Water Framework Directive and the Floods Directive - and have set themselves the task of incorporating adaptation strategies into their management and action plans.

The task of the ÖWAV working committee was to provide a factual presentation of the possible changes to the water management sectors and to differentiate between certain and uncertain effects. This aim is to provide an overview of the current state of research on the effects of climate change on the water industry in Austria and to create a neutral basis for further projects and measures.

This compilation of the changes in the water balance in Austria caused by climate change provides an orientation as to the direction in which this change can go.

Wilfried Schimon

Head of the Water Section at the Ministry of Life

Foreword

In recent years, several major flood events have occurred in Austria (2002 and 2005), of which the 2002 flood event in Lower and Upper Austria in particular caused catastrophic damage. The total damage for Austria alone was estimated at around 3.2 billion€ (BMLFUW, 2003). For the affected countries of Austria, the Czech Republic and Germany, the damage totalled € 15-16 billion.

In contrast, the summer of 2005 was characterised by an intense and prolonged drought across Europe, which caused more damage than the 2002 floods. The number of fatalities as a result of the heatwave far exceeded the number of flood victims. Several storm depressions over Central Europe, such as Lothar in 1999 with an estimated damage impact of USD 6 billion (Swiss Re) and Kyrill in 2007 with economic losses of USD 10 billion (Munich Re), solidified the image among the general public that climate change was already taking place.

The international climate studies documented in the IPCC publications provide a global picture, but only allow indirect conclusions to be drawn for Austria, as the spatial resolution of these models only very limited justice to the complex topography of the Alpine region. In Switzerland and the Federal Republic of Germany, national climate impact research programmes were carried out years ago, which can also be used as an important source of information for Austria, but which cannot directly transferred to Austria. Although a number of individual studies on climate change and its consequences are available for Austria, there is no coherent picture of the possible effects on water management.

For this reason, the ÖWAV up a working group in July 2007 with the aim of compiling the possible effects of climate change on water management in Austria in an interdisciplinary manner and assessing their significance. The topic concerns both the expected changes in climate variables, such as precipitation and temperature, as well as the subsequent effects on glaciological and hydrological processes, which then have an impact on water management usage requirements and the environment. The areas of water supply, energy management, agriculture and forestry as well as the effects on hydrobiology and semi-aquatic flora were therefore also dealt with in the form of presentations. Experts from neighbouring countries were also invited.

The topics, which were jointly defined at the beginning, were dealt with in the form of presentations, the written elaborations of which are included in this brochure. This document the state of knowledge in Austria about possible climate impacts for Austria, while also highlighting the uncertainties in the statements.

We would like to take this opportunity to thank all members of the working group for their active participation over the past year, as well as the Ministry of Life for printing this issue.

Hans-Peter Nachtnebel
Head of the working group

Members of the ÖWAV working committee "Climate Change" and participants in the committee meetings:

Committee head:

o.Univ.-Prof. DI Dr Hans-Peter NACHTNEBEL, University of Natural Resources and Life Sciences, Vienna

Committee members/participants in the committee meetings:

Univ.-Prof. DI Dr Günter BLÖSCHL, Vienna University of Technology
Dr Reinhard BÖHM, Central Institute for Meteorology and Geodynamics, Vienna DI
Frederick M. CATE, ÖKOPLAN Technical Office, Vienna
Dr Andreas DALLA-VIA, Joanneum Research ForschungsgmbH, Graz Prof. Dr Josef
EITZINGER, University of Natural Resources and Life Sciences, Vienna
DI Franz FEICHTINGER, Federal Office for Water Management, Petzenkirchen DI
Dr Robert FENZ, Ministry of Life, Vienna
Baurat h.c. DI Dr Werner FLÖGL, ZT-Büro Dr Flögl, Linz Dr
Martin FUCHS, Pöyry Energy GmbH, Vienna
Dipl.-Geogr. Dr Sven Fuchs, University of Natural Resources and Life
Sciences, Vienna Univ.-Prof. DI Dr Josef FÜRST, University of Natural
Resources and Life Sciences, Vienna Dr Andreas GOBIET, University of
Graz
MR DI Reinhold GODINA, Ministry of Life, Vienna
Dr Klaus GÖTZENDORFER, Jungbunzlauer Austria AG & CO KG, Wulzeshofen o.Univ.-
Prof. DI Dr Dieter GUTKNECHT, Vienna University of Technology
Univ.-Prof. DI Dr Raimund HABERL, University of Natural Resources and Life
Sciences, Vienna Univ.-Prof. DI Dr Helmut HABERSACK, University of Natural
Resources and Life Sciences, Vienna Herbert HEINDL, Vienna University of
Technology
Ass.-Prof. DI Dr Hubert HOLZMANN, University of Natural Resources and Life
Sciences, Vienna OR DI Hubert HONSOWITZ, Vienna University of Technology
Univ.-Prof. Dr Georg A. JANAUER, University of Vienna
Univ.-Prof. DI Dr Klaus KATZENSTEINER, University Natural Resources and Life Sciences, Vienna
HR Univ.-Prof. DI Dr Eduard KLAGHOFER, Federal Office for Water Management, Petzenkirchen
Gernot KOBOLTSCHNIG, International Research Organisation Interpraevent, Klagenfurt MR Univ.-
Doz. Dr Martin KRALIK, Federal Environment Agency GmbH, Vienna
o.Univ.-Prof. DI Dr Dr h.c. Helmut KROISS, Vienna University of Technology DI
Gerhard KUBU, University of Natural Resources and Life Sciences, Vienna
o.Univ.-Prof. Dr Michael KUHN, University of Innsbruck
DI Johannes MOSER, Office of the Carinthian Provincial Government,
Klagenfurt DI Dr Otto PIRKER, Verbund Austrian Hydro Power AG, Vienna
o.Univ.-Prof. DI Dr Wolfgang RAUCH, University of Innsbruck
Dr Ernst RUDEL, Central Institute for Meteorology and Geodynamics, Vienna Dr
Bruno SCHÄDLER, Federal Office for the Environment, Bern (CH)
HR DI Franz SCHILLER, Office of the Upper Austrian Provincial Government, Linz
Univ.-Prof. DI Dr Stefan SCHMUTZ, University of Natural Resources and Life Sciences,
Vienna
MR DI Karl SCHWAIGER, Ministry of Life, Vienna
wHR DI Gerhard SPATZIERER, Office of the Burgenland Provincial Government, Eisenstadt
o.Univ.-Prof. DI Dr Peter TSCHERNUTTER, Vienna University of Technology
DI Mario UNTERWAINIG, University of Natural Resources and Life
Sciences, Vienna DI Hans WEBER, Bavarian State Office for the
Environment, Hof (D)
DI Bertram WEISS, Verbund - Österreichische Elektrizitätswirtschafts AG, Vienna HR DI
Johann WIEDNER, Office of the Styrian Provincial Government, Graz
DI Friedrich ZEMANEK, EVN AG, Maria Enzersdorf
Prof. DI Dr Matthias ZESSNER, Vienna University of Technology
Univ.-Prof. Dr Hans ZOJER, Joanneum Research ForschungsgmbH, Graz

For the ÖWAV:

Andreas GAUL, Consultant for the Water Management Division at ÖWAV, Vienna

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Possible climate impacts for water management in Austria

Reinhard Böhm, Reinhold Godina, Hans-Peter Nachtnebel and Otto Pirker

"Warming of the climate system is unequivocal, as is now evident from observations of increases in global average air and ocean temperatures, widespread melting of snow and ice and rising global average sea level." (IPCC, 2007)

1 Initial situation

The Alpine regions are very sensitive to climate change, as there are seasonal changes in precipitation and is also great spatial variability in the climate and hydrological processes. Studies from Switzerland, Austria and Germany are available for the Alps, which on the one hand document the changes that have already been observed, but which also derive statements for the Alpine region using climate models. As there are various climate models with different emission scenarios that a period up to around the end of this century in their simulations, different results are obtained depending on the forecast period and scenario selected.

2 Climate scenarios and climate models

Climate development can only be estimated using the following chain of effects:

- Assumptions about the social, economic and political development of the humanity, which are presented in the form of 35 IPCC development scenarios;
- the resulting changes in climate forcing, which are simulated by chemical-physical atmospheric models taking into account greenhouse gas concentrations and aerosols;
- The effects of changing climate forcings on the atmosphere, the ocean and the cryosphere (calculation on a finite dense grid using coupled global ocean-atmosphere circulation models (including cryosphere));
- additional consideration of feedbacks between the land and water surface, vegetation and the atmosphere.

Of the IPCC's various lines of development, three are most frequently used, which can be summarised as follows:

- A2 family: Business as usual in a bad sense
- B1 family: full focus on greenhouse gas reduction
- A1B: Mix of strong, globally networked technological development and "socially responsible globalisation".

A1B is to be regarded as an "optimistic globalisation scenario" and is used as the IPCC-2007 lead scenario. B1 serves as an illustration of what could be achieved through extreme reductions in greenhouse gas emissions, and A2 as a deterrent of what would happen if everything got out of hand.

Global circulation models are used to simulate the atmosphere, which depict the development of the climate up to around the end of this century in a coarse-meshed grid. Even with the latest circulation models, the grid size is still several thousand square kilometres. As the earth's surface is parameterised on this scale, the topographical properties are also greatly coarsened, so that local statements for the Alpine region cannot be derived directly.

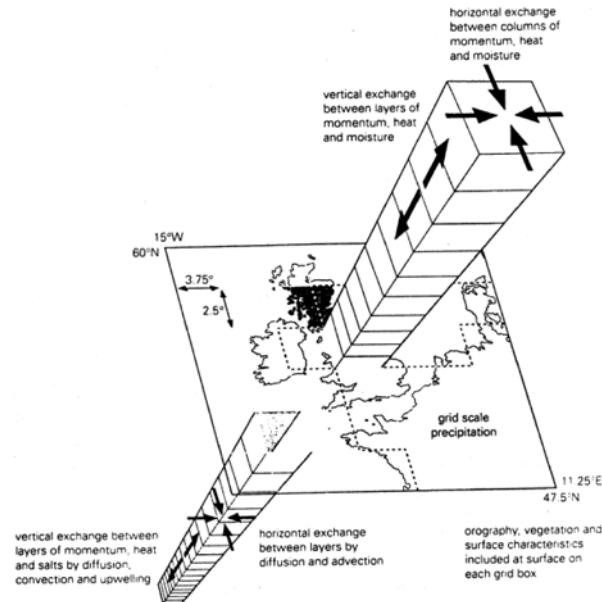


Figure 1: Spatial structure of the Hadley Centre's climate models (HadCM3)

The models consistently show an increase in temperature of up around 4 °C by the end of this century. In general, the simulations point to an increase in global precipitation totals, but the models still show great regional uncertainties.

3 Statements from the climate models for Europe

The global simulations show an increase in the average annual temperature for Europe of around 2.5 °C by 2050 and 4.5 °C by 2100. These values are significantly higher than the specified average global values, which is due to the distribution of land masses and atmospheric circulations. The further we look into the future, the greater the differences in the scenarios, while there are hardly any differences in the simulations for the next three decades due to the inertia of the system. This predicted warming is therefore higher than the temperature increase of around 2 °C observed in the region over the last 150 years (Figure 2).

With this warming, an increase in all types of positive temperature extremes (hot days, absolute maxima, etc.) with a roughly analogue decrease in negative temperature extremes (frost, absolute minima, etc.) can realistically be expected. No change is expected with regard to wind and storm activity over Central Europe. This non-increase in storm activity in the future is consistent with the data findings of the past 130 years (Matulla et al., 2007). For other regions of Europe, there are differing opinions based on the simulation runs.

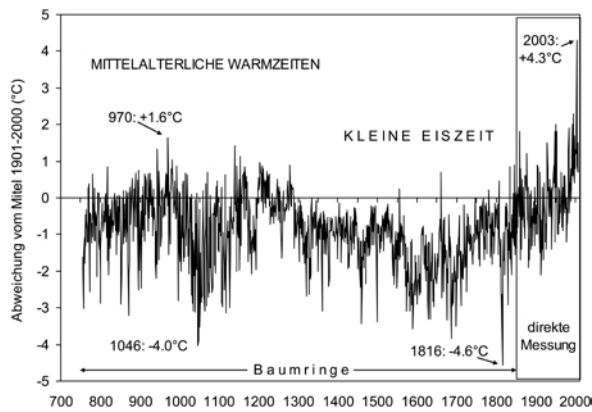


Figure 2: Historical temperature curve in the Alpine region (from tree rings and measurements) (Böhm et al., 2007)

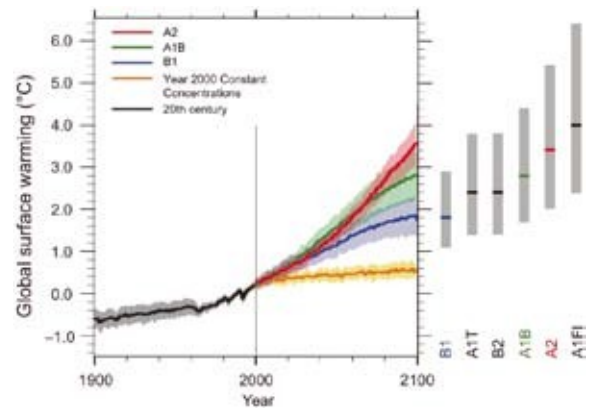


Figure 3: Future global temperature increase (from climate simulations, IPCC Synthesis Report, 2007)

The calculated changes in precipitation are subject to major uncertainties. One can see an increase in annual precipitation with increasing latitude. The Alpine region is located in a transition zone where precipitation increases towards the north, while a decrease is shown in southern Europe. By the end of the century, annual precipitation could therefore increase by more than 25 % over Scandinavia and Siberia and decrease by around 20 % in southern Europe around the Mediterranean. However, *Figure 4 illustrates* the serious uncertainties that still exist using the example of the annual runoff expected for 2020, which is even simulated in a contradictory manner by two "state of the art" models for the Iberian Peninsula, for example.

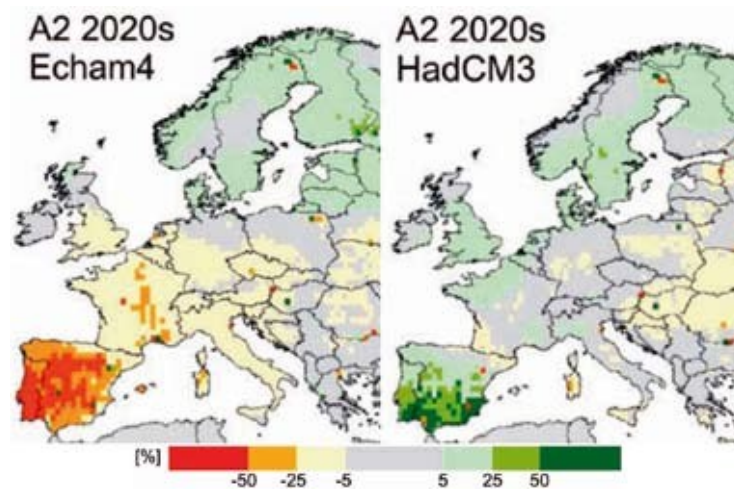


Figure 4: Simulation of the change in mean annual runoff for Europe by 2020 (from IPCC- 2007, WG-2 Report, fig. 12.1)

4 Statements for the Alpine region

The analysis of measurement series provides the following picture of the historical development of temperature (*Figure 5*) and precipitation (*Figure 6*). Shown are anomalies from 124 individual series for the average of the 20th century in the Alpine region, which is delimited by the range 4-19 °E, 43-49°N

was. In addition, the global temperature change is shown, which is significantly more even than that in the Alpine region.

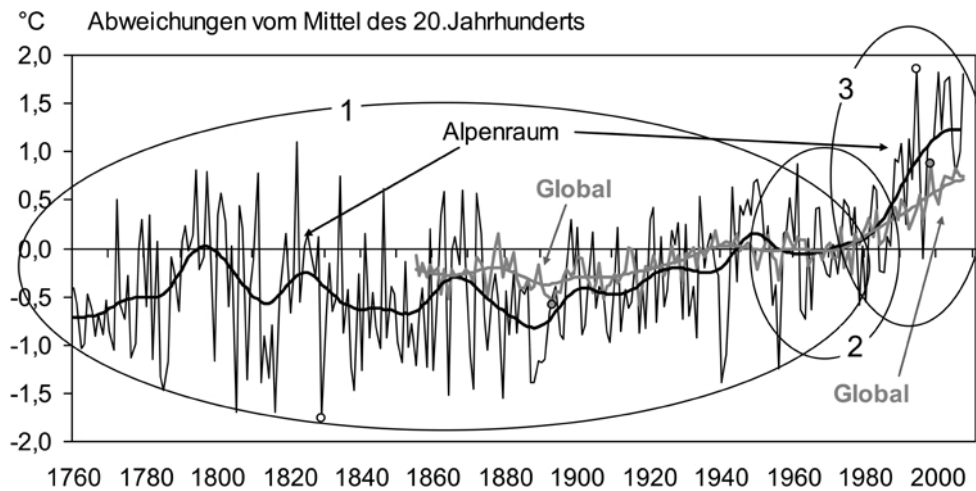


Figure 5: Measured annual average air temperature in the Alps region 1760-2007 (black) and global average 1858-2007 (grey).

1: last natural period - solar and volcanic influence dominant 2: first noticeable human influence - the aerosol period

3: Entering the greenhouse age

(Auer et al., 2007; Jones and Moberg, 2003 (updated))

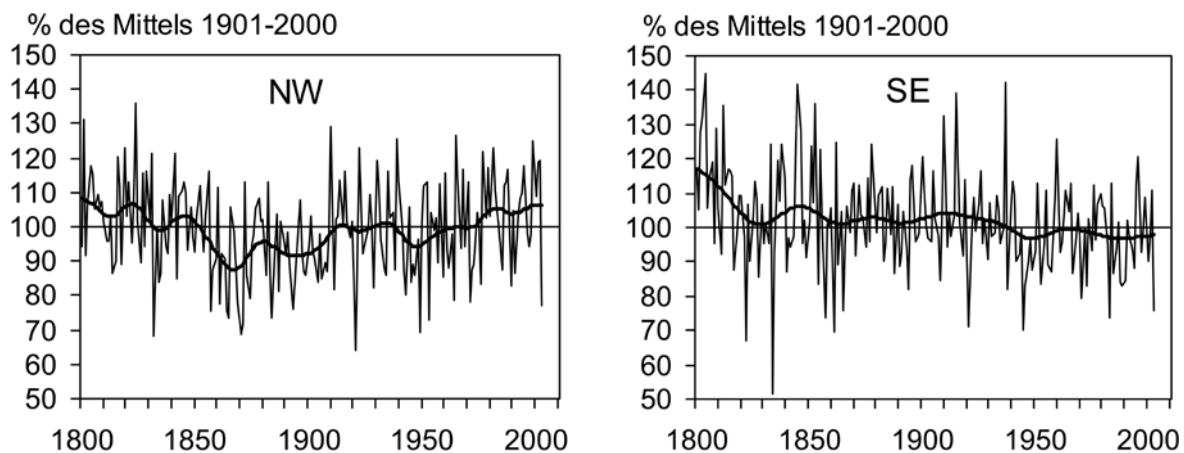


Figure 6: Regional trend differences in precipitation in individual years and for the 30-year average (HISTALP database, Auer et al., 2005)

Left: north-west of the Alpine arc (France, Switzerland, southern Germany, Vorarlberg, Tyrol), right: south-east of the Alpine arc (Hungary, Slovenia, Croatia, Bosnia, Friuli, Veneto, Carinthia, western and eastern Styria).

While the long-term temperature trend in the greater Alpine region is also re-presentative for sub-regions (the +2° applies to Perugia, Vienna, Marseille and Bern, as well as to the high-alpine observatories on the Säntis, Zugspitze and Sonnblick), the precipitation trends in Figure 6 show a striking difference between the long-term upward trend in the north-west of the Alpine arc and the long-term tendency towards drier conditions in the south-west.

south-east. The magnitude of both opposing trends is 10-15 % per century, while the short-term variability is ± 50 %.

For local applications of climate models, higher-resolution simulations are necessary by coupling the information of the coarse-mesh model with regional information on topography, land cover and climate. This is of particular interest if statements are to be made about extreme events.

In "dynamic downscaling", the boundary conditions of the coarse-meshed model are used at the edges of the area and higher spatial resolution is used within the area. As part of the EU project PRUDENCE (2001), four regional models with a spatial resolution of approx. 50 km grid length were applied to Europe and the changes in precipitation and temperature were calculated.

With regard to temperature developments, the model simulations show a clear trend towards an increase in temperature, although the extent of the increase is calculated differently and is therefore between 3-3.5 °C (annual average) in the last quarter of this century and ranges up to 4.5 °C in one model. The effects of the rise in temperature are clearly expressed in the reduction of the active Alpine glacier area to 40-50% (from 1980) by 2050, and to 15-20% by 2100.

The projected change in precipitation varies greatly. While the Swiss model shows a decrease in annual precipitation, the other models also show an increase in precipitation south of the main Alpine ridge. Statements about extreme events are subject to additional uncertainty and cannot actually be derived from this data.

5 Climate change for Austria

First, the observations are analysed and then the expected development is discussed.

The long-term precipitation development in the Danube catchment area upstream of Vienna *is shown in Figure 7*, which shows a period of low precipitation between 1850 and 1870. Overall, the time series of annual precipitation shows a high degree of variability, while the smoothed average is fairly constant. The analysis is based on 28 HISTALP precipitation series. An analysis for Carinthia (Moser et al., 2002) shows a long-term decrease in precipitation (*Figure 8*).

In order to investigate the effects of global climate change on Germany, a regional climate model was embedded in the global climate model and the climate development for the area of Central Europe was calculated with a grid of 10 km side length. The ECHAM5/MPI-OM and REMO model chain developed at the Max Planck Institute for Meteorology (MPI-M) was used for this purpose. The following *Figures 9 and 10* were cut out for Austria from the illustrations in the Federal Environment Agency's report (2008).

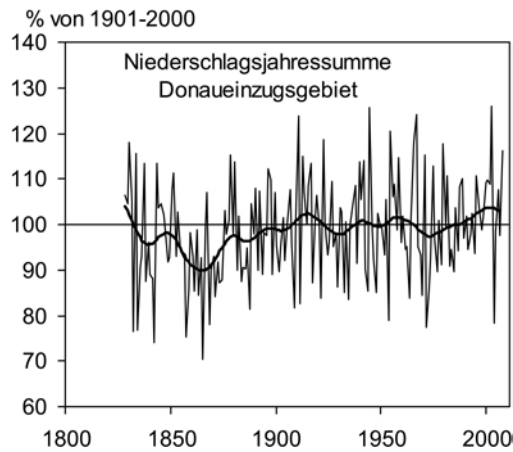


Figure 7: Area average of the annual precipitation totals in the Danube catchment area upstream of Vienna (Auer et al., 2007)

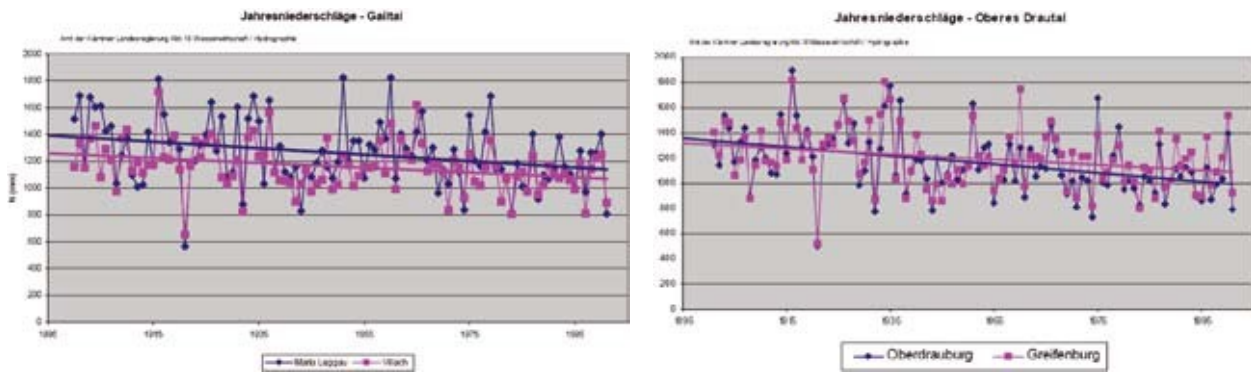


Figure 8: Long-term development of annual precipitation in Carinthia (Moser et al., 2002)

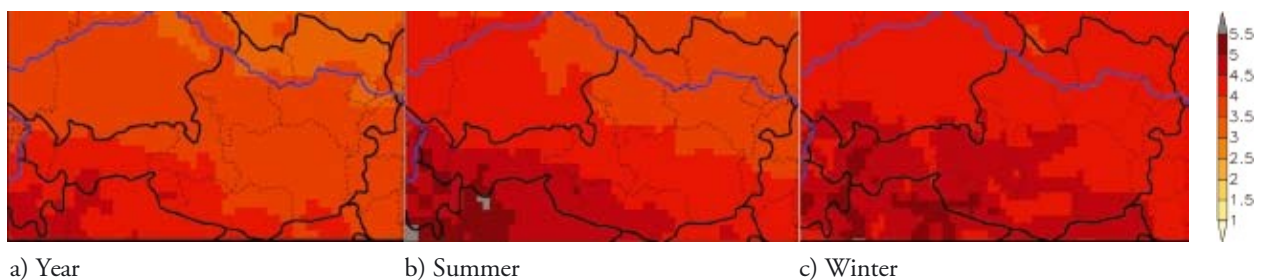


Figure 9: Change in temperatures for the period 2071-2100 compared to the period 1961-1990 (taken from UBA, 2008, pp. 111-115)

Accordingly, a higher increase of up to 4.5 °C should be expected in winter, especially in the south and south-west, than in the rest of Germany, which is given as 3-3.5 °C on an annual average.

The illustrations of the changes in precipitation are shown in Figure 10. On average over the year, only very small changes in precipitation should therefore be expected; in summer, precipitation should decrease, while an increase should be expected in winter. There is a high degree of uncertainty for the Alpine region.

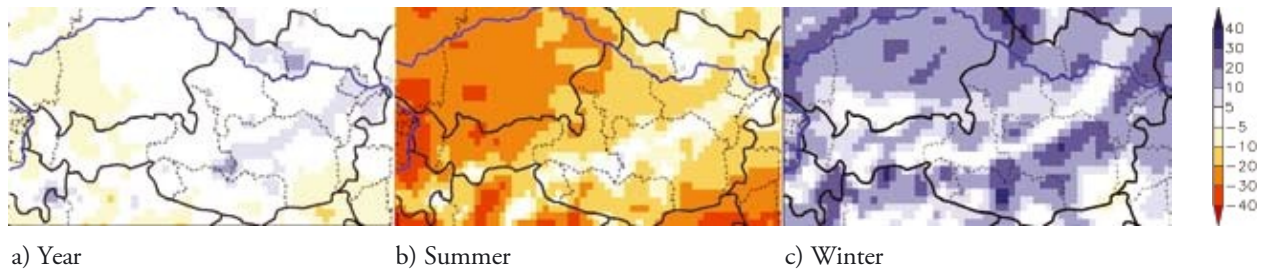


Figure 10: Percentage change in precipitation for the period 2071-2100 compared to the period 1961-1990 (taken from UBA, 2008, p. 111-115)

6 Effects on hydrology and water management

Firstly, the historically observed changes are presented and then the changes obtained through simulations are discussed. In general, it should be noted that the changes observed in the recent past should tend to be comparable with the simulations for the coming decades, but a trend extrapolation should be avoided. Digital data from the Hydrographic Service is available for most monitoring stations from around 1951 and therefore many of the following illustrations refer to the period 1950-2000.

6.1 Water temperature

The water temperature shows a direct correlation with the development of the air temperature. *Figure 11* shows the development of the water temperature of Lake Constance and the River Gail - an example of a watercourse - since 1900 and for Lake Neusiedl since 1951. The moving average over 10 years shows a continuous increase of approx. 1.5 °C in 35 years for Lake Constance since 1970. After a period of decline from 1965 to 1975, the temperature of Lake Neusiedl rose by approx. 2 °C in the following 30 years. The water temperature at the Gail has changed less than at the lake measuring points. However, an increase of approx. 1 °C can also be recognised here since the 1970s.

The increase in air temperature can be seen at all altitudes and therefore also in an increase in discharge at measuring points with a relatively high proportion of glaciers. It is striking that these changes are more clearly recognisable in the summer months of June to August in the past than in the annual mean values in the observation period. *Figure 12* shows the time series of air temperature and discharge at the Vent monitoring station in the Ötztal Alps. While the annual mean values of the temperature and discharge parameters have risen continuously and significantly since the 1960s, the corresponding mean values for the month of June have risen very rapidly over the past 20 years by around +2 °C and +5 m³/s respectively. This result shows that seasonal changes must be taken into account, especially when analysing trends in connection with statements on climate change.

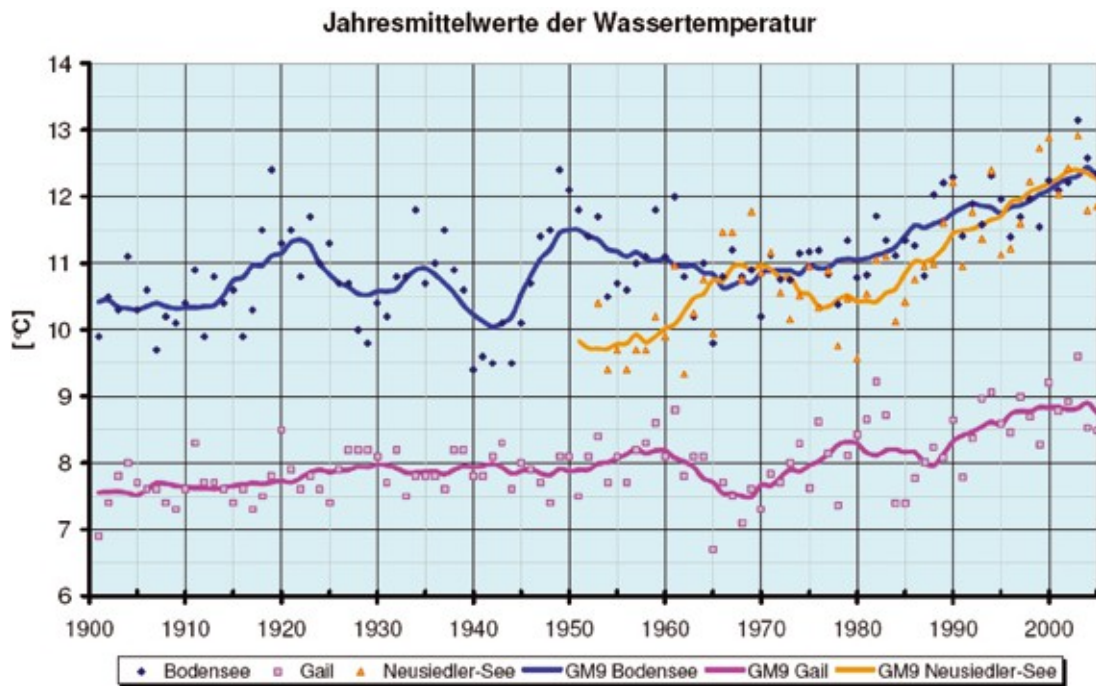


Figure 11: Development of water temperature at selected water bodies in Austria since 1900

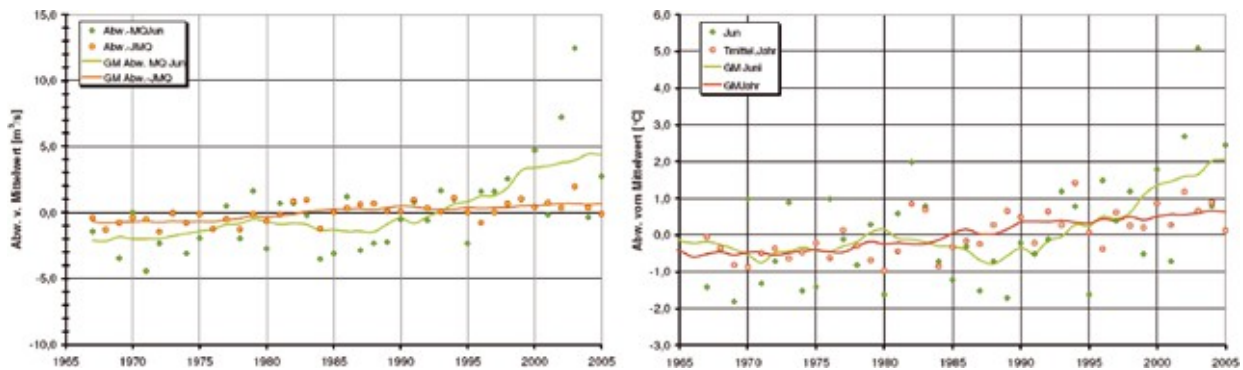


Figure 12: Changes in air temperature and discharge at the Vent measuring point - shown as a deviation from the long-term mean value

6.2 Water balance

The balance components of precipitation, evaporation and runoff form the basis for water management. In a study (Fürst et al., this issue), these components were recorded for all catchment areas taking into account discharges and storage influences. A simulation of all catchments was carried out for the period 1950-2006 on the basis of monthly increments. Some of the results are explained in the Hydrological Atlas of Austria (BMLFUW, 2007) and are summarised below.

A comparison of the current calculation (2007) with the analysis by Kresser (1994) shows only minor differences on an annual average.

Table 1: Comparison of Austria's water balance components

	BMLFUW (2007)	Kresser (1994)
Precipitation	1144 mm	1170 mm
Discharge height	634 mm	654 mm
Act. Evaporation	510 mm	516 mm

Longer-term analyses of the balance components show only minor trends on an annual basis. North of the Alps, there is a slight increase in precipitation (Fig. 13 a), while opposite trends can be seen south of the main Alpine ridge. As a result of the increase in temperature, evapotranspiration increases and the proportion of runoff (Fig. 13b) tends to decrease. This is clearly recognisable in the south of Germany and has also been confirmed by time series analyses of individual stations (see Fig. 8). In some areas, this decline is even statistically significant. As these are simulation calculations, storage and power plant effects are not effective.

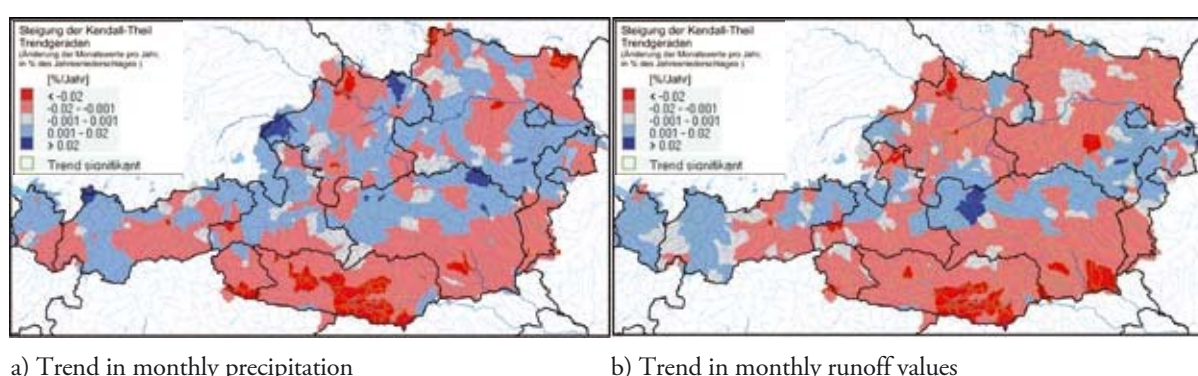


Figure 13: Annual evaluation of trends in monthly discharge heights for the period from 1950 to 2006 (from Fürst et al., 2007)

As part of national and international projects (Nachtnebel et al., 1999, 2001, 2004), several regions of Austria (Gail, Enns and Traisen catchment areas, Marchfeld and Leibnitz field) were analysed using stochastic downscaling methods to determine their reaction to changing climate conditions. This method was used to generate time series of temperature and precipitation with high spatial resolution for various future time periods and to run hydrological models.

In all areas, an annual average temperature rise of 2.5 to 4 °C is expected by the end of this century. The rise in temperature is somewhat more pronounced in the summer months than in the winter months. The temperature increase is lowest in the months of February and March. In the valleys, the temperature increase would be up to 4 °C, while at high altitudes (above 2,000 m) the increase would be 2-3 °C. Evaporation increases significantly and the annual runoff load decreases.

In general, no clear change in precipitation can be recognised from the derived data. Both the precipitation amounts and the seasonal distribution are similar for today and the future. The simulations tend to show a slight decrease in the south of Austria, while increases could even be expected in the northern congested areas.

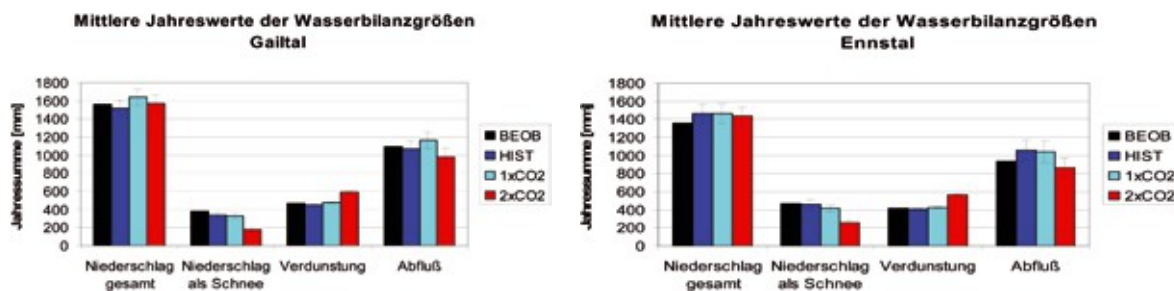


Figure 14: Change in water balance components in the Gailtal valley (Federaun gauge) and in the Upper Enns region (Liezen gauge) (from Nachtnebel and Fuchs, 2001)
black: current state, dark blue: calculated from historical pressure distributions, light blue: climate simulation for 1961-1990, red: climate simulation 2070-2000

The first three bars should show as similar a result as possible, as they relate to the period 1961-1990. The climate signal should be in the difference between the light blue and the red bar. In addition, the uncertainties in the simulations are characterised by fluctuation ranges.

Figure 14 shows a slight decrease in annual precipitation for both areas, which appears somewhat greater in the south. Solid precipitation (snow) decreases by more than half, and the decrease is even greater in the south. As a result of the higher temperatures, evapotranspiration increases and the balance component runoff therefore decreases. A similar picture emerges for the Traisen region, meaning that a decline in runoff can be expected in large areas. For western Austria and in high alpine locations, the picture may be different if precipitation increases at high altitudes and as a result of increased glacier runoff.

Over the course of the year, there is a strong decline in discharge in the spring and summer months and a clear levelling out of the discharge regime overall. The decline is particularly marked in the months of May and June. An increase in runoff is to be expected in the winter months, as a larger proportion of liquid precipitation occurs and snowmelt at lower elevations is also effective in winter. Overall, there is a seasonal attenuation of runoff in alpine areas and a shift in low water periods from winter to late summer. The number of frost days and days with snow cover (Figure 15) decreases by around 50 %. The snow layers will shift uphill by around 400-600 m compared to the current situation.

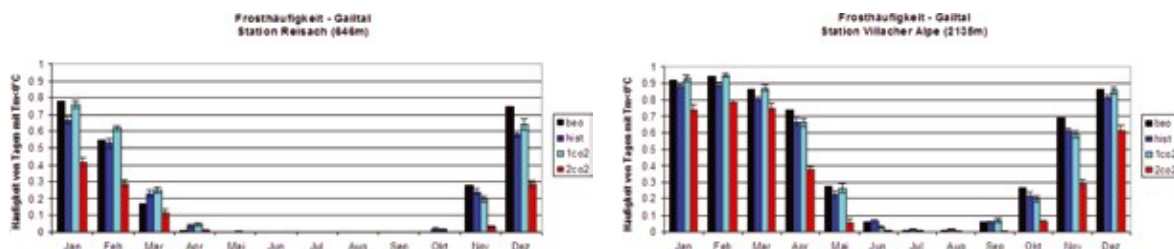


Figure 15: Changes in the number of frost days for a valley and mountain station in the Gailtal for period 2070-2100 (from Nachtnebel and Fuchs, 2001)

6.3 Glacier

The Austrian glaciers represent an important reservoir in the country's water cycle: The volume of ice was estimated at approx. 17 km³ in 1998 (Lambrecht and Kuhn 2007). As a water equivalent of around 16 km³, this corresponds to a water depth of 190 mm or 16 % of average annual precipitation in relation to the total area of Austria. Since the glacier peak in the middle of the 19th century, the Alpine glaciers have noticeably decreased in area and volume. Overall, the following picture emerges for the period from 1960 to 1990: The area of all glaciated areas in Austria has decreased from 567 to 471 km² and has thus decreased by 17 %; the volume has decreased from approx. 23 to approx. 17 km³; this corresponds to a reduction in ice thickness of approx. 10 m (Kuhn, 2007; this issue). As further glacier areas are exposed to melting as temperatures rise, it can be assumed that the proportion of melting will continue to increase over the next forty years the reduction in volume dominates and the contribution declines. By the end of this century, only isolated glacier remnants will remain.

In terms of Austria's water balance, the proportion of runoff from the glacier volume that has melted in thirty years is less than 2 mm/year and is therefore far less than one per cent of the runoff. However, as this melt contribution is only available for a few months of the year and particularly effective in the high Alpine region, the glacier influence is clearly evident in some Alpine valleys.

6.4 Extreme events: Floods and low water

The fact that flood events will be more extreme and occur more frequently as a result of climate change is a frequently heard statement that cannot be substantiated by observations, and the climate signals do not show a uniform effect either. *Figure 16* shows the result of a trend analysis by Department VII/3 - Water Balance for the time series with an observation period of more than 50 years. No uniform regional or temporal behaviour can be seen from this and the identified positive or negative trends and their significance cannot clearly linked to climate change (warming). Direct anthropogenic effects, such as the loss of natural retention areas, straightening and damming of watercourses and the increase in sealed surfaces also have a significant influence on runoff, especially during floods.

The positive trend observed on the Danube is probably related to the increase in events in the order of magnitude of HQ₍₅₎-HQ₁₀. One explanation for this would be the development of the Danube for the utilisation of hydropower and the associated reduction of natural retention areas. An analysis of the annual floods at the Kienstock gauge (Krems-Stein) shows no recognisable change in the extreme floods (Nachtnebel and Debene, 2004). An HQ₁₀₀ was estimated from a sliding series of 50 annual maximum values and this was shown with the confidence range in *Figure 17*.

The changes in low water are much clearer. *Figure 18* shows an upward trend in low water levels in winter since observations began. In summer, on the other hand, a clear reduction in low water levels can be from 1980 onwards. Similar behaviour can also be observed in



Figure 16: Trends in annual maxima for monitoring sites with more than 50 years of flood data (JHQ_{max}) (↑↑ ... significantly increasing; ↑ ... increasing; ↓↓ ... significantly decreasing; → ... constant)

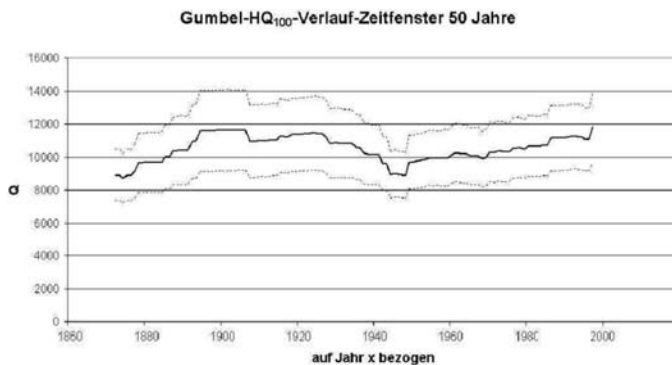


Figure 17: Temporal development of the HQ₁₀₀ for the Kienstock gauge

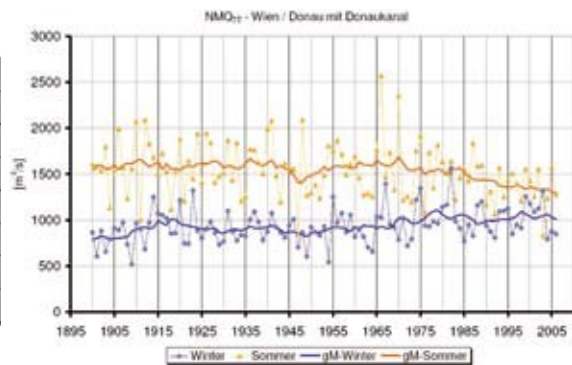


Figure 18: Temporal development of the low water levels for the Danube gauges Reichsbrücke and Danube Canal

This can be observed at many measuring points in the south of Austria (Carinthia and Styria), which show a decrease in seasonal low water discharge in spring and summer.

6.5 Soil and groundwater

The Marchfeld and Leibnitzer Feld were analysed using the same methodology as described for the Alpine catchment areas. The changes in evaporation, soil water balance and groundwater recharge were analysed using a soil water balance model.

Evaporation will increase as a result of the significantly higher temperatures in the future. Although the soil water balance be significantly lower in the summer months in future, no impact on plant growth should be expected in the areas with higher precipitation. In the east and south-east of Austria, a negative effect is certainly to be expected as a result of the decline in soil water.

Figure 19 clearly shows that groundwater recharge in the Leibnitzer Feld currently occurs mainly in the months of October to May, with a maximum in March, although the confidence intervals here are also extremely wide. Towards the end of this century, there would then be a stronger seasonality of groundwater recharge with very low values in summer and relatively high values in winter. Overall, the annual recharge rate would fall by around 25 %. This effect is even more pronounced in individual years. For areas with low precipitation, such as the Marchfeld, a decrease in groundwater recharge of more than 60 % is to be expected. Dalla-Via (this issue) indicates a future decline in groundwater recharge of approx. 40 % for Eastern Styria. The dry phases will also become significantly longer, characterised by low soil moisture. These effects would have to be compensated for by increased irrigation, which would put additional pressure on groundwater resources.

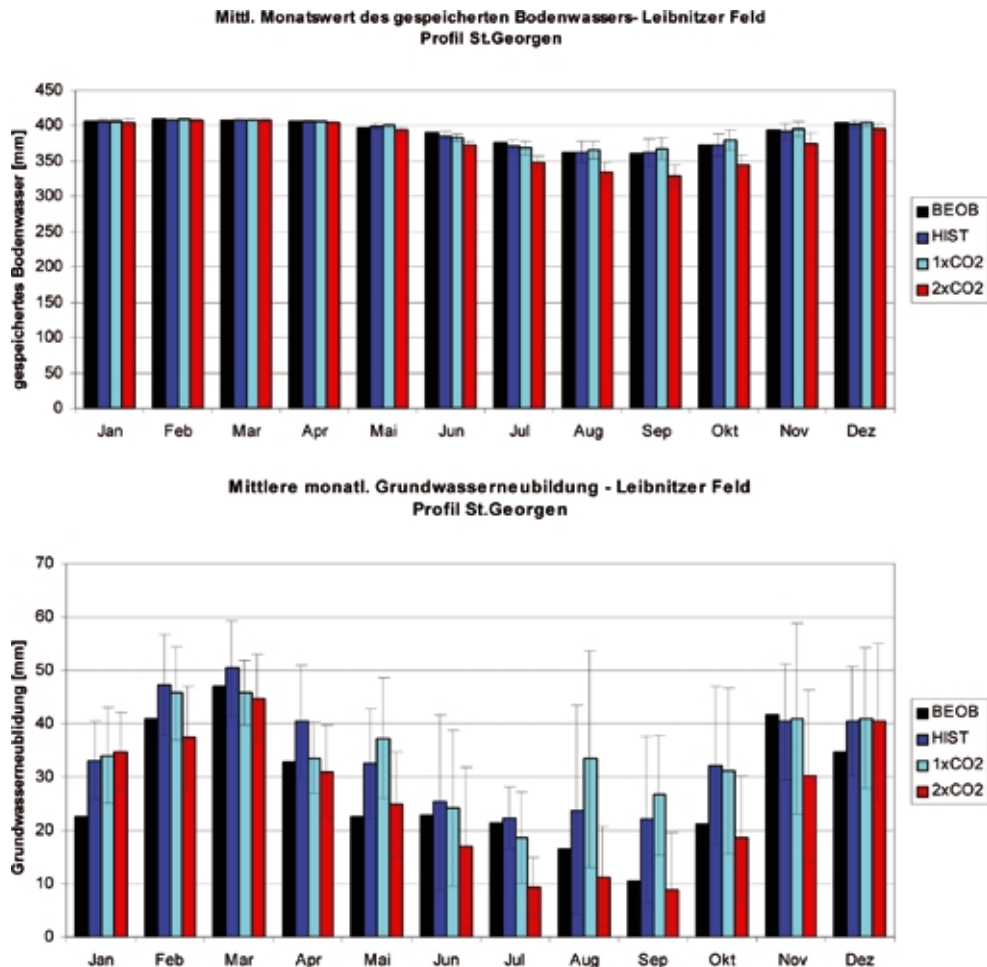


Figure 19: Soil moisture and groundwater recharge in the Leibnitzer Feld (from Nachtnebel and Fuchs, 2001)

6.6 Consequences for hydropower

Climate change is fundamentally changing the framework conditions for hydropower. On the one hand, hydropower as a renewable and pollutant-free energy source is a key factor in the area of electricity production, while on the other hand hydropower itself is characterised by

changes in the water balance due to possible climate change. Possible effects were simulated on the basis of the annual generation coefficients of the run-of-river power plants of Austrian Hydro Power AG (AHP).

The annual series from 1951 to 2007 was used. The generation coefficient is the quotient of the working capacity and the control energy capacity of one or more hydropower plants. The working capacity indicates the energy that the plants under consideration have generated based on the hydraulic supply. The control energy capacity represents the long-term average of the energy actually generated. The generation coefficient is therefore a measure whether generation was above or below average due to the water supply in the period under consideration. *Figure 20* shows that the annual series from 1951 to 2007 shows a slight downward trend, although this is not statistically significant.

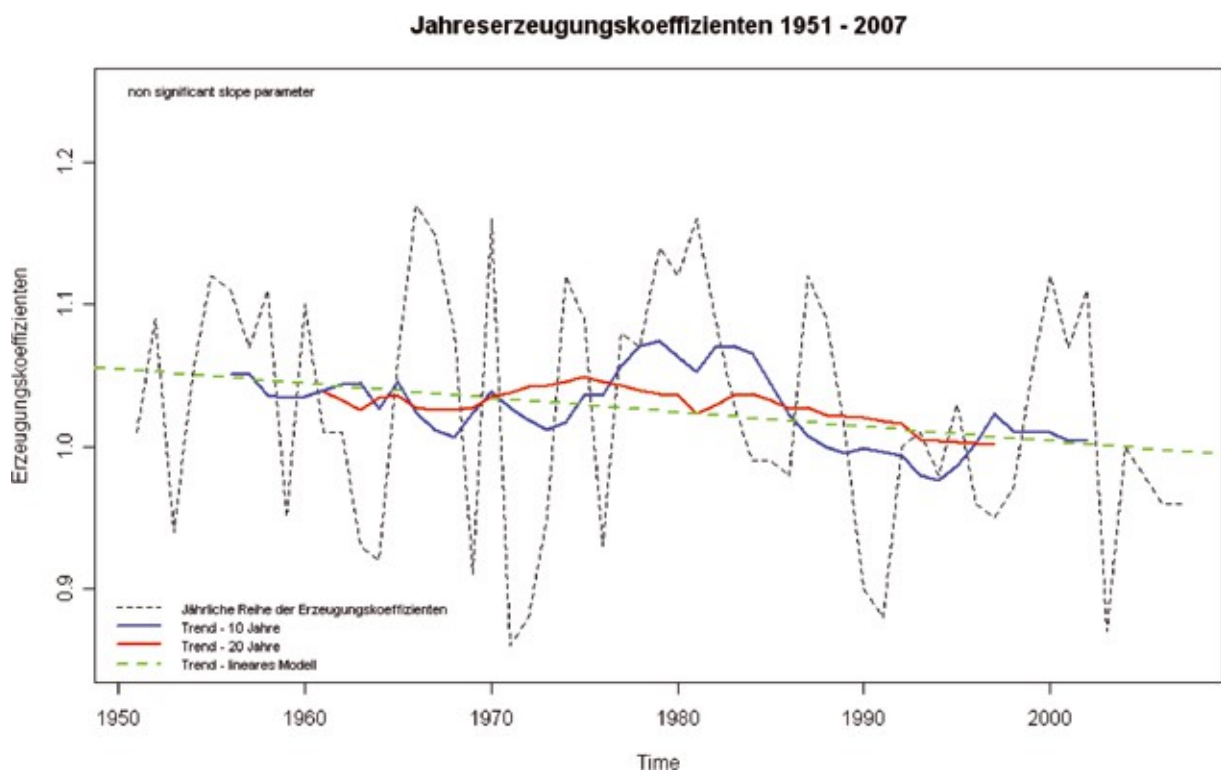


Figure 20: Annual production coefficient 1951-2007

The monthly generation coefficients show a significant increase in the winter months, while there is a significant decrease in summer. A simulation study carried out using boot-strapping continues these trends until around 2030.

These simulations result in the simulated generation coefficients for 2008 and 2028). The shifts in the winter and summer months caused by the significant trend are clearly recognisable: There is a positive shift in the distribution in the winter months and a negative shift in the distribution in the summer months.

What effects can we now expect in the energy industry?

In a study by Nachtnebel and Fuchs (1999), a decline in annual generation of 3-8 % was calculated for hydropower based on existing climate model data. This value is subject to great uncertainty. More probable is the aforementioned effect of a summer/winter shift with approximately constant annual generation. This would even mean a levelling out and thus a positive effect for hydropower in the future. In the next few years, there will be compensation by glaciers at higher altitudes, but the peak will probably be reached in 40 years and after another 30 years the influence of the glaciers can be negated. For water management as a whole, this will mean that the transfer function of the Alpine reservoirs will increase.

Due to the retreat of glaciers and the rise in temperatures, the issue of permafrost will gain new significance in the high mountains. The thawing of previously frozen layers of soil will lead to instability on some slopes and, above all, to an increased input of solid matter into watercourses. The requirements regarding the sediment balance in the operation of hydropower plants will represent a future challenge.

In conclusion, however, it must be said that the need for action is difficult to assess from today's perspective, as the individual models are still quite imprecise and the effects are therefore very difficult to interpret and visualise.

7 Effects on biology

As a result of the increase in temperature, a further invasion of heat-loving aquatic vegetation elements is to be expected (Janauer, this issue). The spread will mainly take place along the Danube, with thermophilic elements from the lower Danube region advancing upstream and macrophytes from the Atlantic region spreading downstream via the Rhine-Main-Danube Canal. Native macrophyte vegetation is particularly at risk in standing and slow-flowing waters or stretches of water. Many of the new species have a strong impact on the infested water system, but especially on the aquatic plants previously growing there. This results in a threat to native biodiversity and impacts on fish and invertebrate fauna and sport fishing.

There is a scientific consensus that global warming will result in a shift in habitat towards higher latitudes and higher sea levels for most cold-loving fish species (Mohseni et al., 2003). It is also assumed that cold-loving species will be replaced by warm-loving species. This assumption is supported by palaeoclimatic studies, which provide evidence of this for the glacial and interglacial periods of the Pleistocene. Applied to Austria, the findings to date would mean that salmonids have been pushed back. Individual species, such as the grayling, are already so endangered as a result of other anthropogenic interventions that additional climate-related impacts could bring them to the brink of extinction. However, it is not just individual species that are affected by climate change, but the entire biotic community. This can be reflected in a shift in the fish region index (Schmutz et al., 2004). In spatial terms, this corresponds to a shift of the fish regions upstream, provided this is still possible in the water body.

8 Impact on agriculture and forestry

The rising temperatures increase the evaporation potential disproportionately, which means an increase in the utilisation of soil water resources by the vegetation. In the low-precipitation growing regions of Austria (Eitzinger et al., this issue), summer crops in particular would be exposed to increasing heat and drought stress. Water-saving crops, cultivation techniques and crop rotations will therefore become more important. An increase in agricultural irrigation requirements is also a logical consequence.

The fertilisation effect of the increasing carbon dioxide content of the air would increase the yield potential of common crops. With climate variability remaining the same, rising temperatures increase the number of hot days and dry periods compared to the previous climate period, which means an increase in stress situations caused by drought and heat for the crops grown to date, especially in warmer and drier locations. Other yield-influencing stress factors that occur in combination with heat, such as z. e.g. yield-reducing ozone damage, could increase.

C3 plants, which include our forest trees, react to increased CO₂ concentrations with reduced stomatal conductivity, resulting in reduced water consumption and increased water utilisation efficiency (Leutzinger and Körner, 2007). This factor thus partly counteracts the increased demand for evaporation due to the rise in temperature. Although climate change forecasts are subject to uncertainties, the risk, i.e. the probability of extreme events occurring, is likely to increase regionally. How the forests react to such disturbances can be seen from the long-term observations of the quantities of damaged timber due to storms and snow breakage as well as bark beetle infestation (*Fig. 21*).

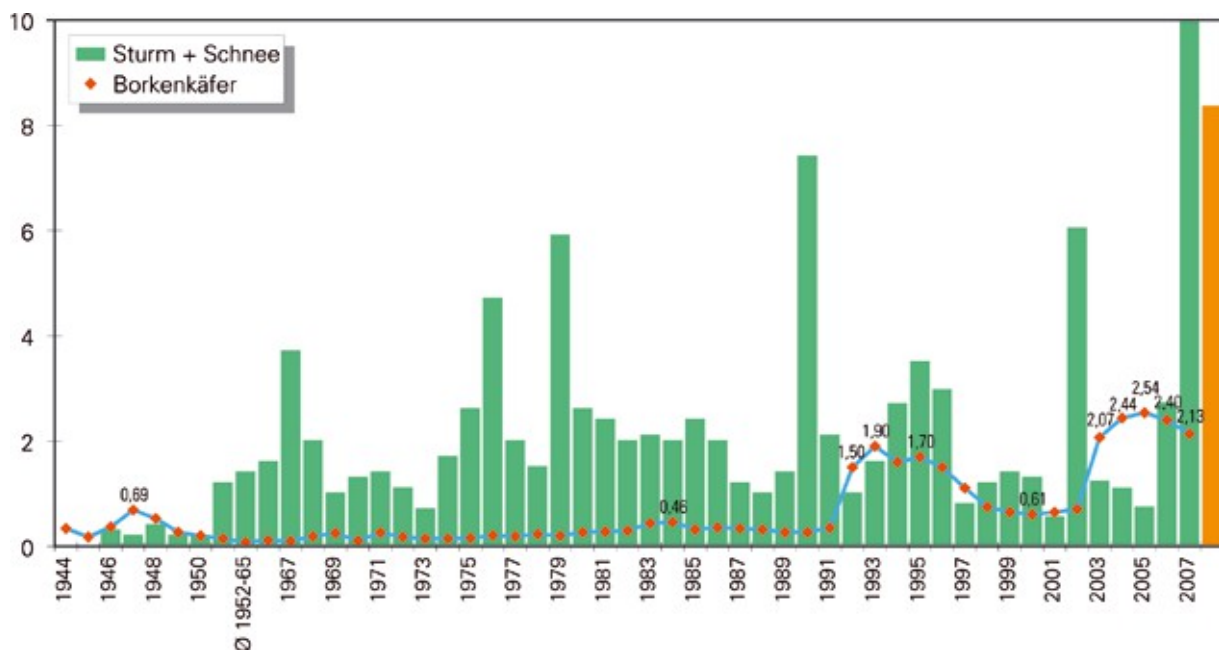


Figure 21: Time series of damaged timber volumes due to bark beetle infestation, storm and snow breakage (Tomiczek and Steyrer, 2003, courtesy of BWF Vienna)

9 Summary

There are no comprehensive climate analyses and climate impact studies available for Austria as there are for Switzerland and Germany. A number of individual studies and analyses in neighbouring countries suggest that annual average temperatures will rise by up to 4.5 °C by the end of this century. The warming should be more pronounced in summer. The statements on precipitation changes are contradictory and are complicated by the fact that the Alps represent a transitional area, with an increase to the north and a significant decrease to the south. Summer precipitation should tend to decrease, while winter precipitation should increase.

Due to the increased temperature, the proportion of snow in precipitation decreases significantly, by up to 50 %, and the climate zones in the Alpine region shift 400-600 metres uphill. The number of frost days and days with snow cover will decrease dramatically in the valley area. The effects on winter tourism can only be partially compensated for by snowmaking systems at higher altitudes.

As a result of the increased evaporation, the annual runoff shows a downward trend, which may amount to 12-18 % by the end of the century. Winter runoff will increase, while summer low flows will in the Alpine foothills. This means that annual hydroelectric energy production will decrease by between 3-8 %, but winter production will increase so that production is better adapted to demand.

The influence of the glaciers will increase the runoff in the melting phases over the next forty years, which is negligible for Austria's water balance, but can have a significant effect on the runoff in glaciated and alpine catchment areas. Since 1960, the glaciers have already lost a volume of 4.9 km³, with a total volume of 17 km³ in 1998.

Due to the retreat of glaciers and the rise in temperatures, the issue of permafrost will gain new significance in the high mountains. The thawing of previously frozen layers of soil will lead to instability on slopes, which will increase rockfall, landslides and mass movements.

There is no uniform picture with regard to extreme events. There is the potential for increased heavy rainfall events, but the simulations show no corresponding signal. Similarly, no such trend has observed in recent decades. In the case of heavily anthropogenically modified rivers, such as the Danube, an accumulation of medium floods can be observed.

In some areas, such as the north-east of Austria, urban water management could be confronted with further demands on the purification performance of wastewater treatment plants as a result of increased low water levels and higher water temperatures. For water supply, a quantitative and qualitative problem is to be expected in the areas with less precipitation, and in particular for small-scale supply systems. The demand for cooling water could also increase. The drinking water supply could also confronted with quantitative problems in some regions, but it should be possible to compensate for this by expanding the network

With the warming, an increase in all types of positive temperature extremes (hot days, absolute maxima, etc.) with a roughly analogue decrease in negative temperature extremes (frost, absolute minima, etc.) can realistically be expected. No change is expected with regard to wind and storm activity over Central Europe. This non-increase in storm activity in the future is consistent with the data findings of the past 130 years. An intensification of dry weather phases is also to be expected, as summer precipitation will decrease and evaporation will increase. In regions where precipitation is already low, this will lead to a shortage of natural water supplies for agriculture, which will increasingly have to resort to compensatory measures such as irrigation systems.

The already observed warming of watercourses, which will intensify in the future, leads to a shift in the species spectrum along the length of a watercourse, which is particularly problematic for rheophilic and cold-loving species.

For forestry, climate change means adapting the choice of tree species, especially for rotation periods of 100 years and more. A reduction in risk in low-lying areas can be achieved by abandoning spruce monocultures. A move away from old-growth forests towards permanent forests with natural regeneration and a wider range of tree species appropriate to the location is seen as one way of the vulnerability of forests. This trend can already be observed in forestry practice. The forest inventory shows a clear decline in pure conifer stands from 70 % in the 1971/80 inventory period to 62 % in the 2000/02 period and an increase in mixed stands and pure hardwood stands. In younger secondary spruce stands, timely thinning measures can reduce water consumption and minimise water stress for individual trees.

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Address of the authors:

Dr Reinhard Böhm
Central Institute for Meteorology and Geodynamics
Hohe Warte 38, 1190 Vienna
Tel. +43-1-36026-2203
E-Mail reinhard.boehm@zamg.ac.at

MR DI Reinhold Godina

Federal Ministry of Agriculture, Forestry, Environment and Water Management Division VII/3 -
Water Balance

Marxergasse 2, 1030 Vienna

Tel. +43-1-71100-6944

e-mailreinhold.godina@lebensministerium.at

o.Univ.-Prof. DI Dr. Hans-Peter Nachtnebel

University of Natural Resources and Life

Sciences, Vienna Department of Water-

Atmosphere-Environment

Institute of Water Resources Management, Hydrology and Hydraulic

Engineering Muthgasse 18, 1190 Vienna

Tel. +43-1-36006-5501

E-Mailhans_peter.nachtnebel@boku.ac.at

DI Dr Otto Pirker

VERBUND Austrian Hydro Power AG Am

Hof 6a, 1010 Vienna

Tel. +43-1-53113-50408

E-Mailotto.pirker@verbund.at

Effects of possible climate change on hydrology and water management in some Austrian regions

Hans-Peter Nachtnebel

Summary

This article an overview of the expected effects of climate change on hydrology and water management in some regions of Austria. These concern the Enns Valley, which part of the central Alpine region, the Gail Valley, which is also influenced by Mediterranean low-pressure areas, and the basin areas of the Leibnitzer Feld and the Marchfeld, where Pannonian climate characteristics are already recognisable.

As the global circulation models have a coarse spatial resolution, stochastic downscaling was applied using ECHAM4 and HadCM3 models. Using the local measured values, time series of precipitation and temperature generated for different climate situations on a daily basis. A continuous runoff model is used for the hydrological analysis, which requires the daily precipitation and temperature series as input.

There is a clear warming in summer, particularly in the higher-lying regions. Precipitation changes only very slightly both in the annual cycle and in the annual total, although its variability increases. As a result of the temperature changes, the snow cover melts earlier, which means that the original low water levels in winter no longer occur. As a result of the increased temperatures, periods of low water now occur in late summer and autumn.

On the one hand, the on hydropower are positive, as winter discharges are increased; on the other hand, the annual discharge load is likely to decrease as a result of increased evaporation while precipitation remains roughly the same. The occurrence times of floods are shifting; with regard to the size of the events, regionally different results can be observed. An increase in the risk of flooding cannot be deduced from the data currently available.

The effects on the groundwater regime are clearly pronounced. Groundwater recharge decreases significantly and exacerbates water management utilisation in the basins, especially in areas with low precipitation.

1 Introduction and problem definition

In recent years, several major flood events have occurred in Austria (2002 and 2005), of which the 2002 flood event in Lower and Upper Austria in particular caused catastrophic damage. The total damage for Austria alone was estimated at around 3.2 billion€ (BMLFUW, 2003), while the damage across Europe € 15-16 billion (Nachtnebel, 2003).

In contrast, the summer of 2005 was characterised by an intense and prolonged drought across Europe, which caused more damage than the 2002 floods. The number of fatalities as a result of the heatwave far exceeded the number of flood victims. Several storm depressions over Central Europe, such as Lothar in 1999 with an estimated damage impact of USD 6 billion (Swiss Re) and Kyrill in 2007 with economic losses of USD 10 billion (Munich Re), solidified the image among the general public that climate change was already taking place. The aim of an ÖWAV working group was therefore to summarise the possible effects of climate change on water management in an interdisciplinary manner. This article is intended to make a contribution to this endeavour.

The Alpine regions are one of the most sensitive areas for climate change, as there are seasonal shifts in precipitation and there is also a high degree of spatial variability in climate and hydrological processes. For the Alps, studies by Baumgartner et al. (1983), Furrer et al. (1980), Kuhn (1988), Kuhn et al. (1993), Auer (1990), Auer et al. (1993), Oeschger (1988), Koch and Rudel (1991), Haiden and Hantel (1991), Lang (1978), Schädler (1987, 1989), Nachtnebel et al. (1995, 1999, 2003), Wanner et al. (2000), Beniston and Jungo (2002), Beniston et al. (2003 a,b), Beniston (2004), Beniston and Stephenson (2004), Schär and Frei (2005), Raible et al. (2006). In this article, several river basins and regions are analysed with regard to hydrological and water management impacts (*Figure 1*). These concern the Enns Valley, which is part of the central Alpine region, the Traisen river in the foothills of the Alps, the Gail Valley, which is also influenced by Mediterranean low pressure areas, and the basin areas of the Leibnitzer Feld and the Marchfeld, where Pannonian climate characteristics are already recognisable.

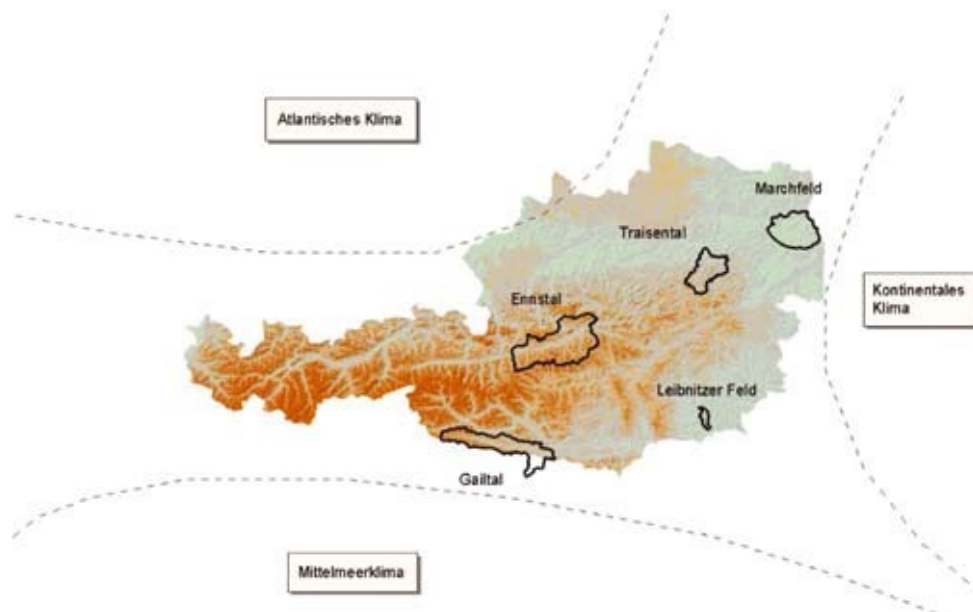


Figure 1: Selected catchment areas and regions

This article summarises the scientific literature and also includes the results of the CCHydro project (Hebenstreit, 2000; Nachtnebel et al., 1999) and the national Hydoklima project (Nachtnebel and Fuchs, 2003; Fuchs, 2005).

Most climate studies concern the global scale, with grid sizes of a few thousand square kilometres being typical (Simmons and Bengtson, 1988; Matyasovszky et al, 1994; Deque et al, 2005; IPCC, 2005, 2007; Jones et al, 2001; IPCC, 1990-2007; Vidale, 2003; Roeckner et al, 2006), and use different emission scenarios for greenhouse gases (IPCC, 2007). The simulation results then provide spatial mean values for the grid elements and therefore cannot be directly transferred to Alpine regions. Few studies refer to the regional scale, as there are a number of scale problems in the methodology. In principle, four methods are available for studying possible regional climate changes, which are based on

- Drawing analogies from historical, observed changes and their transfer to future situations
- the evaluation of longer data series (time series analysis) (Brückner, 1910; Rudloff, 1980; Auer et al., 1988; Bradley and Jones, 1992; Pfister, 1992, 1994; van Storch and Navarra, 1999)
- the regional simulation of climate change (Briffa, 1993; Frei et al., 2003, 2004, 2005; Boroneant et al, 2004; Busuioc et al, 2004; Giorgi et al, 2004; Pal et al, 2004; Kjellström, 2004; PRUDENCE, 2001; Kleinn et al, 2005; Rowell, 2006 and
- the statistical downscaling of large-scale climate change (Bardossy and Plate, 1992; Wilson et al, 1992; Bardossy, 1993; Matyasovszky et al, 1993; Brandsma and Buis- hand, 1997; Wilby, 1997; Matulla et al, 2004)

support.

Similar methodological approaches exist for climate impact research (Gleick, 1986; Lemmelär et al., 1990; Link 2001, Hagemann et al., 2004; Hirschi et al., 2005). The hydrological observation scale requires a higher spatial resolution, especially in the Alpine region with its variable topography. Local, short-lasting events such as thunderstorms and heavy rainfall events are also important. Such processes are not directly reflected in the climate models; even the V-b weather conditions that are important for Austria, caused by lows over the northern Mediterranean, are only partially modelled. For these reasons, the following methods are used for impact studies:

- analogies comparing hydrologically similar areas that are located, for example, in the differ slightly in terms of temperature.
- Sensitivity analyses that show the reaction of the hydrological processes to percentage variations in the Analyse inputs,
- Time series analyses. These are usually very limited in time, so that at best trends in the past can be statistically identified and their extrapolation is problematic (Manabe et al, 1987; Fürst and Nachtnebel, 2007).
- Use of small-scale climate data sets (Bardossy and Gaspary, 1990; Hebenstreit, 2000; Fuchs, 2005). However, it should be noted that the time series should have a corresponding spatio-temporal correlation caused by the topographical conditions. The spatial correlation of heavy precipitation may serve as an example, as well as the altitude dependence of precipitation in congested areas and the formation of inversion layers. All of these processes significantly determine hydrological events and the occurrence of extreme events in small to medium-sized areas.

In this paper, stochastic downscaling (Bardossy and Plate, 1992; Nachtnebel et al., 1999; Hebenstreit, 2000; Nachtnebel and Fuchs, 2003; Fuchs, 2005) is applied, and the daily precipitation and temperature series obtained are used as input for various hydrological models to analyse climate impacts (WMO, 1987; Zehe and Bardossy, 2004). First, the hydrological reactions are investigated with the help of continuous precipitation-runoff models, whereby annual balances, seasonal behaviour and changes in extreme values are examined. When applying the meteorological series in basin locations, a soil water balance model is used to determine the annual soil moisture and groundwater recharge.

2 Methodology

This section briefly discusses the downscaling method used and then describes the two hydrological models,

2.1 Downscaling methodology

The advantage of this method is seen in the fact that, in addition to the climate simulations for past, local observations are also included and thus more information is analysed.

The basis for the various climate models (BMO, GFDL, GISS, NCAR, OSU, ECHAM, HadCMx) are the global circulation models (GCM), which simulate the fundamental climate processes over a large area using a horizontal resolution of hundreds of kilometres. The results of the ECHAM4 model of the Max Planck Institute for Meteorology in Hamburg and the HadCM3 model of the United Kingdom Meteorological Office in Bracknell were used to generate the hydrometeorological time series. *Table 1* shows a list of the most important model data.

Table 1: Model data ECHAM4, HadCM3 (Source: IPCC-TGCI, 1999)

	ECHAM4	HadCM3
AGCM: Horizontal resolution	2.8° x 2.8° (T42)	2.5° x 3.75°
OGCM: Horizontal resolution	2.8° x 2.8°	1.25° x 1.25°
Grid layers AGCM/OGCM	19/11	19/20
Numerical model formulation	spectral	grid
Start time	1860	1860
2 x CO ₂ sensitivity	~2.6 °C	~3.0 °C
Flow correction	Annual average	none

A comparison of the annual variation of the simulated air pressure over time with the observations shows a preference for the HADCM3, as this provides a good match between the observed air pressure and the 1 x CO₂ scenario, while the 2 x CO₂ case is clearly different due to a higher air pressure.

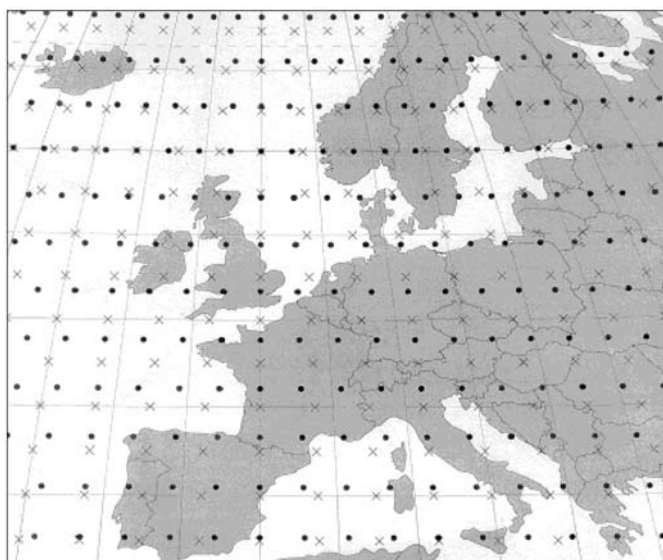


Figure 2 Grid of the ECHAM4 and HadCM3 model

In the usual classification, greenhouse gases are considered as a whole and the recent past is characterised by the "1 x CO₂ case". Model simulations are available for the 1 x CO₂ case as well as for higher greenhouse gas concentrations and for transient processes, i.e. increasing CO₂ concentrations. The 2 x CO₂ case corresponds to a doubling of the total greenhouse gas concentration by around the end of this century, although this still depends on the selected emissions scenario.

Table 2: Development of the concentration of CO₂, CH₄, N₂O (Dehn, 1999; IPCC, 1995)

	CO ₂ [ppm]	CH ₄ [ppm]	N ₂ O [ppb]
pre-industrial	280	0,7	275
1994	358	1,72	312
2100 (IS92a)	700	3,62	417

Due to the lack of spatial resolution of the GZMs and their seasonal precipitation behaviour, which is calculated in opposite directions in some regions, a direct use of these generated data for hydrological questions does not appear to be useful, as the runoff behaviour would also show seasonally opposite characteristics. A direct link between the large-scale pressure distribution, which agrees well with the climate models and is also comparable with the observations (Simmons and Bengtson, 1988), and the local hydrometeorological factors is therefore sought.

In this work, a semi-empirical method (Bardossy and Caspary, 1990; Matyasovszky et al., 1993, Nachtnebel and Fuchs, 2003) was used, which, based on the large-scale pressure distributions, allows a weather situation classification (Baur et al., 1944; Baur, 1947; Hess and Brezowsky, 1969; Lamb, 1972; Kirchhofer, 1973, 1974; Schüepp, 1979; Lauscher, 1985; Rickli, 1995; Salvisberg, 1996; Steinacker, 1999; Gerstengarbe and Werner, 1999), in which the location, extent of the low and high pressure layers and the intensity and direction of the air currents are analysed.

ments must be taken into account. Bauer et al. (1944) define a large-scale weather situation as the mean air pressure distribution of a large area at least the size of Europe over a period of several days, in which certain features of successive weather situations remain the same. The large-scale weather conditions describe atmospheric circulations, which are primarily heat exchange processes. Therefore, both precipitation and temperature are linked to the direction and intensity of atmospheric currents, with topographical factors and cloud cover an influence in addition to seasonal influences. Here, 9 classes (weather conditions) were used separately for the summer and winter half-years:

Group 1: anticyclonic locations

They occur most frequently, whereby the Alpine region lies in the area of a high-pressure foothills, combined with cold-air advances over Eastern Europe. The probability of precipitation and the average amount of precipitation is relatively low in all areas, but slightly higher in the mountain regions (Enns Valley, Gail Valley) in summer due to thunderstorm activity.

Group 2: maritime western locations

Characteristics are (according to Rickli) a large wavelength of the westerly wind drift and the influx of maritime polar air into the Alpine region. In winter, westerly weather conditions mean relatively mild conditions. The probability of precipitation is higher in the areas north of the main Alpine ridge (Ennstal, Traisental) than in the Gailtal or Leibnitzer Feld, for example.

Group 3: North-west and north-facing locations

The conditions are by a north-westerly to northerly flow towards the Alps, combined with frontal passages and a rise in pressure over Western Europe. The controlling low pressure systems move from the North Sea via southern Scandinavia to the Baltic. They cause

The regions in the south have below-average temperatures all year round and have a very high probability of precipitation, especially in the north. The areas in the south are drier in these weather conditions.

Group 4: Anticyclonic easterly and south-easterly conditions

A meridional wedge axis in the area or directly over the Alps and easterly to south-easterly winds close to the ground are decisive for these weather conditions. The Alpine region lies on the south-western edge of a surface high with a core between Poland and Scandinavia. These weather conditions lead to very dry conditions all year round in the regions analysed, with temperatures slightly above the long-term seasonal averages.

Group 5: South facing

It is characterised by a short-wave trough over Western Europe and a south-westerly to southerly flow in the Alpine region, which associated with foehn situations north of the main Alpine ridge. On the windward side in the Gailtal and Leibnitzer Feld, there is high precipitation, especially in autumn. The weather conditions mainly occur in spring and autumn.

Group 6: Continental East

With the exception of SEZ, these large-scale weather conditions are by cold high-altitude air over Central and Eastern Europe. Another characteristic is the decoupling from the North Atlantic circulation. These weather conditions are characterised by a relatively high probability of precipitation in all areas.

The weather is characterised by high precipitation, particularly summer. Temperatures in autumn and winter are below the long-term seasonal average.

Group 7: South-west locations

The two large-scale long-wave weather patterns are characterised by a large-scale south-westerly flow towards the Alps. This brings large-scale subtropical air masses into the Alpine region, which leads to strong positive temperature anomalies. The probability of precipitation is only relatively high in summer due to convective events, otherwise it is rather low.

Group 8: Trough north of the Alps

Low pressure prevails in the North Sea-South Scandinavia-Baltic region. The Alpine region is therefore in the area of a westerly current and thus also in the area of influence of maritime air masses. This is associated with a high probability of precipitation, particularly on the northern side of the Alps, and moderate to high precipitation levels south of the Alps. Temperatures will be slightly below the long-term seasonal average.

Group 9: southern western locations

In these westerly weather conditions, the frontal zone is shifted far to the south. As a result, the cyclonic influence in the Alps is also stronger and there is a higher probability of precipitation in these areas. Otherwise, the differences in the regional characteristics of the weather patterns compared to Group 2 are relatively small.

The sequence of weather conditions is described by a Markov process, whereby the transition probabilities are derived from the historically observed or simulated future pressure distributions. Changes in local events can be caused by changes in the frequency of occurrence of weather conditions, their duration or their time of occurrence.

A conditional statistical distribution is then between a specific weather situation and the local precipitation and temperature observations. It is shown (Nacht- nebel et al, 1999, 2003, 2004) that more frequent and intense precipitation occurs in certain weather conditions, while other weather conditions are characterised by low local precipitation. This relationship is therefore conditioned by the weather conditions and the spatially observed pattern of meteorological variables is also included. This also means that a certain weather situation can have very different regional characteristics, e.g. south or north of the main Alpine ridge. The following assumptions are implicitly made:

- The weather situation classification is valid for the historical as well as for the 1 x CO₂ and 2 x CO₂ case.
- The weather conditions are seasonal and, in addition to the type, are characterised by the characterised by average large-scale pressure head or pressure head differences.
- There is a statistical correlation between the large-scale weather situation and the small-scale weather events (downscaling).
- The statistical correlation applies to all climate scenarios and can therefore be used for simulations.

2.2 Hydrological precipitation-runoff model

A spatially distributed and temporally continuous model is used, the basic stock of which is similar to the HBV model (Bergström, 1976, 1995; Nachtnebel and Fuchs, 2004). The continuous runoff model works on arbitrary time steps, but the daily values of temperature and precipitation are used here due to data availability. To map the spatial variability of the topography and different forms of land use, the entire catchment area is into sub-catchments. These are subdivided into individual altitude zones, land use classes and soil types. By intersecting this information, "hydrologically homogeneous zones" are created (Figure 3), each of which is simulated by a storage cascade (Figure 4). The result for each zone is the individual runoff proportions and the storage content of the snowpack in the soil and groundwater.



Figure 3: Spatial subdivision of catchment areas and designation of "homogeneous hydrological zones"

The sum of the runoff heights from the 3 storages per time unit results in the total runoff volume, which is routed through a fourth linear storage to the catchment outlet. To estimate the various threshold values and storage constants of the model, either longer observation series or spatial information such as soil thickness and soil type can be used. The current system status is determined in the model, which describes the filling levels of the individual reservoirs and the retention in the form of snow density and snow depth.

The potential evaporation from the soil zone is calculated according to Thornthwaite (1948) and the actual value is limited by the soil moisture. In zones with a high storage capacity of the soil, the potential evaporation is then effective, while in dry zones the evaporation decreases linearly towards the wilting point. The infiltration (QVS) from the soil layer into the deeper reservoir (aquifer in the valley area) depends on the soil moisture and is described by a non-linear equation.

If necessary, the snow cover is placed on top of the soil zone. The solid precipitation is distributed normally within a zone. This takes into account wind loads, snow slides and snow

and exposure and means that a zone only gradually tapers off and contributes to the melt runoff. The proportion of meltwater is calculated using the degree-day method, whereby the storage effect of the snow itself is also taken into account. This on the snow depth and density. Furthermore, the refreezing of the liquid part of the snow cover is taken into account, which gradually leads to a compaction of the snow (Braun, 1985). The balance equations must then be drawn up for the individual reservoirs and linked to each other.

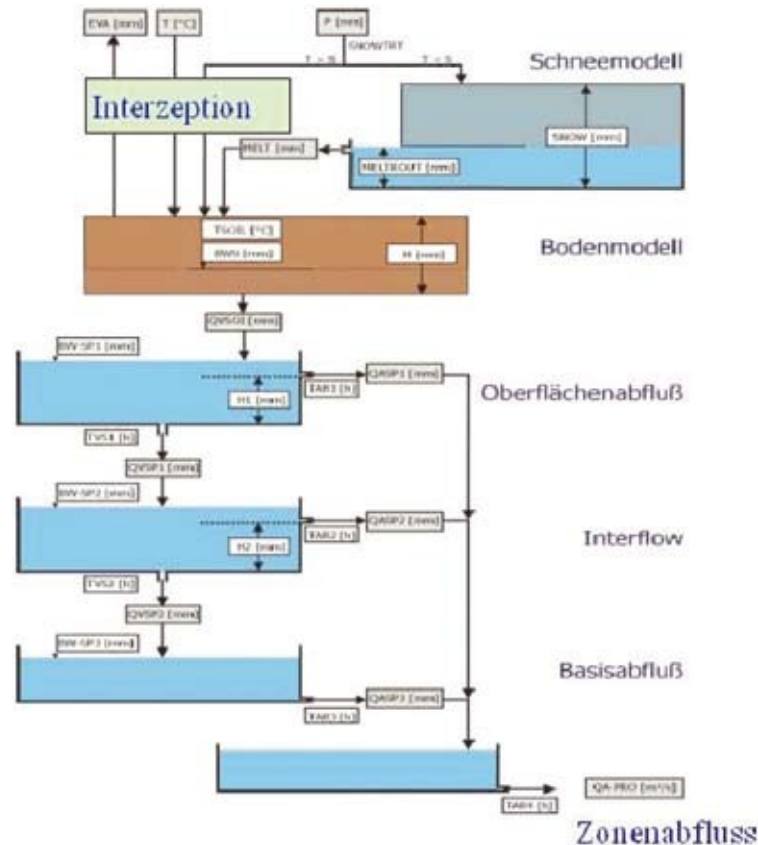


Figure 4: Structure of the runoff model

2.3 Model for the soil water balance

In basin locations such as the Marchfeld or the Leibnitzer Feld, surface runoff is negligible and vertical transport processes dominate instead. In the soil water balance simulation, all components of the soil-vegetation-atmosphere system are described using a numerical simulation model BOWA (Holzmann, 1994). This requires knowledge of the atmospheric boundary conditions (precipitation, temperature, etc.), the position of the groundwater table and the structure of the soil profile. The climatic processes determine the upper boundary condition for the soil water balance simulation.

The soil water balance is calculated on the basis of daily values, although the calculation time steps are much shorter. The vertical distribution of soil moisture in a soil profile is calculated using a one-dimensional finite difference method that approximates the Richards equation (Koorevaar et al., 1983). The soil profile is analysed up to the

gravel top edge is approximated by homogeneous soil layers. Inhomogeneities are taken into account by different soil physical properties of the layers (*Figure 5*).

The upper boundary condition is defined by specifying the infiltration rate. It is determined from the amount of precipitation minus the interception, trough evaporation and surface runoff (if present). The maximum infiltration rate is determined by the saturated permeability coefficient of the uppermost layer.

The lower boundary of the soil profile is formed by a (virtual) gravel layer. It is assumed that hardly any water can be retained in the soil in this layer due to the predominant proportion of coarse pores and that vertical drainage (seepage) into the groundwater takes place. Capillary lift from the gravel body is not possible. If the groundwater table extends into the soil profile, there is a groundwater connection. In this case, the lower boundary condition is characterised by constant (saturated) soil moisture conditions.

Pore water is extracted via the plant roots as transpiration. It is assumed that the root density and thus the transpiration activity decreases linearly with depth. The potential evaporation can be calculated using various formulae, whereby Penman (1956) was applied here.

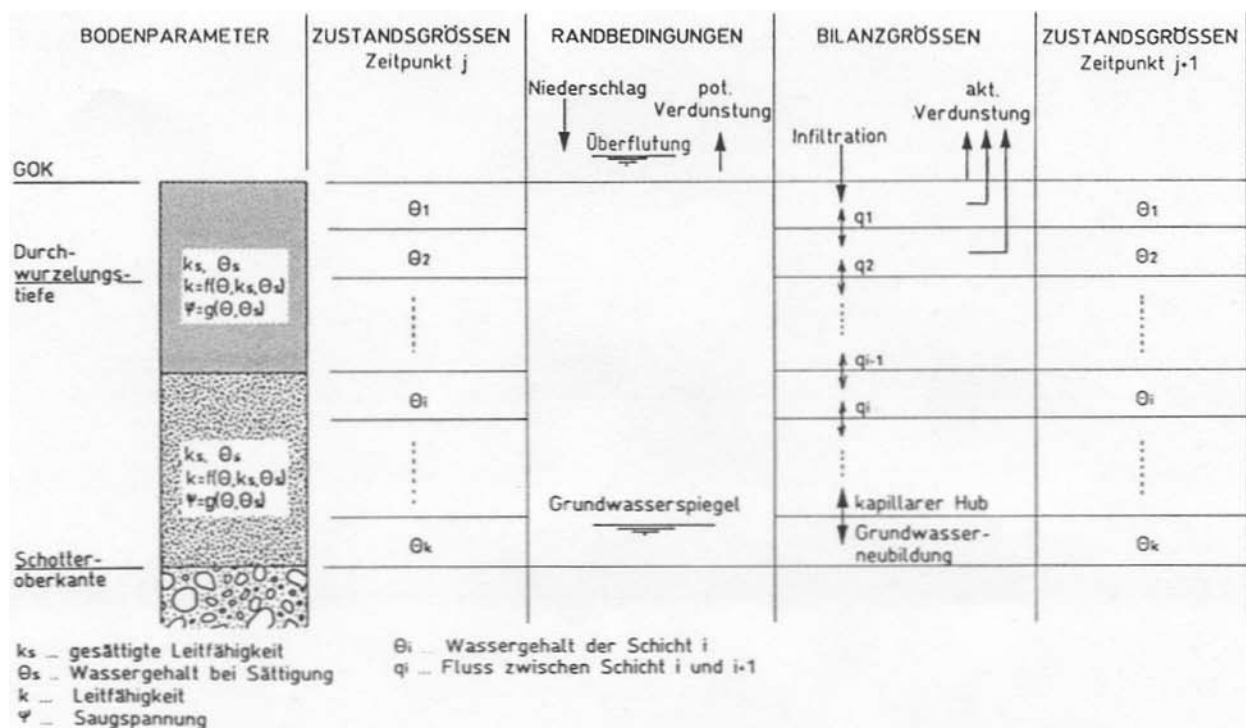


Figure 5 : Soil water balance model

The coupling of the models and the use of different data sets in downscaling are shown in *Figure 6*.

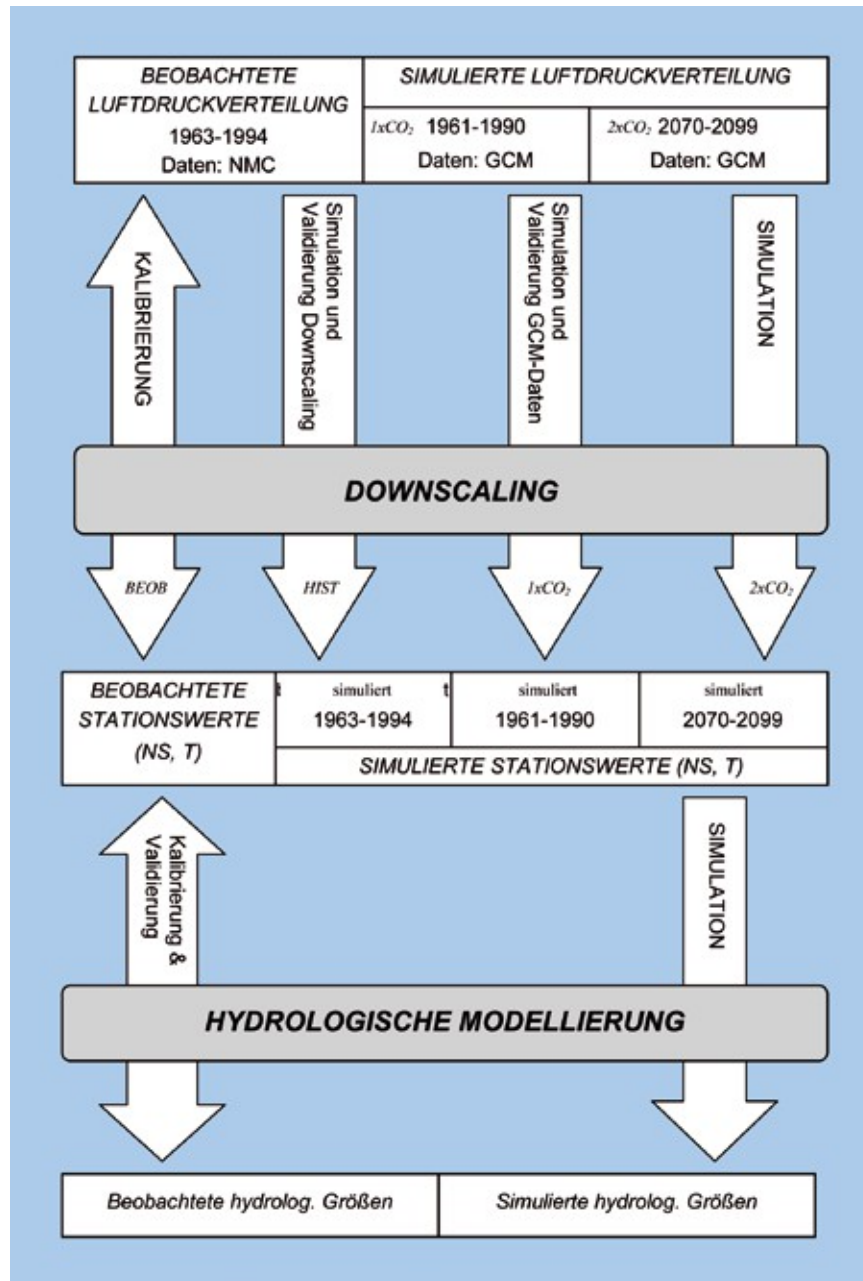


Figure 6: Coupling of downscaling with hydrological model (Fuchs, 2005)

3 Results for the water balance of alpine regions

All representations are standardised: the first bar (black) the observed values, the size of the blue bar was determined from the historically measured pressure distributions, the third bar (light blue) was calculated from the climate simulations for the 1 x CO₂ case and the red bar the results for the 2 x CO₂ case. Furthermore, the uncertainties resulting from the application of the methodology are represented by interval ranges. The first three bars should show values that are as similar as possible, as they relate to the same period but were calculated using different data sets. The difference between bars three and four reflects the actual climate signal.

3.1 Temperature

In all areas, an annual average temperature rise of 2.5 to 4 °C is expected by around the end of this century. In individual months, the change may be even more pronounced. The rise in temperature is somewhat more pronounced in the summer months than in the winter months. The temperature increase is lowest in the months of February and March. In the valleys, the temperature increase would be up to 4 °C, while at high altitudes (above 2,000 m) the increase would be 2-3 °C.

As a direct consequence, the 2 x CO₂ case shows a significant decrease in frost days and days with snowfall and snow cover, especially in the valley areas, where the number of frost days can be reduced by half. Evaporation also increases significantly and the annual runoff load decreases. In glaciated areas, an increased contribution to summer runoff is to be expected due to melting processes. This effect can be significant in small Alpine catchment areas, but is negligible for Austria's overall balance. According to Kuhn (1988, 2008), the melting contribution from glaciers should increase over the next forty years and then decrease continuously as a result of the rapid reduction in glacier area. There was no glacier influence in the catchment areas shown here.

3.2 Precipitation

In general, no clear change in precipitation can be recognised from the derived data. Both the precipitation amounts and the seasonal distribution are similar for the present and the future. A slight decrease would be possible in the south of Austria, while increases could even be expected in the northern congested areas.

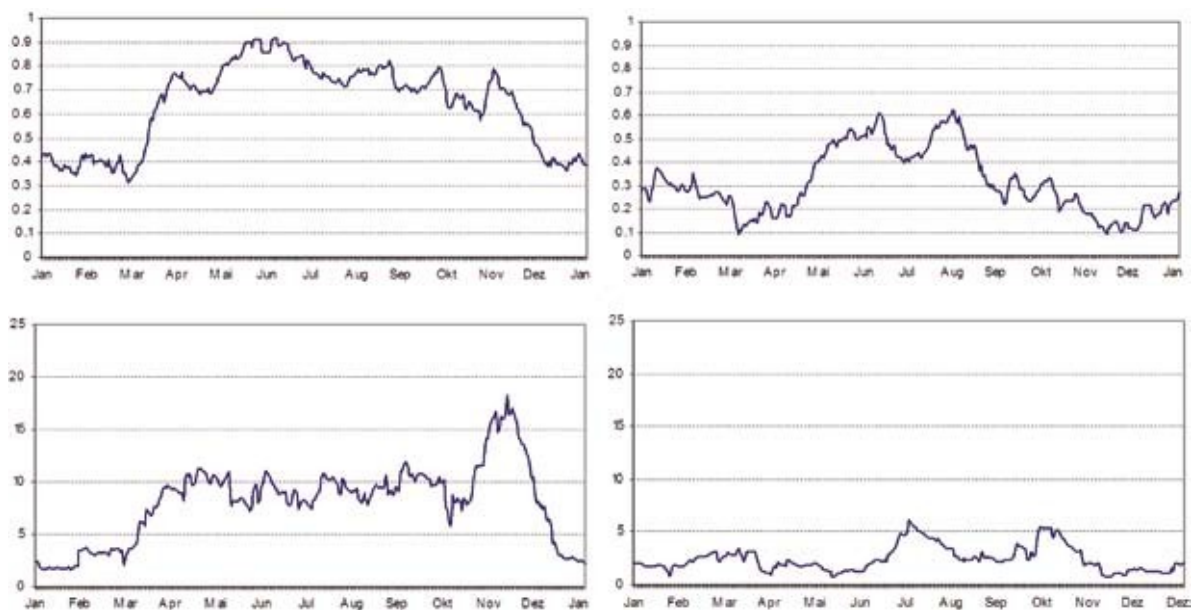


Figure 7: Precipitation behaviour in the Gailtal valley in different weather conditions (WL)
 Top: Probability for precipitation day ($P > 0.1$ mm) left WL3, right WL4 bottom: mean precipitation amount, left WL3, right WL4

To give an impression of the uncertainties in the statement, intermediate steps in the calculation are also explained here for the other areas. *Figure 7* two weather conditions (3 and 4) and the probability of precipitation and the average amount of precipitation for the historical case. WL 3 shows a higher tendency for precipitation, and when this occurs, it also has a significantly higher intensity than WL4, especially in November. The annual variation in precipitation is shown in *Figure 8*. This shows that the errors of the simulation (i.e. the deviations between observed data and data simulated with historical air pressure data) in precipitation downscaling are of approximately the same order of magnitude as the differences between the results for the 1 x CO₂ and 2 x CO₂ scenarios (*Figure 8*).

If we now compare the downscaled precipitation for the past (*Figure 8*), seen that the precipitation frequency tends to be underestimated in the simulation, while the mean N heights on N days are overestimated in almost all months. A clear overestimation occurs above all in autumn. This is mainly due to the smoothing algorithms in the downscaling procedure, e.g. estimation of the parameters including all observed N data within a specific seasonal time series section, but also z. For example, the description of the parameter function using a Fourier expansion of limited order. The cloud cover and precipitation minimum typical for most Austrian regions in October, which is not only caused by the large-scale weather conditions but also by the very stable air layers in this month and is only limited to a few weeks, is therefore only partially reproduced by the downscaling method.

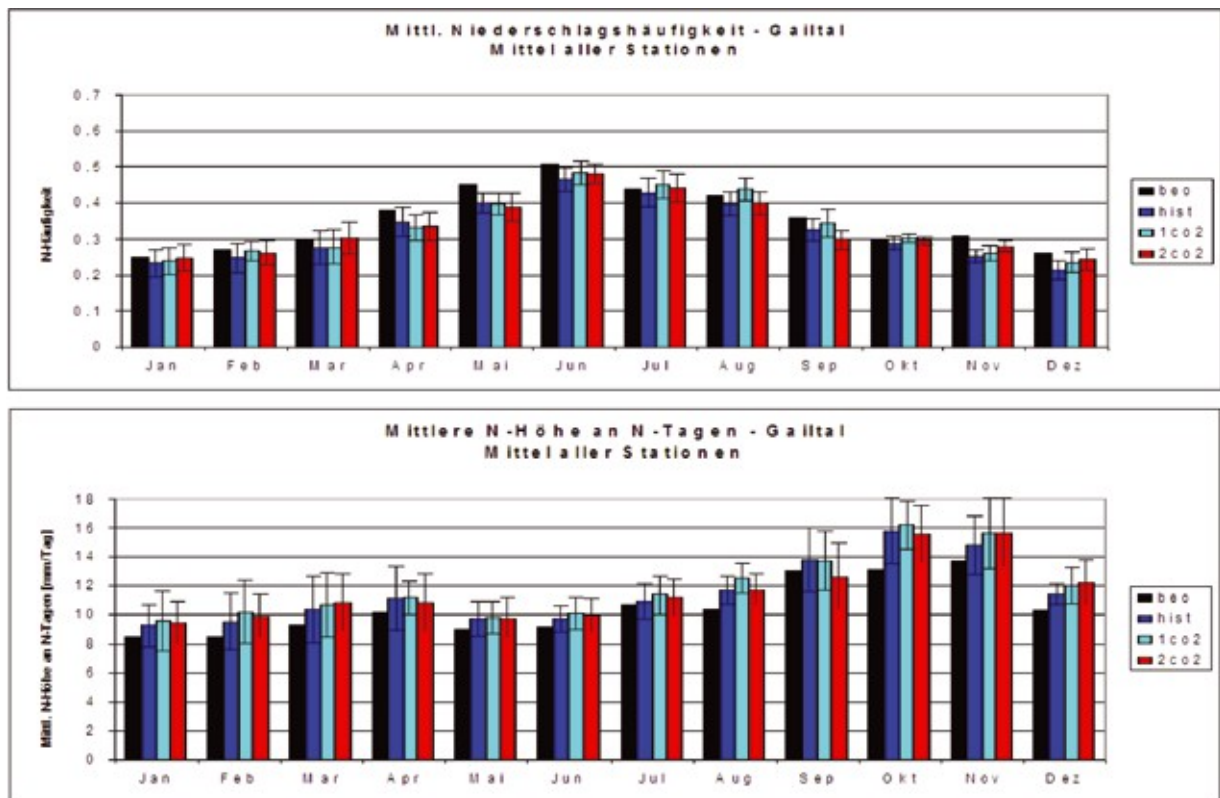


Figure 8: Comparison of the observed and simulated precipitation values over the course of the year

The climate signal (i.e. the difference between the simulation results for the 1 x CO₂ and the 2 x CO₂ scenario) is relatively small overall. The only noticeable difference is a significant decrease in precipitation in the months of August and September. This decrease in N in late summer is mainly due to the increase in anticyclonic conditions (high over Central Europe, high pressure bridge over Europe) in the 2 x CO₂ case. Overall, there is also a trend towards a decrease in annual precipitation.

In addition to the weak climate signal, a significant difference to the results of the temperature downscaling is the large uncertainty induced by the stochastic method.

3.3 Change in runoff and water balance in the Gailtal valley

The influences of the climate change scenarios on the water balance in the Gailtal were investigated using the modelled water balance components precipitation, evaporation and runoff. A summarised representation for annual averages is given in *Figure 9*, while *Figure 10* shows the annual variation in precipitation and runoff in the area.

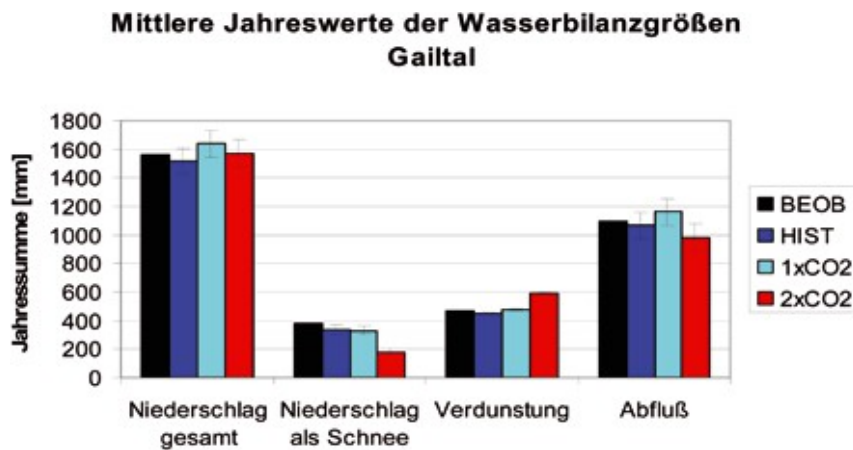


Figure 9: Change in the water balance components in the Gailtal valley

Annual precipitation would therefore decrease by 70-80 mm, which corresponds to around 5 %. Precipitation in the form of snow, on the other hand, would decrease significantly and would only amount to 50 % of the current average value; evaporation would increase as a result of the higher temperature, so that runoff could decrease by up to 16 % compared to the 1 x CO₂ case. If the runoff under changed climate conditions is compared with today's average runoff, a reduction of approx. 11 % can be expected.

There are the aforementioned slight changes in the area precipitation between the 1 x CO₂ case and the 2 x CO₂ case. Slight decreases in precipitation in connection with the increase in anticyclonic weather conditions in the 2 x CO₂ case occur in August and September.

The increase in current evaporation as a result of higher area temperatures is very clear. The strongest increases occur in spring and autumn, although the confidence ranges are very narrow. The current evaporation, especially in the summer, is not influenced by the

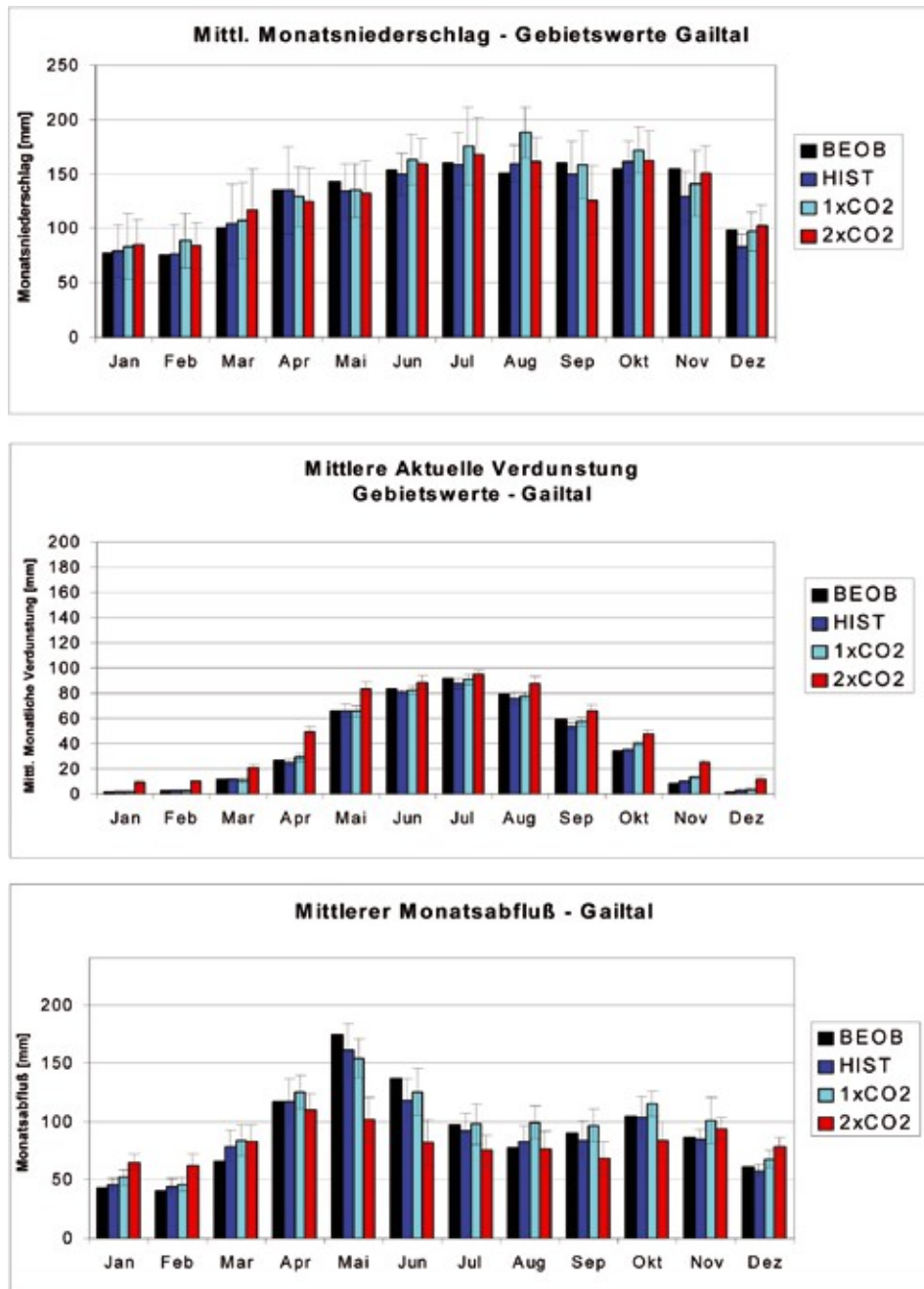


Figure 10: Comparison of observed, historical (downscaled from pressure observations), 1 x CO₂ and 2 x CO₂ simulations for areal precipitation, current evaporation and runoff in the Gaile catchment area

available evaporation energy (potential evaporation) but above all limited by the availability of soil water. Therefore, in the months of June and July, the increase in actual evaporation is less pronounced than that of air temperatures or potential evaporation. Nevertheless, there is a significant increase over the entire year, which has a corresponding effect on runoff.

The runoff (Figure 10) shows a very strong decline in runoff from spring to autumn and a clear overall levelling of the runoff regime in the 2 x CO₂ case. Particularly clear

The decline in the months of May, June, September and October is particularly pronounced, and the confidence intervals also show no overlap in these months. The climate change signal in these months is therefore significantly stronger than the uncertainties caused by the stochastic downscaling model. This is due to the reduced snowmelt contribution and the increase in evaporation. An increase in runoff in the 2 x CO₂ case only occurs in the winter months and is explained by the larger proportion of liquid precipitation.

As neither the standard deviation nor the extreme values of daily precipitation are increasing the flood situation is not expected to worsen.

3.4 Changes in the water balance and runoff in the Enns Valley

The Enns Valley shows similar results, but there are also differences to the Gail Valley in the south. As a result of the temperature increase in the 2 x CO₂ case, the hydrological simulation showed a significant change in the snow conditions in the area. The share of snowfall in the total precipitation in the area is reduced from the current approx. 400 mm in the 2 x CO₂ case to approx. 200 mm. The annual snow depth maxima and the days with closed snow cover will decrease significantly in all altitude zones. In terms of snow conditions, there is a shift upwards by around two altitude zones, which corresponds to a shift in altitude of around 600-900 metres. The area precipitation in the upper Enns region would decrease by around 5% in relation to the Liezen gauge (1 x CO₂ case - 2 x CO₂ case), and a slight increase in the annual average would even be expected when compared with the current precipitation conditions. There may even be a slight increase in the northern reservoir locations and in higher areas. The proportion of solid precipitation will fall by almost half (Figure 11).

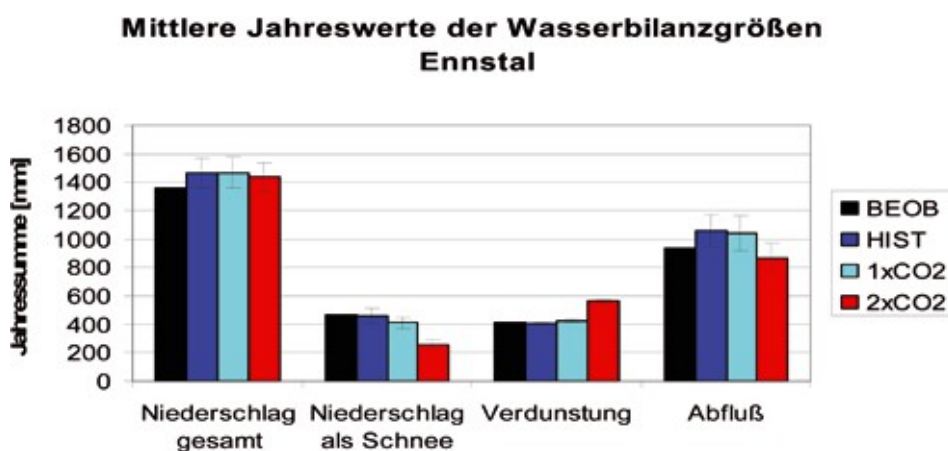


Figure 11: Change in the balance components in the Enns area (Liezen gauge)

The average annual evaporation would therefore increase from 410 mm to 580 mm, which would again result in a reduction in runoff of 12-15 %. However, the confidence intervals for the annual runoff are relatively wide due to the uncertainties in precipitation downscaling.

In the spring and summer months, on the other hand, there is a very sharp decline in discharge and a clear overall levelling out of the discharge regime in the 2 x CO₂-

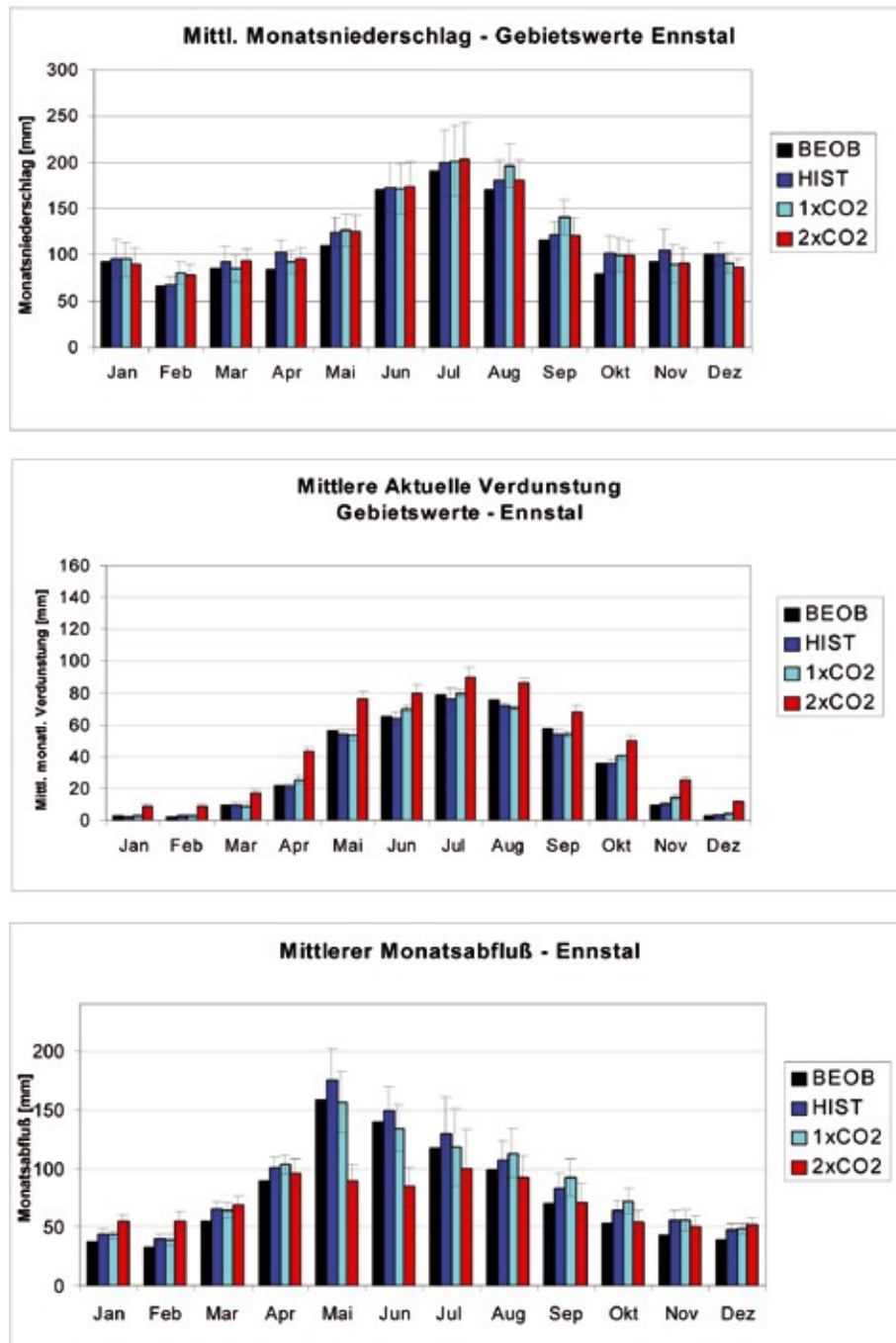


Figure 12: Annual variation of the water balance components in the Enns basin

case. The decline is particularly clear in the months of May and June, and the confidence intervals also show no overlap in these months. The climate change signal is therefore significantly stronger in these months than the uncertainties caused by the stochastic downscaling model. This is due to the reduced snowmelt contribution and the increase in evaporation. An increase in runoff in the 2 x CO₂ case in the winter months and is explained by the larger proportion of liquid precipitation.

The analysis of the daily discharges showed that smaller floods occur less frequently in the 2 x CO₂ case, which leads to a significant flattening of the discharge duration curve. With regard to the ex-

There were hardly any changes in the $2 \times \text{CO}_2$ case, but the data series is probably also too limited to be able to recognise changes in extreme events (Frei, 2003; Ferro et al., 2004; Collins, 2004). However, there is a shift in the seasonality of floods. For example, under the hydrological conditions simulated for future climate conditions, winter floods occur more frequently, while spring and summer floods become significantly less frequent. There is also a shift in the seasonal occurrence of low water in the $2 \times \text{CO}_2$ case. Winter and spring low flows become less frequent. In summer and autumn, the frequency of low water increases significantly.

3.5 Changes in the soil water balance in the Leibnitz Field

The Leibnitzer Feld study area (Holzmann and Nachtnebel, 1995) is a relatively small region with uniform topography and minor differences in altitude. In this region, the runoff components, as calculated the alpine catchment areas, are therefore negligible. Instead, the vertical components of water transport dominate, i.e. a soil model is used, as described in *Chapter 2.3*. The changes in evaporation, the soil water balance and groundwater recharge are therefore analysed.

With good agreement between observed time series and those simulated from historical or $1 \times \text{CO}_2$ data, a clear climate change signal is evident in the $2 \times \text{CO}_2$ case. The extreme values increase by approx. 2 to 4 °C, whereby the increase is somewhat more pronounced in the summer months than in the winter months. The increase in the mean monthly maximum daily temperatures is of the same order of magnitude at both stations, which is not surprising due to the low altitude gradient.

The errors in the precipitation simulation (i.e. the deviations between observed data and data simulated with historical air pressure data) when downscaling are roughly of the same order of magnitude as the differences between the results for the $1 \times \text{CO}_2$ and $2 \times \text{CO}_2$ scenarios. The precipitation frequency tends to be underestimated in the simulation, while the mean precipitation heights on rainy days are overestimated in almost all months. The climate change signal (i.e. the difference between the simulation results for the $1 \times \text{CO}_2$ and the $2 \times \text{CO}_2$ scenario) is relatively small overall. The only striking feature, as already observed in the previous catchment areas, is a decrease in precipitation in the months of August and September. This decrease in precipitation in late summer (Stöckli and Vidale, 2004; Rowell and Jones, 2006) is essentially due to the increase in anticyclonic conditions (high over Central Europe, high pressure ridge over Europe) in the $2 \times \text{CO}_2$ case.

If this data is now used as input for a typical soil profile in the Leibnitz field, changes in the soil water balance and groundwater recharge can be derived. As a result of the significantly higher temperature in the $2 \times \text{CO}_2$ case, evaporation increases, which applies to all components, including interception evaporation. A limitation of the soil water that inhibits transpiration and thus also plant growth due to the increase in interception evaporation obviously does not occur even at the higher temperatures in the $2 \times \text{CO}_2$ case. Nevertheless, the water stored in the soil during the summer months is significantly lower in the $2 \times \text{CO}_2$ case.

Figure 13 clearly shows that groundwater recharge in the Leibnitzer Feld currently occurs mainly in the months of October to May, with a maximum in March, although the confidence intervals here are also extremely wide. During these months, infiltration rates are relatively high due to reduced interception and interception evaporation, while transpiration is very low. The majority of the precipitation in these months can therefore seep into deeper soil areas. There are major deviations between the simulation with observed data and the downscaling scenarios (historical, 1 x CO₂) when calculating the groundwater recharge rates. Between the 1 x CO₂ case and the 2 x CO₂ case, there is a clear decrease, especially in the summer and autumn months when evaporation is high, while groundwater recharge hardly decreases at all in winter. In the 2 x CO₂ case, this results in a stronger seasonality of groundwater recharge with very low values in summer and relatively high values in winter. Overall, the annual recharge rate decreases by around 25 %. This effect is even more pronounced in individual years.

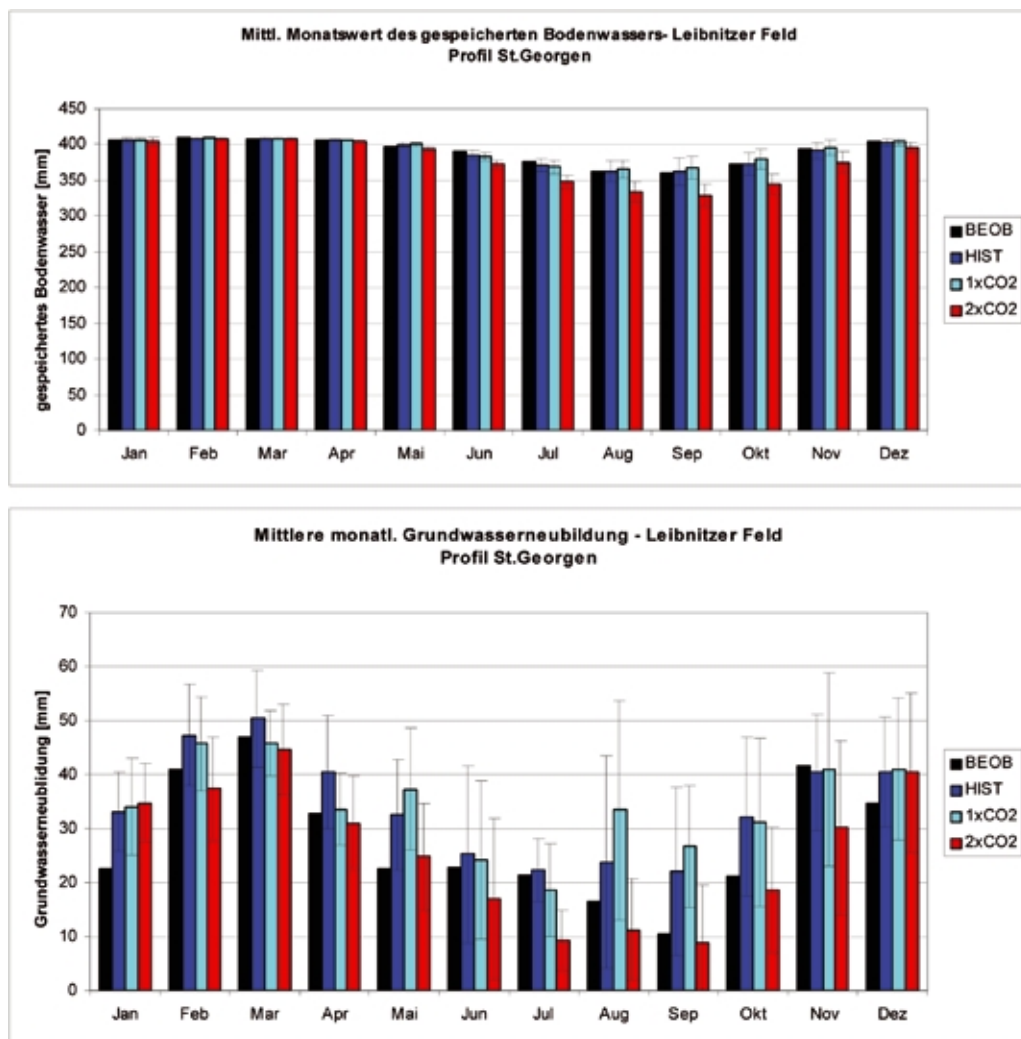


Figure 13: Annual variation in soil moisture and groundwater recharge in the Leibnitz Field

4 Discussion and conclusions

As the currently available simulations of the influence of increasing concentrations of greenhouse gases are mainly on coarse-meshed grids, a stochastic downscaling method was applied to some regions of Austria in order to run hydrological models with spatially higher-resolution and spatio-temporally correlated time series of precipitation and temperature. This was used to calculate the temporal development of the snow cover, the runoff formation and the soil water balance for various meteorological situations. The data for the 1 x CO₂ case, which describes the recent past, and the 2 x CO₂ case, which assumes a doubling of the greenhouse gas concentration, were compared.

In general, it was found that the temperature in all areas will rise by an annual average of between 2.5 and a maximum of 4° C, somewhat more markedly in the summer months than in the winter months. Nevertheless, these changes are to shift the current snow conditions by 600-900 metres in altitude. In the alpine areas analysed (Gail, Enns), the fixed precipitation in the area will decrease by 30-50 %.

No pronounced changes can be recognised in precipitation. A trend towards a decrease was calculated in the south, while precipitation remained almost the same in the upper Enns region; a slight increase was recorded at some higher elevations and in reservoir locations. Some simulations showed a tendency towards higher precipitation in late winter and spring. An increase in extreme daily precipitation not observed, but this may also be due to the limitation to daily values in the calculation.

Due to the higher temperature, evapotranspiration also increased significantly in all regions, resulting in lower soil moisture. In regions with high precipitation such as the Leibnitzer Feld, however, the remaining soil moisture still remains above critical threshold values on average. The greatest and most critical effects are to be expected in areas with low precipitation, such as eastern and north-eastern Lower Austria. While groundwater recharge would decrease by 25 % in the Leibnitzer Feld, it would fall by more than 60 % in the Marchfeld and almost reach zero.

The increased temperature leads to a seasonal shift in runoff. While the months with the lowest runoff have so far in winter, in future these should occur in late summer, possibly early autumn, as precipitation occurs more frequently in winter in the form of rain and thus increases runoff. Snowmelt will occur earlier than at present, although it will also somewhat more muted as less snow is stored overall. In high alpine locations with glaciers, a different picture is to be expected. Although the melt contributions from the glaciers are negligible for the overall Austrian water balance, this contribution can be significant in small catchment areas. Kuhn (2007) indicates a decrease in glacierised area from 567 to 471 km² for the period 1969 to 1999, which corresponds to about 17 %. The corresponding volume from 22 to approx. 17 km³, which corresponds to an average decrease in ice thickness of 10 metres. The maximum melt runoff can therefore be expected in around 40 years, as ever larger parts of the glaciers melt.

For hydropower utilisation in unglaciated catchment areas, a reduction in annual

energy generation of 3-10 %, locally even more, can be expected. This effect is partially compensated for by a balanced water supply, as winter runoff increases by around 5-10 %. In the higher and partially glaciated areas, annual runoff and energy production may increase overall.

The following statements can be made with regard to the frequency of floods and their occurrence times. Based on the simulated data series spanning at least 30 years, no increase in extreme events could be recognised. The increase in winter floods is probable, but these are predominantly not really extreme events.

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Author's address:

o.Univ.-Prof. DI Dr. Hans-Peter Nachtnebel
University of Natural Resources and Life
Sciences, Vienna Department of Water-
Atmosphere-Environment
Institute of Water Resources Management, Hydrology and Hydraulic
Engineering Muthgasse 18, 1190 Vienna
Tel. +43-1-36006-5501
e-mailhans_peter.nachtnebel@boku.ac.at

Hard and soft facts on climate change - an overview

Reinhard Böhm

1 Introduction

"The Earth's climate is out of balance" - this statement has dominated public opinion for some time now. However, as the Earth's climate is a highly complex system that is controlled by numerous external influences and internal feedbacks, and also exhibits great spatial and temporal variability on a wide range of scales, it is difficult to gain an overview, not only for the layman but also for the expert. Questions such as "natural or anthropogenic climate change", "warmer climate= disaster climate" are often treated as "proven facts" in the public debate, from which blame is immediately apportioned and usually surprisingly simple "solutions" are offered, but in reality the matter is usually not that simple - neither in the sense of understanding nor in the sense of "solving the problem".

The contribution of a brief overview attempted here can therefore only be understood as a suggestion to go into more depth if interested. The fact that the guild of climatologists on Earth attempts to provide an overview of the "state of the art" of our science every five to six years within the framework of the IPCC (Intergovernmental Panel on Climate Change) is favourable for this. In 2007, the fourth of this series of "Assessment Reports" was published - 3 volumes totalling 3,000 pages, which can be downloaded free of charge from www.ipcc.ch. As far as the Alpine region is concerned, I can draw on the work of the ZAMG group, of which I myself am a member. On this basis, I will try to take all the necessary shortcuts and give an overview of the vast land of climate change as it appears from a scientific point of view. If you are interested in the political, social and economic aspects, which are difficult to separate, please refer to Böhm, 2008.

2 Air conditioning drives

If you want to understand the past of the Earth's climate and predict its future, the basic prerequisite for this is to know the "climate drivers" that act on the climate system from the outside. Only then can an attempt be made to press the knowledge of the physical laws governing the movements, energy flows and state variables of the atmosphere (air), the hydrosphere (liquid water), the cryosphere (snow and ice) and the lithosphere into a "climate model", which then generates a virtual spatio-temporal image of the Earth's climate. This is then calculated into the past, compared with data from the global climate measurement network and, following this evaluation, also run into the future.

When simulating the past, we can fall back on directly measured or indirectly reconstructed physical climate forcings, the most important of which for the last millennium are *shown in Figures 1 and 2*. On the longer time scale of millions of years and centuries, there are other climate forcings at work than those focussed on here, such as continental drift or the Earth's changing orbital parameters. For these combinations

We will limit ourselves to those factors that been in the foreground in the last and coming centuries.

Figure 1 above shows the changes over time in the formerly

This is the energy supplied by the sun, known as the "solar constant", which reaches the upper boundary of the earth's atmosphere as short-wave radiation. The lower diagram shows the (cooling) effect of volcanic eruptions through the backscattering of "aerosols" hurled into high layers of air by explosive volcanic eruptions. Until about 50 years ago, these two natural factors mainly dominated the climate variations of the last millennium, which began with a "medieval warm optimum", then turned into a "little ice age", from which they finally "recovered" to about the level of the Middle Ages by the middle of the 20th century with increasing solar intensity.

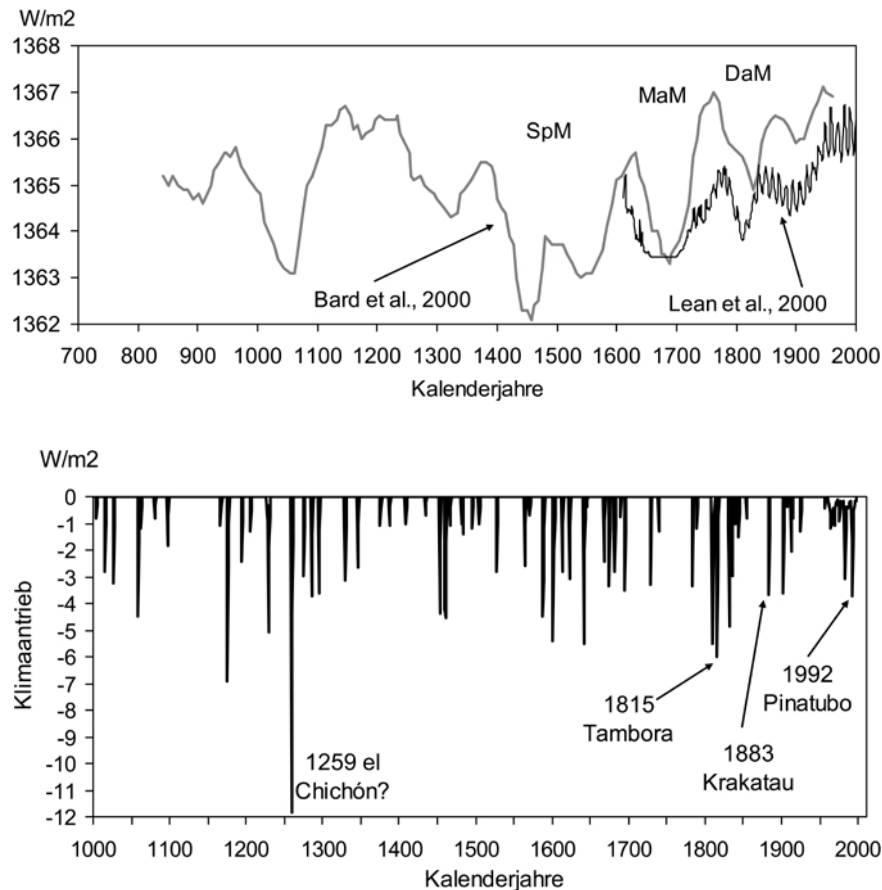


Figure 1: Natural climate forcing in the last millennium:

Above: two reconstructions of the temporal variability of the "solar constants": Black: from direct sunspot observations (Lean et al., 2000)

Grey: derived from cosmogenic tree-ring ^{14}C and ^{10}Be isotope series from ice cores (Bard et al., 2000).

SpM= Spörer minimum, MaM= Maunder minimum, DaM = Dalton minimum

Bottom: 1,000-year reconstruction of climate forcing by volcanic eruptions that sulphur dioxide into the stratosphere, from various ice cores and from direct measurements (Crowley, 2000)

Humans then began to intervene in the climate, albeit in a more ambivalent way than the cliché "anthropogenic global warming". The burning of fossil fuels

energy sources (coal, oil, natural gas) and activities related to food production (wet field cultivation, ruminants, etc.), the two "key greenhouse gases" carbon dioxide (CO_2) and methane (CH_4) in particular were enriched in the atmosphere in ever higher concentrations. However, their warming effect was "masked" in the first decades of extreme economic development after the Second World War, which was initially based on dirty (e.g. sulphur-emitting) technologies. These sulphate aerosols, for example (like those released by volcanoes) had a cooling effect on the earth's climate that lasted until around 1980. Only the major efforts to keep the air clean, in combination with the collapse of the heavy industry and armaments industry in the former Eastern Bloc, brought about a reversal of the trend in this anthropogenic climate forcing, which can be seen in the lower diagram in *Figure 2*.

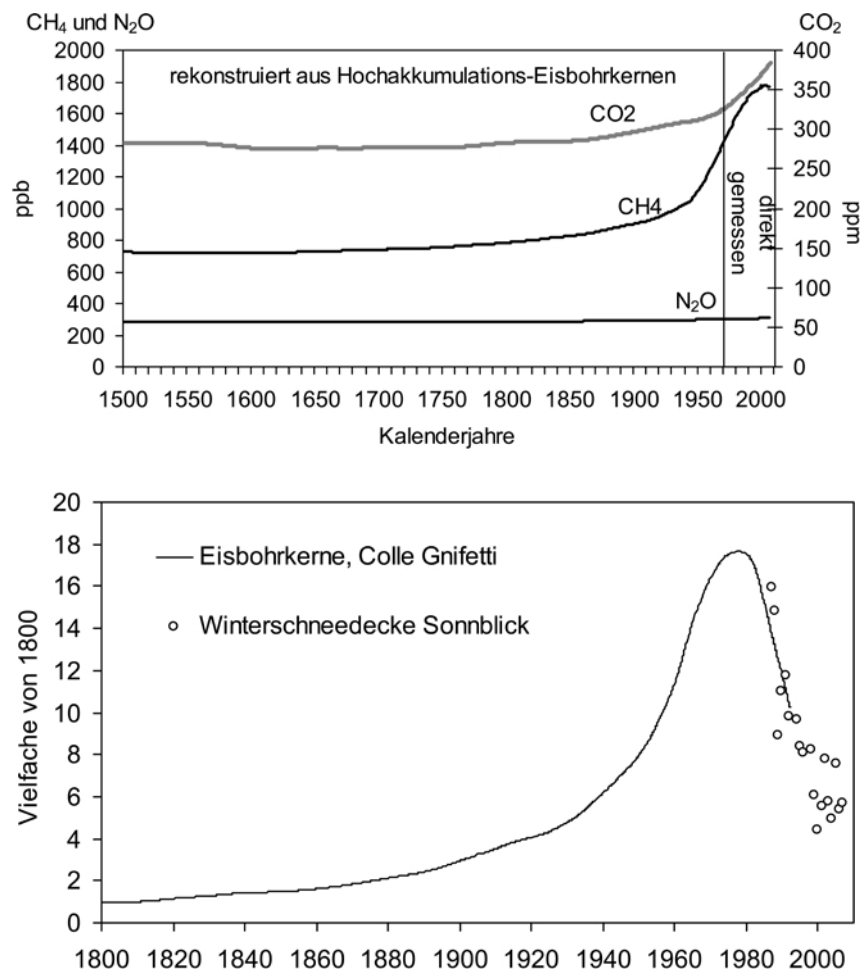


Figure 2: Additional climate forcing by humans:

Above: Development of the three long-lived greenhouse gases 1500-2007 from ice cores and direct measurements, source: Robertson et al, 2001 (updated)

Bottom: Sulphate content in firn and ice in the High Alps. Combined from the highly smoothed data of an ice core (Colle Gnifetti, 4,450 m, Monte Rosa summit zone) since 1800 with annual measurements in winter snow in the Sonnblick area (3,100 m, Hohe Tauern) since 1982

Sources: Wagenbach, pers. Mitt, 2008 and Schöner, 1995 (updated until 2007)

However, it has not been possible to reduce greenhouse emissions, and I do not expect this to happen to any significant extent in the future. The "comfort gas" CO₂, which heats our rooms and powers our machines and vehicles, is probably too closely tied to our civilisation, the need for the majority of the world's population to catch up in this is too great, and methane is probably also too closely linked to feeding humanity. After 1980, this led to the actual entry into the "anthropogenic greenhouse climate", which will probably accompany us for the next few centuries until further technological progress is able to replace the declining reserves of "stored fossil solar energy" (which is what the three fossil fuels are).

The two time series in *Figure 3* show that the long-term and large-scale temperature curve can already be simulated well with coupled global atmosphere-ocean circulation models (AOGCMs, more detailed description in the final section), which are given the climate forcing described above. The characteristics of the climate of the last millennium described above are simulated by the Erik model (named after the Viking Erik the Red) very similarly to what has been reconstructed by hundreds of tree-ring analyses in the Alpine region, for example. In both cases, we find precisely dated individual years, such as the extreme "year without a summer" (1816), we find an almost equally cold summer in the middle of the "climatic optimum" of the Middle Ages, which is typical of an entire cold century embedded in the predominantly warm Middle Ages (the 11th century). We see the gradual cooling in the 14th and 15th centuries and the final plunge into the core period of the "Little Ice Age" in the second half of the 16th century. For the discussion "natural versus anthropogenic climate", it should be noted that until the middle of the 20th century, the hottest summers and also the average temperature level had just reached that of the medieval warm period, and that in the last 20 years the temperature curve has begun to grow out of the "natural range" of the last 1,250 years.

3 Current trends

For the time being, let's stay on the safer ground of measured climate trends. *Figure 4* illustrates the described transition from natural to anthropogenically influenced climate for the greater Alpine region and on a global average, which in this case has already been confirmed by direct instrumental measurements at more than a hundred (Alpine region) and more than a thousand (global) locations in the climate measurement network that has covered Central Europe for 250 years and the entire Earth in sufficient density for 150 years. In the figure we can recognise the still cool last phase of the "little ice age", the naturally induced warming trend that began at the end of the 19th century and finally reached a temporary peak around 1940-1950 - synchronous with a maximum in solar irradiation (see *Figure 2*) (ellipse 1). The first anthropogenic (aerosol) period the climate globally and even more so regionally, which led, for example, to the beginning of a widespread glacier advance period. By 1980, three quarters of Austria's glaciers were advancing again. However, this glacier advance was interrupted very suddenly by the strong warming of phase 3, which is mainly due to the influence of greenhouse gases as described above.

In addition to this described trend, which was about twice as strong in Central Europe as the global average (+2 °C since 1890 compared to +1 °C), the regionally much stronger short-term (high fre-

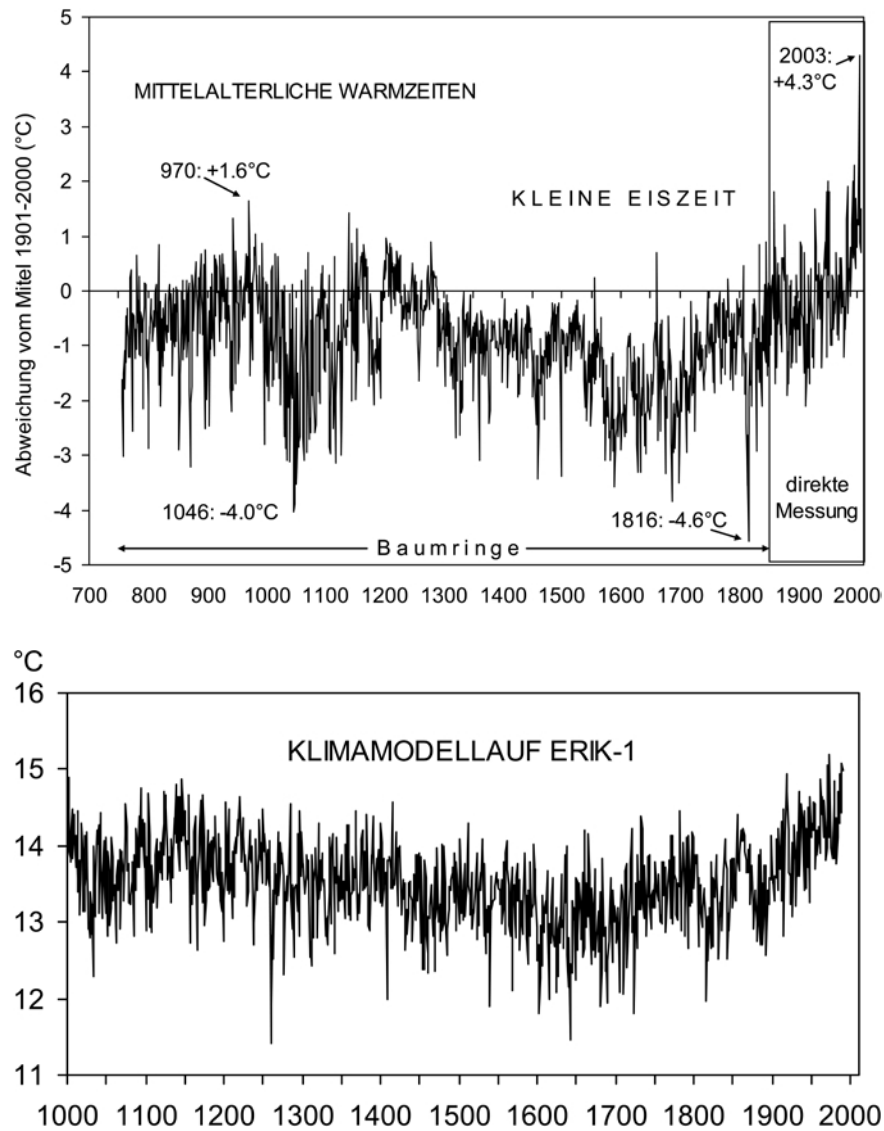


Figure 3: Climate measurement, climate reconstruction and climate modelling for the last millennium: Top: Reconstruction of summer-early autumn temperatures (June-September) in the Alps since the year 755 AD.

755-1850: reconstructed from hundreds of tree-ring density series (Büntgen et al., 2006), 1851-2006: directly measured high alpine HISTALP data (Auer et al., 2007).

Below: Climate model run ERIK-1 for the summer mean temperatures (June-September) for the years 1000-1990 for continental Europe (10°W-40°E, 35°N-70°N) (Zorita et al., 2004)

The more regional and localised we go, the more the long-term background trends are drowned out. This fact should always be borne in mind when another extreme winter (we have experienced both an extremely mild and an extremely cold and snowy one in the Northern Alps in the last three years) or a "millennium summer" (2003) lead to premature conclusions about long-term trends. In climatology, individual cases but also time series that are too short never allow statistically significant trend statements to be made, in the case of precipitation, which is even more variable in terms of space and time, to an even greater extent than in the case of annual mean temperatures.

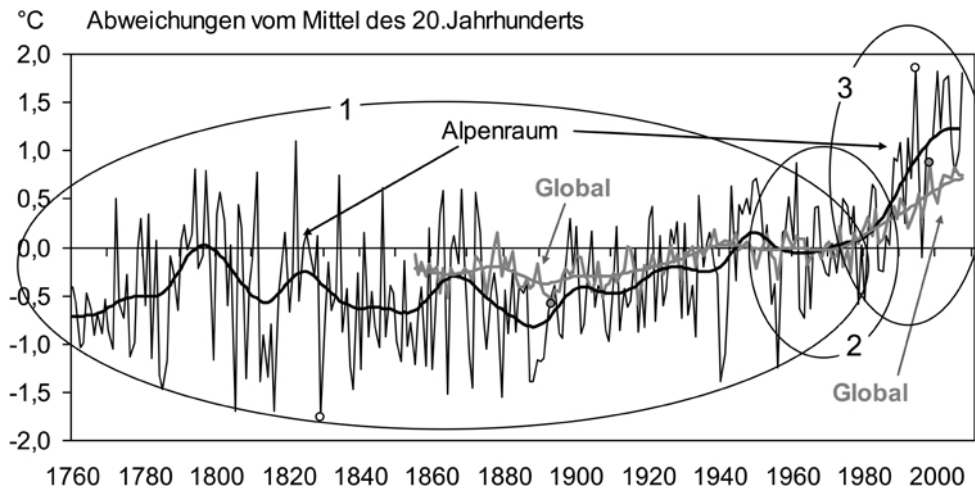


Figure 4: Directly measured annual mean air temperature in the greater Alpine region 1760-2007 (black) and global mean 1858-2007 (grey). Anomalies to the 20th century mean are shown. Alpine region: 4-19°E, 43-49°N, mean of 124 individual series; global land: mean of 2000 mainland and island stations

1: last natural period - solar and volcanic influence dominant 2: first noticeable human influence - the aerosol period
3: Entering the greenhouse age

Sources: Auer et al, 2007 and Jones et al, 1999 (updated)

While the long-term temperature trend in the Greater Alps is also re-presentative for sub-regions (the +2° applies to Perugia, Vienna, Marseille and Bern as well as to the high Alpine observatories on the Säntis, Zugspitze and Sonnblick), the precipitation trends in *Figure 5* show a striking difference between the long-term upward trend in the north-west of the Alpine arc and the long-term trend towards drier conditions in the south-east. The magnitude of both opposing trends is 10-15 % per century, while the short-term variability is ± 50 %.

Without being able to go into the developments of the other climate elements in detail here, it should only be mentioned that a regional increase in sunshine duration could be registered, which was caused by a northward migration of the subtropical high pressure belt, especially in summer, and which also explains the regionally stronger warming compared to the global warming. Over the last 130 years, there has been a slight overall decline in wind, albeit not significant, following stormier conditions at the beginning of the 20th century. In terms of snowfall as a proportion of total precipitation, the warming trend manifested itself again in a decline of around 10-15%, albeit with systematic seasonal differences at different altitudes according to the following pattern: decline in the winter half-year in the valley, in the transitional seasons at medium altitudes of 1,500-2,000 metres above sea level and winter stability in the glacier region, which, however, reacted sensitively in summer.

4 Extreme values

The greatest discrepancy between hard scientific facts and their perception by the public is probably in the area of extreme forms of climate change. Among

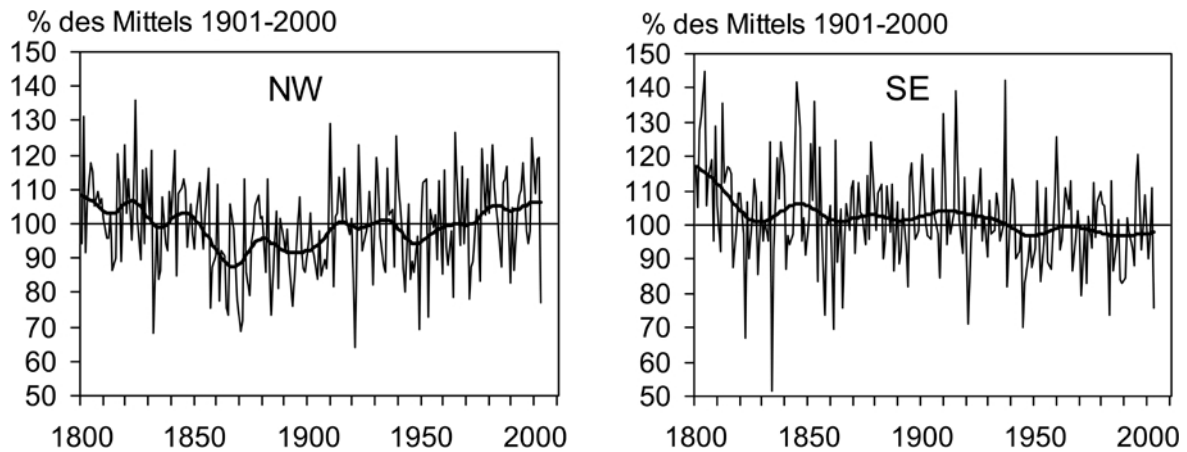


Figure 5: Regional differences in precipitation trends:

Regional annual precipitation 1800-2007, averaged from 50 individual series each.

Left: north-west of the Alpine arc (France, Switzerland, southern Germany, Vorarlberg, Tyrol), right: south-east of the Alpine arc (Hungary, Slovenia, Croatia, Bosnia, Friuli, Veneto, Carinthia, western and eastern Styria).

The individual years and the 30-year smoothed trend are shown Sources: HISTALP database, Auer et al., 2006

The generally accepted catchphrase "the climate is getting crazier and crazier" has given rise to a dogma that is based on the softest facts that climatological science currently (still?) has to offer. Soft facts about extreme values prevail both in the climate reconstruction of the past and in climate modelling - in both cases due to a fact described in Chapter 8.5.4 of the IPCC 2007 Working Group 1 Report (Model Simulation of Extremes):

Because most AOGCMs have coarse resolution and large-scale systematic errors, and extreme events tend to be short lived and have smaller spatial scales, it is somewhat surprising how well the models simulate the statistics of extreme events in the current climate, including the trends during the 20th century. This is especially true for the temperature extremes, but intensity, frequency and distribution of extreme precipitation are less well simulated.

I deliberately left it in the original English text, firstly to counter the accusation of manipulation, but secondly because I am a fan of the nuances that this language offers. This "somewhat surprising" and the "less well" are small masterpieces in themselves. In the original text, before the arduous process of peer reviewing had to be fought through, there was more direct talk of "serious deficiencies in the simulation of precipitation", namely "both in the intensity and the distribution of precipitation".

are two additional challenges when analysing possible trends of extreme values in the past. Firstly, the laws of statistics require long and spatially dense time series in order to get significant results, especially if the very rare and very strong outliers are the aim of the analysis. And it is precisely these long time series that require careful homogenisation of these time series as a sine qua non. As we were able to show (Auer et al., 2007), an average climate time series experiences on average

Every 20 to 30 years there is a discontinuity (relocation, change of instrument or technology, change of environment and numerous other reasons), the strength of which can reach or even exceed the actual climate signal. And the climate time series of daily values (or even shorter time spans) that are particularly needed to analyse "short lived" extreme values show a second special feature of the corresponding field distributions: They are much more spatially decorrelated than longer-term averages or sums. *Figure 7* shows this as an example for the daily to annual values of temperature and precipitation, whereby here the additional special "sensitivity" of the climate element precipitation.

Since highly correlated time series of neighbouring reference series are required for homogenisation of the time series (= adjustment of earlier to the current "state" of a measurement series), *Figure 6* shows the main reason for the fact that, unfortunately, are hardly any homogenised climate time series on a daily basis at present. This applies both internationally and for the Alpine region or Austria. For some subregions of Austria, our group is currently working on the creation of spatially dense daily series for the 20th century, and an international COST action ("HOME" see appendix) is addressing the problem of homogenising daily series, among other things.

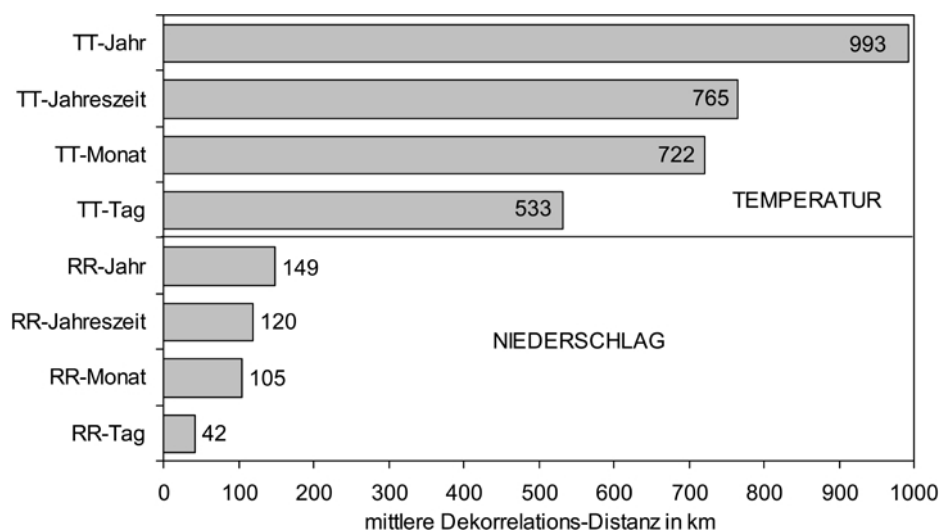


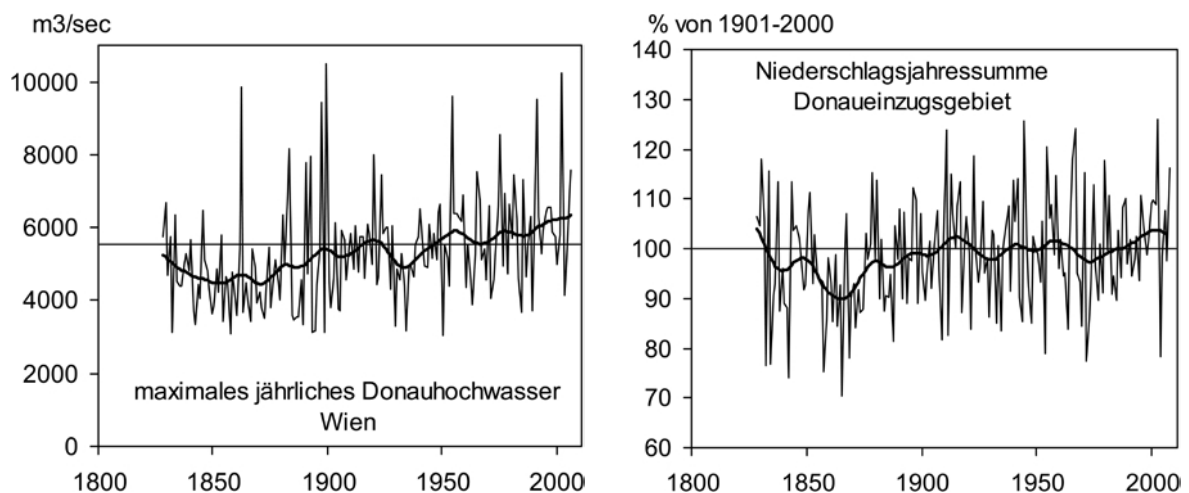
Figure 6: Spatial decorrelation to 50 % common variance in the European climate measurement network of annual, seasonal, monthly and daily values of temperature and precipitation Source: Scheifinger et al., 2003 (internal project report ALP-IMP)

The same problems are mirrored in the trend analysis of extreme values. Their large spatial and temporal variability emphasises local results and obscures an overview of the overall situation. Naturally, this opens up a wide field of dispensable disputes, contradictory results and the osmotic intrusion of ideological argumentation into a topic that has not yet been satisfactorily analysed scientifically. In the patterns of extreme value distributions, which undoubtedly exist in reality but have not yet been analysed in terms of area and time, it is always possible to pick out the event or result that best supports the respective position - "picking" is the existing technical term for this. In order to avoid into precisely this trap, only a few preliminary results are presented below, with the previously

The above should have clarified the necessary restriction with regard to generalisation.

An interesting preliminary result that our ongoing projects in this area have produced or are currently producing (Foralps, A Tale of Two Valleys) is the surprisingly close correlation between trends in extreme value indices on a daily basis and the corresponding trends in mean values, right up to annual averages, particularly in the case of precipitation. *Figure 7* shows an example of extreme daily runoff and precipitation in the large Danube catchment area of the Nussdorf gauge (Vienna), *Figure 8* a comparison of annual precipitation totals and daily value indices climate series in the Hohe Tauern. It therefore does not appear to be the case, at least for the greater Alpine region, that trends in extreme values can also go in completely different directions than those of the mean values. It is therefore not necessarily to be expected that, for example, in areas with a generally decreasing precipitation trend (such as in the south-east of the Alpine arc) an increase in heavy rainfall is to be expected or, conversely, in regions with an increase in precipitation (such as in the north-west of the Alpine arc) dry periods will become more frequent.

This finding also applies to temperature: as the general warming trend increases, so do the extreme values relating to heat and, to the same extent, a decline in the cold indices can be observed. Both not necessarily linear; non-linear correlations are the rule, especially when threshold value overshoots or undershoots are analysed.



Picture 7: Left: Time series of the largest annual flow rates of the Danube near Vienna, 1828-2006, source: Godina R, Hydrographisches Zentralbüro, Vienna, 2008, personal communication Right: Area average of annual precipitation totals in the Danube catchment upstream of Vienna, source: 28 long-term HISTALP precipitation series, Auer et al., 2007 All series for individual years (thin) and smoothed over 30 years (Gaussian low-pass filter)

The preliminary finding of the similarity of the trends of extreme values based on daily values and those of the mean or sum values also gives increased significance a further preliminary result, which was obtained in an as yet unpublished study of the longest HISTALP monthly, seasonal and annual series of air pressure, temperature and precipitation. The longest time series in the region (20 to 30 series each), which all date back to the early instrumental period (before 1840), were analysed for long-term changes in variability using a procedure that can only be briefly outlined here.

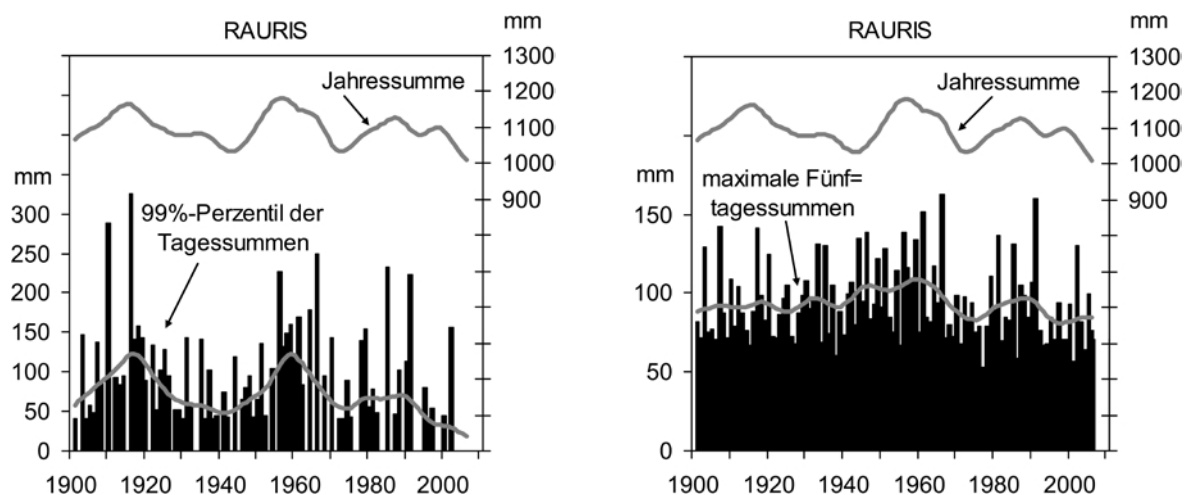


Figure 8: Similarity between the total annual precipitation sum (the 20-year smoothed lines at the top of the figures) and two precipitation extreme value indices for the inner-Alpine town of Rauris (individual years and 20-year smoothed, bottom).

Left: Sum of daily precipitation that is only exceeded by 1% of all cases in the normal climate period 1961-90 (= 99% percentile).

Right: maximum five-day totals for a year Source: Auer et al., 2008a and b

The procedure was applied separately to all individual series in order to the statistically expected increase in variance with a decreasing number of series. In the seasonal and annual analysis, the trends themselves were first eliminated by high-pass filtering (filter width 30 years), then quantiles (from 1 % to 99 %) were calculated in overlapping time windows of 30 years. This allowed distribution-free time series of climate variability to be generated with the inter-quantile ranges. The 80%, 90% and 98% interquantile ranges were analysed in each case. For year-adjusted (by normalisation) sequential monthly series, the analogous procedure was carried out on the basis of 10-year (= 120 monthly) time windows.

For air pressure and temperature, the inter-quantile time series showed very similar regional trends, while for precipitation, the Mediterranean, Atlantic and continental areas of the Alpine region reacted differently regionally and seasonally. Surprisingly, no general increase in climate variability has been observed either in the last 200 years or in the last 50 years, as taken for granted by the public. In the case of temperature in particular, the trends in variability (= the "madness of the climate") are declining in all seasons and in all sub-regions. Figure 9 illustrates this using the example of the sequential monthly series of air temperature, averaged over all 24 long-term series. In the case of precipitation, are different seasonal and regional trends in variability, with the most striking increases in variability typically being observed where the trends in precipitation amounts are also increasing, for example in autumn and winter in the subregion northwest of the Alpine arc. Table 1 shows this as an example for the 90% inter-quantile range and the long-term trend from the 19th to the 20th century.

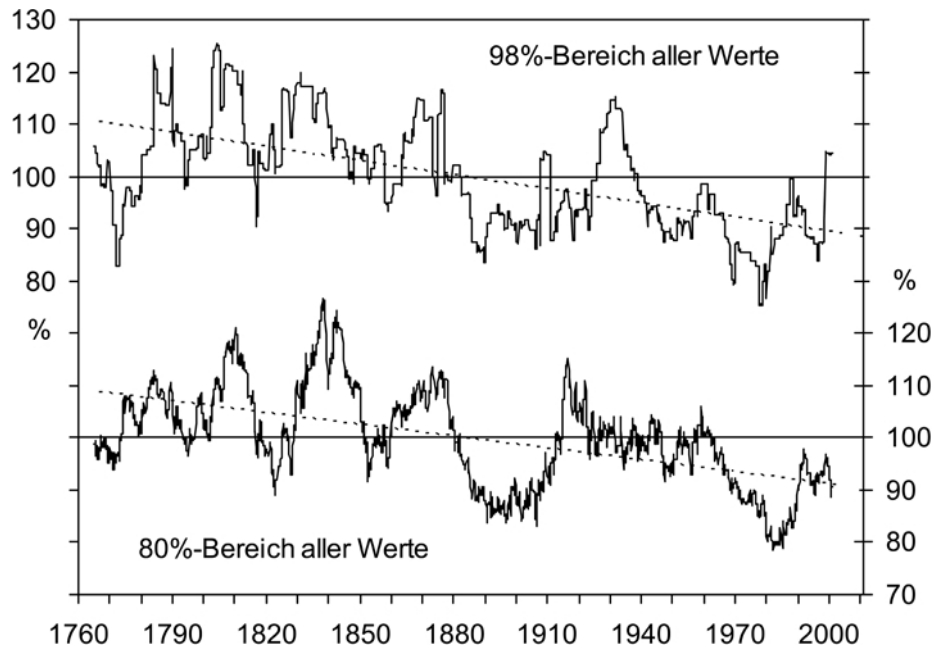


Figure 9: Time series of the variability of the temperature climate in the Greater Alps from 1760 to 2006: 80% and 98% quantile range in time windows of 120 months, which were moved in monthly steps over standardised 30-year high-pass filtered sequential monthly series of the temperatures of the 24 instrumental measurement series of the region beginning before 1840. The mean of all these long-term series is shown here.

Source: Interim result of an ongoing study by the ZAMG Vienna Climate Change Working Group

Table 1: Percentage changes (base value 19th century) in climate variability (inter-quantile range 90 %) from the 19th to the 20th century on a seasonal and annual basis in three sub-regions of the Greater Alps for the climate elements temperature and precipitation

Source: like picture 9

	NE	NW	S	NE	NW	S	NE	NW	S	NE	NW	S	NE	NW	S	
	FRL	FRL	FRK	SOM	SOM	SOM	HER	HER	HER	WIN	WIN	WIN	YEAR	YEAR	YEAR	
Temp.	-16.0	-14.9	-13.2	-20.0	-7.6	-2.3	-3.2	-9.7	+0.3	-7.4	-12.0	-6.2	-28.9	-23.8	-11.5	%
Low.	-6.1	-3.0	-9.0	+1.7	+8.9	-2.4	-0.3	+14.1	-5.8	+8.8	+13.5	-2.6	-4.6	+5.3	-3.7	%

If we want to summarise the characteristics of the trends in extreme climate values, we must first of all point out the major research deficits, which are mainly due to the insufficient density of long-term measurement series. The preliminary results from the greater Alpine region show a general decline in temperature variability with simultaneously rising temperatures. In the case of precipitation, there are regionally and seasonally different trends in variability, but the trends of the extreme values appear to be analogous to those of the sums accumulated over longer periods of time. In any case, the long-term trends of changes in variability and extreme values are significantly lower than the short-term variability itself - a fact that has already been pointed out in the trends of the climate elements themselves.

5 The next 100 years

State of the art for simulating the future climate are so-called coupled global atmosphere-ocean circulation models (AOGCMs), which show a digitised earth including the ocean, air envelope, sometimes even the earth's surface including plant cover, in which the weather of the entire earth actually runs in a certain spatial resolution (currently already less than 100 km horizontally, 20 to 30 ocean layers vertically, and just as many atmospheric layers) and a coordinated temporal cycle frequency (minutes to hours) for the next 100 years or for the past 1,000 years. These model runs thus calculate what the climate system does internally with the external forcing - without any claim to completeness, i.e. what quantities of energy are transported by the oceans from the equator towards the poles, how much liquid water enters the atmosphere through evaporation, how much incident short-wave solar radiation is reflected into space by snow surfaces in Siberia or the Antarctic and also what amounts of energy in the form of long-wave (thermal) radiation are prevented from escaping the Earth's atmosphere by greenhouse gases, what amounts of energy are converted into warming and what into movement (storms) and much more.

Such simulations involve a gigantic computing effort even the fastest computing systems currently available cannot cope with. And these are the fastest computing systems currently in a handful of large international computer centres, and these super-computers need months of CPU time for a typical climate run! It is therefore necessary to make concessions. On the one hand, these are in the direction of the spatial-temporal resolution of the computing grid, on the other hand, the known physical correlations must be simplified in order to be able to calculate them fast enough. All this leads to a varying ability of climate simulation calculations to cope with different forms of climate. On a scale from good to moderate to poor to not at all, modern AOGCMs already provide useful results for global to continental patterns, especially of temperature, over their mean over longer periods of time. They become worse when one goes into the regional to local level and here above all when the water cycle comes into play. Here the facts become increasingly "soft". Difficulties in simulating cloud formation, especially in small-scale convective processes, make statements about heavy rain, hail, thunderstorms or tornadoes only possible to a very limited extent or not at all. However, it is precisely these extreme forms of weather that are at the centre of the climate debate (keyword "the climate is becoming more and more extreme").

The reconstructed climate forcings are available for modelling the past climate, as shown and discussed for the last millennium in *Figures 1 and 2*. *Figure 3* shows a realisation for the last 1,000 years of summer-early autumn temperatures for Europe, and how these quite well the course of a temperature reconstruction based on tree rings from alpine high altitudes. A computational-physical simulation of the future climate course an estimate of the future course of the climate forcing. Typically, AOGCMs are twice for this purpose, once without changes in both natural and anthropogenic external forcing and once under the influence of expected changes in the concentration of anthropogenic greenhouse gases resulting from estimates of future political, economic, social and technological developments. The IPCC has derived numerous "development families" for this purpose, the differences between which are to be analysed.

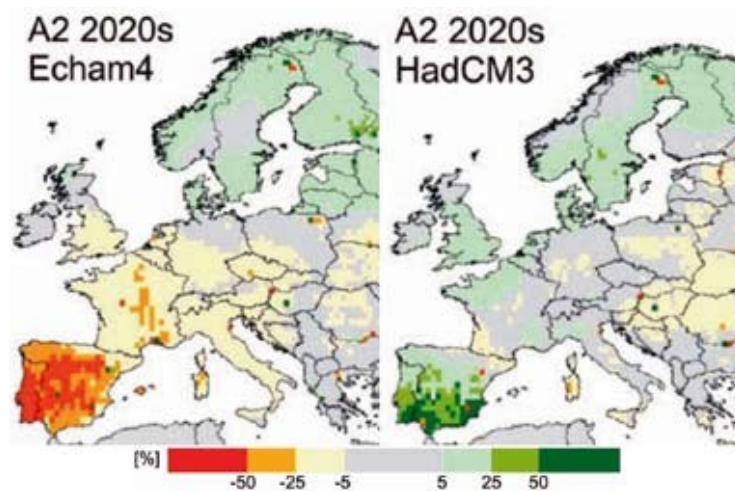
This adds at least as much uncertainty to that resulting from the mathematical and physical modelling uncertainties. The fascinating thing, however, is that it is possible to virtually play through different future worlds in this and thus to estimate the effects of different scenarios on the earth's climate. Table 2 is an attempt to briefly summarise the three main families of scenarios for human development in the 21st century. The IPCC uses the scenario labelled A1B as the lead scenario, which shows the A2 world characterised as "business as usual" or "ugly globalisation", which can be staged out in terms of upward warming. The "brave new world" of planned and globally implemented environmental awareness runs under B1 and leads to the lowest greenhouse gas emissions.

Table 2: Brief description of the three most frequently used IPCC future scenarios of global development in the 21st century according to Nakicenović and Swart, 2000

A1 family	Rapid economic growth in a mobile, globally networked, increasingly interconnected world. Income disparities between individuals and between countries and regions are diminishing. Higher investment in education and research. Longer life expectancy and better health are leading to smaller families worldwide. The earth's population will rise to 9 billion by 2050, then fall to 7 billion by 2100. New technologies are being used quickly. Energy efficiency increases. Very different energy priorities, in the "marker scenario" A1B a balanced mix of different energy sources is used.
A2 family	Little global networking in a regionally differentiated, culturally pluralistic world. Less economic growth overall compared to A1, persistently large income disparities between individuals and between countries and regions. Fertility and mortality rates develop less strongly than in the A1 worlds. The Earth's population continues to grow and reaches 15 billion in 2100. The energy mix is also more differentiated than in the A1 world and is mainly orientated towards regional availability. Large parts of economic development have to be invested in food production, which means that there is less further development in other sectors than in A1, including the energy sector.
B1 family	The main characteristic of the B1 development lines is a high level of environmental and social awareness, which leads to a globally coherent trend towards sustainable development. After decades of turmoil and chaos, community values prevail over individualism, with controlled education and strong political tax mechanisms playing a leading role. Reduced consumption and a strong focus on renewable forms of energy lead to the most significant successes in the sector of curbing greenhouse gas emissions. The earth's population develops in a similar way to the A1 scenarios, but for different reasons, mainly due to increased security as a result of society's social commitment.

Figure 10 illustrates that the implementation of the greenhouse gas forcing derived from the socio-economic-political scenarios in AOGCMs must still be able to deal with major uncertainties, as already mentioned, also due to the still existing inadequacies of the climate models in some areas. It shows the very different results of two of the best models currently available for the expected changes in mean annual runoff on the Iberian Peninsula for the period around 2020. The aforementioned inadequacies of precipitation modelling even lead to contradictory statements

an increase in river flow of up to 50 % in the case of one model and a decrease of up to -50 % in the simulation by the other model.



*Figure 10: Simulation of the change in the water flow of Europe's rivers for the 2020s assuming a "business as usual" greenhouse gas trend (IPCC scenario A2) in per cent. Left: Max Planck Institute for Meteorology, Hamburg
Right Hadley Centre for Climate Prediction and Research, Exeter, UK.
(IPCC-2007, WG-2 Report, extract from fig.12.1)*

There are currently between 10 and 20 "state of the art" global transient simulations of the Earth's climate over the next 100 years. The spatial resolution has been reduced from 100 km to 50 km in the last 10 years. The details cannot be discussed here; a detailed description can be found in chapters 8 to 11 of the Working Group 1 Report of IPCC-2007. The spatial resolution of between 100 and 50 km already achieved by the AOGCMs is already useful for regional purposes and is also recommended in IPCC-2007 over attempts to increase the resolution.

For local applications, the results of the global model calculations must be scaled down (refined), especially if the extreme values are of interest, which are by definition small-scale, particularly in the case of precipitation. The physically more elegant method, and also the one that more useful for understanding the processes, is "dynamic downscaling", which places a finer-meshed model within the framework of a coarser, global realisation and downscales the given structures of the global results. At present, we are moving towards a resolution of a few kilometres, albeit with very different and sometimes contradictory results. The latest development is moving towards compressible calculations, which represents a significant step towards greater realism (CLM models). The currently still rather weak and sometimes contradictory results of dynamic downscaling, especially in the complicated terrain of the Alpine region, make the various methods of "statistical downscaling" interesting, especially when it comes to small catchment areas. Statistical correlations of measured climate data are used for downscaling.

Thanks above all to increasing computer performance, it is to be expected that in the future it will also be possible for climate modelling to take the step that fundamentally improved weather forecasting about 10 years ago, and which enables a quality and thus confidence estimate: the ensemble forecast (a large number of parallel realisations of the future). Currently, the main problem is the insufficient verification of the model results based on the not entirely logical assumption that a good realisation of the present (compared to measurement data) also to the climate future. What the current state of research expects for the climate of the 21st century is now outlined in brief on a regional scale (Central Europe and here especially the facts of interest for Austria):

6 Summarised presentation for Central Europe

On a regional scale in Central Europe, the well-known correlation between the increase in high-frequency climate variability and the long-term climate trends is already noticeable for all climate elements. This means that the following long-term prospects will be "masked" for some time to come by the short-term "noise" of fluctuations from year to year.

Better documented prospects until 2100:

- Sustained temperature rise of 3 °C (A1B scenario), stronger in summer in the Mediterranean region and in winter in the north-eastern part of the region. This is more than the temperature increase of around 2 °C observed in the region over the last 150 years. Scenario variability about ± 1 °C (B1 +2°, A1 +4° and more).
- Increase in all types of positive temperature extremes (hot days, absolute maxima, etc.) with a roughly analogous decrease in negative temperature extremes (frost, absolute minima, etc.)
- Hardly any changes in wind and storm activity (this only applies to Central Europe, in other parts of Europe there is disagreement among the models). The expectation of no increase in storm activity in the future is in line with the data findings of the past 130 years (*Matulla et al., 2007*)
- Continuing decline in the proportion of the solid component of precipitation
- Soil moisture is increasingly influenced by temperature (evaporation) and less by Precipitation supply
- Further glacier retreat of the active Alpine glacier area to 40 to 50 % (from 1980) by 2050, to 15-20 % by 2100

Worse prospects until 2100:

- Precipitation tends to increase north of the Alps, decrease south of the Alps, the boundary between increase and decrease further south in winter, further north in summer. The magnitude of the changes in Central Europe is in the range of -5 to +10 % in winter, with -5 to -20 % in summer. This is slightly more than what we know from the measured regional long-term trends of the last 200 years, which have not exceeded 10 %/100 years.
- Increase in stable fair weather conditions in summer due to further northeastward shift of the subtropical high (according to MPI models, Hadley models do not show this).

Hardly any prospects:

- Changes in convective precipitation, thunderstorms, hail, tornadoes, mudslides, i.e. most of the small-scale extreme forms of climate, which are poorly or not at all included in the model physics of the large-scale models, and which cause difficulties in downscaling. More on this in *IPCC-2007, chapt. 8*
- Beware of interest-driven opinions, especially in the case of extreme values (e.g. presented in *Pielke Jr. et al., 2008*).

6.1 Austria

Better documented prospects until 2100:

- Sustained temperature rise of 3-3.5 °C (A1B scenario), regionally in winter 3° in the west, 3.5° in the east, in summer 3.3° in the north, 3.8° in the south Scenario variability about ± 1 °C (B1 +2°, A1 +4° and more).
- Glaciers: greater decline than the Alpine average, by 2100 only around 10 % of the glacier area will be left. active glacier areas from 1980
- Everything else as described for Central Europe.

Worse occupied prospects:

- The sub-regional differentiation in precipitation tends to be the same as indicated above for Central Europe, only with narrower ranges of numerical values: +5 to +10 % in winter, -7 to -12 % in summer. This seasonal breakdown suggests a dampening of the usual annual precipitation pattern in Austria by 10 to 20 % (lower summer maximum, higher winter minimum). Past measurements suggest a relatively sharp transition from the northern to the southern characteristic along the main Alpine ridge. Essentially, a continuation of the precipitation trend patterns from the last 150 years is expected (*Auer et al., 2007*).

Hardly any prospects:

- Same statement as for Central Europe.
- In individual cases, reliable local estimates can be made "by hand", so to speak, such as an indirect quantification of the future development of the percentage of snow in the total precipitation for the Hohe Tauern (not yet available for the entire area): (internal document of the ZAMG project "A Tale of Two Valleys")
- Decrease in the proportion of snow from the middle of the 20th century to the end of the 21st century (A1B):
 - At 1,000 m above sea level: in winter from 80 % to 58 %, in spring from 38 % to 16 %, in summer unchanged at 0 %, in autumn from 29 % to 6 %
 - At 2,000 m above sea level: in winter from 90 % to 70 %, in spring from 86 % to 66 %, in summer from 17 % to 5 %, in autumn from 59 % to 18 %
 - At 3,000 m above sea level: in winter from 99 % to 97 %, in spring from 99 % to 97 %, in summer from 64 % to 34 %, in autumn from 99 % to 94 %

Acknowledgement

The attempt made here to summarise climate change is necessarily rudimentary due to the required brevity. The overview was only possible thanks to numerous detailed studies, the simple listing of which would also go beyond the scope of this report. It is therefore only necessary to refer to a few basic principles:

For the climate future, reference was primarily made to the broad-based summaries in the IPCC reports 2007, in particular WG-1 and WG-2.

For the other areas, it was possible to draw on basic work from a number of completed and ongoing research projects of ZAMG's Climate Department: CLIVALP (FWF, P15076-N06), ALP-IMP (EU-FP5, EVK-CT-2002-00148), FORALPS (Interreg-Alpine-IIIb Alpine Space, I/III/3.1/21), A Tale of two Valleys (BMWF programme proVision, 133-111), HISTALP-2007 (ZAMG-BMWF 4426/07-9), Ö-Schnee-200 (ZAMG-BMWF 4426/07-21), COST ES project 0601 HOME, HOMDAY (ZAMG-BMWF 4426/07-23).

We would like to take this opportunity to thank them once again for their support.

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Author's address:

Dr Reinhard Böhm
Central Institute for Meteorology and Geodynamics
Hohe Warte 38, 1190 Vienna
Tel. +43-1-36026-2203
E-Mailreinhard.boehm@zamg.ac.at

Climate models, climate scenarios and their significance for Austria

Andreas Gobiet and Heimo Truhetz

1 Introduction

The Earth's climate is a complex system characterised by the absorption of energy from space (solar radiation, visible light), the redistribution of energy through all components of the climate system (atmosphere, hydrosphere, cryosphere, biosphere and pedosphere) and the release of energy into space (thermal radiation from the Earth, infrared radiation). Through industrialisation and the associated emission of carbon dioxide and other greenhouse gases, humans are interfering with this energy balance, as greenhouse gases are permeable to incident solar radiation but retain the energy radiated from the earth (infrared radiation) in the earth's atmosphere. For about 250 years, the change in greenhouse gas concentrations in the atmosphere has been mainly driven by human activities and CO₂ concentrations have increased from about 280 to 379 ppm (2005) during this period - a dramatic change that is unprecedented in at least the last 10,000 years of Earth's history [Solomon *et al.*, 2007]. The effects of the resulting anthropogenic greenhouse effect are now well documented: The global average temperature has risen by around 0.7 °C in the last 100 years, with the trend increasing strongly since 1950 (the last 25 years contributed over 0.4 °C to the trend of the last 100 years) and is more pronounced over the continents than over the oceans (in the Alpine region, for example, the temperature has risen by 1.2 °C in the 20th century [Auer *et al.*, 2007]). Changes have also been observed in precipitation patterns and in the frequency and intensity of some extreme weather events [e.g. Alexander *et al.*, 2006; Schmidli and Frei, 2005], but it cannot be concluded from this that extreme events are increasing in general and worldwide.

Such observations raise the question of the future development of the Earth's climate and, in particular, the effects of expected climate change on society. Nowadays, the basis for such analyses (climate scenarios) is developed using complex climate models. These models are capable of reproducing the most important physical relationships of the climate system.

This article provides a brief overview of the methods and results of modern climate modelling, with a particular focus on climate scenarios for Europe, the Alpine region and Austria. Section 2 with the functioning of climate models and their regionalisation, section 3 presents current climate scenarios for the Alpine region and section 4, after a brief summary, provides an outlook on current developments in climate modelling.

2 Global climate models, regional climate models and downscaling

2.1 Global climate models

The modelling of global aspects of the climate with general circulation models (GCMs) has its roots in the 1970s and has been gaining in importance since the 1990s.

This is because not only pure model results but also assessments of their quality [e.g. *Reichler and Kim, 2008*] and uncertainty [e.g. *Solomon et al., 2007*] are now available.

Climate models are computer programs that divide the atmosphere, oceans and the ground into boxes (typically the side length of these boxes is around 200 km) and calculate the physical state and the mass and energy exchange with the neighbouring boxes in each of these boxes using physical equations (in the atmosphere, a simplified form of the Navier-Stokes equations). The dynamic nature of these makes it possible to calculate not only the state of the climate system at a specific point in time, but also its development over time. In addition, climate processes that cannot be described by the aforementioned physical equations or take place on spatial scales that are too small to be represented by the coarse model grid are integrated into climate models using parameterisations (semi-empirical simplified descriptions of the processes). These include, for example, turbulent processes in the atmospheric boundary layer, convective precipitation ("summer thunderstorms") and radiative transfer.

The way GCMs work is very similar to that of weather models, which are used to forecast the weather over a few days. In contrast to weather models, however, climate models calculate the state of the climate system over many decades, which leads to important limitations in the interpretation of their results. It is well known that weather forecasts only make sense for a few days and that weather models quickly move from the actual state of the atmosphere after this partially predictable period of time. This behaviour is caused by chaotic components of the weather and climate system, which take over the deterministic components after a short time. For climate models, this means that the calculated weather on a certain day far in the future in no way resembles the actual, as yet unknown weather of that day. Even the best climate models cannot produce weather forecasts for the end of the century. The informative value of climate models is therefore limited to the non-chaotic parts of the climate system on a time scale of decades. They describe statistical parameters of the weather that relate to several decades, such as mean temperatures or precipitation totals of a particular season in the last three decades of the 21st century and the typical variability of these climate parameters. Apart from this fundamental limitation, climate models can also be flawed in the description of statistical climate parameters and every climate simulation is subject to uncertainties. The investigation and quantification of these uncertainties (see also section 3) is an important research topic today and makes it possible to assess how much confidence we should have in climate scenarios - an important issue, especially when it comes to decisions on (short-term) costly measures to mitigate or adapt to climate change.

2.2 Regionalisation

The results of different global climate models agree relatively well on a global average, but often show large regional differences [*Solomon, 2007*]. Since regional climate characteristics strongly depend on (from the GCMs' point of view) sub-scale influences such as complex terrain, land use, water-land distribution or snow cover, GCMs, which typically work with horizontal grid point spacings of 100-300 km, are only partially applicable.

They are able to describe regional and local climate changes. For many areas of climate impact research (especially in hydrology), however, it is precisely the regional and local climate characteristics that are decisive. In order to counter this deficit, great efforts have been since the early 1990s to refine global climate scenarios with the help of dynamic and/or empirical-statistical methods.

Dynamic regionalisation methods are based on regional climate models (RCMs), which are embedded in GCMs. RCMs cover a limited region (e.g. Europe), which saves computational resources and allows investment in a refined model grid (typically 50-10 km grid point spacing). RCMs are driven at the boundaries with results from GCMs and recalculate the state of the atmosphere on a regional scale in the centre of the model area. They are therefore independent dynamic climate models which, like GCMs, are based on the Navier-Stokes equations and physical parameterisations. However, due to their finer resolution, many processes that are parametrised in GCMs can explicitly calculated in RCMs. Historically, the method of embedding a regional model in a coarser global model ("nesting") has been used in numerical weather prediction for some time, but was only used for short-term forecasts, as the ability of regional models to provide meaningful results over longer periods of time was doubted. The first RCMs were developed at the end of the 1980s. Pioneering work in this field was done by *Dickinson et al* [1989] and *Giorgi* [1990] when the US government was looking for a suitable site in the Sierra Nevada for the final disposal of nuclear waste with low precipitation over long periods of time. It turned out that regional effects such as the shading effect of mountain ranges, which is partly responsible for the frequency of precipitation, could not be resolved in global models, but could be well described by the first RCM (RegCM, 60 km horizontal grid point spacing). Today, RCMs are used in various areas of climate research, ranging from palaeoclimate research and seasonal predictions to the analysis of anthropogenic climate change. RCMs are particularly important for studies on the effects of global climate change on the frequency of extreme events and on socio-economically relevant sectors such as agriculture, water resource management, energy production and tourism. The spatial resolution of most RCMs is limited to horizontal grid point spacing of 10 km or larger. Climate modelling with an even higher resolution is technically possible and with "non-hydrostatic" climate models are also feasible, but their quality has not yet been well analysed. In this respect, further development of the models is necessary and is also underway.

The great advantage of RCMs is that regional climate processes are described in a physically consistent manner and can therefore be understood in greater depth. Furthermore, RCMs can be coupled directly with other process-orientated models (such as hydrological models). However, the method is very computationally intensive, a problem that is somewhat mitigated by the rapid technical developments in the computer sector.

The methods of empirical-statistical regionalisation are based on the assumption that regional and local climate effects determined by the large-scale state of the atmosphere. With the help of various statistical models, large-scale atmospheric fields from GCMs, which describe the state of the atmosphere in the past ("predictors"), are compared with local climate effects.

climate parameters obtained from measurements ("predictands"). Based on these statistically established relationships, the predictand, i.e. a simulated future measurement representing the local climate conditions, can be derived from large-scale future scenarios generated with GCMs. Statistical regionisation was already used in the early days of numerical weather forecasting to obtain local information from the then still very rough weather forecast models. Since the 1990s, this method has also been widely used in climate research. There are numerous suitable methods such as neural networks, multiple linear regression, canonical correlation or scaling methods, which are not discussed in detail here. Overview articles on this topic were written, for example, by *Wilby and Wigley* [1997], *Zorita and von Storch* [1997] and *Murphy* [2000]. *Von Storch and Zwiers* [2003] also describe many of the statistical models in great detail. A major advantage of empirical-statistical regionalisation is that it is inexpensive in terms of the computing power required. Statistical methods are very flexible and can be adapted to a wide range of climate parameters, regions and time periods, and data are generated (simulated measurements) that are often directly applicable for users in climate impact research. The basic theoretical limitation of the method is the implicit and unverifiable assumption that statistical correlations between predictor and predictand remain unchanged under changed climate conditions. An additional complication is the need for long data series, which is not guaranteed in many areas of the world. Due to the high adaptability and diversity of statistical models, it is also not possible to make general statements about the quality of statistical downscaling methods. Quality analyses must therefore be carried out individually for each application. Nowadays, empirical-statistical methods are increasingly being used to post-process the results of regional climate models. In this case, achieving a finer spatial resolution is secondary and the correction of systematic model errors takes centre stage.

2.3 Current developments in regional climate modelling

For some years now, the focus of research in the field of regional climate modelling has shifted from America to Europe. The EU project PRUDENCE (Prediction of Regional Scenarios and Uncertainties for Defining European Climate Change Risks and Effects, <http://prudence.dmi.dk>), which has set itself the goal of generating a series of high-resolution climate scenarios for the end of the 21st century (2071-2100) for Europe, analysing their reliability and demonstrating their applicability to impact studies, is partly responsible for this. This work is currently being continued in the EU project ENSEMBLES (www.ensembles-eu.org), which aims to quantify the uncertainties and reliability of regional climate scenarios using ensembles of global and regional climate simulations. A similar project is currently being carried out in the USA (NARCCAP, www.narccap.ucar.edu), which is strongly orientated towards PRUDENCE. There are also a number of projects underway in Europe which, building on PRUDENCE and ENSEMBLES, are analysing specific regions in more detail and have a strong focus on the application of regional climate scenarios in climate impact research. The CLAVIER (www.clavier-eu.org) and CECILIA (www.cecilia-eu.org) projects, for example, aim to improve regional climate scenarios for Central and Eastern Europe and to analyse the effects of climate change in these regions. The CIRCE project (www.circeproject.eu) is dedicated to similar tasks in the Mediterranean region.

In Austria, the possibilities of regional climate modelling and its applicability to the Alpine region have been investigated since mid-2003 in the reclip:more project (Research for Climate Protection: Model Run Evaluation, <http://systemsresearch.arcs.ac.at/projects/climate>) [Loibl *et al.*, 2007]. Here, regional climate modelling, induced by the complex study area, was pushed to the limits of the local scale (in PRUDENCE and ENSEMBLES, work is carried out with 50 km and 25 km grid point spacing respectively, in reclip:more the models were run with 10 km grid point spacing), which is particularly important for investigating the hydrological conditions of small areas. The results of the reclip:more project - a climate scenario for the 2040s for the Alpine region - are now available and are presented in Section 3. From autumn 2008, building on the experience gained from reclip:more, a set of continuous climate scenarios from 1951-2050 for the Alpine region will be produced as part of the reclip:century project funded by the new Climate and Energy Fund, with a particular focus on estimating the uncertainties of these scenarios. These scenarios are to form the basis for further climate impact research in Austria.

3 Current climate scenarios for Austria and their uncertainty

3.1 PRUDENCE

An initial estimate of the climate changes to be expected in the Alpine region by the end of the century (2071-2100 compared with 1961-1990) can be derived from the results of the PRUDENCE project [Christensen and Christensen, 2007].

Temperature

A general rise in temperature is expected in Europe, with a west-east gradient in winter (less warming in western Europe influenced by the Atlantic) and a north-south gradient in summer (increased warming in the Mediterranean region). In the Alpine region, warming is less pronounced in winter (+3.5 °C, 0.32 °C per decade) and spring (+3.3 °C, 0.3 °C per decade) than in summer (+5.0 °C, 0.45 °C per decade) and autumn (+4.2 °C, 0.38 °C per decade). These results are derived from an ensemble of 16 climate simulations, which it is possible to roughly estimate the uncertainties. Assuming that the model ensemble covers the uncertainty range, this results in a 99% confidence interval of about ± 1 °C (less in winter) [Déqué *et al.*, 2007]. The results for the expected temperature change in the Alpine region of 3.3 °C to 5.0 °C can therefore be regarded as highly significant.

Precipitation

The pattern of simulated precipitation change over Europe is far more inhomogeneous than the temperature change pattern. There is a strong north-south gradient with decreasing precipitation in the south and increasing precipitation in the north. The zero line moves seasonally from the northern Mediterranean coast and the Alpine region in winter to northern Europe in summer. In the Alpine region, changes of +20 % in winter, 0 % in spring, -26 % in summer and -7 % in autumn are expected. The corresponding 99% confidence intervals

range from 0.01 mm/day to 0.69 mm/day in winter and from -0.27 mm/day to -1.21 mm/day in summer. Although the uncertainties of the precipitation scenarios are considerably greater than those of the temperature scenarios, at least the sign of the change in precipitation in the Alpine region is significant, with the significance of the decrease in precipitation in summer being more pronounced than the increase in winter.

3.2 reclip:more

Since the Alps represent a climatic divide that causes different climatic conditions in nearby regions, it can be expected that climate change in this region will also be spatially very inhomogeneous and will behave differently in smaller sub-regions of the Alpine region than the average climate change in the entire Alpine region. Regional precipitation scenarios that are able to differentiate between small sub-regions of the Alpine region require a higher spatial resolution than the PRUDENCE simulations (grid point spacing of 50 km, which corresponds to an effective resolution of at least 150 km). A first step in this direction was taken in the Austrian project reclip:more (see also section 2.3). Unlike PRUDENCE, reclip:more is aimed at the middle of the 21st century (2041-2050, comparison period 1981-1990) and only deals with the European Alpine region, albeit in greater detail (grid point spacing 10 km). The summarised results of the project can be found in the report by *Loibl et al* [2007] and some detailed results in *Gobiet et al* [2006]. The following results are taken from the report by *Gobiet et al* [2006].

Temperature

The temperature change in the Alpine region up to the 2040s (compared to the 1980s) (see *Figure 1*) is +1.9 °C (+0.32 °C per decade) in winter, +2.2 °C (+0.37 °C per decade) in spring, +2.3 °C (+0.38 °C per decade) in summer and +2.7 °C (+0.45 per decade) in autumn. These results are in good qualitative and quantitative agreement with the PRUDENCE results described above. The spatial patterns of temperature change are not very pronounced, but show a larger temperature trend with increasing altitude, which may be due to feedback processes between snow cover and near-surface temperature. As only three regional simulations were carried out in the reclip:more project, all of which are on the same global simulation, it is only possible to estimate the uncertainties very roughly. The comparison of the three regional simulations shows a range of the temperature change signal of around ± 0.25 °C (slightly different depending on the season). *Déqué et al* [2007] have estimated that the uncertainty of regional simulations in the Alpine region is about 20-25 % of the total uncertainty. If the range given above is interpreted as a measure of the uncertainty of the regional simulation, this results in an overall uncertainty of the reclip:more scenarios of no more than ± 1.25 °C.

The expected changes in precipitation in the Alpine region (see *Figure 2*) are as follows in winter +8.5 % (+0.2 mm/day), in spring +1.4 % (0.0 mm/day), in summer -12.2 % (-0.3 mm/day) and in autumn -13.8 % (-0.5 mm/day). As these values refer to the middle of the century, they are not directly comparable with the PRUDENCE results, but similar basic characteristics: increasing precipitation in winter and decreasing precipitation in autumn.

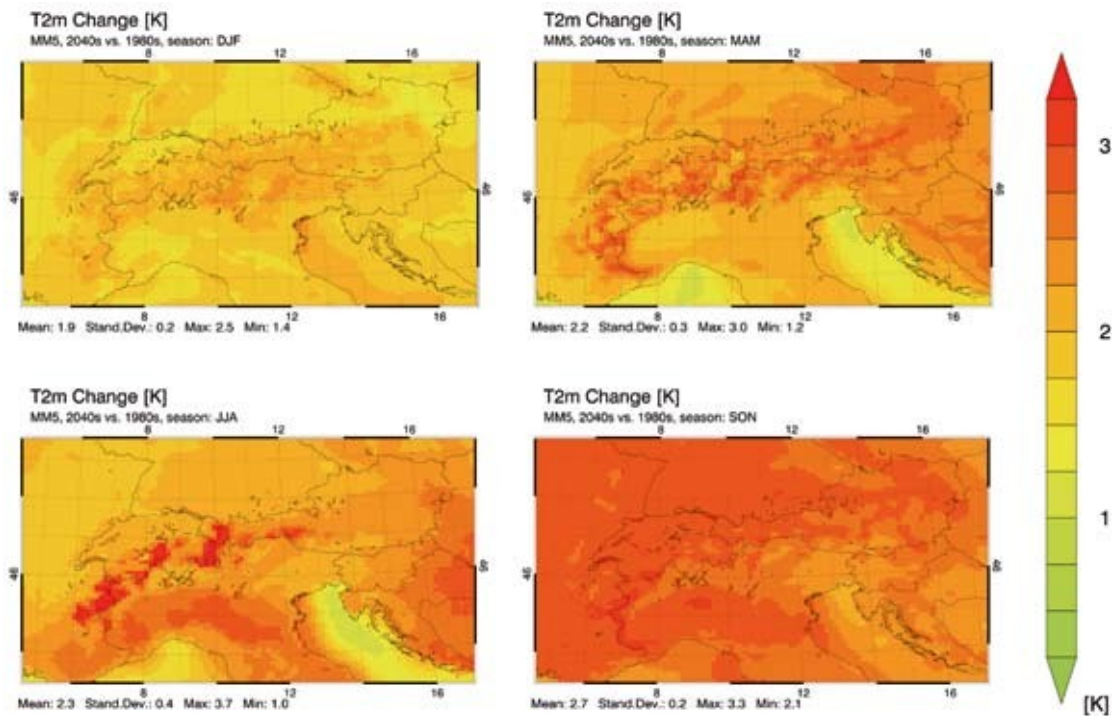


Figure 1: Temperature scenario for 2041-2050 (comparison period 1981-1990). Top left: Winter (December, January, February), top right: Spring (March, April, May), bottom left: Summer (June, July, August), bottom right: Autumn (September, October, November)

in summer south of the Alps. In contrast to PRUDENCE, however, the reclip:more results also show significant decreases in precipitation south of the Alps in autumn. The magnitude of the change in precipitation can be described as consistent, taking into account the different time periods. The range of the three reclip:more simulations is a maximum of ± 5 % depending on the season. According to Déqué *et al* [2007], the regional climate model in the Alpine region 30-50 % to the overall uncertainty, resulting in an overall uncertainty of ± 10 % to ± 15 %. This estimate shows that the climate change signals of precipitation in the Alpine region are only partially significant.

Precipitation changes in particular also show strong regional differences within the Alpine region: In all seasons, except in winter, the main Alpine ridge can be recognised as a clear dividing line between increasing precipitation in the north and decreasing precipitation in the south, which severely limits the significance of the average precipitation changes for the entire Alpine region (positive and negative changes each other out). The advantage of the reclip:more scenarios is that sub-regions around the main Alpine ridge can also be analysed for the first time. Figure 3 shows the precipitation changes in climatologically motivated sub-regions. Depending on the season and sub-region, changes of up to 30 % result. The uncertainties of such small-regional climate change signals are still difficult to assess at present, but initial investigations show that the uncertainties do not significantly as the scale is refined. If these results are confirmed, many sub-regional precipitation scenarios can be regarded as highly significant.

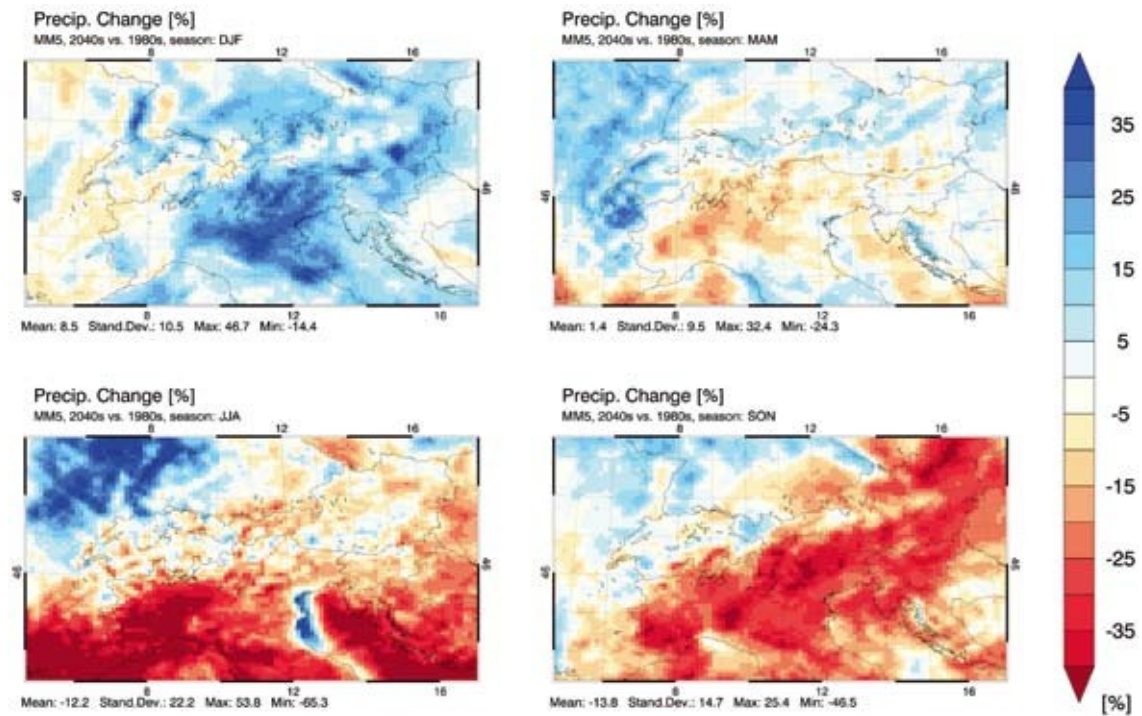


Figure 2: Precipitation scenario for 2041-2050 (comparison period 1981-1990). Top left: Winter (December, January, February), top right: Spring (March, April, May), bottom left: Summer (June, July, August), bottom right: Autumn (September, October, November)

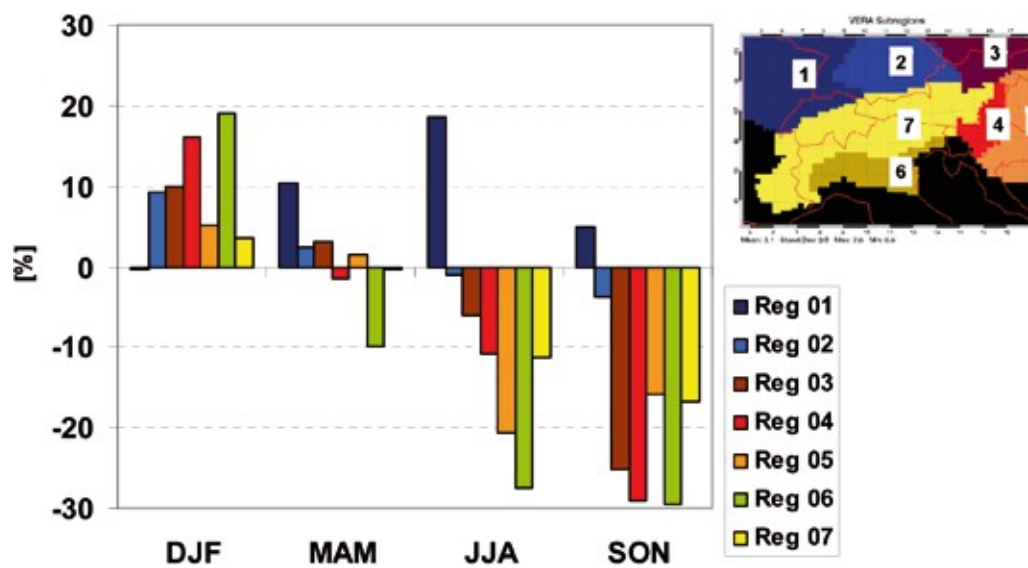


Figure 3: Precipitation scenario for 2041-2050 (comparison period 1981-1990) in subregions of the Alpine region. The results are given in %

3.3 Extreme events

Extreme weather events such as heatwaves, droughts, intense precipitation or wind storms have a high damage potential and therefore have a far greater impact on society than the average climate. When analysing changes in the frequency and intensity of extreme weather events, a regional perspective is particularly important. As with changes in precipitation, it is not possible to draw direct conclusions from global or even pan-European results for extreme events to small-scale regional conditions. This is particularly true for an area as orographically complex as the Alpine region. Climate research is still in its infancy when it comes to small-scale analyses of the effects of global climate change on extreme events and the investigation of their uncertainties, but initial results are available and some rough trends can be presented here. *Beniston et al* [2007] summarises the results of the PRUDENCE project in this regard:

Heat waves

As expected, the climate simulations for the end of the 21st century show a significant increase in the number of hot days (maximum temperature above 30 °C) in Europe in parallel with the general increase in temperature. What is less obvious is that an increase in the frequency and duration of heat waves is also expected and that extreme temperatures in continental regions will rise more than moderate temperatures due to higher temperature variability. Regionally, the clearest signals occur in the Mediterranean region, particularly over southern France and Spain.

Extreme precipitation

Hand in hand with the changes in precipitation totals, the intensity of 5-year precipitation events is expected to increase in northern Europe and decrease in the south. An interesting exception is Central Europe, where increased intensities, especially in the maximum daily precipitation sums, are expected in summer despite decreasing precipitation sums [see also *Frei et al.*, 2006].

Periods of drought

A significantly higher risk of droughts is expected in the Mediterranean region, particularly over southern Spain, the eastern Adriatic coast and southern Greece.

Wind storms

Over the North Sea and Western Europe, a slight increase in winter storm activity (measured at the 90% percentile of the daily mean wind speed) is expected. A decrease in winter windstorms is expected south of the Alps.

Based on the regionally more detailed results of the *reclip:more* project, analyses of the change in extreme events can only be carried out to a very limited extent because the simulation periods (two times 10 years) are very short. However, if one limits oneself to the parameters low

The analysis of precipitation intensity and precipitation frequency shows that areas with decreasing precipitation frequency often have increasing precipitation intensities [Gobiet *et al.*, 2006], which can be interpreted as an indication of more extreme precipitation conditions in the Alpine region.

4 Summary and outlook

Based on the results of the most important projects on future regional climate change in Europe and the Alpine region, the basic characteristics of the climate change to be expected in the 21st century were developed. In the coming decades, a temperature increase of 0.3 to 0.45 °C per decade is to be expected in the Alpine region, with the strongest increases in summer and autumn. In terms of precipitation change, a strong north-south gradient is expected with significantly increasing precipitation in northern Europe (especially in winter) and sharply decreasing precipitation in southern Europe (especially in summer). The Alpine region represents a clear dividing line of climate change, which makes detailed regional analyses necessary in this area. In sub-regions around the Alps, precipitation changes of +20 % in winter (mainly, but not exclusively, in the northwest) and -30 % in summer and autumn (mainly in the south and east of the Alps) are expected. In addition, there are indications of an increased occurrence of heat waves and heavy precipitation, although the reliability of these statements and their small-regional characteristics have not yet been well analysed.

At present, the results of the EU project PRUDENCE are mainly available for analysing future climate change in Europe. More recent results, in particular an extended assessment of uncertainties, are expected from the EU project ENSEMBLES from 2009. Results with a finer spatial resolution for the Alpine region were compiled in the Austrian project reclip:more, although the estimation of uncertainties and the analysis of the change in extreme events are still rudimentary in this case (reclip:more results are available on request for climate impact research, see also Gobiet *et al.* [2008] for an initial application in relation to groundwater recharge in Styria). In autumn 2008, the follow-up project reclip:century will start, which develop continuous climate scenarios of the Alpine region in high spatial resolution from 1951-2050 and will pay particular attention to the bandwidths of uncertainty. These simulations will also be for extreme value analyses and should form the basis for further climate impact research in Austria.

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Address of the authors:

Dr Andreas Gobiet

Heimo Truhetz

University of Graz

Wegener Centre for Climate and Global Change Leechgasse 25,
8010 Graz

Tel. +43-316-380-8448 or -8442

E-mail: andreas.gobiet@uni-graz.at or heimo.truhetz@uni-graz.at

Some thoughts on the question of the increase in extreme meteorological values

Ernst Rudel

1 Introduction

As a result of numerous commentaries and media reports, questions about variations and trends in extreme climate events have also become increasingly important in Austria. In particular, the fact that the economic damage caused by extreme weather events becoming more and more expensive is a reason for many people to consider an increased frequency of such events.

"weather caprices".

Media reports (e.g. in *Der Spiegel*) or "expert testimonies" paint a picture in the public eye. seemingly clear picture of a steady increase in extreme values:

- "Unusually heavy rainfall in Germany, for example, is twice as common today as it was 100 years ago. This can be precisely proven with measurement series in the Alps." (*Der Spiegel*, 2005)
- "Today there is no longer any doubt in science that the climate is becoming more and more extreme!" (Erwin Mayer, climate expert from Greenpeace)
- "Hurricane Katrina with the destruction of New Orleans is the definitive proof of the disasters that threaten us in the future as a result of climate change" (Klaus Töpfer, UNEP, former German Environment Minister).

Assuming that mankind and its activities were really predominantly indirectly and therefore endogenously involved in climate change and the increase in extremes, there would theoretically be a chance to reduce or completely eliminate this influence through active endeavours and to contain the increased danger caused by us.

However, it is extremely difficult to prove this frequency change, as is a worldwide lack of high-quality time series of meteorological data from individual events. Only a few countries have daily data in digital form that goes back more than 50 years and whose quality can be considered reasonably assured through suitable statistical processing. Studies and analyses often show short time series (< 50 years) over which a straight line is laid that shows increasing values. This is often regarded as clear evidence of an increase in weather extremes. However, statements about climate variability and climate extremes precise knowledge and a good understanding of the climate of the past. This knowledge forms the necessary basis for understanding future climate changes. It only becomes a cause for concern when it moves significantly away from the range of past climate variability.

The historical development of extreme events is not as easy to recognise as that of average conditions. In order to prove the statistical significance of changes in the frequency of rare events, much longer data series are required than for frequent events. In addition, the problem of reliable recording also for some events:

For example, storms, floods, avalanches, etc. are generally only recorded if they cause damage or at least fall in areas where they could potentially have caused damage. Extreme events in unpopulated areas were and are generally not or rarely recorded. The statistics are therefore incomplete and contain many sources of error.

2 Definitions

An extreme event in the meteorological sense is an event that deviates strongly from corresponding average values or trends and is extraordinary. The return period is usually significantly longer than ten years. Extreme events are rare events that deviate strongly from a statistical average. They are not necessarily associated with damage. A natural disaster, on the other hand, is a damaging event whose consequences cannot be dealt with by those affected alone. It is not always caused by extreme events. Natural hazards are natural processes and conditions that threaten society and the environment. In this report, extreme events are defined as events that deviate significantly from an average value or trend and are therefore extraordinary. Extreme events are therefore defined by statistical criteria and extreme events often have negative consequences for people and the environment.

3 Climate models and their significance for extreme events

Global climate models make it possible to determine the relative importance of the individual influencing factors on the climate. The following facts have been clearly demonstrated:

- The largest contribution to the observed warming comes from the increased concentration of greenhouse gases, with CO₂ contributing around half and methane a quarter.
- Contrails caused by air traffic and the resulting formation of cirrus clouds do not play a major role - the main effect of air traffic is the increase in greenhouse gas concentrations.
- The contribution of fluctuations in solar activity is significantly less than the effect of the Greenhouse gases.
- Limitation: Scientifically proven knowledge about various variables that control the climate is comparatively limited. Surprises in the climate system can therefore still not be ruled out.

However, climate models do not indicate an increase in meteorological extremes, i.e. climate models do not say that there will possibly be more precipitation and thus also flooding, nor do climate models show that higher wind speeds and storms could become the rule.

However, climate models are forecasting higher average temperatures. This also increases the probability that extreme tropical temperatures (above 30 °C) will occur on more days.

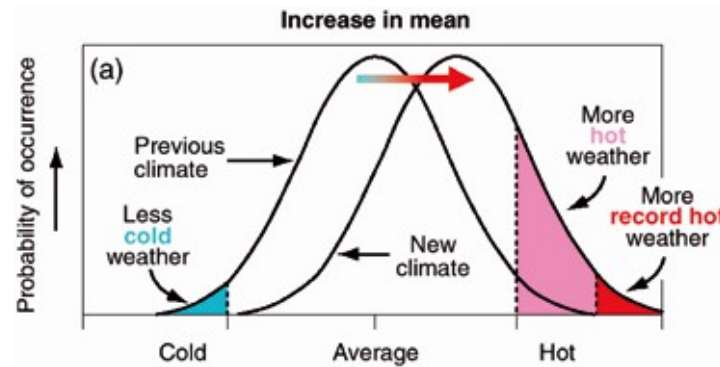


Figure 1: Shift in the probabilities of hot weather with an increase in mean temperature (after IPCC 2007)

This can also be proven with many series of measurements:

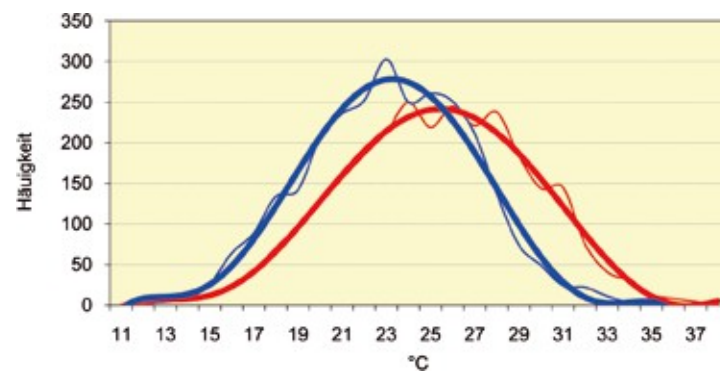


Figure 2: Distribution of the daily maximum temperature in Vienna during the summer months of 1901-1930 (blue) and 1971-2000 (red) (source: ZAMG database)

However, this only applies to temperature, as other meteorological elements such as precipitation or wind peaks do not follow a "normal distribution" and a shift can be recognised in the mean values in the series of measurements, but not in the extremes in the "sea tails"

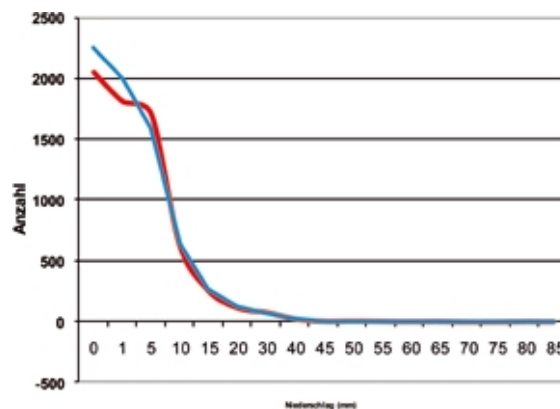


Figure 3: Distribution of daily precipitation values in Vienna in the period 1948-1977 (blue) and 1978-2007 (red) (source: ZAMG database)

4 Data series

The Alpine region has the oldest and most reliable series of measurements, which are of a much higher quality than in most other parts of the world. The oldest Austrian measurement series originate from Kremsmünster Abbey. They date back to the year 1768. Records of days with extreme amounts of precipitation have also been kept there since 1874. However, contrary to general expectations, there has been a downward trend in the number of days with more than 50 millimetres of precipitation in Kremsmünster over the last 130 years (see *Figure 4*).

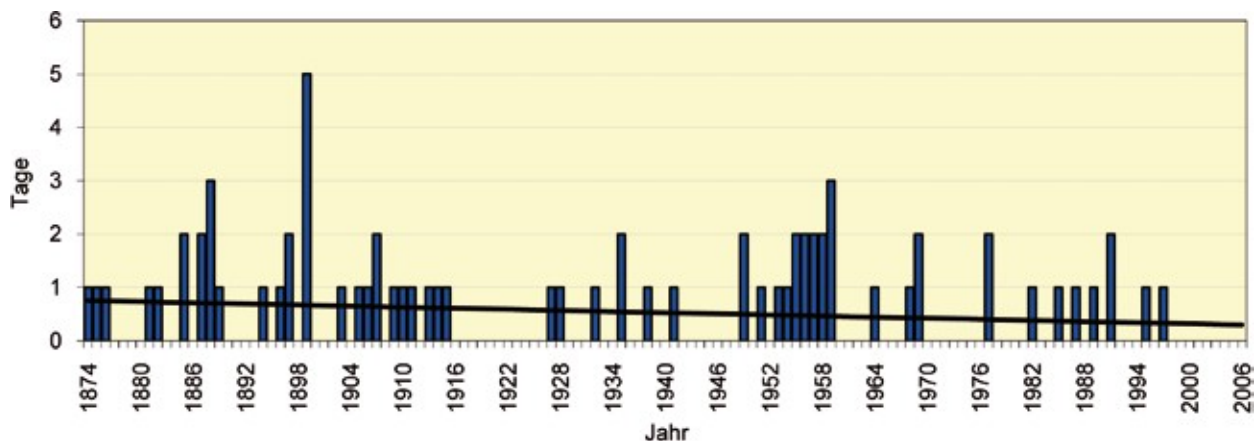


Figure 4: Number of days with more than 50 mm precipitation in Kremsmünster in the period 1874-2006 (Source: ZAMG database)

The records of heavy precipitation in Vienna show a similar picture: between 1903 and 2001, the number of days with more than 30 millimetres of precipitation in Vienna decreased significantly. However, if we only look at the past 40 years, we see the opposite trend in the same category.

The list of examples could be continued indefinitely. And of course with other meteorological data series such as extreme storms and hurricanes, thunderstorms, hail etc.

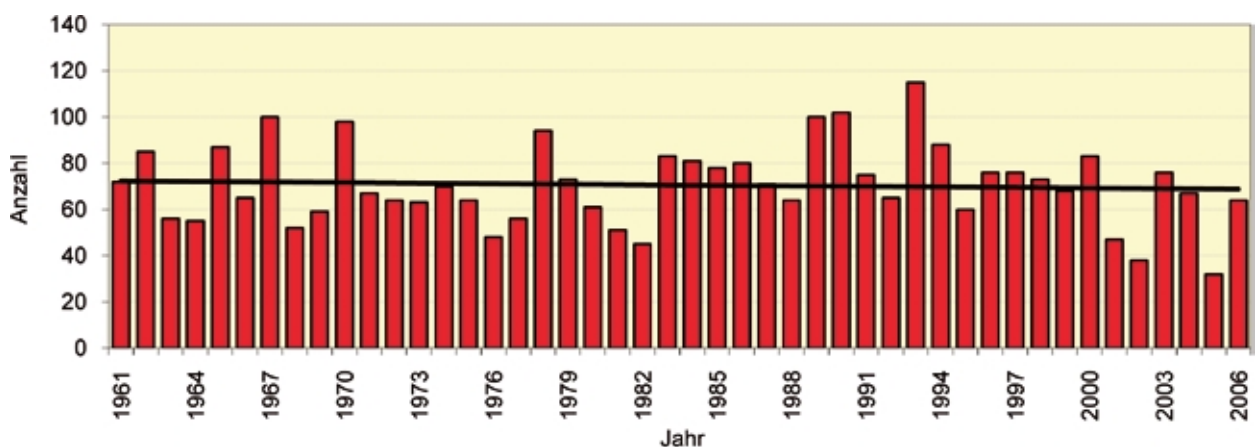


Figure 5: Number of hailstorms in Austria. Sum of reports from 44 stations (source: ZAMG database)

The probability of hail occurring in Austria varies greatly from region to region. South-eastern Styria and the Danube valley in particular are considered to be at risk of hail. A long-term study of the frequency of occurrence and intensity of hailstorms is not available for Austria.

There is a publication from Slovakia: R. Drinka, I. Bašták Ď., M. Gera: *Climate Changes and Their Impact on the Statistical Structure of Atmosphere Stability Diagnostic Parameters Needed FOR Downscaling*

Assuming an A1B scenario, the probability of a stabilisation of the stratification of the atmosphere decreases and thus also the probability of thunderstorms and hail. The results can be easily transferred to Austria.

Worldwide, there are hardly any measurement series recognised by science that allow directly measured statements about the changes in this climate element. This is because the series of measurements are usually far too short and are also inhomogeneous due to the change in measuring instruments, especially at non-comparable locations (urbanisation). The statements circulated in the media such as "strongest storm ever measured" are therefore hardly surprising against this background. Earlier "stormy times" are either not available in the climate archives or are not comparable with modern measurement series are simply not comparable.

Depending on the message you want to convey, trend analyses can be presented in a popular way and conclusions can be drawn. But even these "short-term analyses" do not show a uniform trend worldwide. In some areas, there does indeed appear to be an increase in extreme events, while other data sets seem to document a decline. In short, there is no uniform trend for an accumulation of extreme weather events that could be applied to the entire globe.

Studies from the USA show that media activities in highly developed industrialised countries such as North America, Western Europe, Australia and Japan draw much more attention to extreme weather events than in other countries. As a result, the perception of these events appears to be increasing.

There is no doubt that our industrialised society as a whole has become much more vulnerable to extreme weather events than it was a few decades ago. Infrastructural progress in particular makes us susceptible to damage caused by floods or storms, for example. Damage to power plants, wastewater treatment plants, oil tanks, etc. and, as a result, environmental pollution on a massive scale naturally drive economic losses to enormous heights, even in the event of minor extreme weather disasters. It is therefore no wonder that the insurance industry is trying to show us exponentially increasing expenses to cover damage caused by natural disasters and is doing so successfully.

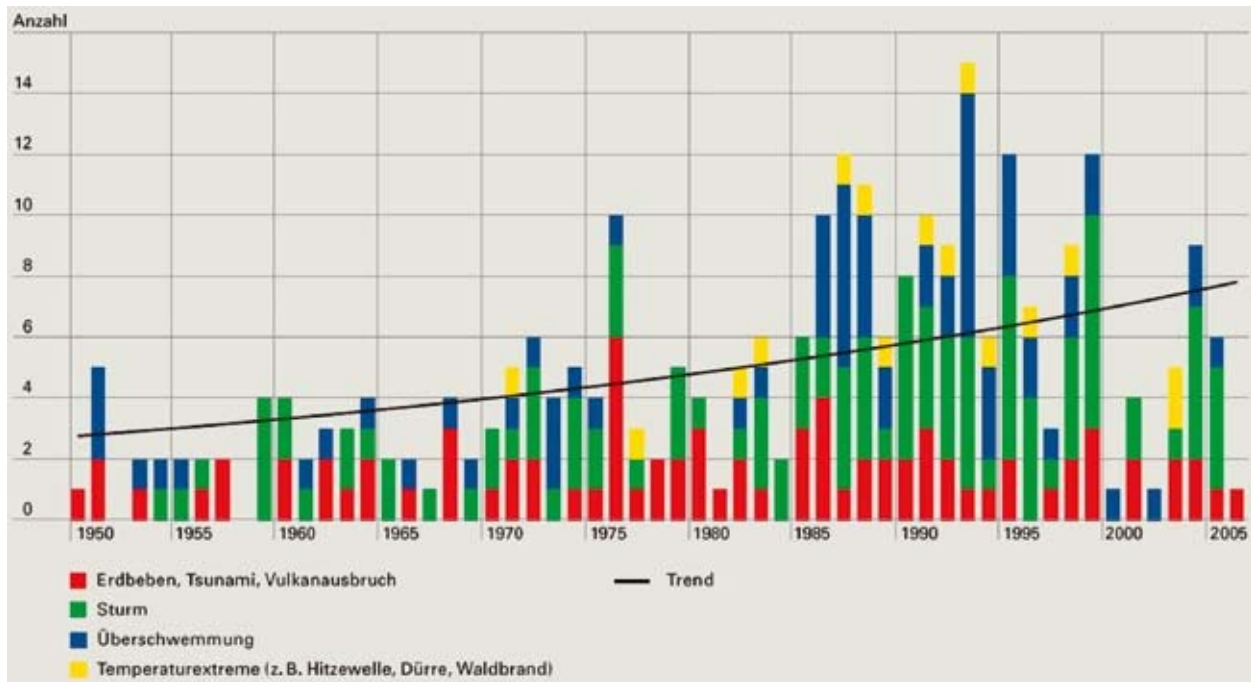


Figure 6: *The view of the insurance industry (source: Munich Reinsurance Company)*

Figure 6 shows the raw data, which is strongly influenced by factors such as settlement structure, population growth, social structure, concentration of population and capital in conurbations, settlement and industrialisation in "exposed" areas. A study from the USA shows that if all these factors are taken into account, the extreme trend of the upper curve in its "normalised" form, as the study describes it, has disappeared. Although the great variability can be seen from year to year, no long-term trend can be identified. The correlation with warming has disappeared.

A similar picture can be seen in the following illustration:

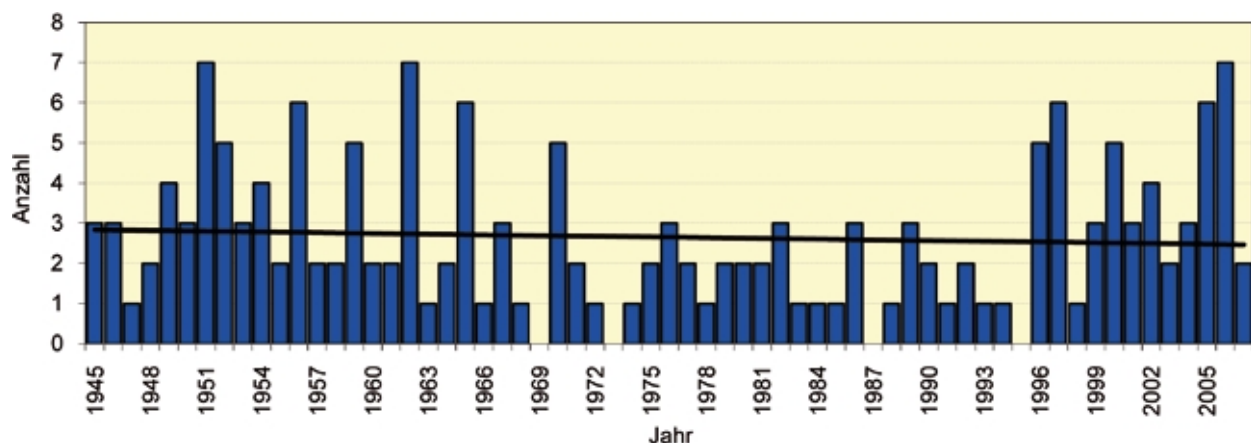


Figure 7: *Intensity 3-5 hurricanes that made landfall in the US (Source: NCDC pers. comm. Communication)*

5 Outlook

Analyses of historical sources in Central Europe show very clearly that weather disasters were a constant companion. This applies to all the variants mentioned, such as thunderstorms, storms and floods. Studies of long-term trends of various extreme weather events (storms, precipitation, thunderstorms, hail) in Central Europe, which are currently being carried out at ZAMG, show that there is no evidence of a significant trend towards an increase in these events in the Alpine region over the last three hundred years. The statement that, in addition to a rise in temperature, changes in precipitation and the occurrence of extreme weather events such as storms and floods can also be observed cannot be substantiated. On the contrary, many sources clearly show that in some phases of climate development over the last 1,000 years, disasters have occurred more frequently than in the last 100 years. Individual events even considerably exceed the extreme cases that afflict us today.

Here, however, it is necessary to point out the problems with the sources, as extreme weather events were usually only described descriptively in history and an evaluation in the modern sense is not available.

Weather extremes are an integral part of the weather, which can sometimes lead to very unpleasant side effects for people, but which have always had to be reckoned with. Inter-annual variability has been, is and will continue to be high in the future. Years with many extremes will be followed by "calm" years. What has changed over the course of history, however, is above all the settlement behaviour of humans. The populated area has expanded considerably in recent centuries, and risk zones have been and continue to be built on. Avalanches, mudslides and localised flooding have devastating effects for this reason alone. Huge halls with storm-prone flat roofs are erected in the landscape, usually disregarding the maximum possible snow pressure, because we live under the mistaken assumption that hardly any snow is to be expected in the future anyway. In recent decades, housing estates and industrial plants have increasingly been built in flood zones due to a lack of space and the trend is continuing.

Author's address:

Dr Ernst Rudel
Central Institute for Meteorology and Geodynamics
Hohe Warte 38, 1190 Vienna
Tel. +43-1-36026-2201
e-mail: ernst.rudel@zamg.ac.at

Climate change and water balance in southern Germany - results of the KLIWA co-operation project

Hans Weber

1 Introduction

It is now undisputed among climate researchers worldwide that the observed global temperature increase in recent decades is essentially not attributable to natural climate influences and climate fluctuations, but is to be categorised as a consequence of the anthropogenic greenhouse effect. In order to estimate the further development of global climate change, climate researchers are calculating global climate projections using increasingly complex climate models. All of these global models predict progressive warming; their quality and suitability are demonstrated by the fact that they can reproduce past temperature trends quite well with corresponding emission scenarios. Depending on the climate model and emissions scenario, the calculated global temperature increase over the next 100 years is most likely to be between 1.8 °C and 4.0 °C. The United Nations Intergovernmental Panel on Climate Change (IPCC) reconfirmed this trend in its Fourth Assessment Report (2007) and further substantiated it scientifically.

Current climate change is often reduced to global warming. However, due to the physical interrelationships, it is to be expected that the global water cycle will also intensify as a result of the higher heat energy available. This is then likely to have a considerable impact on the water balance at various spatial and temporal scales and, as a result, on the following areas of water management in particular:

- technical flood protection and flood prevention due to possible changes in the height, duration and frequency of floods
- the water supply due to the change in groundwater recharge with effects on the availability and quality of water supplies
- water protection through the change in seasonal discharge and water temperature conditions with effects on the material balance of rivers and lakes
- the management of water bodies by changing the mode of operation, in particular of Water reservoirs and hydropower plants as well as other water uses
- the planning and dimensioning of municipal drainage systems through a stronger consideration of impacts due to the possible intensification of heavy precipitation.

A lack of knowledge about the extent and magnitude of the effects of regional climate change was the reason for the co-operation project "Climate Change and Consequences for Water Management (KLIWA)" in 1999. Since then, this long-term project has been carried out by the states of Baden-Württemberg and Bavaria together with the German Weather Service (DWD), and the state of Rhineland-Palatinate has also been involved since 2007. The aim of the KLIWA project is to obtain information on the possible regional effects of climate change on the water balance and the systems dependent on or influenced by it for the coming decades (up to 2050) in order to be able to develop suitable adaptation strategies on this basis.

2 Conception of the KLIWA co-operation project

The project currently comprises 5 main topics with the associated areas of investigation and work (Figure 1), which are bundled in the process concept. Each project area is made up of a series of sub-projects.

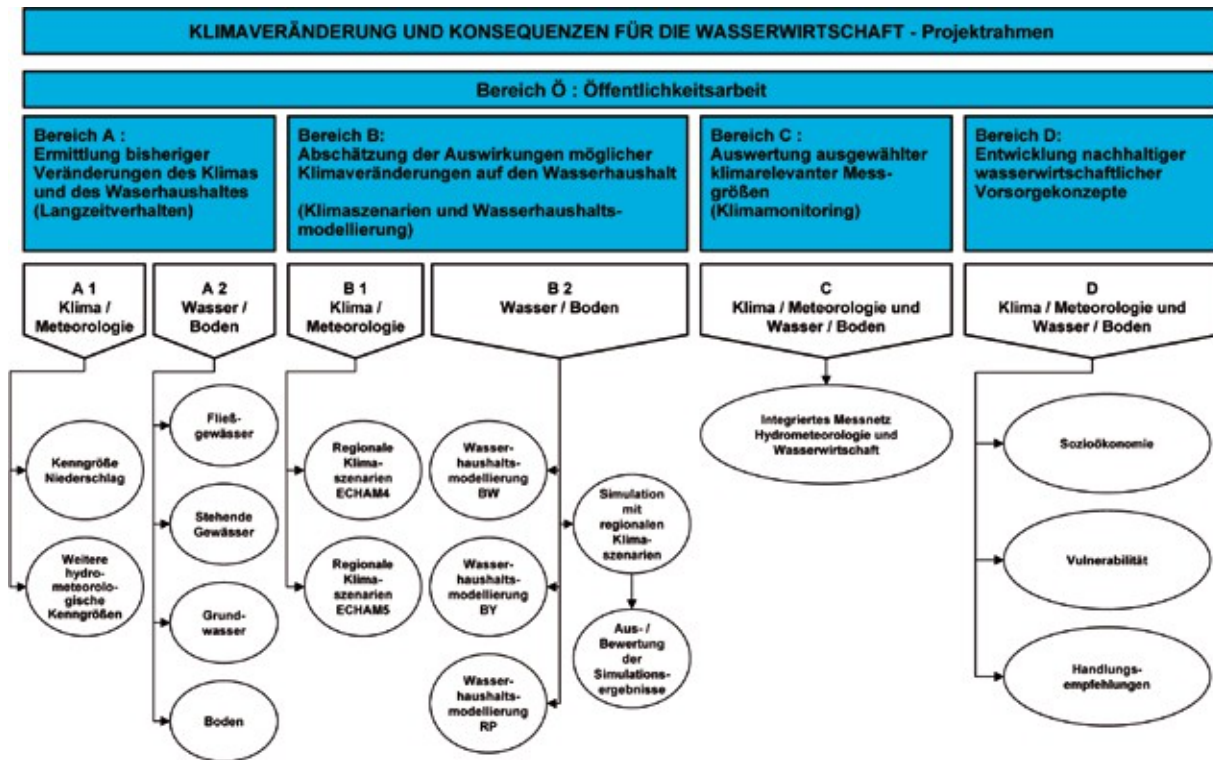


Figure 1: KLIWA project - project structure and areas of work

Project area A is dedicated to analysing previous changes in the climate and water balance; the available long measurement time series were statistically evaluated according to agreed procedures. Project area B is concerned with determining the hydrological effects to be expected in the future, which are estimated with the help of regional climate projections as data input and subsequent water balance modelling. Project area C is designed as a longer-term monitoring programme of the water balance for southern Germany via an integrated hydrometeorology (DWD) and water management (Bavaria, Baden-Württemberg and Rhineland-Palatinate) measuring network and represents climate monitoring in its future application.

Based on the knowledge gained about the effects of regional climate change, possible impairments of water management utilisation and protection functions will be investigated. Recommendations for appropriate adaptation measures will then be developed to minimise negative impacts. Project area D has so far focussed on the problem of possible flood aggravation and the consequences for flood protection. The expected changes in low flows and the development of groundwater recharge are currently at the centre of the investigations.

The problems of climate change, project activities and results are presented to decision-makers and the population in project area Ö (public relations). Raising awareness of the problem of climate change is seen as a basic prerequisite for necessary adaptations and climate protection measures, so this area of work is of particular importance within the project. The project's website (www.kliwa.de), on which the research results of the cooperation project are comprehensively documented and published, is seen as an important information medium.

3 Climate development in southern Germany in the 20th century

The investigations of long measurement time series of hydrometeorological and hydrological (usually more than 70 years) provide information on the natural fluctuation ranges observed to date and any systematic changes (trends) in the observed values that may already be recognisable. This allows objective conclusions to be drawn about regional climate development in the 20th century on the basis of the existing measurement series. To this end, some of the measurement series are summarised on a regional basis and average representative statements are made for the KLIWA study areas and the KLIWA regions (33 river basins and 9 regions for the joint study area of Baden-Württemberg and Bavaria). Some of the key findings are presented below.

3.1 Air temperature

Air temperature plays a special role in the water cycle, as it significantly determines evaporation and the capacity to absorb water vapour in an air mass. The air temperature therefore has a considerable influence on precipitation patterns. Knowledge of the temperature development over the past decades is therefore of particular interest. The results were analysed and presented on an area-specific basis and also chronologically for the hydrological half-years and calendar months. The annual mean temperatures of the study areas are between 6.4 °C and 9.7 °C and are characterised by significant increases. The calculated trends of the two half-years for the period from 1932 to 2005 are between 0.6 °C and 1.5 °C (*Figure 2*). The seasonal and region-specific characteristics are also noteworthy: The highest increase in mean monthly temperatures occurs in the winter months, particularly in December and for the months of August and October, while hardly any changes can be detected for April and September.

3.2 Snow cover duration

Due to the observed increase in temperature in the winter months, solid precipitation takes a back seat to sleet or rain. The studies in southern Germany on the long-term behaviour of snow cover duration therefore show a trend towards winters with less snow and less permanent snow cover. At low (< 300 metres above sea level) and medium altitudes, the duration of snow cover is decreasing significantly. This decline is more pronounced in the western parts of southern Germany than in the eastern parts. In higher-altitude areas (> 800 m above sea level), there is only a slight decrease in snow cover duration or even a slight increase in some cases.

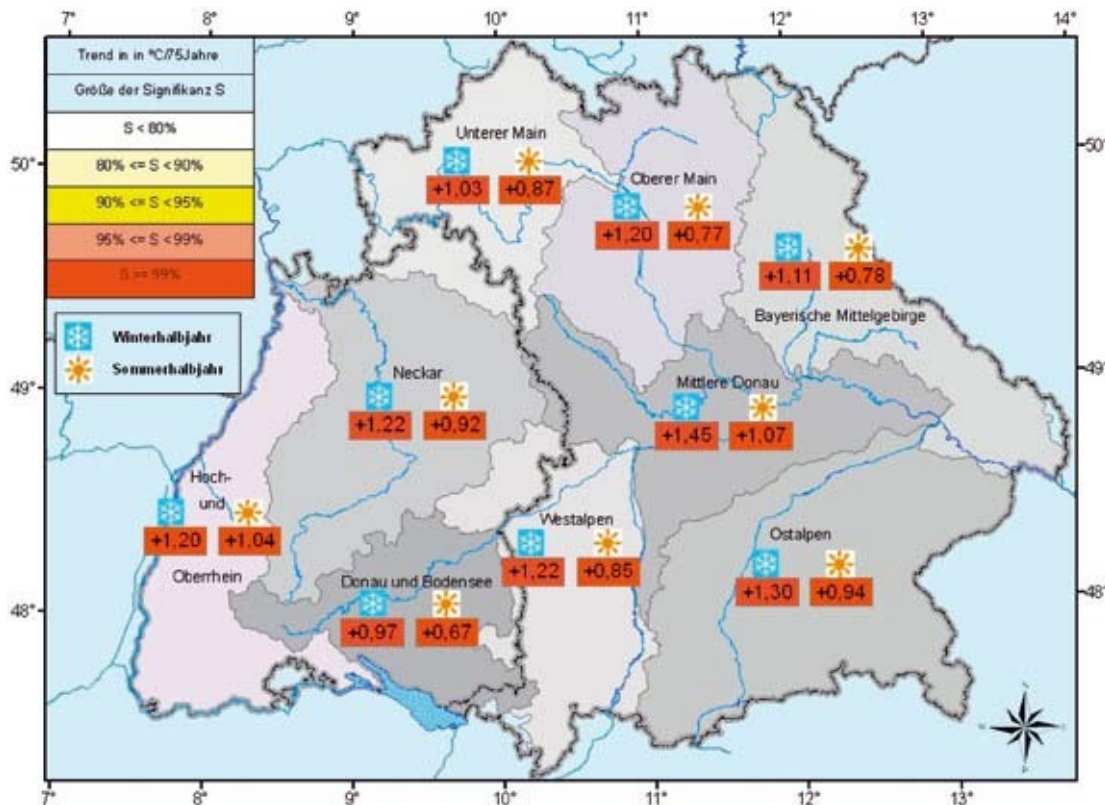


Figure 2: Development of area mean temperatures in the hydrological half-years / absolute trend in the period 1931-2005 (trend in $^{\circ}\text{C}$) with indication of significance

3.3 Behaviour of precipitation, heavy precipitation

The annual precipitation levels in the individual regions of southern Germany have remained more or less the same in recent decades or have even increased slightly. However, the intra-annual precipitation regime has changed: there has been a shift in precipitation from the summer to the winter half-year; in winter, precipitation has increased significantly. This development is shown as an example for the half-yearly precipitation of two river basins in northern and southern Bavaria, namely the Regnitz and the Isar (Figure 3).

The behaviour of heavy precipitation, which sometimes leads to flood discharges in conjunction with snowmelt events, has been studied in depth due to its relevance to water management. Heavy precipitation refers to a precipitation event that has a high precipitation intensity in relation to its duration and therefore occurs infrequently, e.g. on statistical average no more than once a year. In the KLIWA studies, only the heavy precipitation of higher durations (24 to 240 hours) and the respective maximum values of a time period (year, summer and winter) have been considered so far due to the data situation. Long series of measurements from over 400 precipitation stations have been statistically analysed for this purpose. In addition to an initial station-related evaluation, an area-related analysis was also carried out as part of the first climate monitoring programme for the time series extended up to 2005.

In the winter half-year, heavy precipitation heights in southern Germany increase significantly at the measuring stations - albeit with regional differences - and the trends are also predominantly significant.

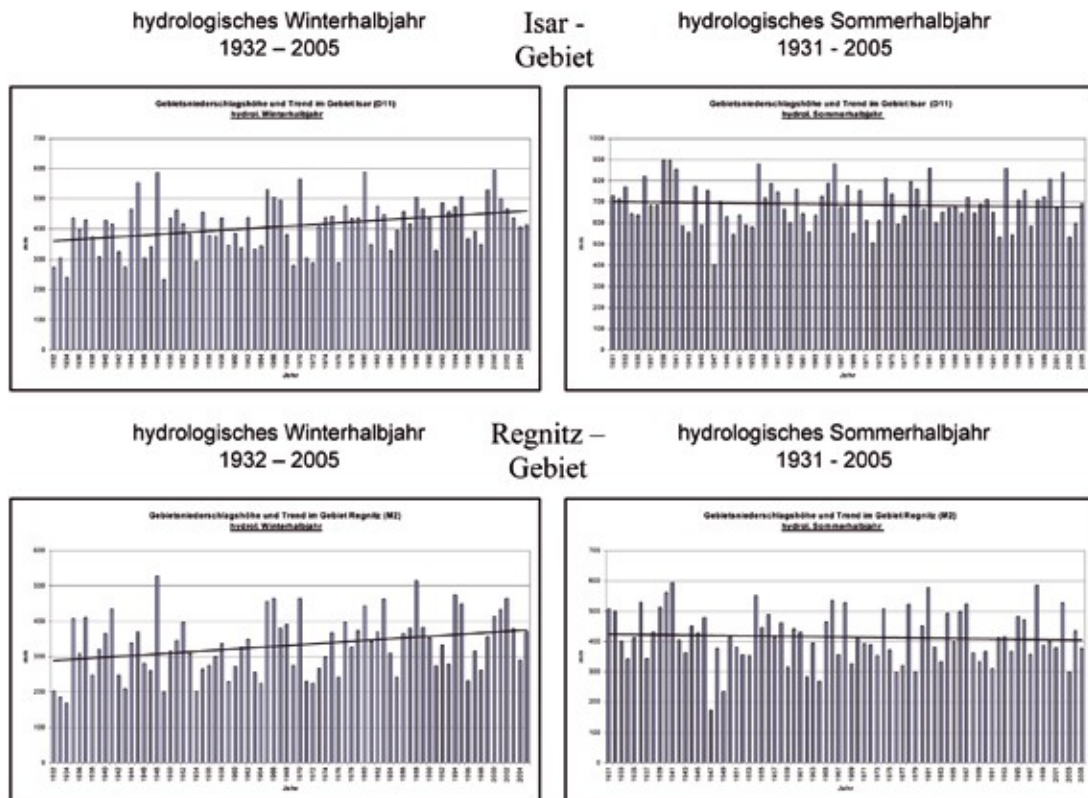


Figure 3: Trend behaviour of the half-yearly regional precipitation series in the river basins of the Regnitz (northern Bavaria) and the Isar (southern Bavaria)

This also applies to the time series of the regionally aggregated measurement series (Figure 4). In contrast, no striking uniform changes can be observed in summer. The cause of the increase in winter heavy precipitation levels is seen in the more frequent occurrence of westerly weather conditions, combined with the influx of warm, humid air masses.

3.4 Change signals in the discharge time series

In connection with water management issues, the discharge regime of our water bodies and its possible changes are of interest. The long-term behaviour of flood discharges was initially the focus of KLIWA, as public opinion often equates climate change with the occurrence of large floods, such as those that have been recorded more frequently in recent times. Conventional statistical methods such as linear trend analysis and the Mann-Kendall significance test were primarily used for the statistical analyses of the annual series and the annual cycle (Fig. 5). The analysis of a total of 90 long gauging station time series with the annual maximum discharge values (observation period 1932-2005) initially shows: 73 of 90 time series show increasing trends. Using the significance test, however, only around 40 % of the trends are shown to be weakly to strongly significant. The hydrological winter half-year shows comparable behaviour with regard to the evaluation criteria of trend and significance. In the hydrological summer half-year, the number of gauging stations decreases slightly as the trend in flood discharges increases. Overall, however, the discharge series with a positive and significant trend clearly predominate (see Table 1).

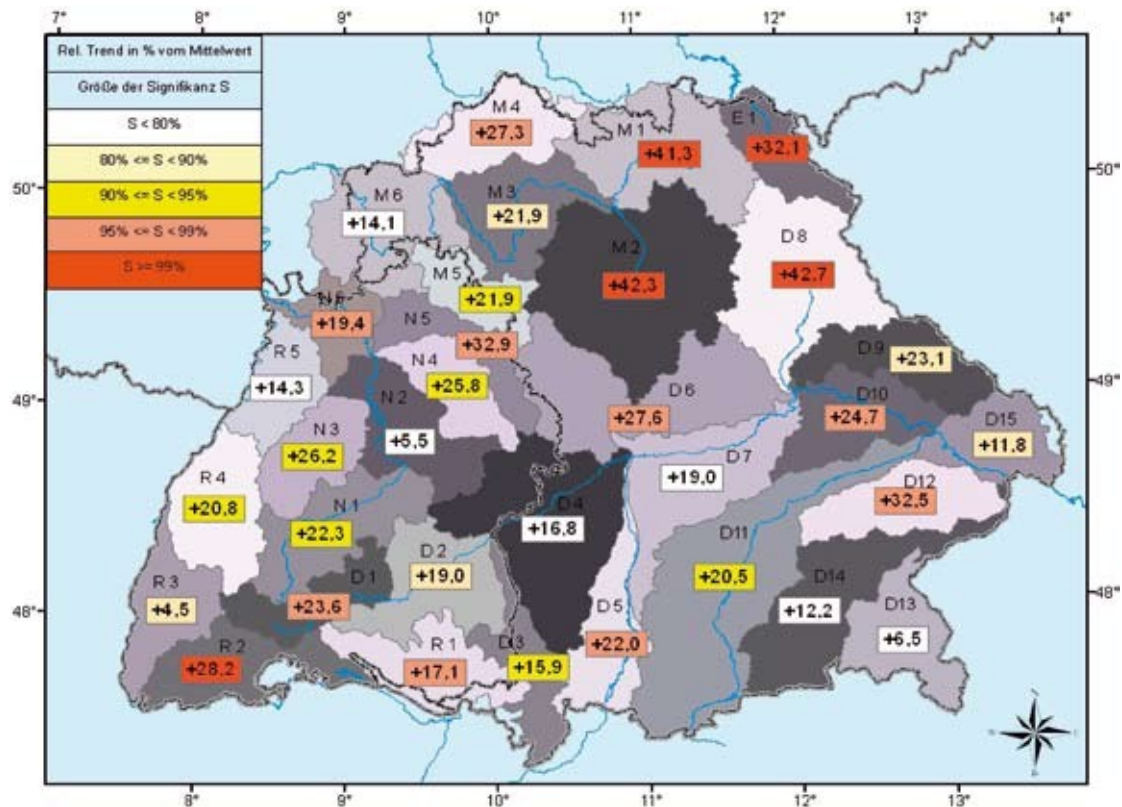


Figure 4: Trend behaviour of regional heavy precipitation (duration 24 h) in the hydrological winter half-year (relative trend as a percentage of the mean value for the period 1932-2005)

Table 1: Overview of the trend behaviour of flood discharges at 90 gauging stations in Baden-Württemberg and Bavaria for two different time periods

Tendencies	Investigation period from 1932-1998*	Investigation period from 1932-2005*
Hydrological year as a whole (Nov. - Oct.)		
↘ Level with decreasing trend / significant	26 / 3	17 / 2
↗ Level with increasing trend / significant	64 / 21	73 / 41
Hydrological winter half-year (Nov. - Apr.)		
↘ Level with decreasing trend / significant	16 / 3	13 / 3
↗ Level with increasing trend / significant	74 / 32	77 / 44
Hydrological summer half-year (May - Oct.)		
↘ Level with decreasing trend / significant	38 / 5	26 / 3
↗ Level with increasing trend / significant	52 / 25	64 / 30

A comparison of the survey results for the periods 1932-1998 and 1932-2005 reveals clear shifts in the results. Due to the extended investigation period up to 2005, an increase in water levels with a trend towards flood aggravation can be observed, while less downward trends can be observed ana logically. The major flood events have an effect here.

the last few years in Bavaria. In addition, the significance of the trends has increased. In the analysis up to 1998, only around 25 % of the trends were significant on average; by 2005, this figure had risen to 40 %. Looking at the partial time series from the 1960s of the last century onwards, there are also clear tendencies for increasing trends in both maximum discharges and the frequency of winter floods. However, these changes also show the influence that the length of the time series can have on statistical analyses.

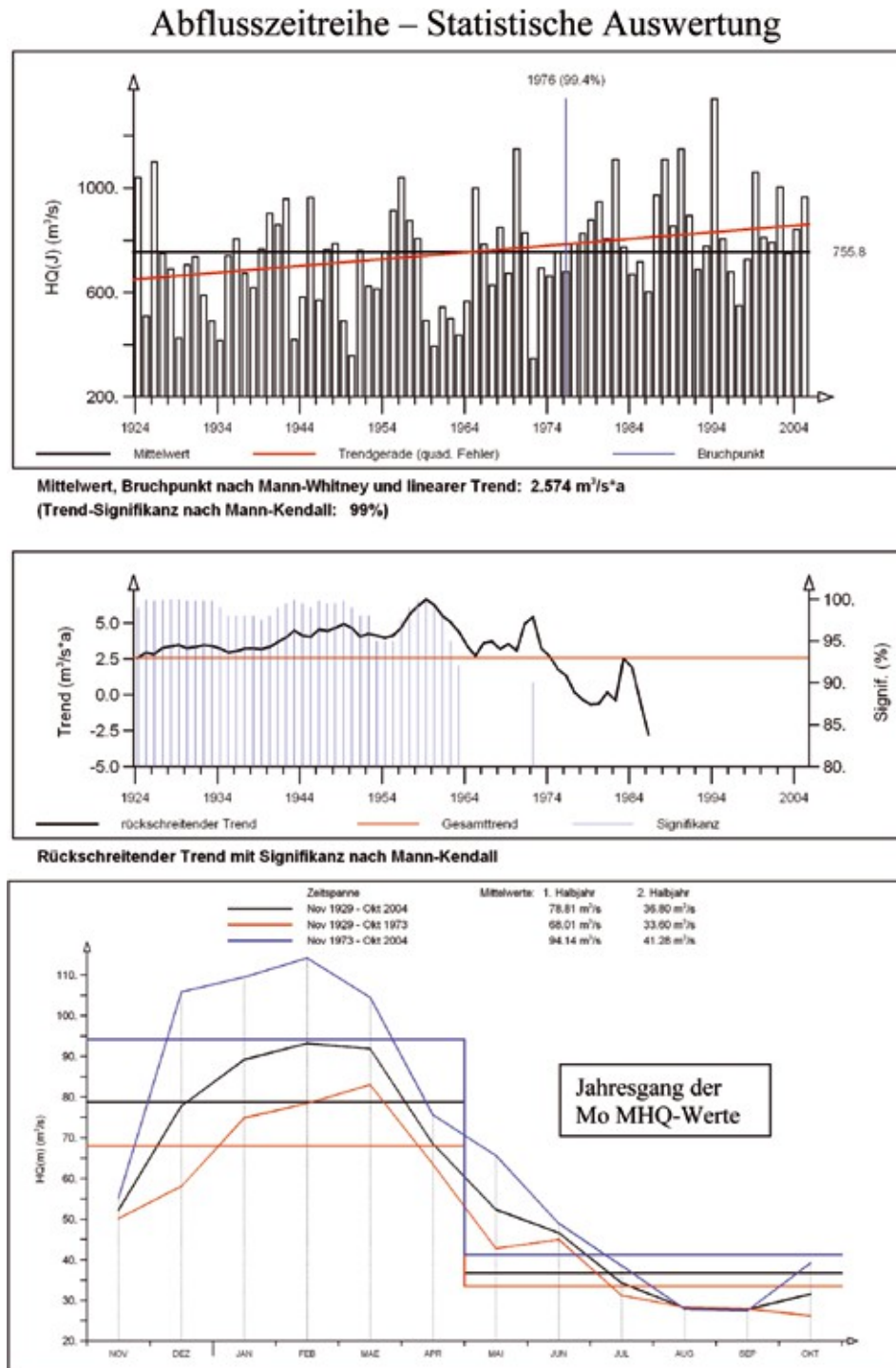


Figure 5: Example of the results of statistical analyses of the long-term behaviour of discharge time series (note: different gauging stations)

These findings on the development of the annual and semi-annual values are also supported by the evaluation of the annual course of the monthly maximum discharge values and the detectable changes. The average monthly flood discharges in the winter half-year have been higher at the majority of gauging stations in recent decades than in the previous period (*Figure 5*, lower diagram). This observed development is consistent with the evaluation of the trend behaviour, according to which the time series of the last decades predominantly show a tendency towards increasing trends, even if the statistical findings are not as clear as in the case of heavy precipitation.

3.5 Evaluation of the KLIWA results

The changes observed in the meteorological parameters of temperature and precipitation, together with the changes in flood discharge, are interpreted by KLIWA as the first clear effects of the human-influenced climate change that began a few decades ago.

The above results show that the effects of global climate change have also been making themselves felt in southern Germany for some time and have intensified in some cases in recent years. Firstly, this is the sharp rise in temperature, but also the change in the precipitation regime. Both have a significant impact on the regional water cycle and, according to current knowledge, lead to an increase in the risk of flooding, but also to a temporary reduction in water availability in connection with longer, hot dry periods. This raises the question of how the effects of climate change will develop in the future.

4 Modelling climate development up to 2050

The trends in climatological and hydrological parameters determined in the previous measurement time series cannot be extrapolated directly into the future, as the climate processes and their complex interactions non-linear and may change over time. In order to obtain statements about possible changes in the climate in southern Germany over the next few decades, regional climate scenarios therefore had to be developed. As there as yet no optimal method for this, various institutions were commissioned as part of KLIWA to create regional climate scenarios using three different methods. The data sets of the ECHAM4 global model were used as the basis.

In order to obtain comparable results, largely identical boundary conditions were specified: Measurement data 1951-2000, verification period 1971-2000, global model ECHAM 4 with the IPCC emission scenario B2, scenario period 2021-2050. After comparing and evaluating the results from three downscaling methods - one dynamic and two statistical - which, as expected, provided a certain range in the results, further analyses were primarily based the results of the WETTREG method. This modelling approach provides the following statements on temperature and precipitation trends for the period 2021-2050 compared to 1971-2000 for southern Germany.

4.1 Temperature development

The air temperature in southern Germany will continue to rise significantly in the future. The climate scenario used determines an increase in the annual average air temperature of approx. 1.7 °C for a possible climate development by 2050.

Winter half-year:

The increase in the mean air temperature of almost uniformly approx. 2 °C is more pronounced than in summer, which is associated with, among other things, a decrease in the average number of ice days (days with $T_{\max} \leq 0$ °C) by almost half.

Summer half-year:

With a significant increase in the mean air temperature of approx. 1.4 °C, the number of hot days ($T_{\max} \geq 30$ °C) is expected to double in large parts of southern Germany.

4.2 Precipitation conditions

For the selected future scenario, the increases in annual average precipitation values amount to approx. 8 % with a range of 0 % to approx. 17 %. Large-scale precipitation will predominantly decrease slightly in summer in southern Germany. On the other hand, winter precipitation is expected to increase significantly (*Figure 6*). Depending on the region the increase can be up to 35 %. A striking feature of the annual precipitation totals is an area with relatively high changes in the south-west-north-east half-arc of the Black Forest-Odenwald-Spessart-Rhön-Frankenwald. These changes will also have a significant impact on water runoff in the future. Quantitative reliable

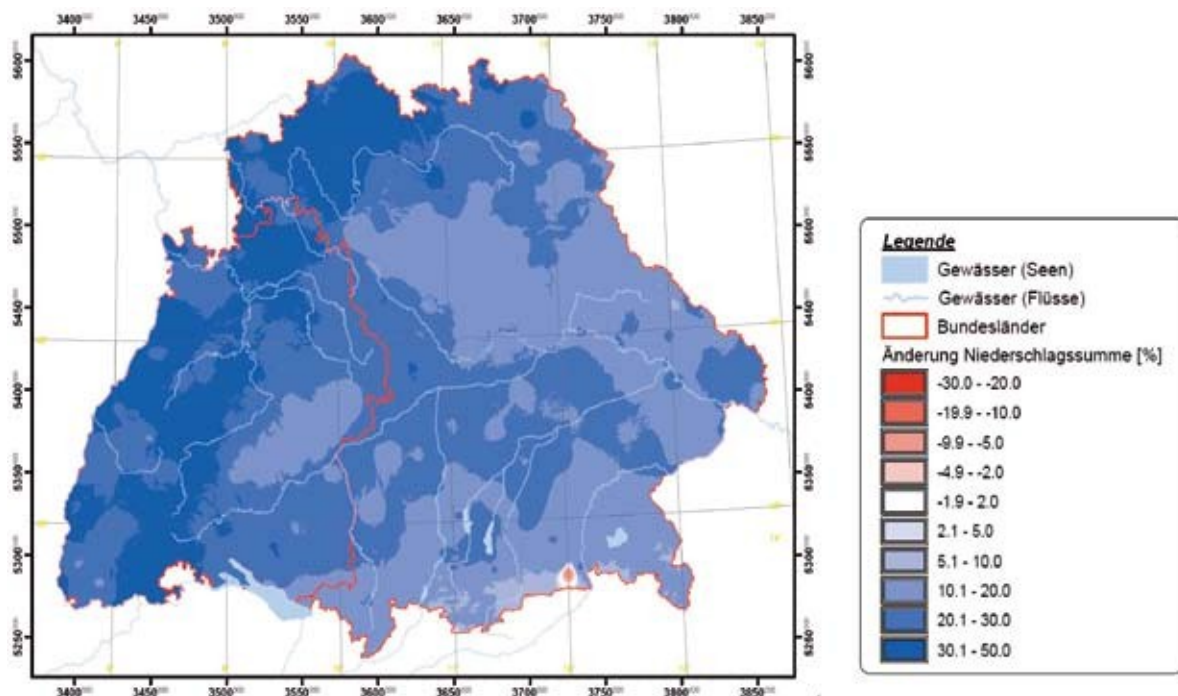


Figure 6: Change in future precipitation levels in the winter half-year compared to today: difference between scenario (2021-2050) and actual state (1971-2000)

However, statements on the future development of convective short-term precipitation (thunderstorms), which are important for urban drainage and flooding in small catchment areas, cannot yet be provided by the available regional climate models.

4.3 General conclusions

The results of the further development of climate change based on regional climate models can generally be summarised as follows:

- The warming continues. The average air temperature will continue to rise, with stronger in winter than in summer.
- Precipitation will increase in winter, while in the summer months there is a tendency for a slight decrease is to be expected.
- An increase in the duration and frequency of rainy westerly weather conditions (especially westerly cyclones), which are important for the development of flooding in winter, is also to be expected. Overall, more frequent occurrence of extreme weather events is to be expected.

These changes will also have a significant impact on the water balance in the future, especially on water runoff.

5 Effects of climate change on runoff conditions

5.1 General information on the procedure

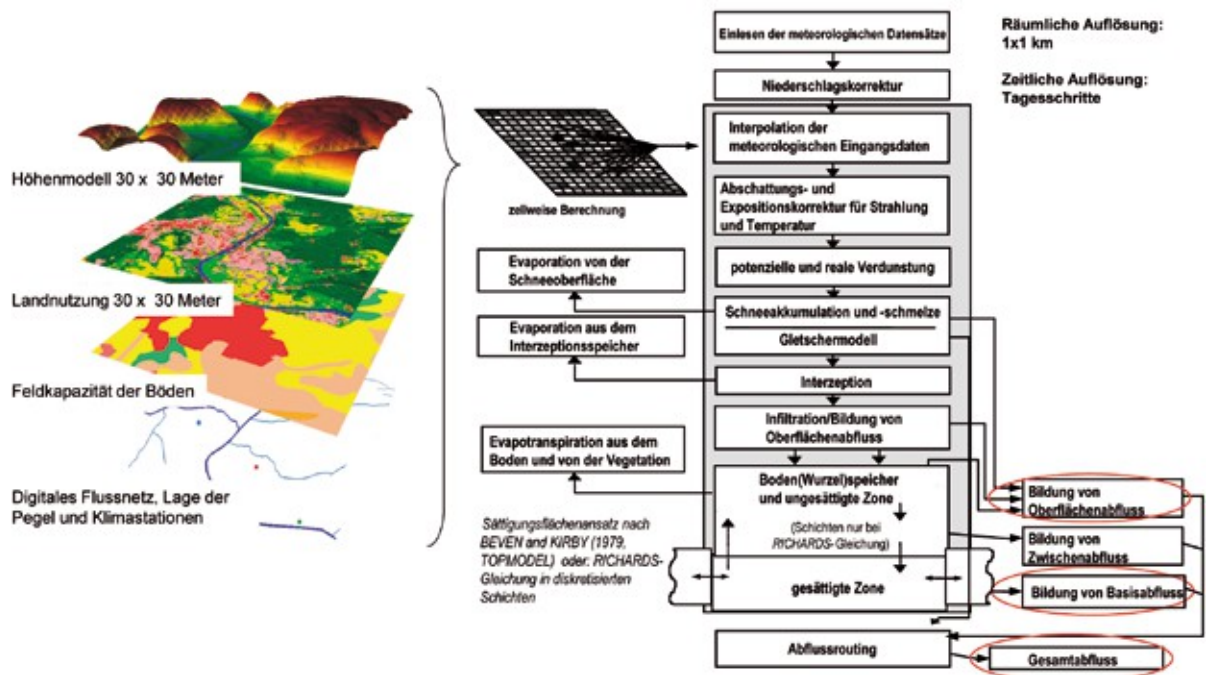
The data from the regional climate scenarios are used as input variables for the water balance models (WHM) in order to be able to make statements about the effects of climate change on the water balance (e.g. runoff in water bodies). However, as experience has shown, it is essential to critically examine the input data in advance because the climate models still have problems with the modelling of precipitation and can deliver implausible results. This applies in particular to the Alpine region.

In Bavaria, water balance models based on the ASGi/WASIM modelling system are currently for the river basins north of the Danube; the southern areas are still under . The WASIM-ETH model depicts the physical processes of the water cycle in high spatial and temporal resolution (*Figure 7*). This applies in a similar way to the LARSIM model, which was used to model the river basins of Baden-Württemberg. The water balance modelling focused on possible future changes in runoff. From the simulated runoff continuum, the first focus was on future flood discharges and their statistical changes; the discharges determined in the process were also analysed in more detail using methods of extreme value statistics.

5.2 Change in flood discharges

The results of the simulation calculations show a significant increase in average flood discharges, but also in extreme discharges. Even if the results from the emergency

Wasserhaushaltsmodell: GIS-basierte Simulation



Verwendetes Wasserhaushaltsmodell: Bayern - ASGi mit Modellkern WaSiM-ETH

Figure 7: Structure of the water balance model WASIM-ETH - Instrument for the river basin-related determination of the discharge continuum for different conditions

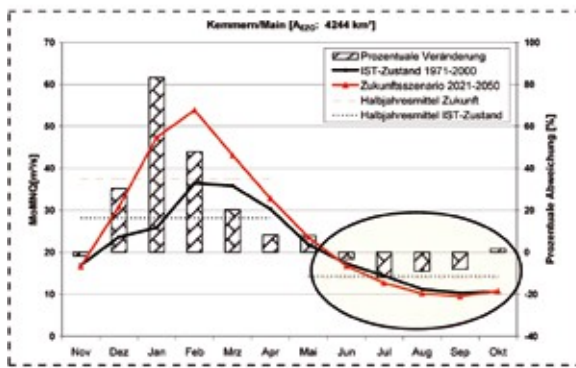
Although the model chain (global model - regional climate model - water balance models) and the model assumptions are subject to uncertainties, the regional climate models used confirm this statement, so that it can be assumed that climate change will exacerbate flooding for the period under consideration up to 2050. The results of the simulation calculations on the effects of the selected regional climate scenario in the Main area are shown as examples for two gauging stations (Figure 8). In addition to the change in the mean monthly maximum discharge, the evaluation of the simulated time series also shows how the mean monthly low discharge values can change.

5.3 Adaptation of flood protection planning

Against the backdrop of an expected worsening of floods, it was necessary to develop an adaptation strategy for the area of flood protection for precautionary reasons, which takes into account the possible development of the coming decades, but also takes into account the existing uncertainties. The key points of the provisions should therefore be that they can be adapted in the long term without adverse effects and at the same time if necessary (e.g. in the event of new findings in climate research) ("flexible and no regret" strategy).

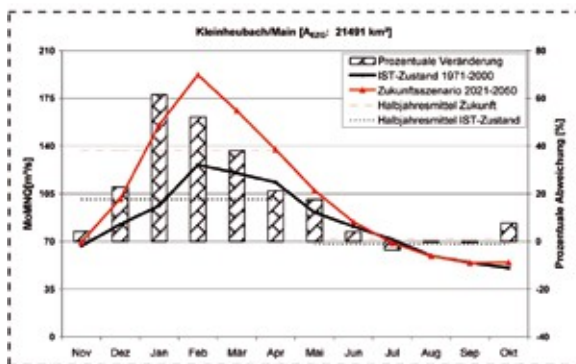
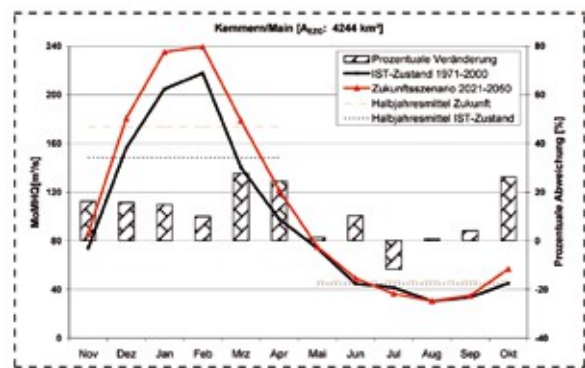
The analyses gave reason to modify the previous procedure for planning flood protection measures and to consider a "climate change load case" in the future. Based on case studies from practice, it was demonstrated that consideration of the

MoMNQ [m³/s]



Pegel Kammern / Main (A_{EZG}: 4.244 km²)

MoMHQ [m³/s]



Pegel Kleinheubach / Main (A_{EZG}: 21.491 km²)

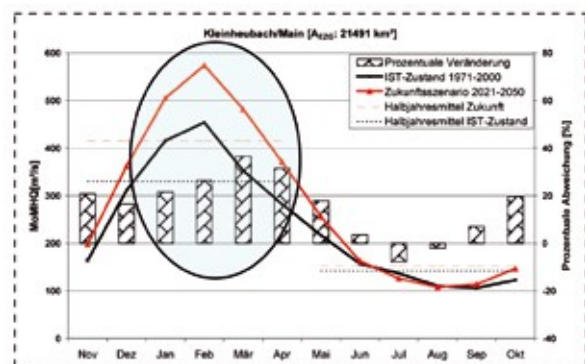


Figure 8: Change in discharge parameters (MoMHQ and MoMNQ) of the future scenario (2021- 2050) compared to the simulated actual state (1971-2000) - illustration for two gauging stations in the Main area

The effects of climate change on technical flood protection measures would have led to relatively moderate cost increases if this load case had already been taken into account during the planning stage and at least appropriate precautions for later adaptation had been taken during construction. Subsequent adaptations without at least prior planning precautions, on the other hand, are usually associated with very high costs.

In future, the **climate change load case will** therefore be taken into account when assessing flood protection measures. This is done in the planning process by adding a "climate change factor" to the currently valid statistical assessment basis (e.g. HQ100). In Bavaria, a flat-rate surcharge factor of 1.15 been introduced since the end of 2004 for flood discharges with an annuality of up to HQ100, although this can also modified regionally in justified cases. Due to the uncertainties that still exist in the climate models and the modelling assumptions, it is planned to update the climate change factor when better information becomes available. The application of the climate change load case for flood protection concepts with an addition of 15 % to the HQ100, or 25 % for some river basins, has been introduced analogously in Baden-Württemberg with a technical guideline.

6 Outlook

Based on the findings of the investigations of measured and modelled hydrometeorological and hydrological parameters in southern Germany, the broad-based KLIWA investigation programme has already led to concrete consequences, not least from a precautionary point of view. These include the improvement and inclusion of future climate changes in the determination of the design flood for technical flood protection systems, in particular through:

- the application of the updated KOSTRA Atlas in KOSTRA-DWD-2000 and
- the introduction of a "climate change load case" in flood protection planning.

As global climate research progresses and modelling tools improve, the findings to date will continue to evolve. For example, new regional climate scenarios for Germany have recently become available, which were developed on behalf of the Federal Environment Agency, are based on the ECHAM5 global model and various emission scenarios and the period up to 2100. The KLIWA partners are in the process of analysing these scenarios in more detail and transferring them to runoff, particularly via the water balance models, in order to rapidly develop the existing knowledge with more recent specifications. Furthermore, the establishment of a long-term monitoring programme (climate monitoring) and the investigation of the effects of climate change on the future development of low water runoff and groundwater recharge to safeguard water resources are already underway. The results of these investigations will then be used to develop further adaptation measures as required to mitigate the expected hydrological challenges posed by climate change - more often too much and more often too little water (*Fig. 9*).



Figure 9: Hydrological effects of climate change on the water balance: more often too much, more often too little of the precious commodity water

Water management is thus to changing conditions as a precautionary measure. Ultimately, however, climate change will not only have a major impact on water management, but also on society as a whole.

In addition to the development of adaptation strategies for the affected areas, climate protection with the aim of reducing greenhouse gas emissions must also be significantly strengthened. To this end, in November 2007 Bavaria adopted the

"Climate Programme Bavaria 2020" was adopted. This broad-based programme of measures the main activities in the three long-term fields of action on climate change up to 2011, namely the reduction of greenhouse gas emissions, the development of adaptation measures and the intensification of research and development. The KLIWA project will increasingly provide the necessary data basis for further adaptation measures.

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Author's address:

DI Hans Weber
Bavarian State Office for the Environment
Ref. 81 - Climate Change, Climate Impacts and Water
Resources Hans-Högn-Straße 12
95030 Hof (D)
Phone +49-9281-1800-4810
e-mail hans.weber@lfu.bayern.de

Trends in hydrological variables and in Austria's water balance

Josef Fürst, Harald Kling, Hans Peter Nachtnebel and Thomas Hörhan

1 Introduction and objectives

In the discussion on the hydrological consequences of global climate change, changes in extreme values, in particular the size and frequency of floods, but also dry periods, have so far taken centre stage (e.g. Kundzewicz et al. 2004, Svensson et al. 2004, Nobilis & Lorenz 1997). In terms of hydrology and water management, however, it is also of interest to know whether and how the components of the water balance - precipitation, evaporation, runoff and changes in reservoir levels - have changed. Findings on long-term changes in annual precipitation and their seasonal distribution - an essential basis for water management measures. Both the temporal change and the spatial pattern must be taken into account (Dore 2005, Xu et al. 2003, Birsan et al. 2005). A seasonally and geographically differentiated analysis and visualisation of the trends improves the basis for water management compensation measures.

The aim of the study was to carry out trend analyses on the basis of monthly precipitation and runoff data at the Hydrographic Service (HD) measuring stations, for which long time series (at least from 1951) are available, and to prepare them for presentation in the Hydrological Atlas of Austria (HAÖ) (BMLFUW 2007). The point-based time series analyses are supplemented by simulations with a water balance model in order to achieve improved area-based statements.

2 Data basis

The following information is used as a data basis:

- Observed monthly precipitation and monthly mean discharge at the monitoring stations (generally from 1951). In total, the HD provided discharge time series from 132 stations and precipitation time series from 641 stations. There are no gaps in the time series from 1951 to 2000. Some of the discharge data also refer to gauges that are significantly influenced by inflows and outflows or reservoirs.
- Calculated monthly precipitation from the Skoda/Lorenz model (Skoda & Lorenz 2003) from 1951.
- Data from the HAÖ (monitoring network maps, elevation model, catchment area classification, maps of the mean water balance figures 1961-1990).
- Monthly average air temperature from 1951 onwards at representative measuring points.

3 Methodology

Suitable methods for analysing trends in the time series of monthly averages must take into account that the series of monthly values exhibit a clear autocorrelation and seasonality and are not normally distributed. For these reasons, the classic linear trend analysis is ruled out. As

The Mann-Kendall trend test (Helsel & Hirsch 1992) is a parameter-free method that can recognise monotonic trends without assuming normally distributed data. The seasonal Mann-Kendall trend test is used to take into account the seasonality of the data. This approach has been used for recent studies of the trend behaviour of hydrological variables in Switzerland (Birsan et al. 2005) and Canada (Burn & Hag Elnur 2002), among others.

The Mann-Kendall trend test (MK test) uses a rank-based procedure that is particularly suitable for non-normally distributed data with outliers and non-linear trend behaviour (Helsel & Hirsch 1992). The null hypothesis for the MK test for trend in the random variable x is that the probability $\Pr(x_j > x_k) = 0.5$, for $j > i$. The MK statistic S is the number of positive signs of all differences $(x_j - x_k)$ where x_j and x_k represent the data values at time points j and k , with $j > k$.

The seasonal Mann-Kendall trend test (SMK) takes seasonality into account by first calculating the MK test separately for each of the m seasons then adding the result together. For monthly data ($m = 12$), January data is only compared with January data, February only with February, etc. **Kendall's S statistic** S_k for the entire series is obtained by summing the statistics S_i of the individual months.

The MK test is used to test the significance of a monotonic relationship between two variables. The Kendall-Theil method (Helsel & Hirsch 1992) provides a robust estimate of the slope for a parameter-free (distribution-independent) fitted trend line. The Kendall-Theil slope β is the median of the slopes between 2 data points.

The spatially distributed, conceptual water balance model COSERO (Eder et al. 2005, Kling et al. 2006) is applied for the entire Austrian territory for the period 1951-2000 in monthly steps and provides time series of the spatially distributed water balance components as a result. The water balance model was developed as part of a project in the "Hydrology of Austria" of the Austrian Academy of Sciences (HÖ 27/2003, Kling et al. 2005). A slightly modified version of the model was used to calculate the seasonal water balance for the period 1961-1990. This version has now also been used to calculate the water balance for the period 1951-2000. This model divides Austria into 16 internal calculation areas, which in turn are divided into sub-areas that generally correspond to sub- and intermediate catchments (a total of 188 sub-areas). The model zones of the sub-basins are based on a 1-x-1-km grid.

The results of the water balance model are time series of the water balance components in monthly increments from 1951-2000 for each model zone. These high-resolution data were first aggregated for catchment areas for further visualisation and analysis, corresponding to the catchment area boundaries in the HAÖ water balance maps. The trend analysis described above was applied to the resulting hydrographs for 1,044 catchment areas. The calculated parameters can be assigned to a catchment area in the GIS and visualised as an area.

4 Results

The trend analyses were applied both to the time series of precipitation and runoff observed at the measuring points and to the water balance variables calculated with the model. As a very good agreement was achieved, the further presentation of the results is only given for the areal analyses of the model results. Detailed documentation can be found in HAÖ (BMLFUW 2007) and in Fürst et al. (2007). For the trend analysis, the model results were first aggregated for 1,044 catchment areas (average size of approx. 86 km²), so that a time series in monthly steps was created for each of these catchment areas and each of the balance variables. The methods described for analysing trends were to these time series. To determine the slope of the Kendall-Theil trend line, the time series were rescaled as a percentage of the mean annual precipitation in the respective area.

4.1 Annual evaluation of trends in monthly precipitation and runoff heights

The first step was to analyse the annual trends in precipitation and runoff depth. It shows a clear spatial pattern with decreasing precipitation south of the main Alpine ridge. North of the main Alpine ridge, both areas with increases and decreases are shown. However, it should be noted that the trends are only statistically significant in a small proportion of the areas (probability of error 10 %). The trends in runoff heights show a similar spatial pattern, but are statistically verified in a larger area (approx. 55 % of the national territory).

4.2 Evaluation of winter and summer half-years

A seasonal trend analysis was carried out separately for the months of the winter half-year (November to April) and the months of the summer half-year (May to October). trend behaviour differs significantly from year to year and geographically, which is statistically verified in many areas. A positive trend in precipitation north of the main Alpine ridge very striking in the winter months, which contrasts with an equally clear negative trend in the south, especially in a large part of Carinthia. This spatial pattern can also be seen to a lesser extent in the discharge heights.

In the summer months, the spatial separation by the main Alpine ridge is less pronounced. Precipitation strongly negative trends, especially in the north (Alpine foothills, Mühlviertel), but the trend in the south is also largely negative, although not statistically significant. A striking feature is an extensive zone of positive precipitation trends extending from the Mürz Valley across the Vienna Basin and northern Burgenland. The spatial pattern of the precipitation trends is essentially also in the discharge heights, with the focal points of strongly negative trends in the Upper Austrian Alpine foothills and in southern Carinthia.

4.3 Evaluation of seasons (quarters)

The geographical differences in the trends become even more pronounced when the analyses are based on four seasons. For better comparability and readability, the maps

The trends in precipitation and runoff depths for each season can be easily compared by summarising the graphs at the end of the description.

4.3.1 Seasonal trends in precipitation

The trends in precipitation show clear differences, even opposing trends, in the four seasons and a pronounced spatial separation by the main Alpine ridge.

The contrasting trends in the autumn and summer months are particularly striking. In the autumn months, the area north of the Alps is characterised by strongly positive, statistically significant trends (*Figure 2*). On around 32 % of the national territory, the monthly precipitation shows a trend of 0.025-0.05 per cent of the mean annual precipitation. Over the last 50 years, the three autumn months have produced an increase of 3.75-7.5 % of the annual precipitation, i.e. additional precipitation heights of 45 to 90 mm with an average area precipitation of around 1,200 mm. It is noteworthy that this additional precipitation does not lead to a corresponding trend in runoff heights in the autumn months (*Fig. 3*), but *instead* stored. South of the main Alpine ridge, the autumn months show no or only slight trends, with slightly negative trends in Carinthia, East Tyrol and the southern part of Tyrol.

Winter precipitation (*Figure 4*) shows a slightly negative, statistically insignificant trend over 58% of the federal territory. The eastern part of Carinthia, on the other hand, forms a larger area with significant negative trends. Large parts of Upper Austria and western Lower Austria, as well as the Vienna Basin, show no trend or a slightly positive trend.

During the spring months, no distinct spatial pattern is recognisable in the trends (*Figure 6*). Overall, slightly positive trends predominate. Eastern Carinthia and southern Styria form an area with a partly clearly negative trend. Parts of the upper Enns and Mura regions, up to the Schwarza and Leitha regions, show positive trends, albeit only partially significant.

Finally, the map of trends in the summer months (*Figure 8*) shows an opposite picture to the autumn trends north of the Alps. Upper and Lower Austria show strongly negative trends, most of which are also statistically significant. The Salzach region, on the other hand, only a weakly negative trend. There is also a noticeable area of slightly positive trends in western Tyrol up to around the Ötztaler Ache.

4.3.2 Seasonal trends in discharge

The maps for the trends in discharge heights consistently show larger areas with statistically significant trends than precipitation, even if the magnitude of the trend is smaller.

The discharge heights in the autumn months (*Fig. 3*) show the lowest trend. There is a larger coherent area with a negative trend south of the main Alpine ridge, starting in the Ötztal valley and extending into southern Burgenland. The northern Alps are characterised by neutral to slightly positive trends, with clearly positive trends in the upper Enns region.

can be observed. The northern Alpine foothills and the areas north of the Danube show no or slightly negative trends.

If one compares the map with the trends in autumn precipitation (*Fig. 2*), it is noticeable that the clearly positive trends in precipitation do not lead to corresponding trends in runoff in the autumn months, but only cause a reaction in the runoff trend in winter. The additional autumn precipitation is therefore increasingly stored in the soil and groundwater.

The map of runoff trends in the winter months (*Figure 5*) largely reflects the spatial pattern of trends in precipitation levels in the autumn months: clearly positive trends occur north of the main Alpine ridge, while weakly negative trends can be observed south of it.

The runoff trends in the spring months (*Figure 7*) show a similar spatial pattern as in the autumn months, although the negative trends from East Tyrol via Carinthia to southern Burgenland and the positive trends from the Enns region to the Vienna Basin are more pronounced. Positive are also noticeable in the Lower Mühlviertel.

In the summer months (*Figure 9*), 80 % of the national territory shows a negative trend, which corresponds very well with the precipitation trends north of the main Alpine ridge, but is much more pronounced in the south than the precipitation trend.

The temporal development of seasonal precipitation is easily recognisable in a representation of the cumulative deviation from the mean value. *Figure 1* shows the course for one area north of the Alps (Rodl) and one south of the Alps (Obere Gail).

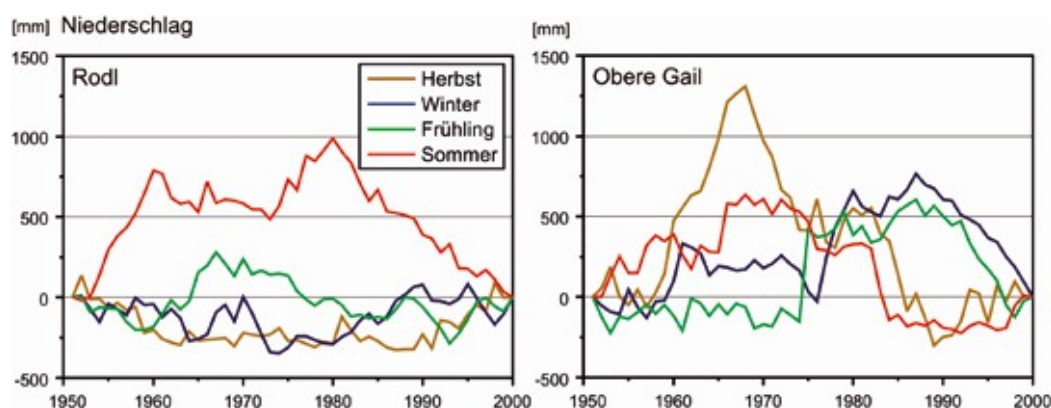


Figure 1: Cumulative deviation from the mean value of precipitation in the catchment areas of the Rodl (Upper Mühlviertel) and the Upper Gail (Carinthia) from 1951-2000

In the Rodl region, a largely monotonous positive trend in autumn precipitation can be seen over the entire period. Summer precipitation, on the other hand, is characterised by a clearly negative trend. In the upper Gail area, negative trends predominate, with a particularly negative trend from around 1975 in winter and spring. Summer and autumn precipitation, on the other hand, show the strongest negative trend in the period up to 1980.

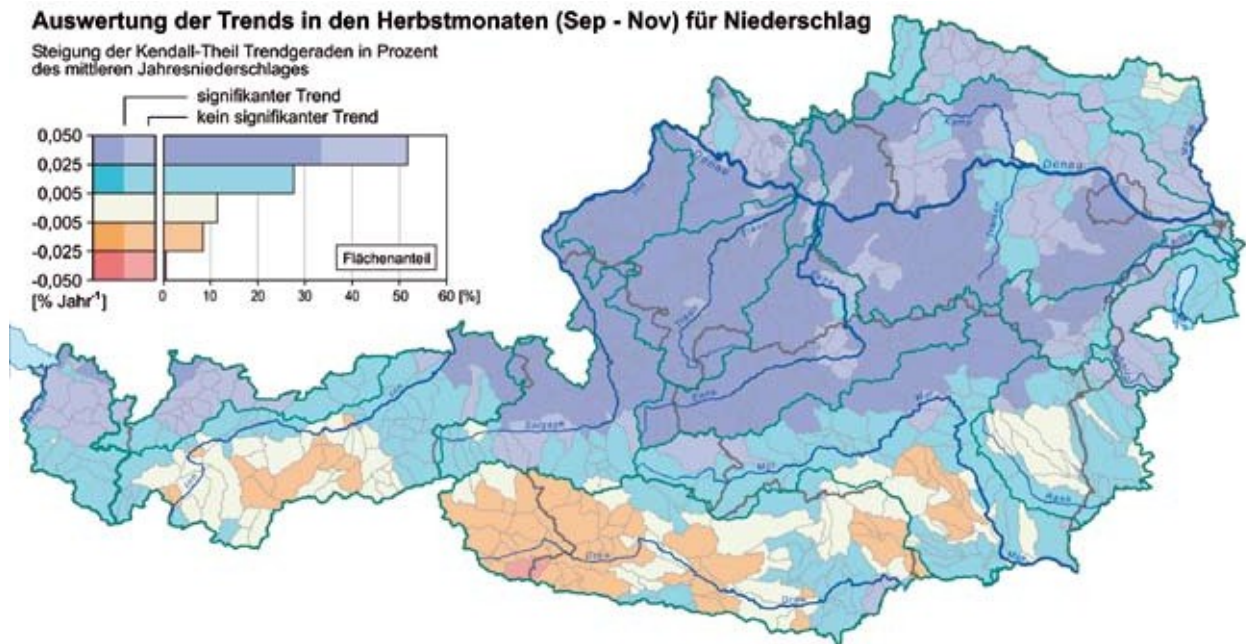


Figure 2: Evaluation of the trends in the autumn months for precipitation

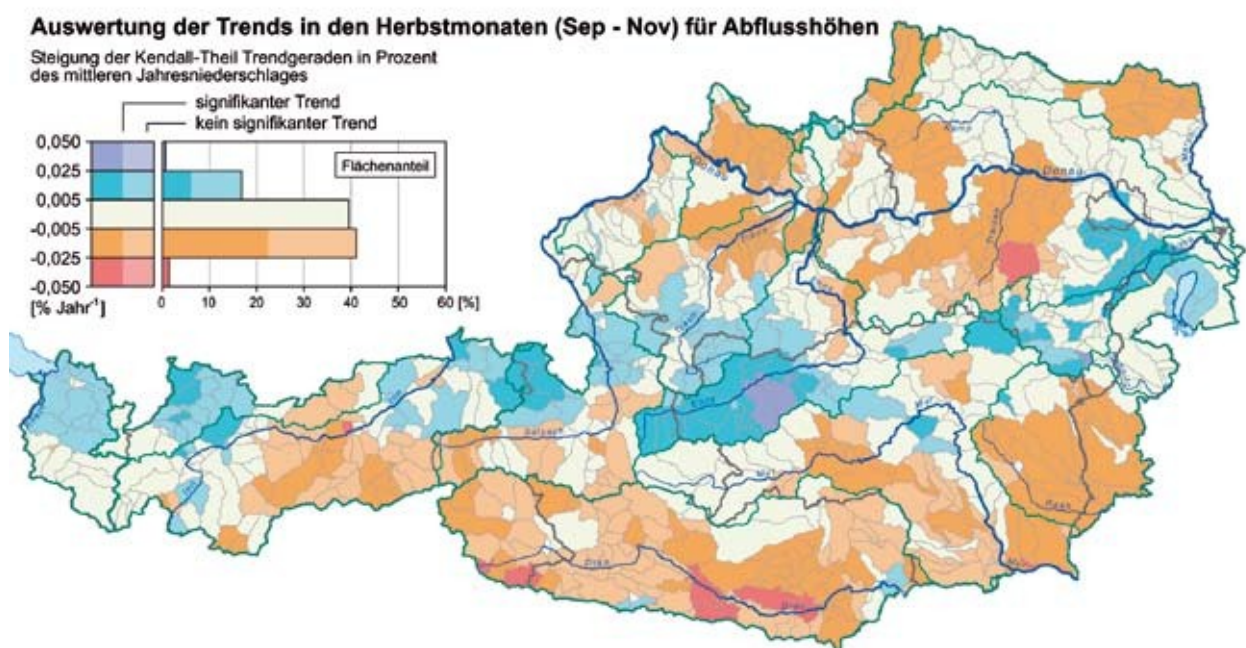


Figure 3: Evaluation of the trends in the autumn months for discharge heights

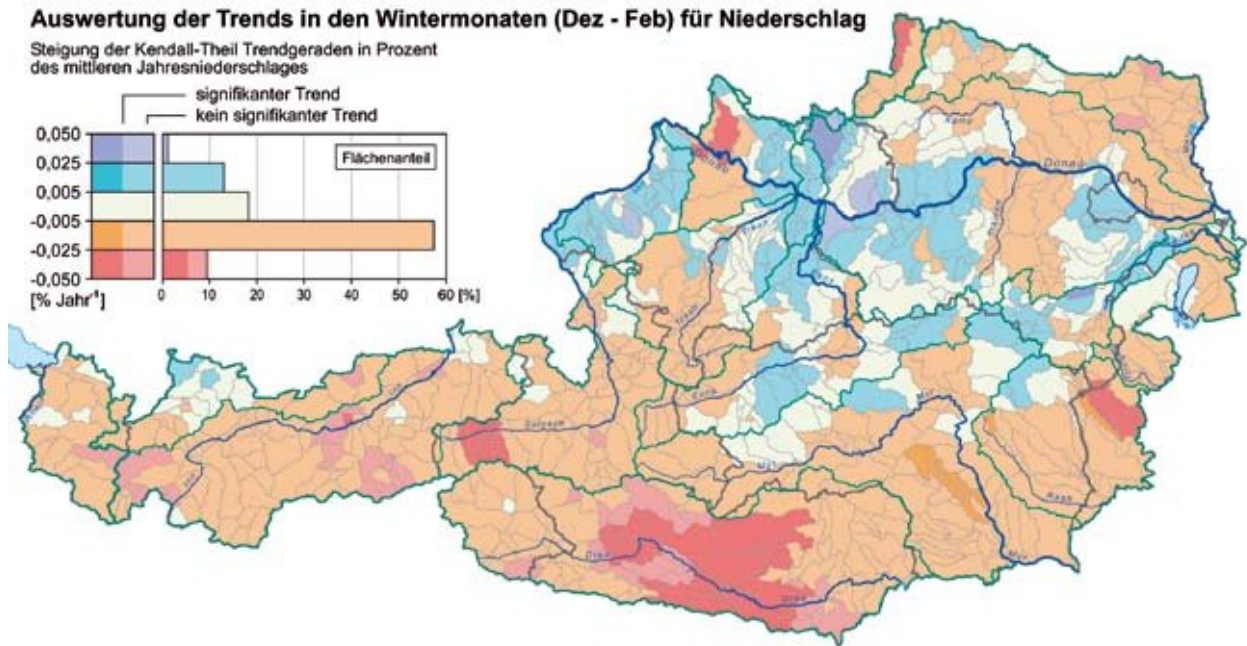


Figure 4: Evaluation of the trends in the winter months for precipitation

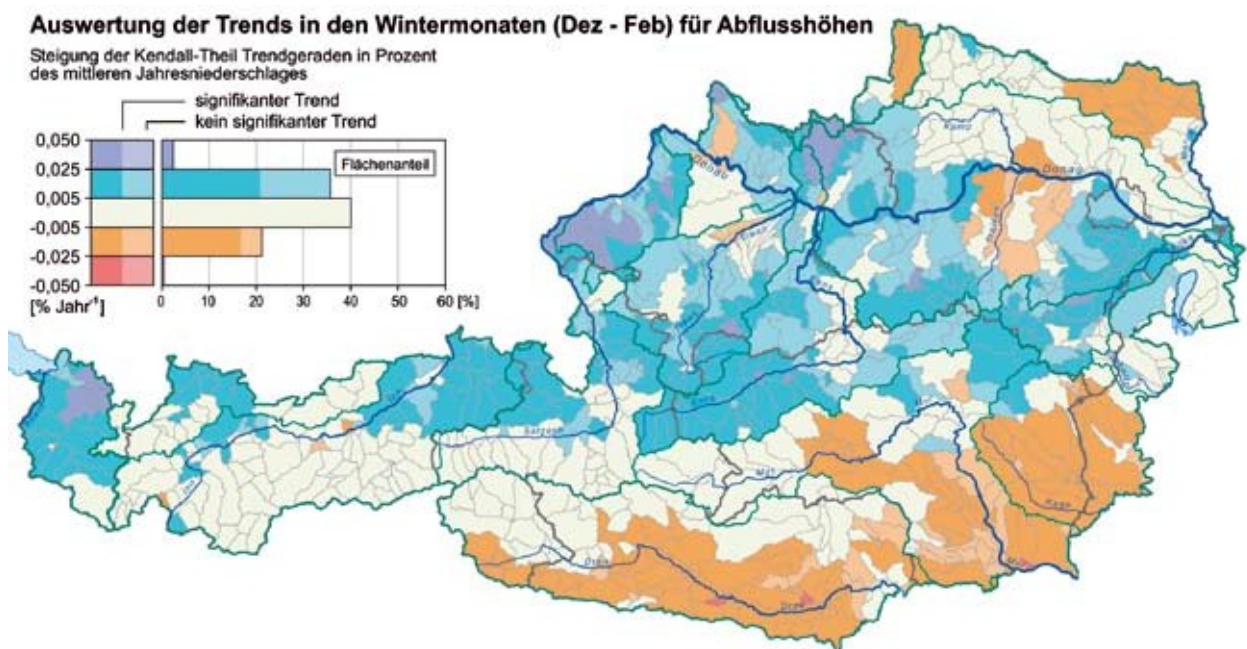


Figure 5: Evaluation of the trends in the winter months for discharge heights

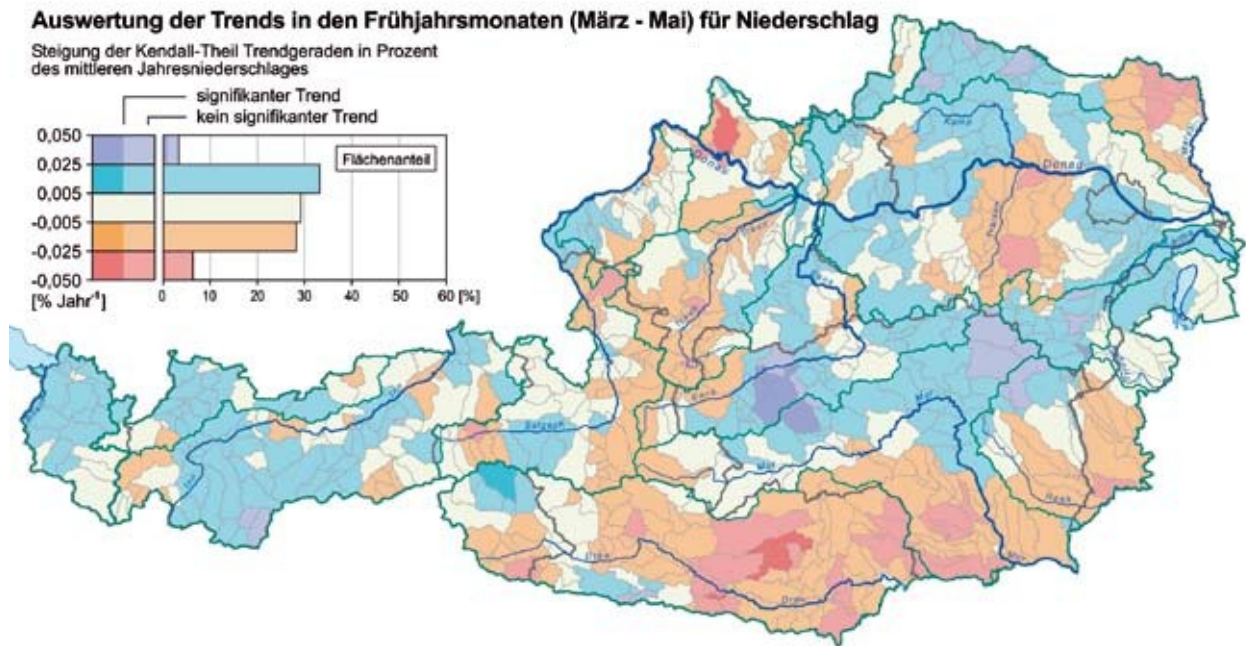


Figure 6: Evaluation of the trends in the spring months for precipitation

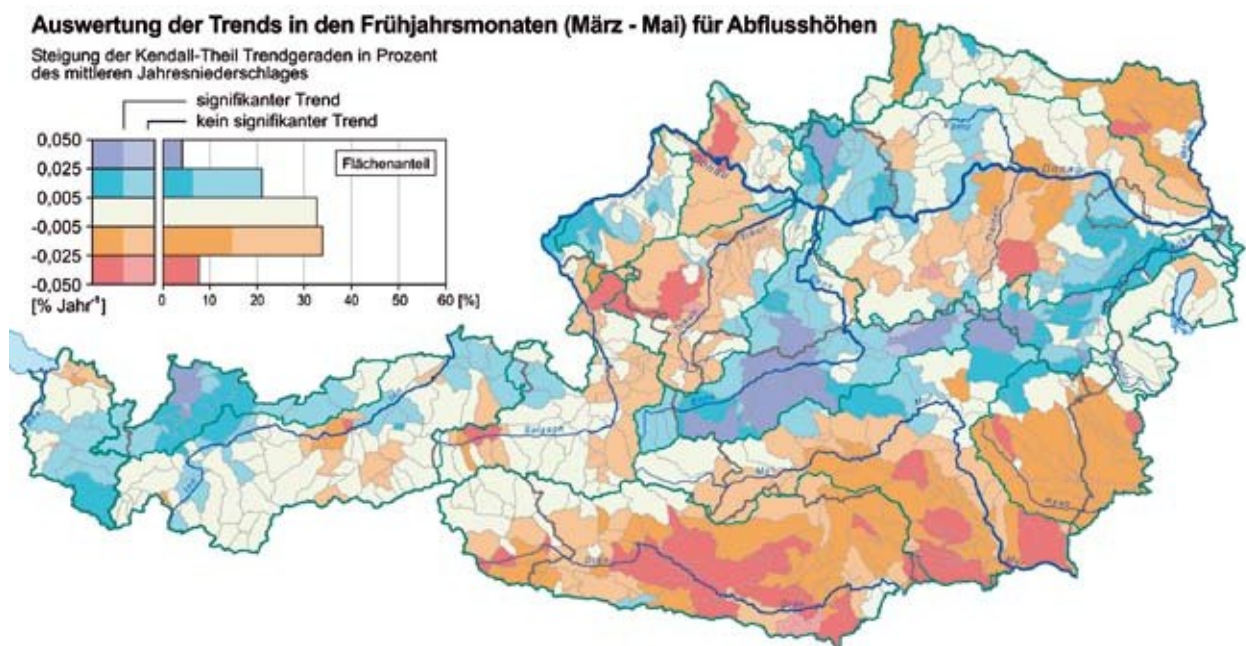


Figure 7: Evaluation of the trends in the spring months for discharge heights

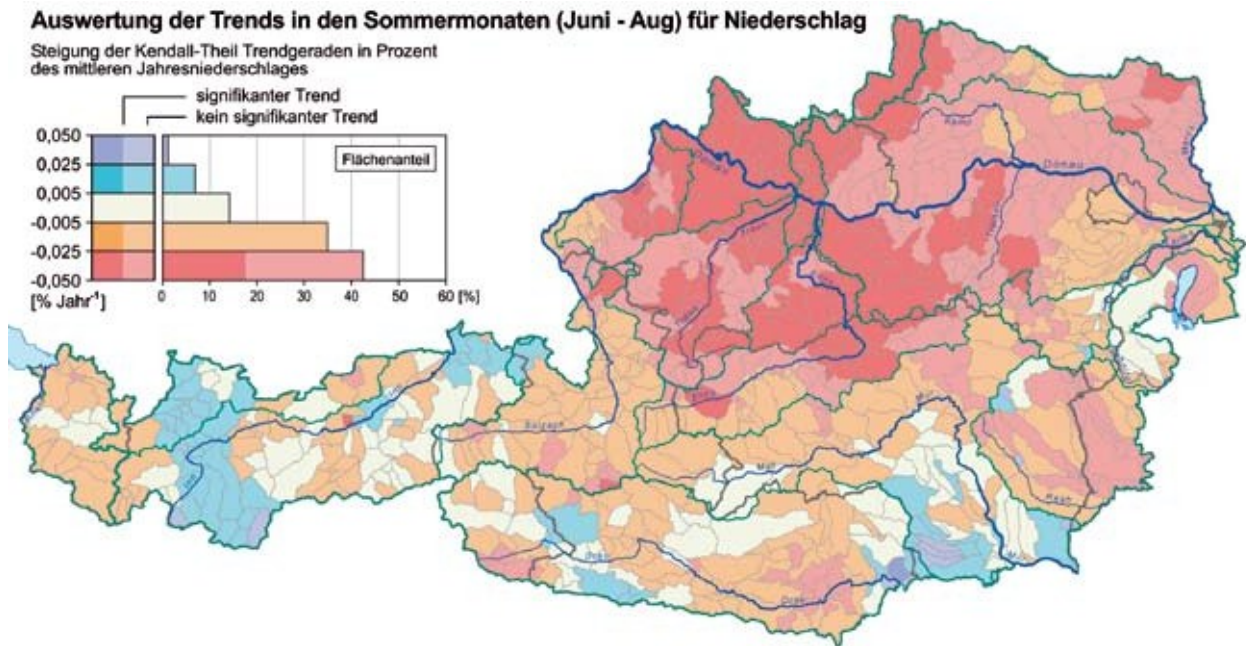


Figure 8: Evaluation of the trends in the summer months for precipitation

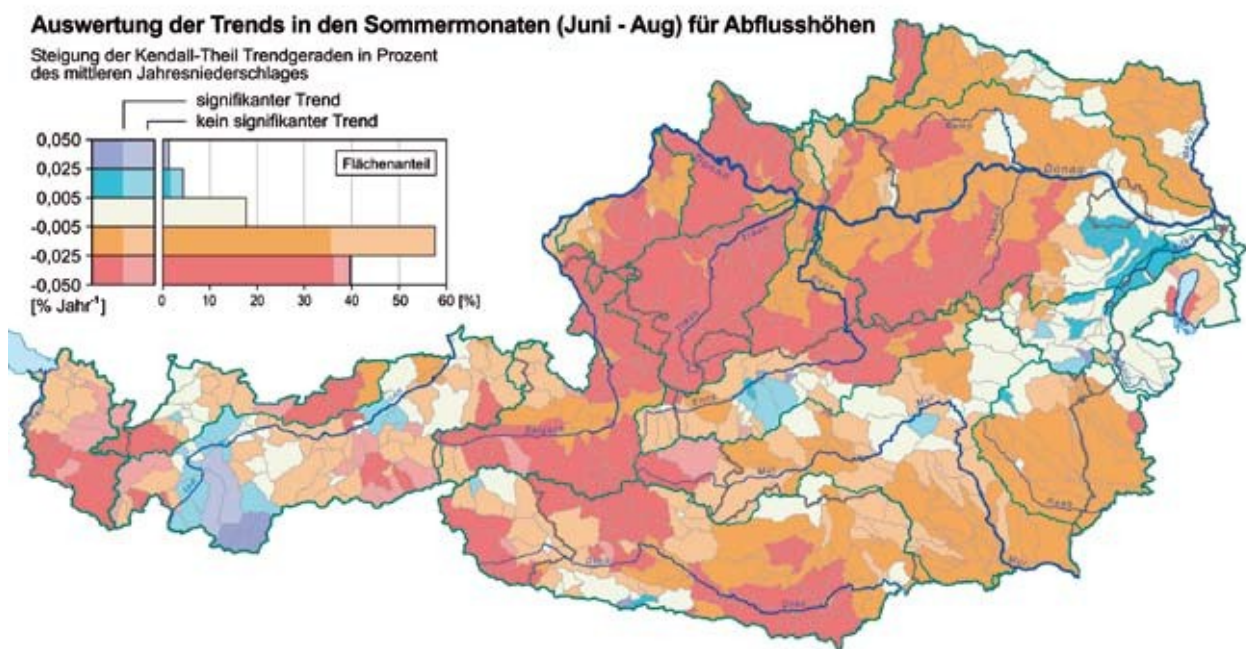


Figure 9: Evaluation of the trends in the summer months for discharge heights

5 Summary

The trend analysis based on the results of the water balance model shows a good correlation with the measurement point-based analysis. The main difference in content is that the discharge heights from the water balance model correspond to a hydrological situation unaffected by inflows and outflows and other hydraulic engineering measures.

- In a year-round view, trends in precipitation and discharge levels are only weak and hardly statistically .
- Only when analysed over time for half-years or quarters do clear differences emerge. Trends vary greatly over time and space.
- The spatial patterns of the trends in precipitation and discharge heights are generally similar for a one-year period. Only the strongly positive trends in precipitation in autumn are only reflected in the runoff height trends in winter.
- The main Alpine ridge forms a clear spatial boundary. Clearly positive trends in autumn/winter north of the main Alpine ridge contrast with negative trends in the south.
- The area south of the main Alpine ridge - especially Carinthia - is characterised by negative trends throughout the year, which are most pronounced in the spring and summer months.

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Address of the authors:

Associate Professor DI Dr Josef
Fürst DI Dr Harald Kling
o.Univ.-Prof. DI Dr Hans Peter Nachtnebel DI
Thomas Hörhan
University of Natural Resources and Life Sciences Vienna
Institute of Water Resources Management, Hydrology and Hydraulic Engineering
Muthgasse 18, 1190 Vienna
Tel. +43-1-36006 5511
E-mail: josef.fuerst@boku.ac.at_
iwhw.boku.ac.at

Climate change and socio-economic changes - challenges in dealing with natural hazards in mountain regions

Sven Fuchs

1 Introduction

Mountain areas are particularly sensitive to changing environmental parameters. With regard to natural hazards, climate change a double challenge for them, firstly from the perspective of the natural environment and secondly from the perspective of a densely populated living and economic space. Changes in the natural environment affect the atmosphere, hydrosphere, lithosphere and biosphere. Of particular importance here are the changes in the environment caused by climate change, such as changes in the temperature regime (possible increase in temperatures) or precipitation distribution (accentuation of precipitation events), which in turn can have an impact on geomorphological erosion and erosion processes.

An increase in temperatures leads to a melting of the glaciers and a thawing of the permafrost soil with a time lag of years to decades. This relationship can be clearly established for the Alpine region for the period since the end of the so-called "Little Ice Age" in the middle of the 19th century. While the glaciers have lost more than a third of their surface area in the entire Alpine region (Maisch et al., 1998), the data on the reduction of the permafrost body varies from 10 % (Haeberli et al., 1998) to around 40 % (Stötter et al., 1996). This is accompanied by the successively increased availability of unconsolidated erodible loose material and a general shift of potential process fracture areas to higher altitudes. In combination with a change in precipitation distribution, it is not unlikely that torrential processes and geomorphological mass displacement processes will increase in intensity and frequency in the future.

Such natural processes become a hazard when they overlap with human utilisation requirements, as the space available for economic activities and settlement purposes is naturally limited in mountain areas. A change in process behaviour can therefore lead to an increased risk potential for exposed values, especially if the average fluctuation range of these processes exceeds a certain threshold value based on the respective technological and economic conditions.

Over the last 60 years, there have been significant changes in the utilisation of mountain areas. The Eastern Alps are favoured over some regions in the (Southern) Western Alps. Economic activities have increased, the settlement structure has changed and settlements have expanded, and the amount of infrastructure required for this, such as roads and supply lines, has increased. This has been accompanied by a change from a society originally characterised by agriculture to a service society, with the tourism sector in particular experiencing considerable growth rates. The parallel transition to a leisure-orientated society is contributing above all to an increase in the number of people spending time in areas at risk.

Possible developments in risk resulting from these changes are shown in *Figure 1* in nine (at least theoretically) possible trends.

		Exposed values and/or number of people at risk		
		Decrease (-)	no change(o) /	Increase ()+
Magnitude and frequency of natural processes	Increase ()+	A ∅	B +	C ++
	no change(o) /	D -	E ∅	F +
	Decrease (-)	G --	H -	I ∅

Figure 1: Illustration of possible risk developments based on the intensity and probability of occurrence of natural hazard processes and the value and probability of presence of endangered objects and persons (Fuchs et al., 2004a)

The intensity or magnitude and frequency of a process can increase, decrease or remain at the current level. Based on recent findings, it must be assumed that all displacement processes with water as an agent will experience an accentuation (e.g. Houghton et al., 2001; Solomon et al., 2007), and thus an increase in the processes cannot be ruled out at the very least. The damage potential affected by these processes is also subject to variability, with large parts of the Eastern Alps showing an increase in the concentration of values due to the socio-economic development of recent decades.

Against the backdrop of the individualisation of society that began at the end of the 19th century, dealing with natural hazards was increasingly seen as a state task and was subsequently carried out, for example in Austria, by the Forest Engineering Service for Torrent and Avalanche Control and today's Federal Hydraulic Engineering Administration. Until the second half of the 20th century, permanent and temporary damming measures to prevent processes in the catchment areas took precedence, but from the 1960s onwards there was an intensive discussion about the possibility of passive protection measures, which was reflected in Austria with the creation of the first hazard zone plans in Tyrol.

However, the investment of considerable resources in the construction of relevant areas of potentially dangerous processes and in the flood protection of watercourses, combined with the creation of hazard zone plans, should not obscure the fact that complete protection against damage caused by natural processes is not possible. This has been particularly true since the 1990s, with considerable damage caused by avalanches (winter 1998/99), torrential processes (1999, 2002, 2005) and floods (2002, 2005, 2006) in the Alpine region.

Analysing and evaluating natural hazards has been the subject of research with natural hazards for decades. Hazard assessments are carried out, particularly by engineering scientists, in order to determine the ex ante effects of hazardous processes on the settlement and urban environment.

economic area. As long as the relevant system boundaries are stable and the framework conditions for processes do not change, this methodology can be used to minimise future damage.

However, changes in the natural environment, which lead to an increase in hazard potential, and changes in the cultural environment, which result in an increase in exposed values, lead to a significant increase in risk in mountain areas. As changes in both the natural and cultural environment are subject to change over time, a temporal approach is of great importance when assessing natural hazard-induced risks, a fact that is evident at least in the area of hazard zone planning. An increased risk of natural hazards is offset by a lower acceptance of natural hazard events by the population, a reduced financial scope of the public sector for active protection measures as well as limited spatial reserves and consequently conflicts with hazard zones.

2 Changes in the natural environment

Numerous long-term changes in climate have been observed at the scale of continents or ocean basins (Houghton et al., 2001; Solomon et al., 2007). These include changes in temperatures, as well as widespread changes in precipitation, ocean salinity, wind patterns and aspects of extreme weather events such as drought, heavy rainfall, heat waves and the intensity of tropical cyclones.

For many large-scale regions, the precipitation amounts show long-term trends from 1900 to 2005. Significant increases in precipitation were observed in parts of Northern Europe. Precipitation varies greatly both spatially and temporally. The westerly winds in the mid-latitudes have become stronger in both hemispheres since the 1960s. The frequency of heavy precipitation events has increased over most land areas, in line with warming and the observed increase in atmospheric water vapour. Palaeoclimate information supports the interpretation that the warmth of the last half century is unusual for at least the last 1,300 years. Mean temperatures in the northern hemisphere in the second half of the 20th century were very likely higher than during any other 50-year period in the last 500 years and probably the highest in at least the last 1,300 years.

A direct forecast for the Alpine region proves to be difficult to realise based on the available information from the IPCC, because the Alps are too small on a northern hemispheric scale and therefore do not appear directly due to the large mesh size of the models. However, a growing body of evidence suggests that the following impacts in hydrological systems are likely (Parry et al., 2007; Solomon et al., 2007), which have a direct influence on the natural environment in European mountain areas:

- Increased runoff and earlier onset of maximum runoff volumes in spring for numerous rivers fed by glaciers and snow
- Increased instability of the soil in permafrost areas and mass displacement processes in Mountain regions

- For settlements in mountainous regions - due to the melting of glaciers - there is a Increased risk of flooding after glacial lake outbursts.

By the middle of the 21st century, the average annual runoff in rivers and water availability at high latitudes is projected to increase by 10-40 %. In addition, the amount of water stored in glaciers and snow cover is expected to decrease. As a result, the availability of water in regions that are supplied by meltwater from the large mountain ranges will decrease. Where extreme weather events become more intense and/or more frequent, the associated economic and social costs will increase. This increase will be significant in the areas most directly affected. Due to far-reaching and complex interdependencies, the effects of climate change will spread from directly affected areas to other regions.

The negative effects include an increased risk of inland flooding and increased erosion (mainly due to increased thunderstorm activity as a result of increasing temperature differences).

The impacts are very likely to change due to changes in the frequency and intensity of extreme weather and climate events (Houghton et al. 2001; Parry et al., 2007; Solomon et al., 2007).

With regard to the Eastern Alps, the main expected changes are summarised in *Table 1*, with higher temperatures and an accentuation of precipitation events representing the key parameters with regard to the natural hazard situation.

Table 1: Significant expected climate changes in the Alpine region, higher temperatures and an accentuation of precipitation events are key parameters with regard to the natural hazard situation (based on data in Parry et al., 2007; Solomon et al., 2007)

Safety of observed changes (second half of the 20th century)	Changes in appearance	Safety of projected phenomena (during the 21st century)
Probably	Higher maximum temperatures and more hot days over almost all land areas	Very likely
Very likely	Higher minimum temperatures, fewer cold days/freezing days over almost all land areas	Very likely
Very likely	Reduced day/night temperature fluctuations over most land areas	Very likely
Probably, over many land areas of the middle and higher latitudes of the northern hemisphere	Increase in more intense precipitation, accentuation of precipitation events	Very likely, over many areas

Changes can also be expected for the processes taking place in the natural environment due to the projected or forecast changes in the climate. Due to the accentuation of precipitation events, there will primarily be an increase or rise in the frequency and magnitude of those geomorphological processes where water is the agent or the triggering factor (see *Table 2*). Apart from landslides, for which no change is currently detectable, avalanches are the exception; based on a detailed analysis of system behaviour in the period from 1950 to 2000, no change can be assumed here (Laternser & Schneebeli, 2002).

Table 2: Projected or predicted changes for natural hazard processes in the Alpine region (according to data in Bader & Kunz, 1998; Wanner et al., 2000; Solomon et al., 2007)

Process	Trend development
Avalanche	Consistent
Debris flow	General increase
Rockfall	Constant to increasing
Landslide	No detectable change
Landslide	Increase
Flood	Increase

Nevertheless, such changes, for example with regard to torrential processes in Austria, cannot yet be directly proven. *Figure 2* the results of an analysis of torrential events in Austria for the period 1972-2004. Although some years with a significantly higher number of events and correspondingly higher damage totals are conspicuous, a general trend towards an increase cannot be confirmed.

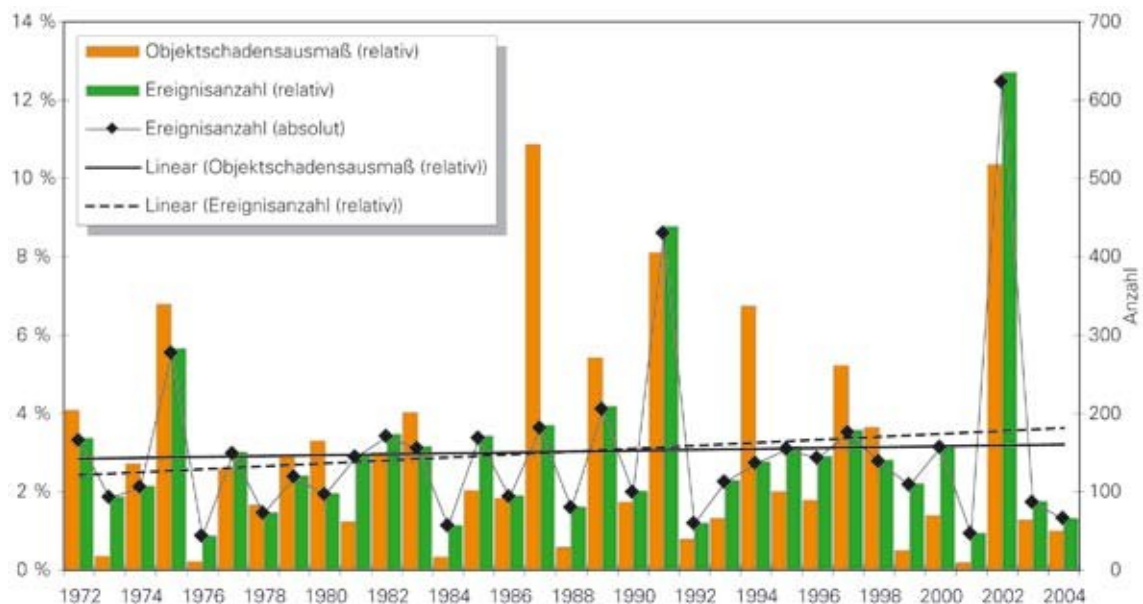


Figure 2: Number and damage of torrential events in Austria for the period 1972-2004, N = 4894 (from Oberndorfer et al., 2007)

Nevertheless, dealing with the expected climate change in the Austrian Alpine region is conceptually interesting, as the designation of hazard zones is based on the so-called design event. In the area of torrential and avalanche hazards, this represents a rare event with a recurrence probability of around 150 years (Republic of Austria, 1976). A change in the underlying time series may lead to a change in the delimitation criteria and may subsequently lead to a spatial expansion of the hazard zones. This in turn has a direct impact on the number of exposed objects and thus on risk management.

3 Changes in the cultural area

Socio-economic changes to the structures in mountain areas result in changes to the damage potential. These changes can be divided into long-term and short-term variability, with different patterns coming into play in each case.

3.1 Long-term aspects

Based on the concept of regional development types, a first systematic and quantifying analysis of socio-economic structural change in the Alpine region shows strong spatial disparities (Bätzing 1993), which are associated with long-term population change. These small-scale phenomena are also reflected at meso- and microscale levels. *Figure 3* illustrates this using two photographs of the town of Davos (Switzerland), clearly showing the densification of the settlement structure and the expansion of the settlement area. The corresponding increase in exposed values has been documented for this example by Fuchs & Bründl (2005) and is shown in *Figure 4*. The long-term increase in the number of buildings and the associated values has also been demonstrated for other Alpine settlements, for example in Keiler (2004) for the village of Galtür (see *Fig. 4*), and in Keiler et al. (2006) for other villages in the Austrian Paznaun Valley.

In Davos, the number of endangered buildings almost tripled between 1950 and 2000, the value has risen by a factor of 4, with the majority of the increases occurring in the category of residential buildings. In Galtür, the number of buildings at risk more than doubled in the same period under investigation, and the corresponding value increased by a factor of 5, with the highest increases being observed in the category of accommodation establishments.

If the development of the damage potential is also subjected to a detailed spatial analysis, remarkable differences emerge. In the yellow hazard zones in Galtür, the building values at risk increased by a factor of 9 during the period under investigation, which is also higher than the increase for the entire municipality (factor 8). If a 10-metre area directly adjacent to the yellow zone is considered - i.e. in the officially "safe" area - an increase in building values by a factor of 15 can be observed. This increase occurred after the hazard zone plan was finalised. Accordingly, this reflects the consideration of the hazard zone plan in the zoning of the municipality (Keiler & Stötter, 2004). As a result of this development, the areas directly adjacent to the zones



Figure 3: Settlement development in the Alpine region using the example of Davos (Switzerland); left: 1940s, right: 2002 Photos: J. Trauffer (left), S. Fuchs (right)

The number of buildings in the narrow area that closes the danger zone is roughly the same as in the danger zones. A similar development can also be observed in Davos, where an extension of the currently valid hazard zone plans by 10 metres increase the number of buildings at risk by around a third and the number of people at risk would double (Fuchs & Bründl, 2005).

The number of tourists staying in mountain regions has also risen significantly over the long term in recent decades, as illustrated in *Figure 5* for the municipality of Galtür based on the number of guest beds. According to a spatially differentiated analysis of the increases in the number of buildings, there are generally far more people staying in endangered areas today than in previous decades, which is particularly evident during the winter months (*Figure 6*).

3.2 Short-term aspects

Parallel to these long-term changes in damage potential, there are short-term fluctuations that primarily affect mobile assets and people. These fluctuations can be quantified and are further subdivided into seasonal, weekly and hourly variations on different time scales (see *Figure 7*, Keiler et al., 2005). Maximum values in the presence of people in endangered settlement areas of Alpine tourist destinations are found during the typical winter high season around Christmas, the end of February and Easter. These periods often coincide with periods of increased avalanche activity.

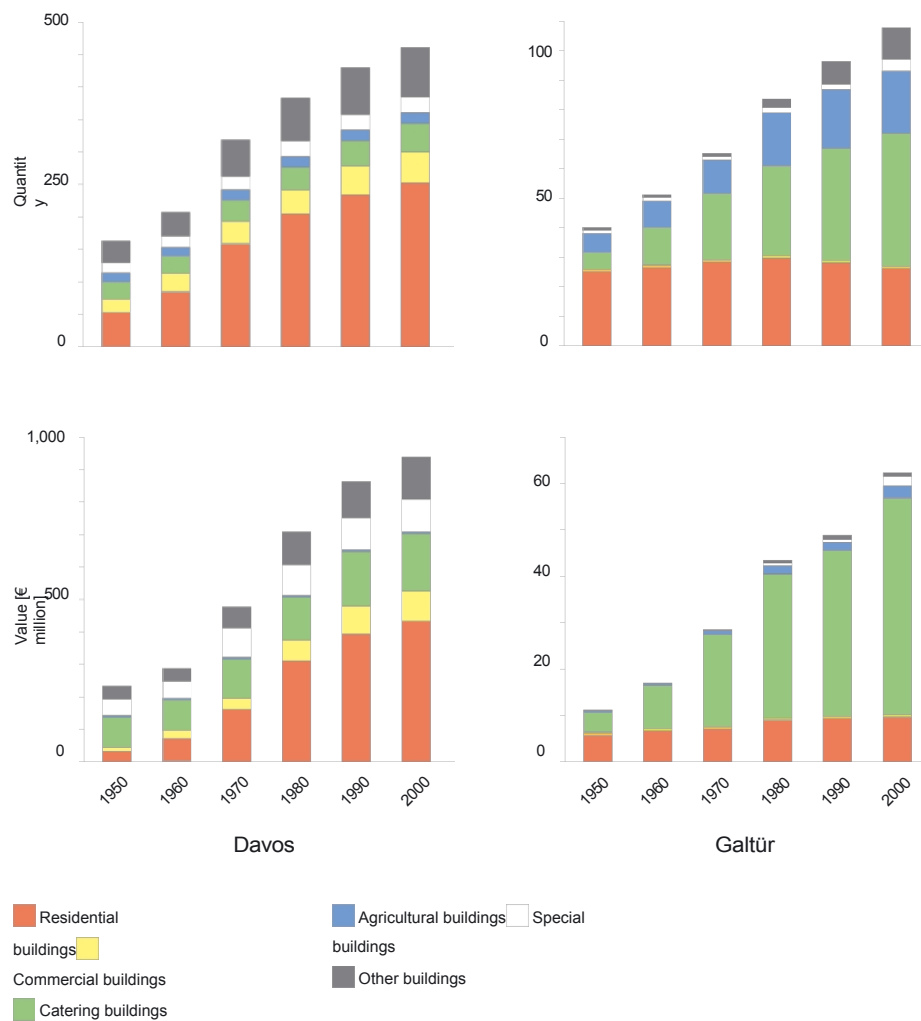


Figure 4: Increase in the number and values of buildings at risk in Davos (CH, left) and Galtür (A, right), from Fuchs & Keiler (2008)

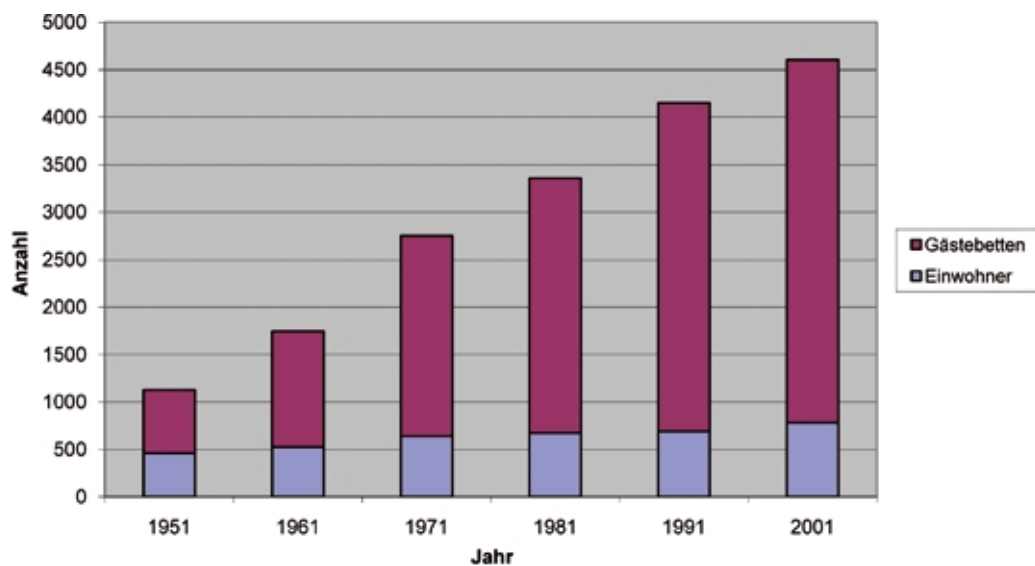


Figure 5: Development of the number of inhabitants and guest beds in Galtür for the period 1951-2001, from Keiler (2004)

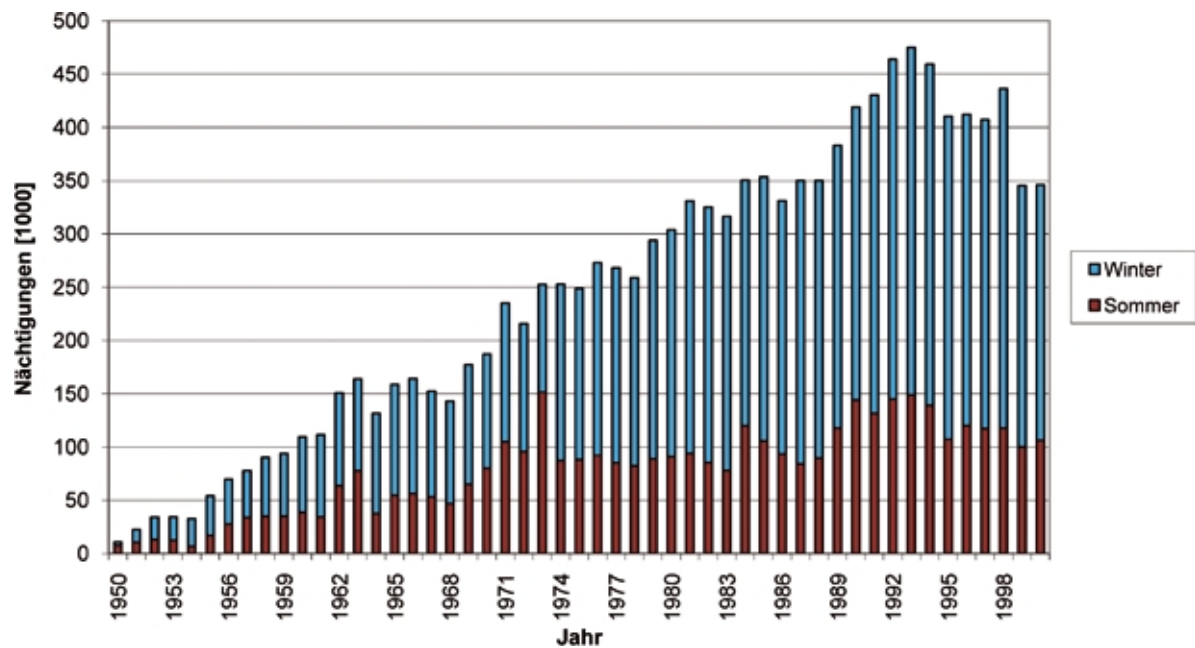


Figure 6: Development of the number of overnight stays in Galtür in the period 1950-2000, from Keiler (2004)

The weekly rhythm of the tourism movements is evident and is characterised by daily fluctuations with a factor of between 4.4 (Neighbourhood) and 6.5 (Season). Over the entire observation period, the number of people in the study area changes by a factor of 6. The weekly rhythm of tourism movements is evident and is overlaid by daily fluctuations with a factor of between 1.4 (low season) and 3.4 (high season) (Figure 7).

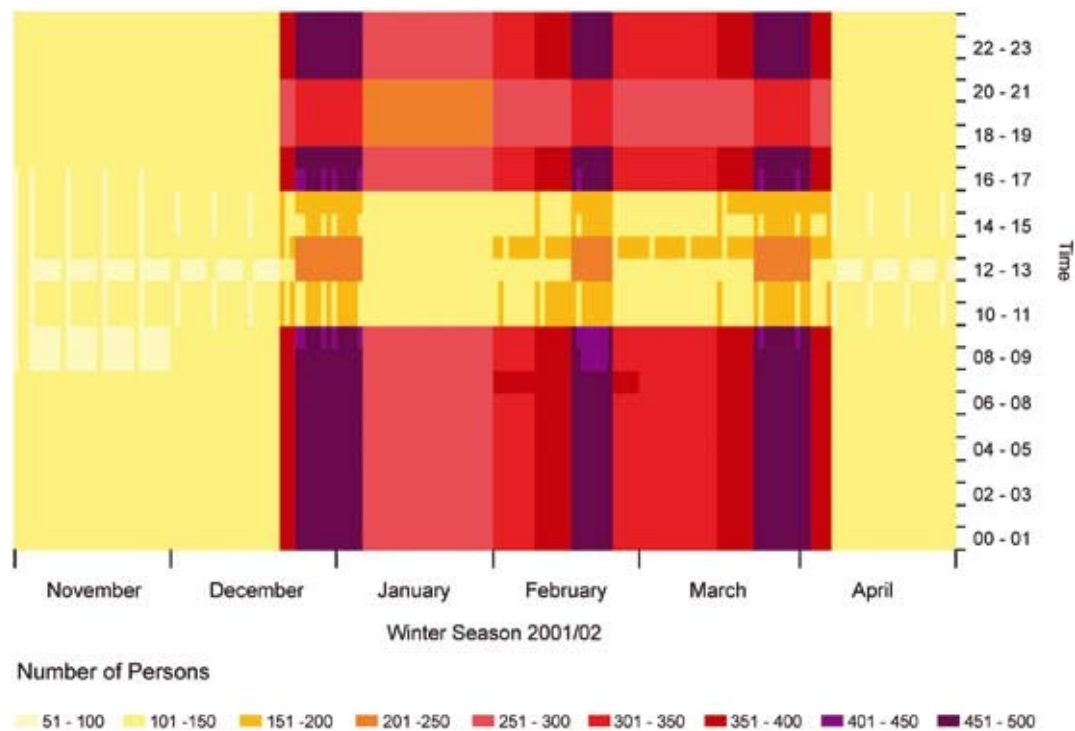


Figure 7: Seasonal, daily and hourly variations of exposed persons in Galtür (Keiler et al. 2005)

4 Summary

Processes of global change (defined in the Human Dimension Programme as changes to the natural and cultural environment) have manifested themselves through intensive changes, especially in the Alpine region in the 20th . In the future, a certain increase in the process intensity or probability of occurrence of processes involving flowing water (floods, mudslide and landslide processes, Solomon et al., 2007) can be assumed. Equally significant, however, are simultaneous processes in the cultural space, especially changes in the presence of people and mobile objects as well as the associated exposed values due to changes in economic and social structures. Long-term changes in the process and socio-economic area are thus overlaid by short-term fluctuations that can lead to risk peaks (Fuchs & Keiler, 2008).

A multi-temporal view in terms of risk analysis records changes in several factors in a study area over a defined period of time. The analysis is linked on several time scales in order to be able to assess the main parameters influencing the risk with the underlying influencing factors. The temporal component in the assessment of natural hazard-induced risks is thus multitemporal and combined on a long-term and short-term scale.

Scale dependence is a basic property of all processes taking place on the land surface. The problem of scaling has two essential components: Direct measurements are generally limited to small spatial and temporal sections and are not directly transferable to the respective higher level, as they differ in their properties. Conversely, large-scale forecasts, such as those currently published at regular intervals by the IPCC, are only able to reflect regional differences to a limited extent. For this reason, direct up- or downscaling is not possible; instead, a combined analysis and assessment must be carried out on different scales. This applies to climate change as well as to changes in socio-economic structures, which can lead to an increase in the exposed damage potential, and has a direct influence on risk management strategies.

Risk analyses carried for natural hazards are generally static approaches, but natural hazard-induced risks are subject to temporal variability, as risk-influencing factors (processes, property values, vulnerability) are variable in the long and short term. In this respect, high losses cannot be ruled out in the event of extreme events, as risk reduction always relates to the calculation of the corresponding design events. These in turn are subject to variability due to the processes of climate change. Extreme events such as the 1999 avalanche winter and the floods in 2002, 2005 and 2006 have shown this.

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Address of the author

Dipl.-Geogr. Dr Sven Fuchs
University of Natural Resources and
Life Sciences, Vienna Institute of
Alpine Natural Hazards Peter-Jorden-
Straße 82, 1190 Vienna
Tel. +43-1-47654-4373
e-mailsven.fuchs@boku.ac.at

Climate change and glacier retreat

Michael Kuhn

1 The relationship between climate and glaciers

Glaciers in all mountain ranges around the world react very sensitively to changes in the climate. Their growth and shrinkage is controlled by several climatic variables: snowfall and distribution by the wind determine the accumulation, temperature, solar radiation and humidity determine the energy balance and thus the melting or evaporation of the ice. In the firn basin or accumulation area of an Alpine glacier, even under today's warm conditions, a surplus remains each year, which sinks into the glacier, transforms from snow to firn to ice and is brought down by the flow of the ice into a zone there is enough energy to melt it.

The reaction of a glacier to climate change therefore also depends on the ice movement, i.e. the steepness and thickness of the ice and the shape of the glacier bed, as well as its length, altitude and exposure to the sun. Different glaciers therefore react differently to a uniform climate change.

There were two small glacier advances in the Alps in the 20th century, around 1920 and around 1980, which is remarkable in that atmospheric CO₂ increased continuously during this time and shows that glaciers react to more than just the greenhouse effect. In general, however, increasing temperatures have taken their toll on glaciers in recent decades: since 1952, for example, the Hintereisferner in the rear Ötztal valley has lost an average of around 28 metres of ice thickness over its entire area and the terrible summer of 2003 cost the Alpine glaciers between 2 and 3 metres of ice thickness. Radar measurements show that the largest Austrian glaciers still have over 200 metres of thick ice in reserve, while many smaller ones are already perforated and some have disappeared completely.

The mass balance of the ice establishes the link between climate, glacier and meltwater runoff. As with the balance sheet of a commercial enterprise, the mass balance of a glacier is the comparison of income and expenditure, which in glaciology is referred to as accumulation (+) and ablation (-).

Accumulation includes snowfall, wind drift after precipitation and the transport of snow by avalanches. Wind is therefore an important climatic factor for glacier formation: after precipitation, snow is removed from the ridges and steep slopes of the glacier surroundings and deposited in the glacier hollows by wind and avalanches - in many cases this doubles the actual precipitation, and it is clear that the shape of the terrain is an important boundary condition for the existence of glaciers. However, as the shape of the terrain does not change over time, glacier changes are entirely due to climate change.

The ablation of ice consists mainly of melting, with only a small amount of evaporation or wind erosion. Melting requires energy (330 joules per gram of ice), which is supplied by the atmospheric energy.

spherical energy flows must be supplied:

Solar radiation, infrared radiation, sensible and latent heat flows.

Solar radiation varies greatly depending on the time of day and season; it can reach 1,000 watts/m² in summer, but a large proportion of this is reflected again. The reflectivity (the so-called albedo) of the surface is around

- 80 % with fresh snow, i.e. only 20 % is absorbed,
- 60 % for summer firn,
- 40 % with snow-free glacier ice, i.e. 60 % is absorbed.

A fresh snowfall in summer therefore reduces the energy of solar radiation available for melting to a third compared to snow-free ice, which shows that even just a few centimetres of fresh snow in summer are much more important for the glacier than the same amount in winter. Alpine glaciers therefore react more to summer weather than to winter weather. The situation is different for glaciers in maritime climates such as Norway, which are more dependent on winter snow.

In cloudless skies, the infrared radiation from the surface of the ice is stronger than that from the atmosphere to the surface, so that the ice can remain frozen on clear nights, even if the air temperature is above zero.

The turbulent flows of so-called sensible heat are directed from the air to the ice when the air temperature is positive; those of so-called latent heat depend on the air humidity (negative in the case of evaporation, i.e. dry air). As measured values of all these energy flows are very rarely available, but as the temperature correlates very well with solar radiation (lots of sun = high temperatures), the daily average air temperature can be used as a parameter for the melting of ice: 1 °C corresponds to approx. 5 mm of ice, for example 4 cm of ice can melt in the course of the day on a summer's day with an average temperature of 8 °C.

2 The relationship between climate and glacier size at first glance

There are therefore two complexes in the mass balance that characterise the relationship between glaciers and climate well: Precipitation and wind promote glacier growth, temperature and radiation promote melting. This means a fundamental difficulty in the climatic interpretation of past glacier sizes, because a glacier advance can either follow a period with lower temperatures or higher precipitation. Without additional information, this decision cannot be made, although pollen or tree rings can be used for glacier advances that occurred longer ago.

Temperature measurements have been available in the Alps for just over 200 years, and precipitation measurements for 150 years, which objectively provide the information required above. At first glance or as a first approximation, the Alpine data match the observed glacier retreat: since the glacier peak of 1850-1860, annual precipitation has decreased by around 10 %, while the annual mean temperature in the Alps has risen by more than 1.5 °C.

In the same period, for example, the well-documented Hintereisferner in the Ötztal has declined as follows:

Length	ca. 1850: 10.3 km	today: 7.3 km
Area	14.1 km ²	8.8 km ²
Volume	1.5 km ³	0.5 km ³

3 Details of the archive - non-linear correlations

The long-term trend in glacier retreat therefore corresponds well with the long-term rise in temperature. At second glance, on closer inspection on a scale of decades or years, the connection between climate variables and glacier behaviour is no longer so clear.

With this time scale, it becomes clear that the climate/glacier relationship does not depend on individual climate parameters such as temperature or precipitation, but on combinations of these parameters, such as the aforementioned snowfall in summer, and that not just two, but a whole series of parameters (precipitation and wind, temperature and humidity, radiation and clouds and albedo, etc.) have an effect.

Since the glacier peak, which occurred in various areas between 1850 and 1860, there have been several episodes in which the majority of Alpine glaciers have either advanced or increased their flow velocity:

Highest level	1850-60	clear moraines
Isolated advances	1890-1900	
General advance	1920	clear moraines
Accelerated movement	1943	
General advance	1980	Moraines in medium-sized glaciers
Heavy ice loss since 1982.		

The reconstruction of the climate from the archive of glacier levels is further complicated by the fact that the speed of ice movement can increase very rapidly during advances, and to an extent that can only be approximately calculated with the current state of knowledge. Examples of average velocities and extreme values can again be given for the Hinter eisferner, for which annual measurements of ice movement have been available since 1894.

Before 1915:	15-30 m/year
1919	125 m/year
1930	7 m/year
1943	55 m/year
1965	15 m/year
1978	45 m/year
today	5 m/year.

It appears that in the accelerated phases the ice not only deforms more quickly itself, but also slides more quickly over the ground because the glacier bed deforms differently under increased pressure. The extent of these accelerations cannot be precisely calculated with the current state of knowledge, so that there is a second question mark in the causal chain climate - glacier size:

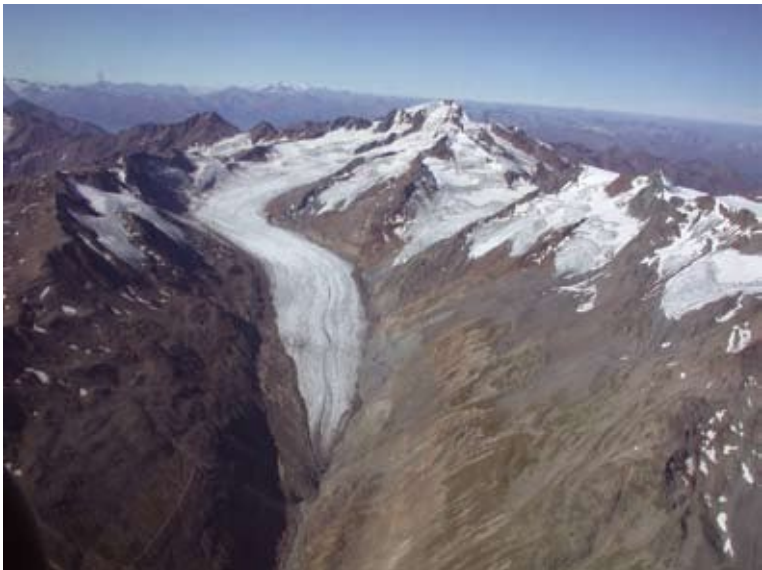
Temperature & precipitation → Mass balance → Ice dynamics → Glacier length, moraine.

This chain can be calculated from left to right within certain limits, but if you read the archive of glacier lengths from right to left in order to reconstruct an earlier climate, then you have fundamental difficulties with the ice dynamics and with the decision of temperature or precipitation.

The long-term development can therefore not be calculated quantitatively, but the strong glacier shrinkage of the last 150 years fits very well qualitatively with the temperature rise, which was about twice as strong in the Alps as the global average. According to measurements and model calculations, a warming of 1 °C corresponds to a melting of 1 m of ice per year on average over the glacier area.

4 Examples of glacier retreat over centuries and decades

After this overview of the climate-glacier relationship, the development since 1894, the beginning of international co-operation in glacier research, will be illustrated with examples and facts (see the pictures on the following pages).

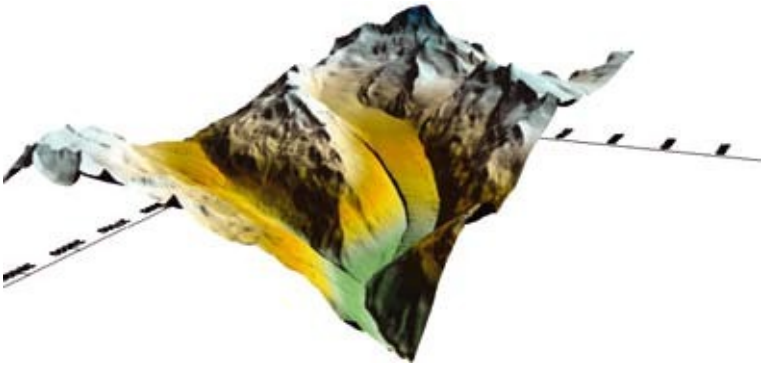


1894

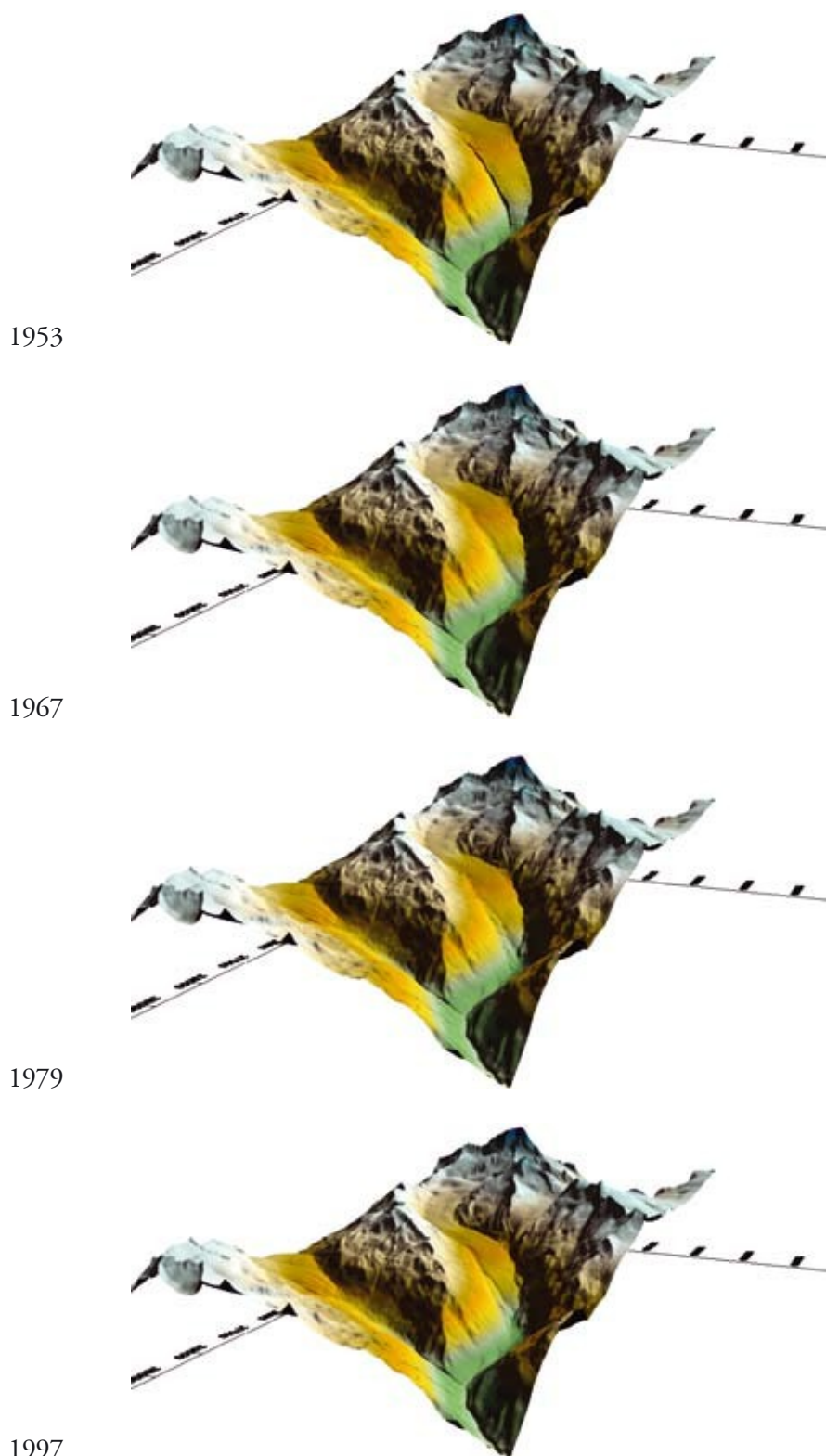
(Norbert Span, 2003)



1920



1939



Figures 1 to 8: The Hintereisferner in the southernmost Ötztal with the summit of the Weißkogel in the hot summer of 2003. From the same perspective, a series of terrain models digitised from precise map representations. The rapid decline of the tongue and the comparatively stationary behaviour of the firn basin are striking. The colours indicate the height above sea level (Processing: Dr. Norbert Span)

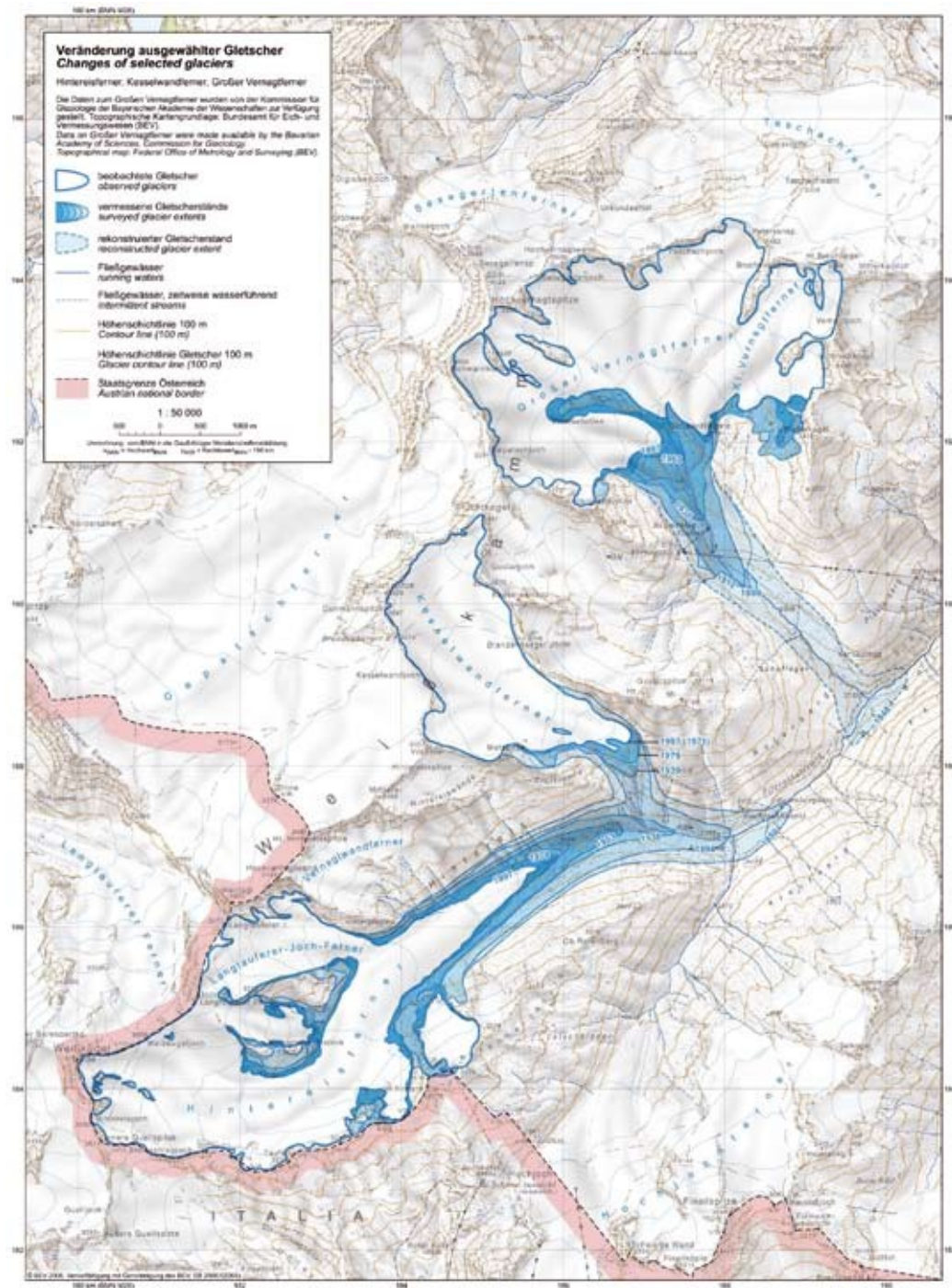


Figure 9: The changes in the three particularly well-documented glaciers Hintereisferner, Kesselwandferner and Vernagtferner since their peak in 1848. From the Hydrological Atlas of Austria

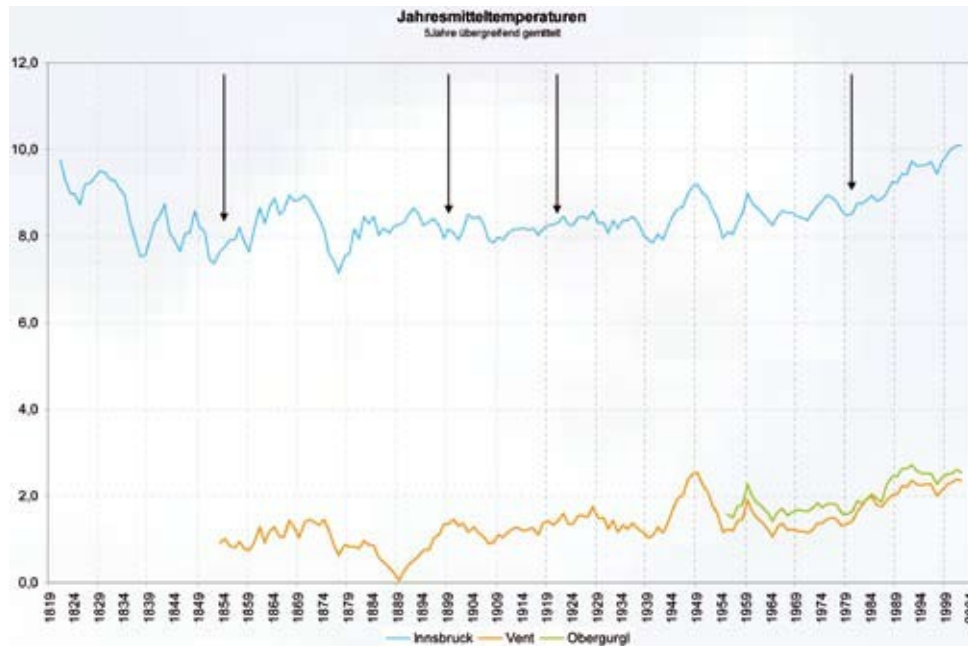


Figure 10: The annual mean temperatures of Innsbruck (580 m), Vent (1,900 m) and Obergurgl (1,940 m), shown as a five-year average. The timing of the fluctuations and trends do not always coincide in Innsbruck and the mountain stations. The arrows indicate the times of general glacier advances; they cannot be derived from the temperature alone (Processing: Fritz Pellet)

Höhe, Größe und Topographie bestimmen individuelle Massenbilanzen

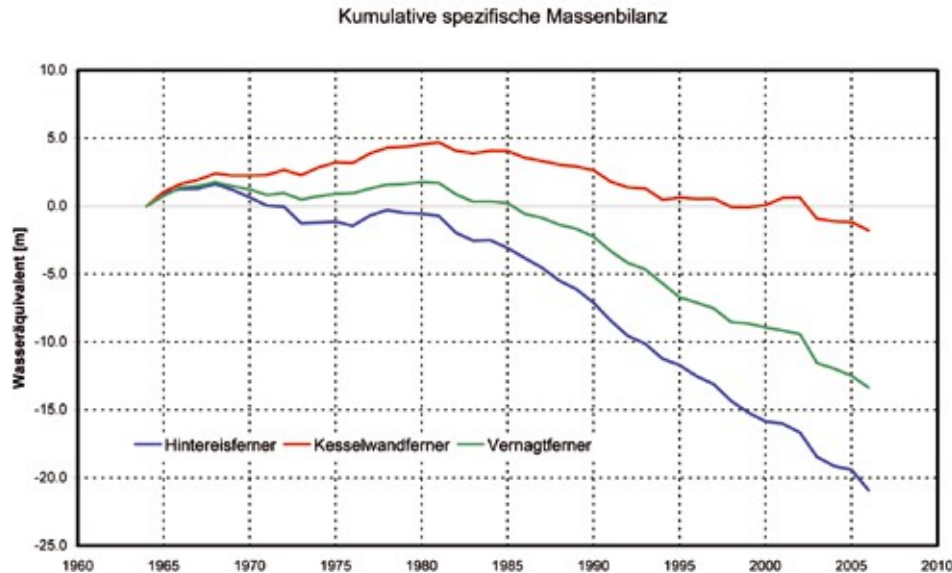


Figure 11: The relative change in the mean specific mass balances of the three glaciers in Figure 9. From autumn 1964 to autumn 2007, the Hintereisferner lost 21 m of water equivalent, i.e. approx. 23 m of ice thickness averaged over the entire glacier. Its tongue has lost more than twice this amount (see also Figure 16). Different height, size and topography have led to different reactions to the same climate change in these three neighbours (Editing: Fritz Pellet)

5 Documentation of recent changes for all Austrian glaciers: the Austrian Glacier Inventories 1969 and 1998

In the summers around 1998, all Austrian glaciers were photographed from the air, analysed photogrammetrically and processed into digital elevation models in 5 and 10 m grids. At the same time, the images from 1969 were according to the current state of the art, so that the changes in height (ice thickness), area and volume from 1969 to 1998 are known. The absolute values of the volumes were determined by radar measurements of the ice thickness on 50 glaciers to date, which approx. 40 % of the total area of the Austrian ice surface. *Figures 13 to 16* show examples of the products of these analyses.

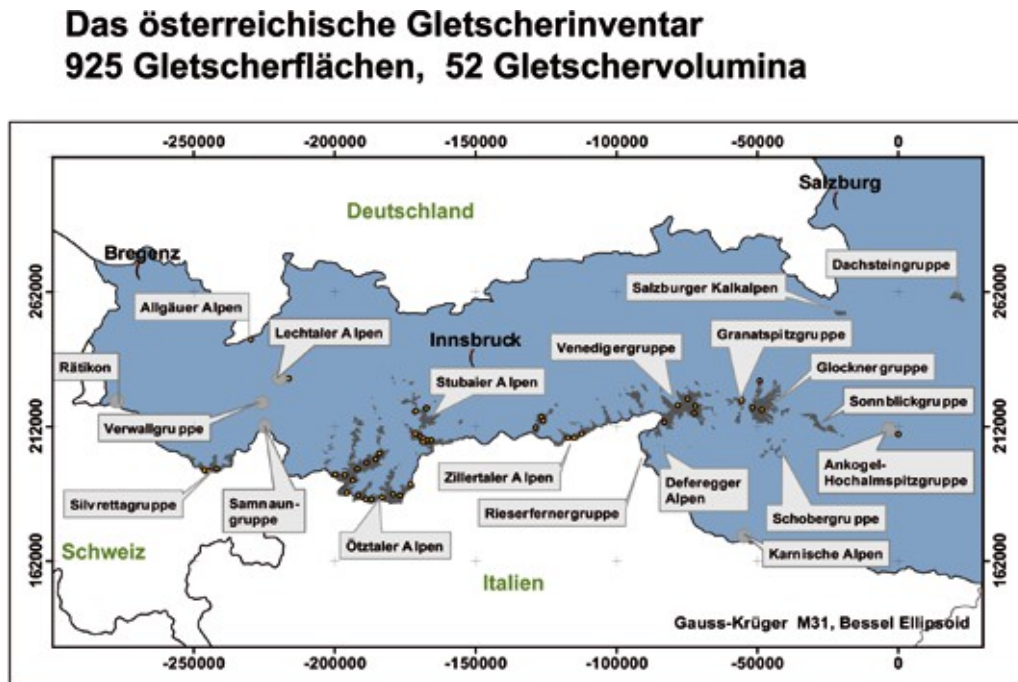


Figure 12: Overview of Austria's glaciers in the 1969 and 1998 inventories. The yellow dots show the glaciers whose ice thickness was measured with radar (Editing: Dr Astrid Lambrecht)

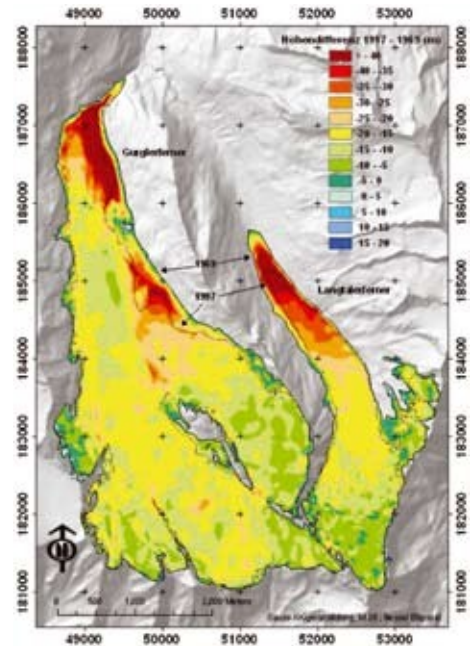
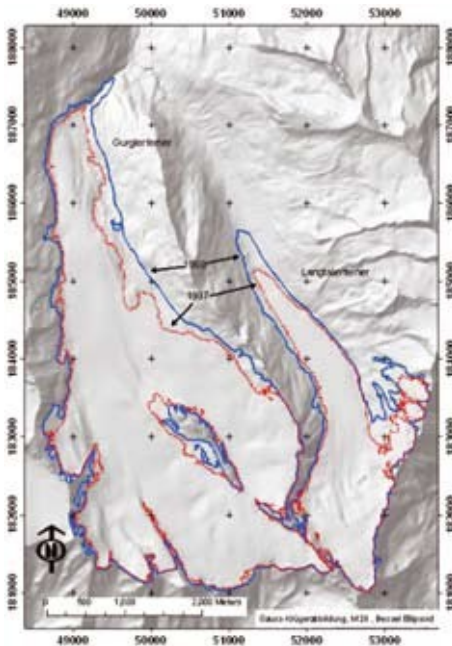
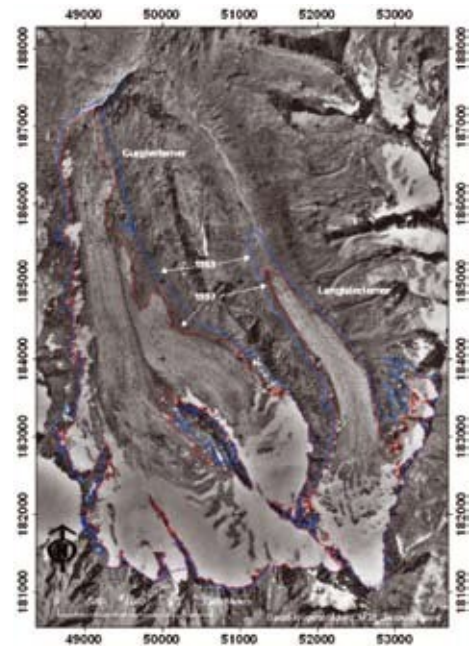


Figure 13: Aerial view of the Längentaler Ferners (left) and the Gurglerferners on 20.9.06, the centre of the picture is at 11° 00' E, 46° 48' N, looking south

Figure 14: Orthophoto of the two glaciers in Figure 13, their extent in 1969 is in blue, that of 1997 in red. The light-coloured firn can be distinguished from the dark glacier ice even better here than in Figure 13 (Processing: Dr Astrid Lambrecht)

Figure 15: Surface model with shading (hillshade) with incidence of light from the north-west (Processing: Dr Astrid Lambrecht)

Figure 16: Surface model with colour code for the change in surface elevation from 1969 to 1997 (Processing: Dr. Astrid Lambrecht)

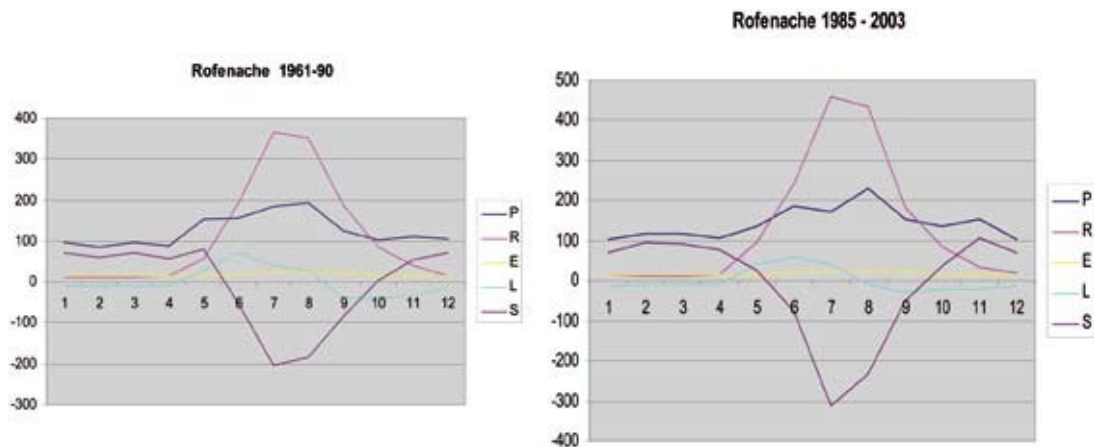
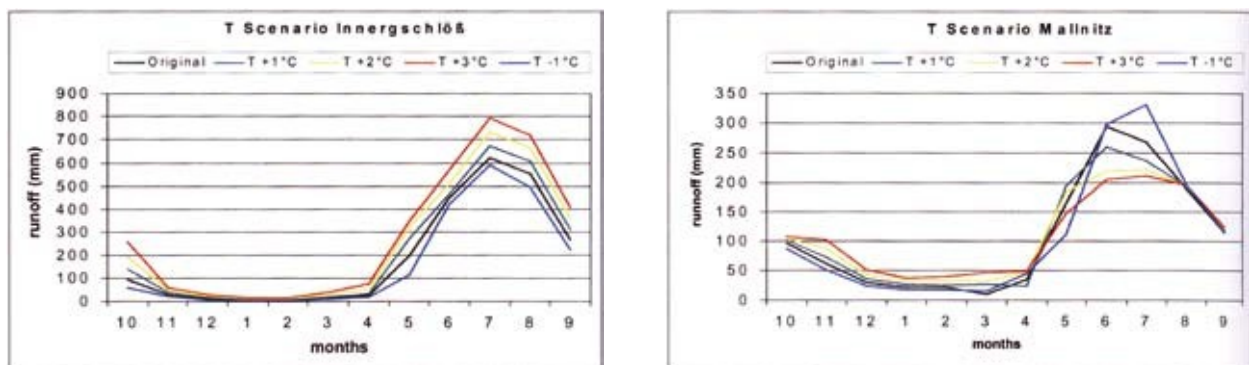


Figure 17: Change in the water balance of the catchment area of the Vent-Rofenache gauge ($10^{\circ} 54' 40''$ E, $46^{\circ} 51' 27''$ N, approx. 1,900 m), in mm per month in the periods 1961-90 and 1985- 2003. Precipitation P, runoff R, evaporation E, storage S, of which liquid storage L. The strong increase in summer runoff by approx. 100 mm per month is mainly due to the melting of glacier ice (the "glacier donation") and only partly to an increase in precipitation. Changes in evaporation and liquid water storage are not significant



Figures 18 and 19: Mean values 1961-1990 of the monthly discharges of the Innerschloß gauge ($12^{\circ} 27' E$, $47^{\circ} 07' N$, 1,686 m) and scenarios for changed temperatures. In this heavily (approx. 38 %) glaciated catchment area, the peak discharge remains in July and August even with a warming of 3 °C, and there is practically no discharge in the winter months. This is different in the Mallnitz area ($13^{\circ} 10' E$, $46^{\circ} 59' N$, 1,174 m), which is only 2 % glaciated, where the runoff peak is significantly flattened and the winter runoff is doubled

Address of the author

o.Univ.-Prof. Dr Michael Kuhn University of
Innsbruck
Institute of Meteorology and Geophysics Innrain
52, 6020 Innsbruck
Phone +43-512-507-5150
e-mail michael.kuhn@uibk.ac.at

Climatic developments and their impact on the water supply in Eastern Styria

Andreas Dalla-Via

1 Introduction

The following article briefly summarises the chapters on climate, groundwater hydrology and water supply in the book "Wasser & Wirtschaft im Klimawandel - Konkrete Ergebnisse am Beispiel der sensiblen Region Oststeiermark" by Prettenthaler & Dalla-Via (2007). The economic aspects are briefly discussed in Chapter 8.

According to the IPCC (2001), the global average air temperature rose by around 0.6 °C in the 20th century. In the Alpine region, temperatures rose by around 1.8 °C. In addition to the trend towards higher temperatures, which can also be observed Eastern Styria (see *Figure 1*), there has been a trend towards lower precipitation in large parts of the Eastern Styrian hill country since the beginning of the 1990s. At the Sinabelkirchen station, for example, precipitation in the years 2000 to 2003 was on average 21 % below the long-term average (1900-1990). Depending on the reduction measures for greenhouse gases, a global temperature increase of 1.3-5.8 °C can be expected in the next 100 years according to the IPCC (2001).

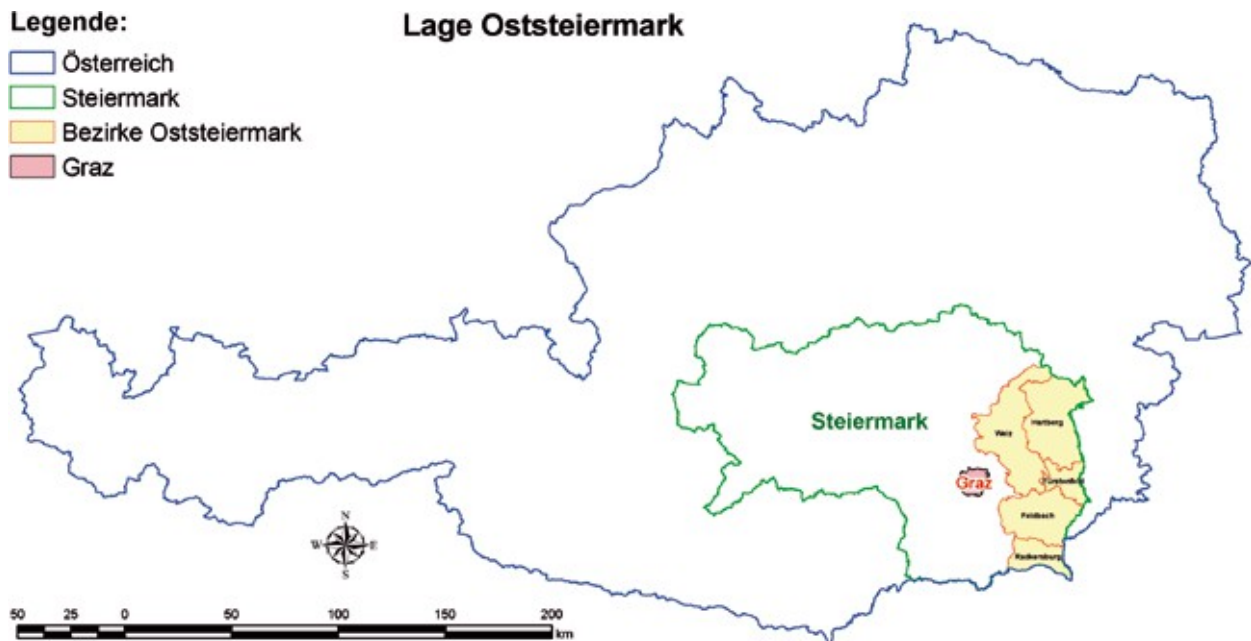


Image 1: Location of Eastern Styria

For Austria, regionalisations of global climate models indicate temperature increases of up to 3 °C over the next 50 years (Formayer et al., 2004 in Kromp-Kolb, 2004). These negative climatic developments groundwater recharge and water demand from the perspective of water management have led to measures being taken by the municipal water suppliers and the water management authorities of the province of Styria in Eastern Styria.

2 Water supply structure

The three main pillars of private and municipal water supply in Eastern Styria are shallow pore groundwater, deep groundwater and springs. Their importance for the water supply in the study area varies greatly depending on their location.

Due to the very unfavourable climatic and hydrogeological conditions (comparatively low and declining groundwater recharge, low permeabilities and groundwater thicknesses as well as inadequate water quality in many places), the shallow, mostly unconfined groundwater bodies of the main valleys in Eastern Styria are generally only suitable for private individual water supplies and not for municipal water supplies.

In contrast, more favourable hydrogeological conditions prevail in the Lower Mur Valley. On the basis of extensive investigations in the Lower Mur Valley - summarised in Fank et al. 2002 and Fank et al. 2003 - new locations for municipal well systems were proposed and the planning basis was developed. However, taking into account groundwater quality, ecology and agriculture, no further extensive abstraction is possible in the Lower Mur Valley.

The availability of spring water in the Eastern Styrian Randgebirge was analysed in a study by Harum et al., 2001. According to this study, although there is still sustainably usable spring water in the Eastern Styrian Randgebirge, the practical usability of these resources is very limited due to the long distances and the low water availability per unit area.

In the NANUTIWA project, "Fundamentals for the sustainable utilisation of deep groundwater in the eastern Styrian and Pannonian basin" were collected in the Raab catchment area (Domberger et. al., 2004). Deep groundwater is also used for municipal water supply south of the Raab catchment area. Analyses show declining supply trends at some wells (Dalla-Via & Fank, 2006). However, an overall assessment of the future development and utilisation possibilities of deep groundwater in the study area is extremely difficult due to the heterogeneous subsurface structure, the difficult to quantify recharge mechanisms, the wide-ranging hydraulic interrelationships and the currently still inadequate database.

3 Observed climatic developments

According to climate data from the Styrian Hydrographic Service, there are no significant trends towards a decrease in precipitation in the eastern Styrian Randgebirge and Lower Murtal. However, significant decreases in the annual precipitation totals can be in the Riedelland region of Eastern Styria. At the Sinabelkirchen station, an average of 11% less precipitation fell between 1991 and 2003 than the long-term average (1900-1990). With regard to air temperatures, an increase can be observed at all the stations analysed from the Bergland to the Lower Murtal, with this being strongest in the Riedelland region of Eastern Styria.

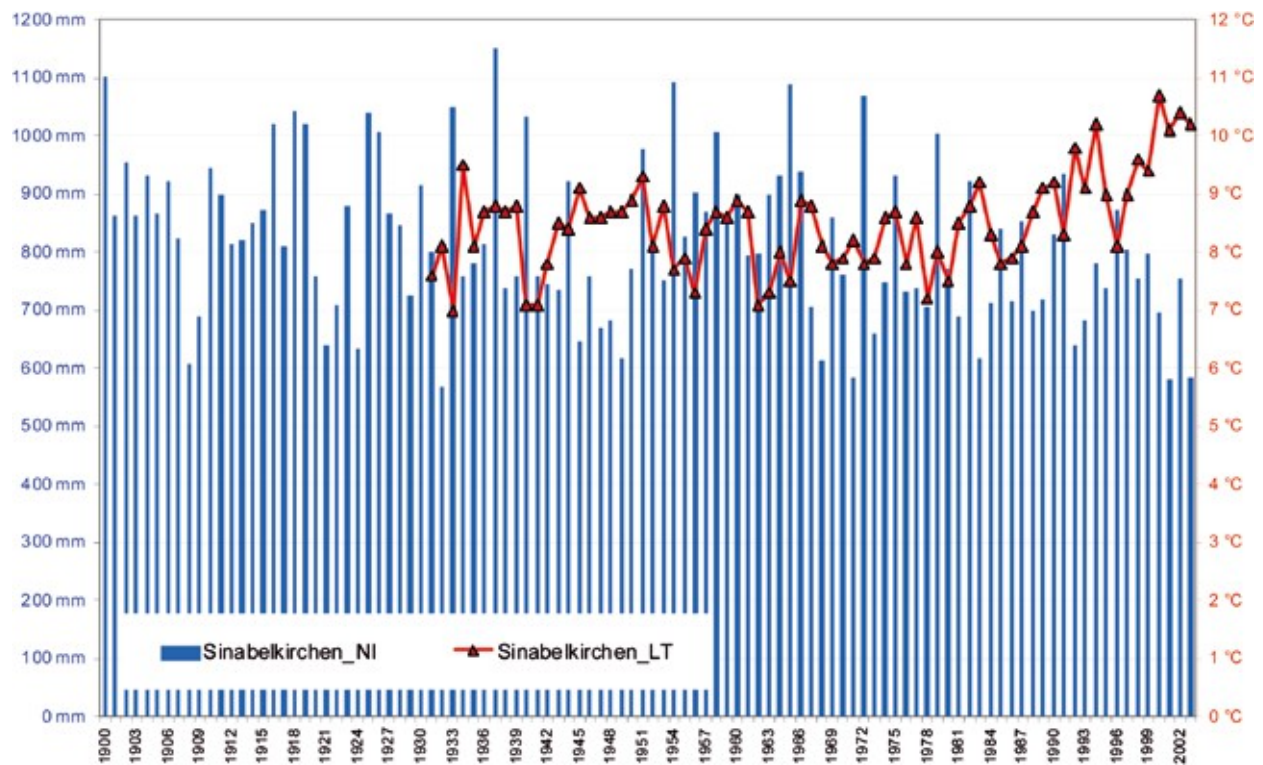


Figure 2: Annual sums of precipitation (1900-2003) and annual mean air temperature (1931- 2003) at the Sinabelkirchen station in the eastern Styrian hill country

Table 1: Comparison of mean air temperatures and mean annual precipitation totals for the period 1991-2003 with the period 1961-1990 (data basis: climate data from the Styrian Hydrographic Service) *as of 1983

Cf. 1961-1990 / 1991-2003	Sea level	Location	Air temperature annual average	Precipitation Annual total
Vorau	690	Bergland		- 1 mm
St. Jakob im Walde	922	Bergland	+ 0.85 °C*	+ 66 mm
St. Johann b. H.	410	Bergland/Riedelland	+ 0,48 °C	- 1 mm
Sinabelkirchen	350	Riedelland	+ 1,33 °C	- 58 mm
Tenting	200	Lower Mur Valley	+ 0,83 °C	+ 1 mm

4 Climate scenario for the 2040s for Eastern Styria

In Austria, the possibilities of regional climate modelling and its applicability to the Alpine region have been investigated since mid-2003 in the project reclip:more (Research for Climate Protection: Model Run Evaluation) (Gobiet et al., 2004; Loibl et al., 2004; Truhetz et al., 2005). For the project "Water & Economy in a Changing Climate" - Prettenhaler & Dalla-Via (2007), Gobiet & Truhetz created an analysis of the climate for the period 1981-1990 that was as realistic as possible by embedding the RCM in so-called re-analyses of the European Centre for Medium-Range Weather Forecasts (ERA-40).

was analysed. This climate analysis can be compared with corresponding climate observations and therefore also provides the basis for the method evaluation. On the other hand, the RCM is driven by boundary conditions from the global climate model ECHAM5 (Roeckner et al., 2003) in order to generate a regional scenario for 2041-2050 based on the emission scenario IS92a (Leggett et al., 1992) and the corresponding control run.

The climate change signal is derived from differences between two regional climate simulations driven by boundary conditions from a global climate model. The first simulation describes the reference period (1981-1990, "control run") and the second a decade in the future (2041-2050, "scenario"). The climate change signals can then be added to the climate of the reference period (1981-1990) to generate a climate scenario for the 2040s.

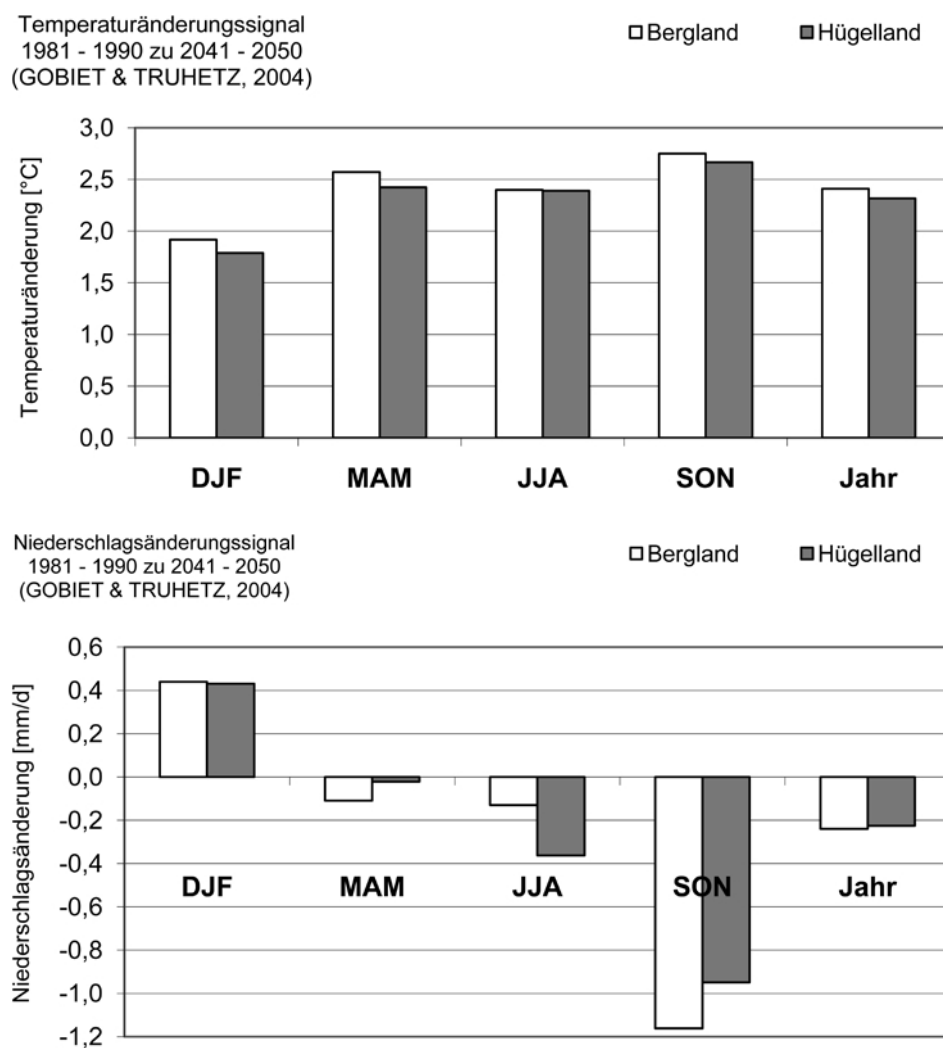


Figure 3: Seasonal temperature (top) and precipitation change scenario (bottom) 1981-1990 compared with 2041-2050 DJF: Dec, Jan, Feb; MAM: March, Apr, May; JJA: June, July, Aug; SON: Sep, Oct, Nov; Year: annual mean value

The results show a significant temperature increase of 1.8-2.8 °C with less warming in winter and peaks in autumn. In general, the simulation shows a slightly stronger

warming in the mountains (~ 0.1 °C), although this difference should not be considered significant. The simulated temperature increase of around 2.4 °C is well above the error limits of the method and can be regarded as a well-supported prediction.

The expected change in precipitation varies greatly from season to season, with more precipitation expected in winter (0.4 mm/d, ~ 30 %) and significantly less in autumn (up to -1.2 mm/d, ~ 45 %). A slight decrease in precipitation was simulated as an annual average (-0.2 mm/d, < 10 %), although this value cannot be significant. The simulated precipitation changes in winter and autumn are above roughly estimated uncertainties of the method and should be taken seriously (by analysing their impact on water supply), but they should be considered as a possible scenario and not as a prediction in the strict sense.

5 Effects on groundwater recharge

Groundwater recharge from infiltrating precipitation is the most important input factor for groundwater. In addition to climatic parameters, the amount of groundwater recharge is also strongly determined by land use and the storage capacity of the soil. Based on climate scenario calculations for the 2040s (change signals compared to the 1980s) by Gobiet & Truhetz, see Chapter 4, the extent of the reduction in groundwater recharge compared to the reference period was analysed using a number of simple empirical methods.

On maize fields with winter fallow, for example, the reduction in groundwater recharge (1980s compared to 2040s) would be between 18 and 46 % according to the method of Renger & Wessolek (1990). For maize areas without winter greening, the reduction in groundwater recharge calculated using the Haude single-layer balance model would be between 31 and 45 % or 87 and 118 mm depending on the usable field capacity (nFk) (see Table 2). For grassland, the reduction in groundwater recharge is between 28 and 66 % depending on the nFk (see Table 3).

Table 2: Reduction of groundwater recharge at the Fürstenfeld station for maize fields without winter greening with selected usable field capacities in the 2040s compared to the 1980s according to the Haude single-layer balance model

Land use	nFk	GWNB_JS_1980s	GWNB_JS_2040s	Reduction	Diff_mm
Maize - winter fallow	70	285	198	31 %	87
Maize - winter fallow	100	273	172	37 %	102
Maize - winter fallow	135	267	150	44 %	117
Maize - winter fallow	170	263	145	45 %	118
	MW:	272	166	39 %	106

Table 3: Reduction of groundwater recharge at the Fürstenfeld station for grassland areas with selected usable field capacities in the 2040s compared to the 1980s according to the Haude single-layer balance model

Land use	nFk	GWNB_JS_1980s	GWNB_JS_2040s	Reduction	Diff_mm
Grassland	70	146	105	28 %	41
Grassland	100	114	72	37 %	42
Grassland	135	83	43	49 %	41
Grassland	170	63	21	66 %	42
MW:		101	60	45 %	41

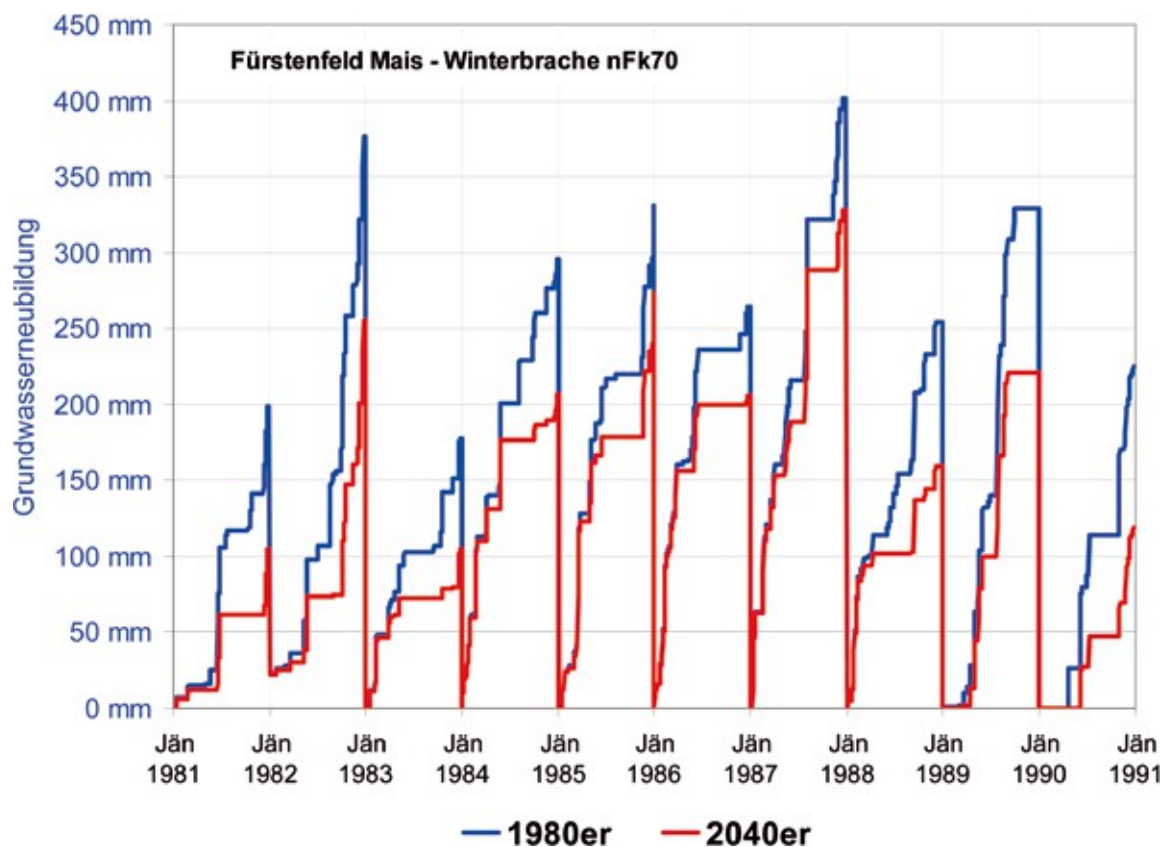


Figure 4: Temporal development of groundwater recharge at the Fürstenfeld station for maize areas without winter greening at a usable field capacity of 70 mm (method: Haude single-layer balance model) in the 1980s and 2040s in comparison

6 Effects on groundwater resources

The Styrian Hydrographic Service has been operating a more or less dense network of groundwater measuring points in Eastern Styria and the Lower Mur Valley since around the mid-1960s, depending on their significance for water management. Due to the individual boundary conditions (groundwater recharge, location in relation to surface waters or influencing groundwater bodies), the groundwater monitoring network has been

The groundwater dynamics at each measuring point are more or less different depending on the type of water abstraction.

An analysis of groundwater levels at selected groundwater measuring points in the periglacial main valleys of Eastern Styria (gauge 5269 in the Raabtal, gauge 5791 in the lower Ilztal) in the low water area shows that between 1991 and 2003 the lowest groundwater levels since the mid-1960s (start of recording) were reached. The lowest groundwater levels were recorded in 1992, 1993, 2000, 2001, 2002 and 2003. Although the groundwater levels in the last few years up to 2003 were not significantly lower than at the beginning of the 1990s, the accumulation of extremely low groundwater levels and the comparatively long duration of the low groundwater phases (especially in 2002) is striking.

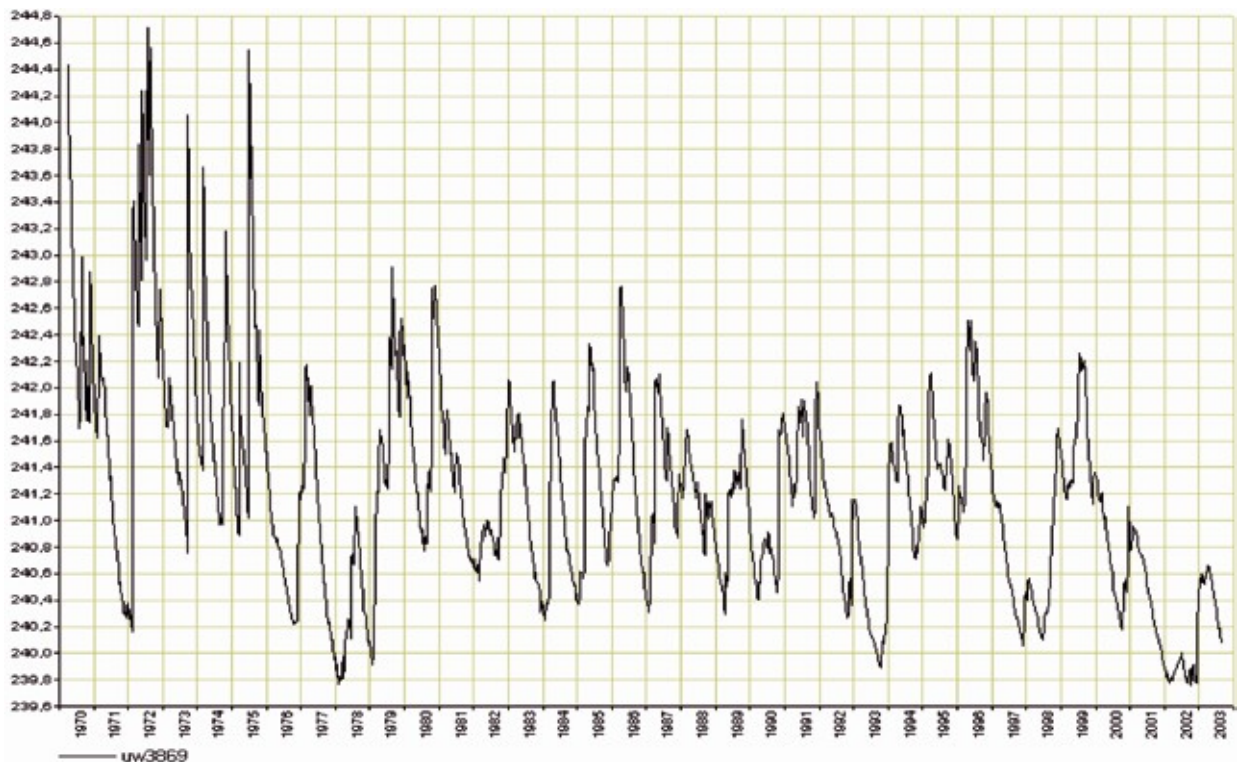


Figure 5: Groundwater hydrograph at gauge 3869 (on the lower terrace north of Lichendorf) from 1970 to 2003 in the Lower Mur Valley

Groundwater measuring point 3869 of the Styrian Hydrographic Service is located on the lower terrace of the Lower Mur Valley, which is only slightly influenced by the discharge of the Mur compared to the Auzone, and where groundwater levels have been recorded continuously since 1970. An analysis shows that the groundwater levels here in 1977 and 1978 were about as low as in 2002 and 2003, but here too the increasing frequency and duration of low groundwater levels has been noticeable in recent years.

In view of the future climatic scenarios (see Chapter 4) and their effects on groundwater recharge (see Chapter 5), an accumulation of low and falling groundwater levels in the eastern Styrian Riedelland can be expected in the future.

7 Impact on water demand and water supply

Extremely hot and dry summers such as those between 2000 and 2003 led to a sharp drop in groundwater levels in Eastern Styria, with record lows being reached in parts of the study area since records began in the 1960s. At the same time, the peak demand for water rose sharply in the summer months. The reason for the increased demand for water on the part of the municipal supply that numerous private wells, which draw shallow pore groundwater, have comparatively low average water levels and low permeability. This means that the same amount of lowering of the groundwater level in the periglacial valleys of Eastern Styria generally has a much greater effect than in the Grazer Feld, for example. As a result, numerous households and businesses could not make do with their private wells and had to connect to the municipal water supply. It was only possible to maintain the supply through additional external water and the construction of new well systems in the Lower Mur Valley.

Surveys of water suppliers (2005) on the supply situation in 2003 (Dalla-Via in Prettenthaler & Dalla-Via, 2007) show that the majority of water suppliers reached the supply limit in 2003.

Based on the climate scenario for Eastern Styria (see Chapter 4) and the resulting reduction in groundwater recharge (see Chapter 5), a further increase in connection densities can be assumed.

Future "dry years" will have a particularly negative impact from a groundwater hydrological perspective because the average groundwater levels would fall in the current climate scenarios and therefore lack the necessary reserves for such extreme events. This affects both municipal and smaller private well systems or springs in the same way. This means that there will be further new connections to the municipal water supply network during such dry periods.

In addition, many households with a public water connection also have a domestic well and use it primarily for service water purposes. During dry periods in particular, these private water resources are available to an even lesser extent (further drop in the groundwater table), which leads to a further temporary increase in peak demand despite a possibly already high connection density (> 90 %).

There is a lack of comprehensive basic data for an exact quantification of future water demand in the study area. Nevertheless, a rough estimate of future additional water demand from private households is possible:

The connection densities of all 192 municipalities in the study area are known from the census results (2001). Due to the developments mentioned above, a further increase in the minimum connection densities is to be expected.

Of course, due to the different natural and settlement-related in

There will continue to be a certain range of connection densities in the future. However, if, for example, a minimum connection density of 75% is , this results in an additional average water demand of 88 l/s or an additional peak demand of 158 l/s for the hilly region, including the Lower Mur Valley, but excluding municipalities above an average altitude of 600 m above sea level. If the highlands were included, the peak demand would rise to 220 l/s with a minimum connection density of 75 %.

Assuming that there will not only be an increase in the connection density in municipalities with a low connection density today, but that the connection rate and consumption will increase in all municipalities, the additional peak demand in the hilly region alone, including the Lower Mur Valley, will be in the order of 200 l/s (see *Figure 6*).

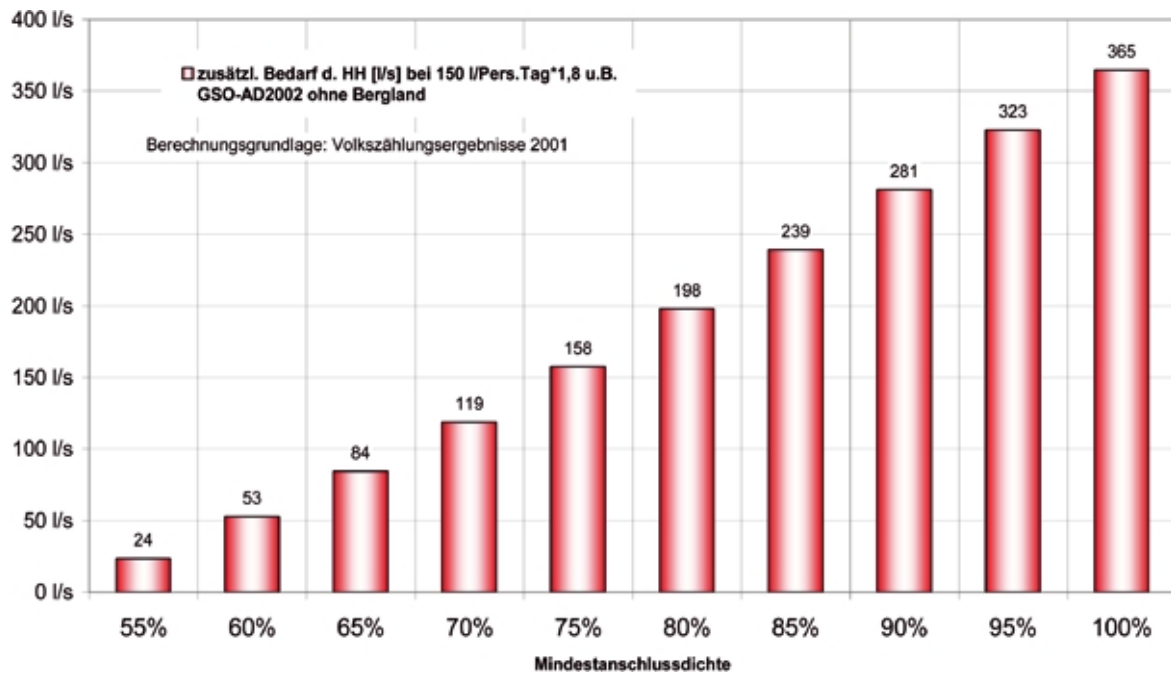


Figure 6: Additional peak water demand of households (150 litres/person and day times 1.8) based on the 2001 census results, taking into account the connection densities of the GSO without Bergland

Increased recourse to the groundwater reservoir of the Lower Mur Valley has actually been achieved through the realisation of corresponding large well locations and the closing of gaps in the supra-regional supply network. The inclusion of drinking water from long-distance pipelines is now also in the implementation phase (transport pipeline from Graz via Gleisdorf to Hartberg - start of construction in autumn 2007).

8 Economic aspects

The economic aspects of water scarcity in Eastern Styria were analysed under the direction of F. Prettenthaler (RTG - Joanneum Research) and are described in detail in Prettenthaler & Dalla Via, 2007. In a nutshell, it can be stated that around 20,700 people work in businesses in Eastern Styria for which water plays an indispensable role.

The accommodation sector, which employs around 8,000 people, is a key component of the production process and, as was assumed before the start of the study, is the most prominent. All of these sectors together generate a production value of around € 2.7 billion per year or a gross value added of around € 1 billion. If there were to be a loss of production (or a lack of overnight stays) for even just two weeks due to water shortages, this would result in economic losses of € 40 million and turnover would fall by € 105 million.

Looking at the future development of these sectors, a further increase in employment of around 4,000 to 6,000 people is forecast - assuming an uninterrupted water supply - with the catering and accommodation sector continuing to be the most dynamic. If the water supply cannot be secured beyond the current level, investments in the water-intensive sectors and, as a result, the aforementioned employment opportunities will be jeopardised.

The "Eastern Styria transport pipeline" with the inclusion of drinking water from long-distance pipelines and the networking of as many regional water suppliers as possible to reduce the individual vulnerability of the regions in the event of drought ("closing gaps in the regional supply network") lead to investment costs of around € 15 million for the transport pipeline (if constructed with plastic pipes) and € 47 million for closing gaps. If we compare this with the economic damage resulting from the loss of added value in the event of a two-week production stoppage in water-intensive industries if the status quo in the drinking water supply is maintained, as mentioned above, an economic justification for the measures is quickly found.

With regard to the question of the extent to which demand-generating measures such as an active pricing policy or the increased use of water-saving technologies could represent possible alternatives to the construction of the transport pipeline, the study comes to a negative conclusion: even if only 50 % of the capacity of the transport pipeline were to be saved, the price of water, which would reduce the quantity demanded in this way, would have to be around two and a half times the current level, a scenario that hardly seems realistic due to the lack of political acceptance by the population. However, water-saving technologies also have too little potential overall to be seen as a serious alternative to infrastructure investment.

9 Summary

A downward trend in precipitation and the upward trend in air temperatures, particularly in the central area of the East Styrian hills since the early 1990s, meant that the majority of municipal water suppliers reached their supply limits in the extremely dry year of 2003. Due to the comparatively unfavourable hydrogeological conditions, many domestic wells were no longer able to supply the affected properties and the connection densities rose sharply. If the climate scenarios for Eastern Styria for the 2040s materialise (Gobiet & Truhetz, 2004), groundwater recharge would be reduced more or less significantly depending on the soil and land use conditions. The expected higher temperatures and the additional people whose private domestic wells and springs will no longer be able to meet the water requirements of the

The need to cover the entire year will lead to a further increase in connection densities. Based on existing groundwater hydrological studies in the areas of spring water in the mountains, deep groundwater and pore groundwater, there are no major additional water resources in the study area.

If there to be a loss of production (or a lack of overnight stays) for even just two weeks due to water shortages, the economic damage would amount to around 40 million€ (F. Prettenthaler in Prettenthaler & Dalla-Via, 2007). The potential for savings through increased water prices or water-saving technologies does not appear to be reasonable or sufficient.

For the reasons mentioned above, the ongoing construction of a water transport line for the external supply of Eastern Styria and the further networking of water suppliers is considered to be very sensible.

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Address of the author

Dr Andreas Dalla-Via Joanneum
Research
Institute for WaterResourceManagement
Elisabethstraße 16/II, 8010 Graz
Phone +43-316-876-1375
E-mail: andreas.dalla-via@joanneum.at

Switzerland in 2050 - Effects of climate change on the water cycle and water management

Bruno Schädler

Summary

Climate change, which has been observed for over a hundred years and is becoming increasingly apparent in the future, has an impact on the water cycle and on all areas of water management. Based on the available observation data and the latest scientifically compiled climate scenarios for the year 2050, expert groups led by the "Advisory Body on Climate Change OcCC" have summarised how the individual sectors and user groups, e.g. water supply, energy production, flood protection or agriculture, could be affected.

1 Overview of changes to date

Thanks to the measurement networks of the national meteorological, hydrological and glaciological services established in the 19th century, we now know that the temperature in Switzerland has risen by 1.6 degrees in western Switzerland, 1.3 degrees in German-speaking Switzerland and 1.0 degrees on the southern side of the Alps in the 20th century (SAEFL, 2002). The precipitation regime has also changed. Annual precipitation has increased by around 113 mm (8 %) in the 20th century. As evaporation from the land surface, vegetation and watercourses also increased by 99 mm as a result of warming, the total annual average discharge of our watercourses - the result of the water balance - has remained practically unchanged (*Fig. 1*).

Similar to the air temperature, the temperature of Swiss watercourses has risen by up to 2 °C in recent decades, with the increase in water temperatures being greater on the Central Plateau than in the Alpine regions (Jakob et al., 2007). However, these watercourses with catchment areas influenced by glaciers do not show a significant increase in water temperatures. The increase in water temperature in lower-lying areas is primarily reflected in the increase in periods with higher temperatures (*Figure 2*). It is these peaks that particularly troublesome for cold-water fish (trout, grayling) and displace them to higher regions.

The changes in climate have had a direct impact on the glaciated areas and the mass balance of the Alpine glaciers (*Fig. 3*). Since the glaciers peaked in the middle of the 19th century, the Swiss glaciers have lost around 45 per cent of their area and more than half of their volume. In comparison, the losses in the entire European Alpine region have been even greater because the average height of the glaciated regions there is lower than in Switzerland. Although the losses in Switzerland amount to around 55 billion cubic metres of water, they only correspond to just under one year's precipitation and, at an average of around 1 %, represent a very small proportion of the annual Swiss territorial runoff. For a small or medium-sized watercourse in the mountains, however, the glacier melt in spring and summer can make a considerable contribution, especially in warm years.

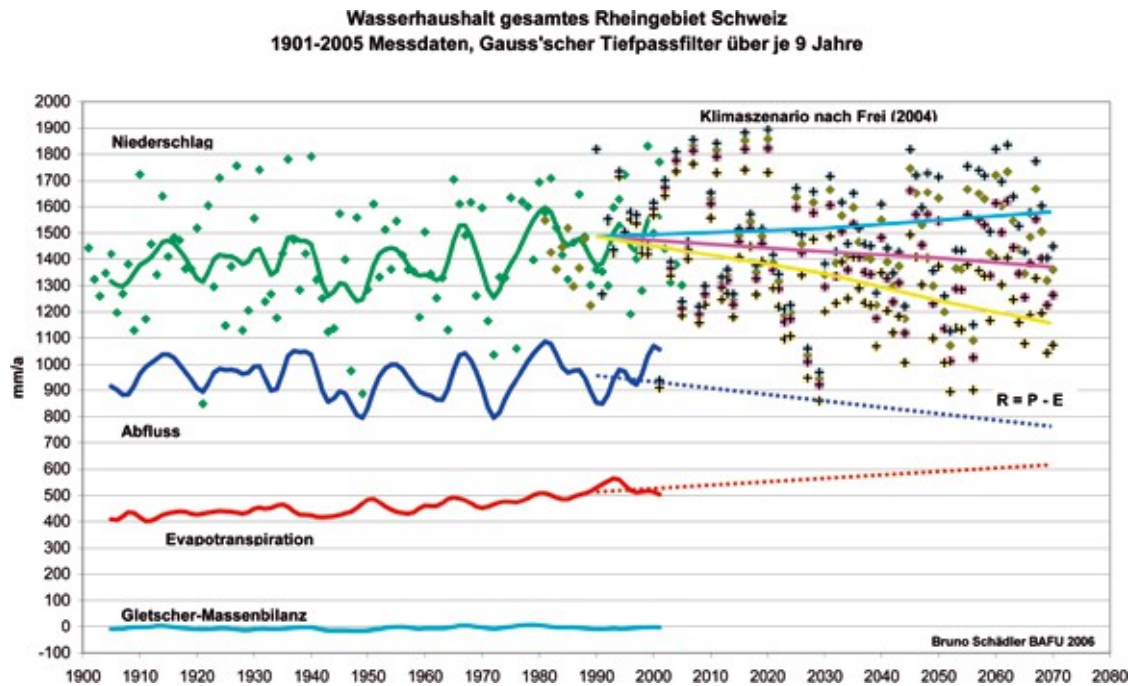


Figure 1: The water balance in Switzerland measured since 1900 (Schädler et al., 1992; FOEN, 2007) and as a scenario up to 2070. The annual precipitation values and the low-pass filtered values over 9 years each for precipitation, runoff and evaporation are shown up to 2005. The 5%, 50% and 95% quantiles from Frei (2004) are shown for the precipitation scenarios (analogue Figure 4). In accordance with these scenarios, the annual values from the 20th century have been inserted. 21st century have been transferred and modified. Evaporation has been extrapolated linearly and runoff has been calculated from the water balance using the 50% quantile of precipitation

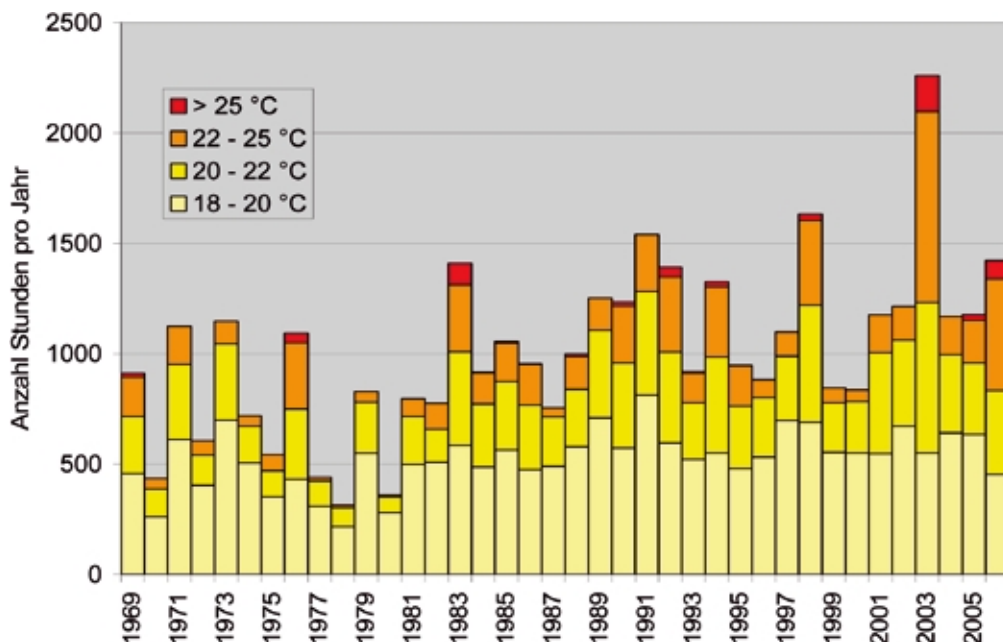


Figure 2: Duration in hours of the water temperature for certain temperature ranges in the Thur in Andelfingen (catchment area 1696 km², mean altitude 770 m above sea level) for the period 1969-2006 (after Jakob et al., 2007)

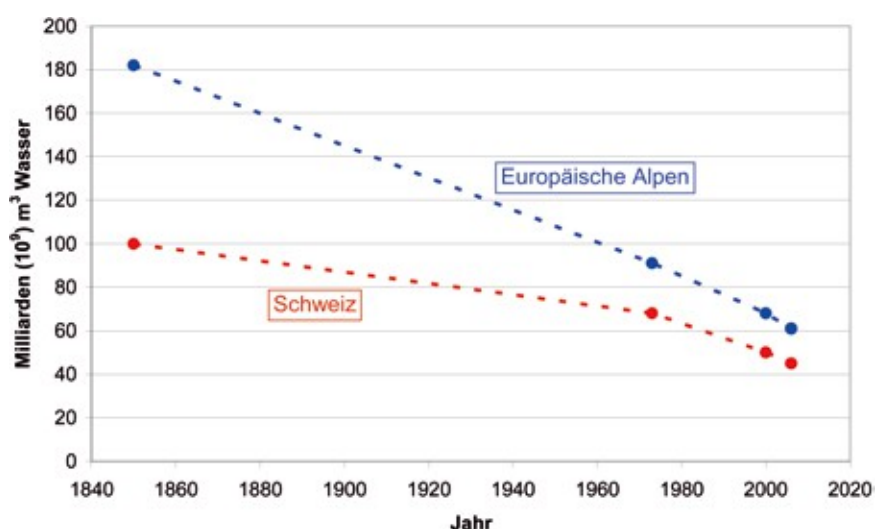


Figure 3: Water volumes stored in glacier ice in Switzerland and the European Alps time from 1850 to 2005 (after Maisch et al., 2004; Huss et al., 2008; Zemp et al., 2006)

2 Future scenarios

How will global climate change affect Switzerland in 2050? What are the possible consequences and vulnerabilities for the environment, economy and society in Switzerland? Are new measures and adaptation strategies necessary? These questions were discussed by the "Advisory Body on Climate Change (OcCC)" together with over 100 experts from a wide range of disciplines in nine working groups and the answers were set out in a comprehensive report (OcCC, 2007).

The OcCC was set up by the Swiss government in 1996 and is tasked with formulating recommendations on climate and climate change issues for the attention of politicians and the administration. Around 30 personalities from research, business and the federal administration are involved in this specialised committee.

2.1 The climate scenario up to 2050

To ensure that all nine working groups assumed the same future prospects, the OcCC provided climate scenarios that correspond to the current state of knowledge (IPCC, 2007). The scenarios described here are based on extensive simulations with global and regional climate models in European climate research projects (PRUDENCE, Christensen et al., 2007; STARDEX, Goodess, 2003), and their specific evaluation for the Alpine region (Frei, 2004; Frei et al., 2006; Schmidli et al., 2007). The inclusion of various scenarios on the future emission of greenhouse gases allows a quantitative description of the uncertainties involved. The scenarios describe the changes up to the middle of the 21st century (2050) compared to the end of the 20th century (1990).

The **temperature** will therefore rise by 1 to 3.5 °C. The seasons differ only slightly and regional differences in warming are smaller than the estimated uncertainty.

range (Figure 4). As a result of the warming, the vertical temperature structure in the atmosphere is shifting. As a result, the mean snow line and the permafrost line are also expected to rise by 150 to 600 metres.

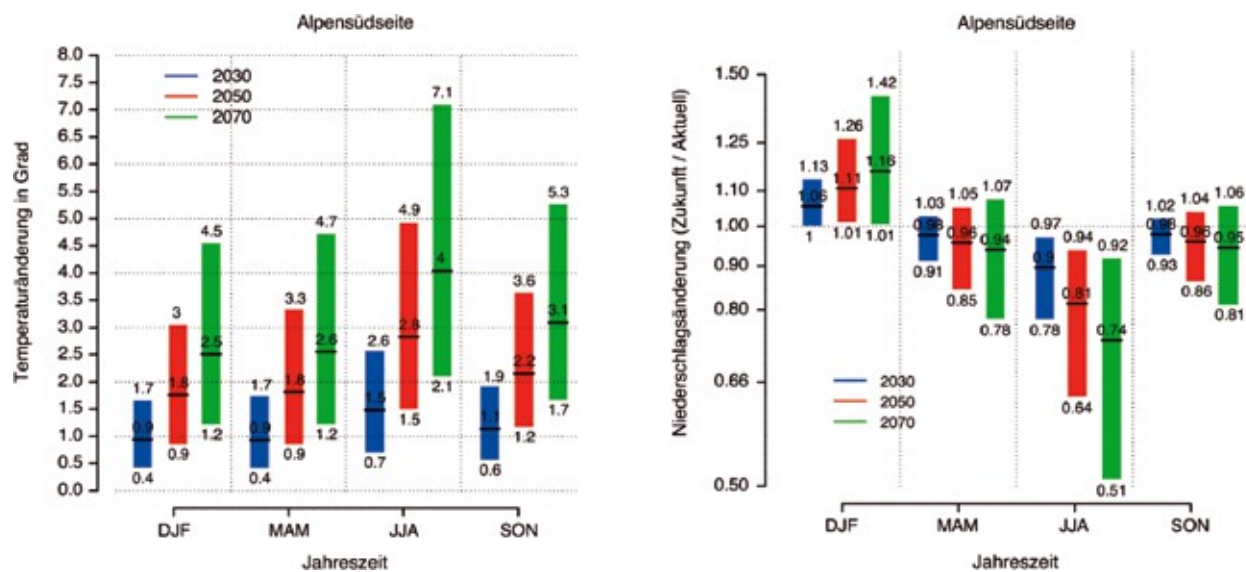


Figure 4: Change in mean temperature (left, in degrees) and mean precipitation (right, future/present ratio) in the four seasons for southern Switzerland. The bars show the uncertainties and the lines the best estimate of the change in each case. The changes are shown for the periods 2020-2040 (blue), 2040-2060 (red), 2060-2080 (green) compared to 1980-2000 (from Frei, 2004)

In contrast to the temperature, the mean annual variation in **precipitation** varies considerably, with increases of 0-20 % in winter (December-February) and decreases of 5-30 % in summer (June-August). In spring and autumn, the changes are in between. The shift towards drier summers dominates the change in annual precipitation amounts, for which a decrease of 5-10 % is estimated. (Figures 1 and 4). On average, these changes result in a decrease in mean annual precipitation of 5-7 per cent (correspondingly in the north by around 75 mm and in the south by around 140 mm).

The analyses of the climate simulations allow a rough quantification of the **precipitation extremes** for annual periods between 5 and 50 years. There are only slight differences for the extremes with a duration of between 1 and 5 days. A distinction must be made between the seasons: In autumn, an increase in extreme values of up to 10 % is expected on the northern side of the Alps and 20 % on the southern side of the Alps. In winter and spring, the increase is between 0 and 20 % on both sides of the Alps. In the worst-case scenario, a 100-year event today could become a 20-year event in the future (Figure 5; Frei et al., 2006). In winter and spring, an increase in precipitation volumes is expected as a result of the combination of higher peaks and longer duration.

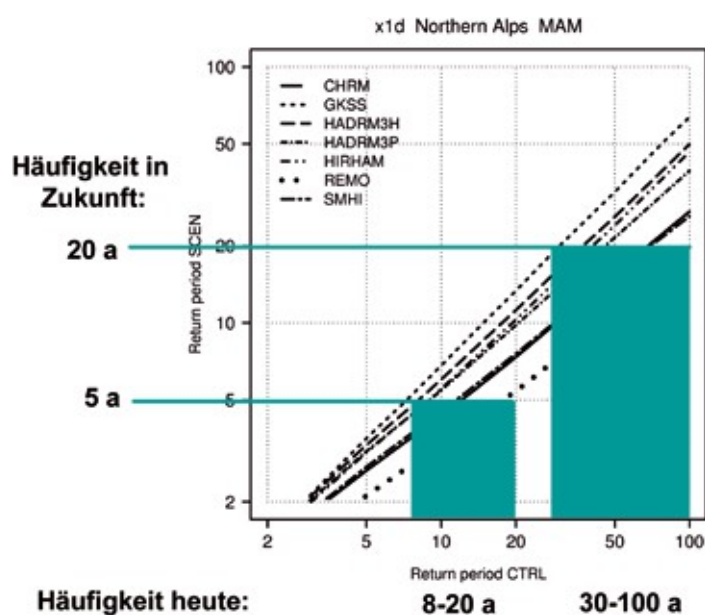


Figure 5: Change in the frequency distribution of 1-day heavy precipitation in spring for the northern Swiss Alpine region. Comparison of the present (CTRL) with the 2050 scenario (SCEN) for various regional models (from Frei et. al., 2006)

The great variability of the results and the limited confidence in the model simulations do not allow any conclusions to be drawn for the summer. At best, tendencies for an increase in the northern Alps and a decrease in the southern Alps can be recognised.

2.2 Effects on the water balance

The water cycle is characterised by the climate and current weather patterns. Runoff is indirectly linked to precipitation. Over several years, they follow the changes in precipitation with relatively constant evaporation. However, only a small proportion of precipitation flows off directly; the majority is stored in the snow cover and glaciers, in the soil, in groundwater and in natural and artificial lakes, for example. In the short term, runoff is influenced by the release of water from all of these reservoirs.

Based on the degree of glaciation and snow cover, different runoff types can be distinguished, which exhibit different seasonal runoff patterns (Aschwanden and Weingartner, 1985). Watercourses that are mainly fed by melting glaciers and snow (glacial type) exhibit the greatest fluctuations in discharge. Here, the mean monthly discharge can easily change by a factor of 30 between winter and summer. Discharge fluctuations are lowest in watercourses that are mainly dependent on rainfall (pluvial type).

Based on the present climate scenario, the following general changes in the water cycle must be expected by 2050:

- Evaporation will generally continue to increase as a result of warming. Due to the drying out of the soil, evaporation can also be limited locally and temporarily.

and consequently be reduced

- At lower and medium altitudes, less precipitation falls in the form of snow due to the warming. The snow line, which separates snow-covered areas from lower-lying areas, rises by around 360 metres with average warming. With average warming, the glacier area in the Alps will decrease by around three quarters by 2050. As a result of the change in the build-up and degradation of snow cover, the rise in the snow line and the melting of glaciers, the runoff regimes will shift downwards by around one to two regime levels. As an example, the change in the discharge regime of the Lütchine is shown in *Figure 6*. A shift of the seasonal peak towards spring, i.e. towards a nival regime is clearly recognisable. At the same time, the total discharge volume decreases due to the decrease in precipitation and the increase in evaporation
- With the reduction in precipitation and the increase in evaporation, the average annual runoff volume is likely to decrease by 7 to 12 % (*Fig. 1*). At less than 1 % of runoff in large rivers, the contribution of glacier meltwater is comparatively small and cannot compensate for the decline, even temporarily. However, the contributions of meltwater from glaciers in small streams close to glaciers are important in summer and are likely to be largely absent later on
- Groundwater recharge will decrease in summer and autumn in all non-glaciated areas. However, this will be largely compensated for by the additional precipitation in winter, so that groundwater levels will not fall overall.

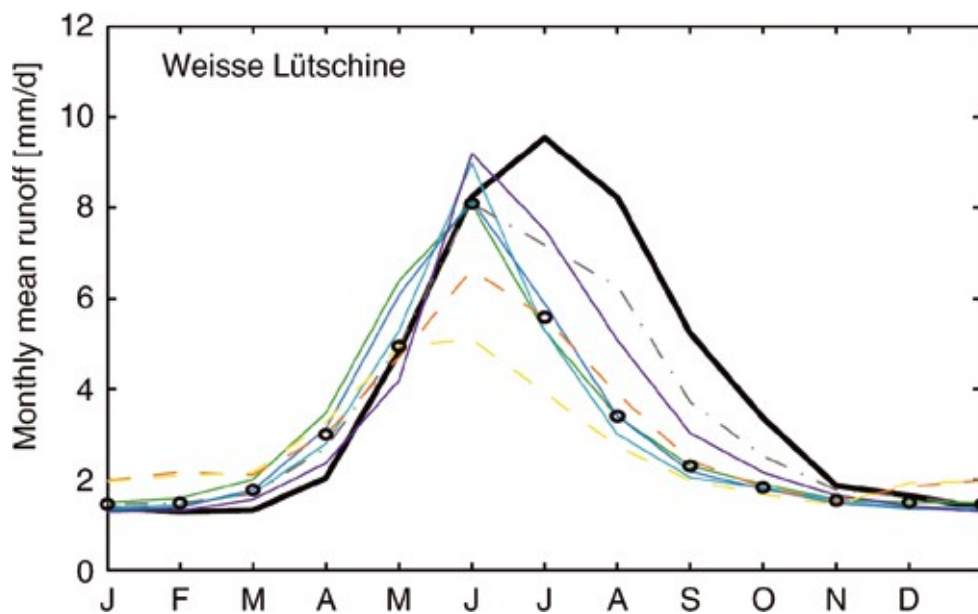


Figure 6: Discharge regimes of the Lütchine, a tributary from the Jungfrau region to Lake Brienz in the Bernese Oberland (catchment area 164 km², mean altitude 2,170 m a.s.l., glaciation 17.6 %). Shown as coloured lines are the calculated discharge regimes based on the scenarios for 2077-2099 from the PRUDENCE project (Christensen et al., 2007). The red curve is the current state. The small circles show the median of all model runs (from Horton et al., 2005)

3 Effects on important areas of water management

3.1 Utilising water to generate electricity

Switzerland covers around 60 % of its electricity demand, or 1/8 of its total energy requirements, with hydropower. Climate change is an important factor influencing electricity production. On the one hand, hydropower is heavily dependent on the water supply (precipitation and meltwater), while on the other hand, thermal power plants, such as nuclear power plants, are reliant on sufficient quantities of cooling water.

In the short term, a greater supply of water is locally for hydropower in summer due to the melting glaciers. In the long term, however, the water supply and therefore electricity production in summer will decrease. Decreasing precipitation and increased evaporation are responsible for the reduction in runoff volumes. The lower water discharge is likely to lead to an average 7 % reduction in hydropower production by 2050.

Due to the higher water temperatures, nuclear power plants can draw less cooling energy from the rivers in summer and electricity production will decrease. In summer 2003, for example, the output of nuclear power plants in Switzerland had to be reduced by 25 % for 2 months. This reduces the amount of electricity produced per year by 4 %. Water temperatures in rivers will continue to rise until 2050. As a result, the technical facilities for cooling thermal power plants will have to be adapted and expanded in order to avoid production restrictions as far as possible.

3.2 Heat and drought

Dry soils and summer droughts will also increasingly affect agriculture in Switzerland. Irrigation would then be necessary in many places. Accordingly, the necessary equipment will have to be purchased in future and farmers will have to be trained in efficient irrigation techniques. Where possible, other plant varieties that require less water must also be cultivated.

Navigation on the Rhine is an important supply route for Switzerland. Every year, 9 million tonnes of goods (15 % of foreign trade by volume) are transported to Switzerland on the Rhine. Increased low water levels in summer and autumn would massively restrict transport capacity in some cases. The result would be an undesirable switch to more expensive and energy-intensive means of transport.

The summer heat also leads to an increase in water temperatures. This reduces the habitats of many native fish (e.g. trout), as they are unable to reproduce or live at all when water temperatures are only slightly higher. Other fish such as carp or exotic fish could migrate.

Overall, there is new competition for water in the small and medium-sized watercourses on the Central Plateau, especially during dry periods: agriculture wants to pump irrigation water, the demand for cooling water from power stations and industry is increasing due to declining groundwater levels, and the water supply is becoming increasingly scarce.

As water levels rise, more water seeps out of the watercourses, drinking water extraction takes more water and ecosystems on and in the water need sufficient and not excessively heated water to survive. In addition, riparian states further downstream also have legitimate claims to a sufficient water supply. It is quite conceivable that neighbouring states will make claims regarding the increase in water levels during low water and the supply of larger quantities of drinking water.

3.3 Water supply with drinking and process water

Extreme dry periods will occur longer and more frequently in summer - in line with the decrease in average precipitation and the number of precipitation days. The combination of a decrease in precipitation and higher evaporation can lead to a regional decrease in the moisture content of the soil. Irrigation of agricultural crops could become necessary in these regions. In addition to the 50,000 hectares of irrigated land today, more will be added in the future.

With the decrease in snow reserves in the Alps, rivers that are currently fed by meltwater in summer will also dry out more frequently and seasonal water storage in the Alps will decrease. Groundwater recharge will tend to increase in winter and decrease in summer and autumn. Groundwater levels will fall slightly overall. The reduced water supply will lead to a competitive situation between different uses and users:

- Drinking water utilisation
- Utilisation for cooling industrial plants
- Demand agricultural irrigation
- Water ecology's demand for sufficient water
- Claims from neighbouring residents.

Water will therefore increasingly scarce commodity in summer - albeit limited in terms of time and location. The need for appropriate management will increase. This will have an impact on utilisation priorities, utilisation rights and utilisation prices. In addition to management approaches, measures must also be taken on the demand side, in particular in agriculture (efficient use of irrigation techniques through to the choice of crops planted) and in terms of service and drinking water requirements.

3.4 Aquatic ecology

Climate change has a direct impact on water temperatures. In Switzerland, water temperatures in rivers and in the surface layer of lakes on the Central Plateau will rise by around 2 °C by 2050 compared to 1990. As a result of future warming, living conditions may deteriorate in lakes that currently have a sufficient oxygen content.

The warming of water bodies has an impact on cold-water fish: their suitable habitats are reduced and the species composition . Cold- and warm-water fish

benefit from the warmer winters. Estimates show that with a warming of 2 °C by 2050, the habitats of salmonids in Switzerland will shrink by 1/5 to 1/4 compared to today. Both cold- and warm-water fish will benefit from warmer winter temperatures: the growth phases will last longer and the fish will grow faster. As a result, the waters will become more suitable for carp (cyprinids) and exotic fish species. Diseases such as the parasite infection PKD (Proliferative Kidney Disease) will spread at warmer water temperatures.

3.5 Flood

Statements on the development of flood discharges must be differentiated seasonally, regionally and also in relation to the size of the catchment area (KOHS, 2007).

Northern Alpine areas below 1,500 metres above sea level:

In winter, higher flood peaks are to be expected due to increased soil water content and higher precipitation. Despite warming, snow cover and thus combined melting/rainfall events remain possible on the Central Plateau. As before, the influence of melting events with the altitude of the catchment areas in the altitude range under consideration. Annual floods already in these areas in winter and spring. They may continue to occur into spring in the future, but will be higher depending on the climate.

In summer, floods will be lower, especially at lower elevations, because the runoff disposition of the soil is significantly reduced due to reduced precipitation and higher evaporation. However, it should be noted that convective heavy precipitation is always to be expected in summer, which can lead to flooding, especially in small catchment areas.

Northern and inner Alpine areas above 1,500 metres above sea level:

The regimes in the Alps change from glacial to nival in character. In winter, runoff rises slightly due to occasional rainfall. However, this does not result in any real floods. In spring, small melt floods are possible, the peaks will be larger than today. However, the annual floods will occur in summer, as they do today, and are not expected to become any larger. Hardly any changes are to be expected in autumn.

South side of the Alps:

In winter and spring, the stronger increase in precipitation also suggests a corresponding increase in flooding. In summer, decreasing precipitation indicates smaller discharge peaks. The hydrometeorological conditions in autumn remain decisive for the annual floods. Here, the higher precipitation levels lead us to expect an increase in floods.

3.6 Flood protection

Both the expected increase in damage potential and the possibility of more frequent flooding require greater protection against flooding. Switzerland's flexible flood strategy includes spatial planning measures to limit the damage potential,

Object protection to minimise susceptibility to damage, structural protection measures and emergency measures. In any case, the increased degree of uncertainty in the flood protection assessment must be minimised by taking the case of overload into account. A constant review of the risk situation is a prerequisite for the long-term effectiveness of the implementation of the strategy, as both the damage potential and the hazard situation are constantly changing. In the future, alpine reservoirs could also be increasingly used for the retention of flood peaks. Their use as multi-purpose facilities will gain in importance.

3.7 Solids transport

Due to the retreat of the glaciers and the thawing of permafrost, the solid material potential increases significantly in areas between approx. 2,300 and 2,800 metres above sea level (*Fig. 7*). The new loose debris collects in glacier forefields, in ditches and stream beds and can be swept away during floods, triggered by possibly more intense and liquid precipitation up to higher altitudes, and travel down to the valley and into inhabited areas as **debris flows**. Erosion, bedload transport and solid deposits are very often responsible for major damage during floods. The potential for such events will increase significantly, but only in mountainous regions.



Image 7: The Tschierwa Glacier (Engadin, Val Roseg) has melted considerably since its peak in the mid-19th century. The exposed old lateral moraines and the ground moraine form an increasingly large solid material potential. Today, part of these large debris masses is still mostly frozen (permafrost) and therefore less exposed to erosion. However, individual erosion gullies are already visible. In the background centre left Piz Bernina with Biancograt and far right Piz Roseg (Photo: B. Schädler 2006)

In the foothills of the Alps, the future hydrometeorological conditions mentioned above (precipitation, snow cover, evaporation) in winter and spring allow for longer periods of water-saturated soil.

soils. Wet soils on steep slopes can cause **landslides** in the event of more intense precipitation. Overall, this increases the risk of damage to infrastructure such as transport links and tourist facilities in mountain regions.

4 Conclusion

The expected consequences of global warming on the water cycle up to the year 2050 seem manageable for Switzerland from today's perspective and assuming that the warming remains within the expected range. A study commissioned by the Federal Office for the Environment (Ecoplan/Sigmaplan consortium, 2007) has shown that the expected economic damage in 2050 will be in the region of 0.15 GDP% per year, but could then - with a large margin of uncertainty - develop to around 0.5 GDP% per year by 2100.5 GDP% per year. This observation should not obscure the fact that the longer-term development in the second half of the 21st century is crucially dependent on the emission reduction measures taken in the coming years and decades and that the consequences will be much greater in the event of a "business-as-usual development". In addition, there will be many countries around the world, especially poorer developing countries, which on the one hand will face much more serious consequences and on the other hand will not have sufficient financial resources to adapt. The resulting geopolitical developments could also have major consequences for Switzerland (Arbeitsgemeinschaft INFRAS/Ecologic/Rütter, 2007).

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Author's address Dr

Bruno Schädler Federal

Office for the

Environment Hydrology

Division 3003 Bern (CH)

Phone +41-31-324-7758

e-mail bruno.schaedler@bafu.admin.ch

Climate change and aquatic macrophytes

Georg A. Janauer

1 Introduction

For around 20 years, the measurement of climate-relevant data has shown that a trend towards higher CO₂ levels in the atmosphere together with its warming and the increase in heavy precipitation and the corresponding consequences in the runoff behaviour of watercourses has become manifest in large parts of the world (IPCC 2008). With the usual delay for bodies of water, this influence is slowly leading to a staggered increase in water temperatures in surface waters and to changes in hydrological boundary conditions. This results in a gradual change in the living conditions for aquatic organisms, of which aquatic plants (*commonly known as "climbing plants"*) are discussed in this article.

2 The water body as a habitat for aquatic plants

Macrophytes are all "*aquatic plants that can be identified to the species level with the naked eye*" (quote from Westlake, 1973), whereby two aspects must be taken into account:

- (i) This characterisation does not apply to all aquatic plants, as in many cases anatomical features that can only be seen with a magnifying glass or microscope are also used for final identification (the microscopically small algae are excluded here a priori as "microphytes").
- (ii) in relation to their specific habitat, different "growth forms" occur in the - and on - and - the water.

In general, the density of water, which is almost 1,000 times higher than that of air (1.2041 kg/m³ at 20 °C at sea level), influences the physical and chemical properties of many environmental parameters:

- Reduction of gas diffusion (especially CO₂, O₂), therefore poorer supply of plants with these vital gases
- Attenuation of the light and spectral change with water depth (blue light can be penetrate deepest in pure water)
- Daily and seasonal temperature amplitude significantly lower than in the air
- Stronger mechanical effect of the movement of the ambient medium water (flow), in contrast to the movement of air (wind), at the same speed
- Fluctuations in the water level, thus drying out or waterlogging, possibly also in the daily rhythm, with a corresponding effect on the aquatic plants.

3 Macrophytes in the aquatic ecosystem

On the one hand, aquatic plants must adapt to the aquatic habitat; on the other hand, they influence the characteristics of the water body and thus the living conditions for other organisms.

Macrophytes produce oxygen during the day and thus support all respiring organisms. At night, however, they breathe like animal organisms.

As green plants, they produce biomass by means of photosynthesis, but the amount of biomass depends heavily on the growth form and climatic conditions. Submerged plants produce less biomass, floating-leaf plants and pleustophytes (floating freely on the water surface) up to approx. 5 times more, and reed plants can produce up to 8 times the biomass of submerged plants, with production rates reaching higher values in warmer climates.

More important than these two influences, however, is the structural function of macrophytes: "Water has no beams", but the stems and leaves of aquatic plants, like the trees in a forest, form those structures in the structureless water body that are indispensable as a habitat for many aquatic animals, but also for algae, bacteria and fungi.

All those species that can cover large parts of the water surface with their stands (e.. pond and water lilies, duckweed) reduce the radiation of light and heat into the water body and thus lower the water temperature under this plant cover. The lowering of the temperature influences, among other things, the heat-dependent development of other aquatic organisms (e.g. insect larvae). Several growth forms can be distinguished:

- Submerged macrophytes (submerged aquatic plants)
These plants usually only reach above the water surface with their inflorescences:
Pondweed, waterweed, milfoil (*picture 1*)



Image 1: Crested milkwort (Zaya, Lower Austria), all photos:© G.A. Janauer 2007

- floating leaf plants
Examples: Water lily, water lily



Picture 2: Water lily (Danube oxbow lake, Romania)

- Amphiphytes (amphibious plants that thrive under and above water or on the shore)
Examples: Berle, swan flower, pennywort



Picture 3: Berle, submerged and protruding from the water (centre)

- Pleustophytes ("water floats", free swimming, not rooted in the sediment)
Examples: duckweed, water fern, water lettuce



Picture 4: Duckweed cover (drainage ditch, Hungary)

- Helophytes (reed plants with only the lower part of the shoot in the water)
Examples: bulrushes, reeds, sedges



Picture 5: Reedbeds (Danube Delta, Romania), photo

- Fringing herbs, from the shore with runners over the surface of the water growing)
Examples: Water speedwell, Ludwigia



Figure 6: Riparian weeds (Ludwigia, Senegal)

Further information on the function of macrophytes in the aquatic ecosystem can be found in Janauer & Dokulil (2006).

4 Comparative data

In order to arrive at a concrete statement in terms of the research question, a series of studies were used that contain quantified data on the occurrence of macrophytes in various water bodies in the Danube corridor (*Fig. 7*). These data sets from the pilot study (Janauer et al. 2003) and the final report of the MIDCC project (MIDCC 2005 and www.midcc.at). Data for the Freudenau reservoir comes from the Interreg III A project FIDON (Chovanec et al., 2006).

Table 1: Partner states in the MIDCC project

Austria (project management)
Germany (Baden-Württemberg, Bavaria)
Slovenia
Slovakia
Hungary Croatia
Serbia Bulgaria
Romania

Figure 7 shows why comparative data from the Danube region is particularly suitable for estimating the consequences of climate change:

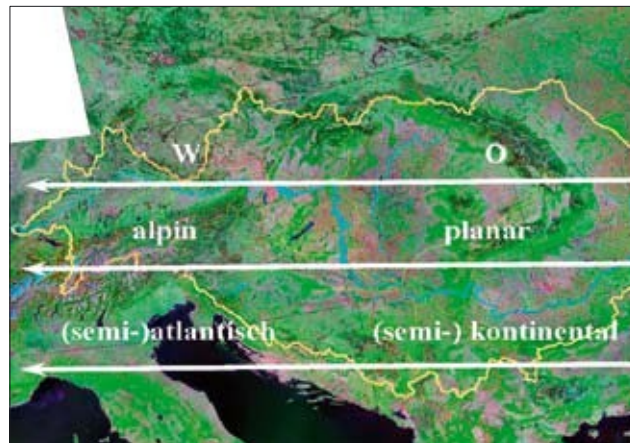


Figure 7: Gradients in the Danube catchment area

are significant gradients in the Danube catchment area, which are reflected in climate zones, altitude levels, terrain morphology and - with regard to the distribution of plant and animal species - in a general west/east differentiation.

5 Consequences of climate change

Considerations on the effects of climate change are usually focussed on the increase in the average annual temperature. However, the extension of the period of higher temperatures into autumn and the occurrence of less cold winters without extreme frost periods must also be taken into account in our geographical location and in relation to aquatic vegetation.

Under these conditions, the following changes can be expected or are already recognisable:

- **Warmer summers**

Increased summer temperatures promote heat-loving plant species that are characterised by a higher temperature optimum or a greater temperature amplitude and increase their chance of more biomass. Subsequently, it has been shown that a permanent establishment of cold-sensitive species is possible in protected spatial niches with less cold influence (e.g. water fern in the Mauthnerwasser).

- **Warmer winters**

As there is a positive correlation between the biomass that does not die off in autumn and the growth success of many macrophyte species in the following spring (Janauer & Wychera 2000, chapter "*Inter-annual biomass relationship*"), mild winter temperatures are the cause of particularly strong development of persistent populations. This relationship has also been observed again recently (e.g. development of milfoil in gravel ponds).

and in the Old Danube). This also increases the "starting advantage" for certain species, which can then more easily dominate other species.

5.1 Migration and neophytes

Migrating species are aquatic plants that migrate from their original distribution area to a new area, whereby in many cases the cause of this migration can only be speculated (e.g. passive transport of distribution units such as shoots), but climate change cannot be ruled out as one of the possible causes. This applies in particular to species that migrate from warmer parts of Europe to previously colder regions (e.g. east-west migration upstream along the Danube). A recent example is the water fern, which has been found in a body of water in the Praterauen since 2006 and can form large populations. Its former upstream boundary was in Slovakia.

On the other hand, the Nuttall's water star has been found in Austrian tributaries of the main Danube since around 1975 and since found its way to Romania. Nuttall's water plague, which has also spread downstream to all neighbouring countries, has taken the same route in roughly the same period.

A similar development has recently been shown by genuine newcomers, the neophytes. Ornamental plants from mostly subtropical regions are readily available from specialist retailers and via the internet and are released into native waters when they become too abundant in aquariums or garden ponds. As a result of warmer winters, they can easily establish themselves in the Danube region, for example. For example, an aquatic species of the thistle family, *Crassula helmsii* (related to the stonecrop) has spread across the British Isles and large parts of Western Europe (Fig. 8), sporadically even as far as Germany. It is not unlikely that this species will appear in Austria in the next decade as it migrates along the shipping canals. The high competitive power of this plant could lead to a displacement problem for some native species in standing and slow-flowing waters.



Image 8: *Crassula helmsii* (Kirk) Cockayne

The European and Mediterranean Plant Protection Organisation (EPPO, 44 member states) deals with the introduction of cultivated and wild plants and has developed a concept for limiting and/or combating these invasive alien species ("plants as harmful organisms"). In the UK, the UK Non-Native Species Secretariat NNSS is active in the same direction (www.nonnativespecies.org).

The problem has also been recognised in Germany and a list of neophytes has been published (*Table 2*). Further information is available on the Internet at www.aquatischeneophyten.de.

*Table 2: List of neophytes in Germany (source: lana-plan / van de Weyer) www.lanaplan.de/public/index/rubrik/Makrophyten/artikel/Aquatische%20Neophyten
Species that already been recorded in the Danube region are shown in italics. Year in brackets: first occurrence in Germany*

<i>Azolla filiculoides</i>	(1980)	<i>Lemna aequinoctialis</i>	(1980)
<i>A. mexicana</i> c.f. (?)	(?)	<i>L. minuta</i>	(1990)
<i>Crassula helmsii</i>	(1988)	<i>L. turionifera</i>	(1965)
<i>Egeria densa</i>	(1914)	<i>Myriophyllum aquaticum</i>	(1988)
<i>Eichhornia crassipes</i>	(?)	<i>M. heterophyllum</i>	(1960)
<i>Elodea canadensis</i>	(1965)	<i>Pistia stratiotes</i>	(?)
<i>E. nuttallii</i>	(1953)	<i>Sagittaria latifolia</i>	(1952)
<i>Hydrocotyle ranunculoides</i>	(2004)	<i>S. subulata</i>	(1980)
<i>Hydrilla verticillata</i>	(1907)	<i>Salvinia auriculata</i> agg.	(?)
<i>Hygrophila polysperma</i>	(2005)	<i>Shinnersia rivularis</i>	(1992)
<i>Lagarosiphon major</i>	(1970)	<i>Vallisneria spiralis</i>	(1900)

Another highly invasive plant genus *Ludwigia* (hayweed), whose species originate from South America, among other places, and have spread from Western Europe to the British Isles and further onto the continent. Its ability to completely overgrow large areas of standing or slow-flowing water, as well as damp meadows, with shoots floating on the water has led to major problems in the utilisation of fish ponds in France and elsewhere. Occurrences have also been reported in Spain, Italy, Belgium, the Netherlands and Switzerland.

In general, submerged macrophytes pose less of a threat than those that can colonise the entire water surface of bodies of water. Spread rates more than 100 hectares per year have been reported. Direct consequences include

- Covering even large areas of water in a very short time
- Under the plant cover, a strong reduction in oxygen concentration and considerable cooling of the water body (possibly a problem for the development of aquatic insects or fish)
- Slowing down the water runoff
- Massive suppression of all other, especially useful native submerged macrophytes
- Rapid acceleration of siltation (new sediment formation) of water bodies and in
The result is accelerated sedimentation.

Figures 9 and 10 show that native plant species can also have similar effects if, for example, long periods of drought create favourable conditions for the mass reproduction of certain macrophytes.



Figure 9: Duckweed cover (oxbow lake on the March)



Figure 10: Mass growth of water nut due to low water levels (Csatloi Holt Tisza/Theiss oxbow lake; devastating fish mortality due to lack of oxygen)

5.2 Increased runoff as a result of climate change

Changes in the runoff behaviour of rivers as a result of heavy rainfall events lead to more severe short-term flooding of rivers. Since on the regulated rivers of Europe, therefore

As the backwaters of the Danube in have also become the last refuges of still-water plants in a cultivated landscape that has been completely dominated by the expansion of settlements and intensive agriculture, such flooding conditions jeopardise the aquatic plants that survive there. This has been clearly demonstrated by scientific studies in the near-stream areas of oxbow lakes (Janauer, unpublished). However, even more intensive negative effects are by the rehabilitation of oxbow loops that are not locally and regionally coordinated, as a result of which large parts or even the entire discharge of a watercourse are constantly channelled back through the oxbow system.

6 Summary

In conclusion, the following scenario for climate-induced changes in macrophyte vegetation can be presented:

Further invasions of heat-loving aquatic vegetation elements are certain to follow, some of which are already manifesting themselves in the form of migrations of known species. The spread is mainly along the Danube, with thermophilic elements from the lower Danube region advancing upstream and macrophytes from the Atlantic region spreading downstream via the Rhine-Main-Danube Canal. Where neophytes are affected, they are often introduced via anthropogenic dispersal, e.g. from ornamental or swimming ponds. In protected habitats, they can easily survive the winter cold period: for example, the occurrence of the originally tropical-subtropical water screw *Vallisneria* in a basin of the Danube-Oder Canal has been known for over 40 years.

The time horizon for at least partial climate-induced spread of migratory species and neophytes is not in the future - individual newcomers are already part of our aquatic flora today.

Native macrophyte vegetation is particularly at risk in stagnant and slow-flowing watercourses or sections of watercourses. As this particularly affects the riparian areas of the Danube, which are extremely worthy of protection, special attention is required here. However, even in the main channel of the Danube, there was mass growth of duckweed in the Szent Endre area (Hungary) in 2007, and the mass occurrence of Cabomba in Hungarian irrigation channels has been known since 2002.

Many of the new species have a strong impact on the infested water system, but especially on the aquatic plants that previously grew there. Threats to native biodiversity, effects on the fish and invertebrate fauna and sport fishing, the obstruction of various recreational uses (swimming, boating), an increase in the sedimentation rate, but also changes to the reference conditions for recording the ecological status of water bodies in accordance with the European Water Framework Directive are some of the effects that are at least partially caused or strongly supported by climate change. However, the extent to which it is possible to intervene in these processes has not been conclusively clarified.

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Address of the author

Univ.-Prof. Dr Georg A. Janauer University of
Vienna
Department of Limnology and Hydrobotany Althanstraße 14,
1090 Vienna
Tel. +43-1-4277-543 60
E-Mail georg.janauer@univie.ac.at

Effects of climate change on fish fauna

Stefan Schmutz

Summary

Current and predicted changes in the habitat conditions of aquatic ecosystems due to climate change imply changes in organismic colonisation. As the final links in the food chain, fish integrative indicators of changes in aquatic ecosystems. The rise in water temperature and changes in the hydrology of water bodies and their consequences are the most important factors influencing fish fauna. Fish are by climate change in a variety of ways. Overall, a shift in species composition towards warmth-tolerant species and a reduction in cold-loving species is to be expected. A progressive degradation of the already highly endangered fish fauna is to be expected, primarily due to the interaction with other pressures such as continuum disruptions, regulations, hydropower plants and other thermally effective influences. While intensive research has been carried out on this topic in the USA since the 1990s, there is a serious lack of relevant studies in Austria and Europe.

1 Consequences of climate change for the habitat conditions of fish

The heat balance along the course of watercourses, which is essential for fish fauna, depends on many factors. The spring temperature (groundwater temperature) correlates clearly with the mean annual air temperature. As the distance from the source increases the watercourses warm up in summer and cool down in winter, whereby the degree of warming/cooling depends on the dimensions of the watercourse. With regard to a possible temperature change, it is also important whether the water bodies currently freeze over in winter and thus interrupt the coupling between air and water temperature. Locally, this temperature continuum can be interrupted by lakes or groundwater inputs. Shading by riparian vegetation is also a significant factor. Small bodies of water generally exhibit greater diurnal and seasonal fluctuations than large bodies of water.

As Austria's largest watercourse, the Danube integrates the changes in temperature conditions throughout the entire catchment area very well. The average annual water temperature of the has risen by 1.3 °C in the last century, from 8.9 °C to 10.2 °C (*Fig. 1*). The four highest values were all in the last decade. The trend has also continued in recent years.

This development appears to be significantly intensified in the future. Morrison et al. (2002), for example, predicted an average increase in summer temperatures of 1.9 °C for the Fraser River (BC, Canada) for the period 2070-2099 based on a doubling of CO₂ emissions.

Temperature changes also influence other factors that characterise the fish fauna. These, above all, the discharge pattern, but ultimately also changes in the solids content and the morphological structure of the water bodies. Last but not least

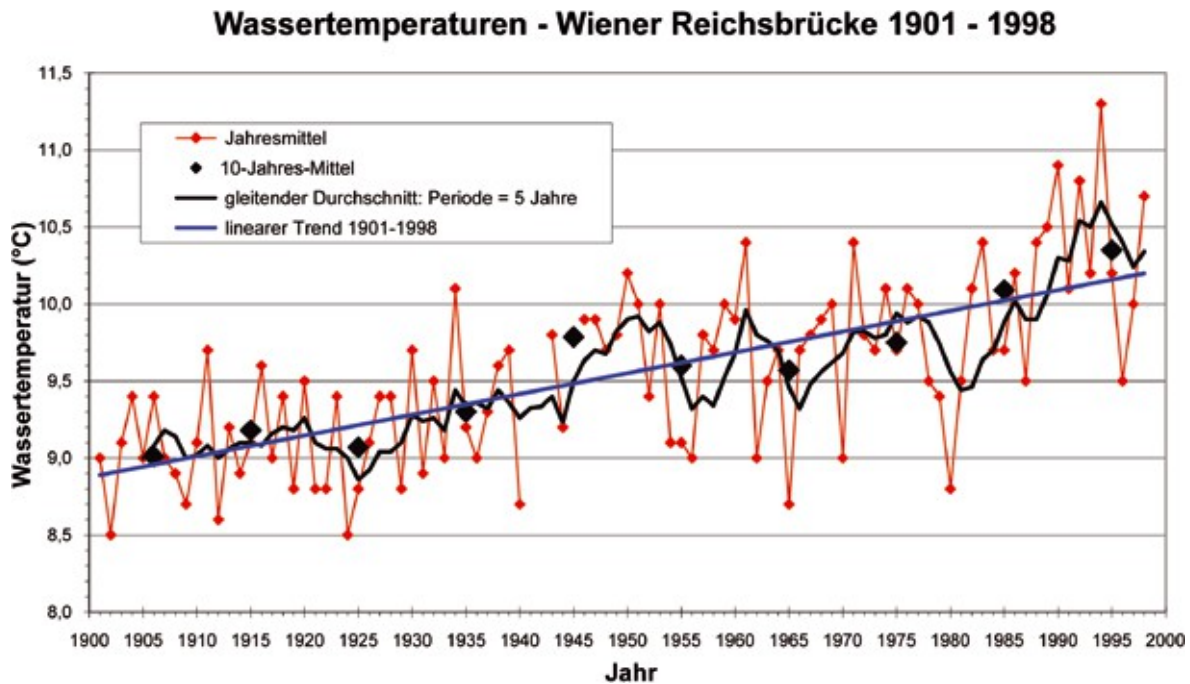


Figure 1: Increase in the mean annual water temperature of the Danube near Vienna in the 20th century (unpublished data compilation by S. Hohensinner based on various HZB publications)

secondary effects, such as an increase in the sensitivity of fish to toxic stress caused by climate change.

2 The importance of water temperature for fish

Water temperature is one of the most important factors for aquatic organisms. As warm-blooded organisms, fish are particularly sensitive to changes in temperature. All areas of life are strongly influenced by the water temperature. The water temperature regulates migration behaviour, egg development, spawning, fertilisation and growth rates, inter- and intraspecific competition, but of course also the metabolism, respiration and the tolerance of the animals to parasites and pollutants (Fig. 2). Even minor changes often limit the survival of individual species or life stages (Jungwirth & Winkler, 1984). In zoogeographical terms, temperature plays a decisive role. Water temperature is one of the most important factors for regional and altitudinal distribution limits of individual fish species and communities.

The influence of temperature on fish is divided into three areas: (1) lethal temperature leads to death, (2) sub-lethal temperatures lead to physiological impairments (growth reduction, reduced reproductive capacity, etc.), (3) optimum temperatures ensure the best viability. Fish species have different temperature preferences. Cold-loving species are opposed to warm-loving species. Fish comparatively narrow thermal niches in terms of temperature niveau, temperature optima and temperature limits for fish cenoses. Interventions in the temperature regime of a body of water can have disadvantages for one species and advantages for the other. An increase in temperature can

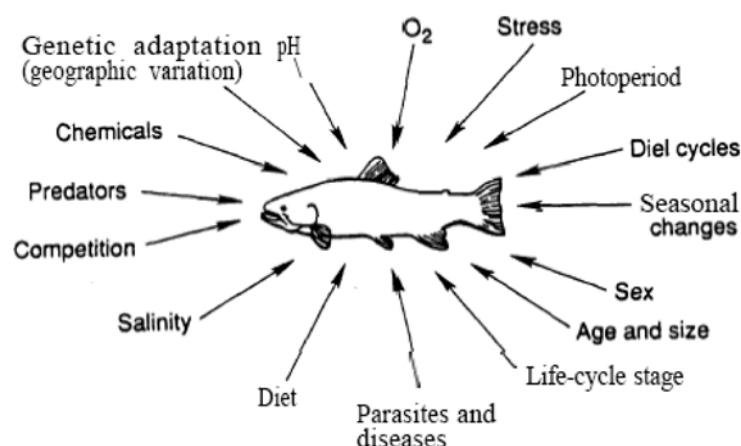


Figure 2: Interactions between water temperature and other factors influencing fish (after Hutchinson, 1976)

disappearance of salmonids (trout-like fish). On the other hand, cyprinids (carp-like fish) are bound to certain minimum temperatures with regard to their reproduction.

In order to be able to estimate the influence of changes in the temperature regime on the respective fish species, it is essential to know the temperature preferences and limits for the various life stages. Table 1 examples of temperature optima and limits for the different life stages of brown trout, grayling, bullhead, aitel, barbel, nase and minnow. The different life stages of fish have different temperature tolerance ranges. In most cases, the egg stage has a narrower temperature range compared to the adult stages. Reproduction place within a particularly narrow range. The tolerance ranges also vary considerably between the different species.

Table 1: Temperature optima and limits [°C] for some fish species occurring in Austria and their life stages (Küttel et al., 2002)

		Brown trout <i>Salmo trutta</i> <i>fario</i>	Grayling <i>Thymallus</i> <i>thymallus</i>	Coupler <i>Cottus</i> <i>Gobio</i>	Aitel <i>Leuciscus</i> <i>cephalus</i>	Barbel <i>Barbus</i> <i>barbus</i>	Nase <i>Chondostroma</i> <i>nasus</i>	Minnow <i>Phoxinus</i> <i>phoxinus</i>
Adult stage	Maximum	25-30	26	33	39	30		31
	opt. optimum	19	18	27	25			25
	unt. Optimum	4	4	4	8			13
	Minimum	0	0	0	8			0
Juvenile	Maximum	28		33				23
	opt. optimum	14		27				
	unt. Optimum	7		5				
	Minimum	0		0		14		
Egg development	Maximum	14	16		30	21	19	16
	opt. optimum	9	14		24	21	17	16
	unt. Optimum	1	7		16	16	12	6
	Minimum	0	0		16	16	10	6
Reproduction	Maximum	13	15	14	18	29	16	22
	opt. optimum	10	10	14	18	20	16	22
	unt. Optimum	1	6	7	13	14	10	11
	Minimum	1	4	7	13	8	7	7

3 Longitudinal zonal classification of watercourses according to biocenotic regions

Watercourses have been categorised into fish regions from the source to the estuary for around 140 years. Today we also speak of biocenotic regions (Illies and Botosaneanu, 1963). The classic regions include the spring, the spring stream, the upper and lower trout region, the grayling region, the barbel region, the bream region and the ruffe-flounder region. In the context of global warming the fish regions play an important role as bioindicators, as the characteristics of the fish communities strongly influenced by the water temperature. In this way, the effect of temperature changes on the aquatic ecosystems can be comprehensibly documented in comparison to known reference conditions. The affiliation of a fish community to a fish region can be described and quantified using the fish region index (Schmutz et al., 2000). Increases in the fish region index signify a "potamatisation", decreases a "rithralisation" of the communities. Changes in species composition caused by global warming therefore lead to potamatisation phenomena and can be quantified using the fish region index (Schmutz et al., 2004; Matulla et al., 2007).

Table 2: Fish regions, biocenotic regions and associated fish region indices (Schmutz et al. 2000)

Fish region	Biocenotic region	Fish region index
Upper trout region	Epirhithral	3
Lower trout region	Metarhithral	4
Grayling region	Hyporhithral	5
Barbel region	Epipotamal	6
Bream region	Metapotamal	7

4 Changes in the distribution of individual species and species communities

While studies on climate change and its consequences for freshwater fish have been underway for around 15 years, particularly in North America, there is hardly any research on this topic in Austria (Schmutz & Jungwirth, 2003). However, there is a scientific consensus that global warming will result in a shift of habitats towards higher latitudes and higher altitudes for most cold-loving species (Jackson and Mandrak, 2002; Rahel, 2002; Mohseni et al., 2003). It is also that cold-loving species will be replaced by warm-loving species (Daufresne et al., 2004). This assumption is supported by palaeoclimatic studies, which provide evidence of this for the glacial and interglacial periods of the Pleistocene. However, this is only possible if climate change takes place long enough for adaptation to be possible. A further prerequisite is that the temperature change is synchronised with other important habitat factors.

In a broad-based study in the USA (764 measuring points), Mohseni et al. (2003) predicted that, assuming a doubling of CO₂ in the atmosphere, 36 % of the under

The change in the distribution range is associated with a northward shift in the distribution area. These changes are associated with a northward shift in the distribution area. In contrast, the number of suitable sites for warm-water fish will increase by 31 %. According to prediction models by Casselman (2002), an increase in temperature leads to an increase in the density of warm-water species (e.g. for black bass by a factor of 2.5 at 1 °C warming, by a factor of 6 at 2 °C and by a factor of 14.7 at 3 °C). In contrast, the density of species with a preference for colder water decreased (e.g. pike by a factor of 2.4 with 1 °C warming and by a factor of 17.9 with 2 °C warming). Similarly, an increase in temperature results in a reduced reproductive rate in cold-water fish (e.g. in American charr, hatching success decreases by a factor of 1.5 with a 1 °C temperature increase in autumn and by a factor of 20.1 at 2 °C). Jackson and Mandrak (2002) forecast for Canada that the expected temperature increases will lead to a northward expansion of the distribution area of warm-water fish. It is predicted that 24,000 populations of 4 non-competitive cyprinid species will as a result of the spread of the black bass, a predator. Rahel (2002) summarised studies that use existing, temperature-related distribution limits of certain species predict distribution shifts as a result of temperature increases. He described a habitat loss for cold-water fish in the Rocky Mountains of 17 to 72 % as a result of a 1-5 °C temperature increase (July average).

The extent to which fish are able to ensure their viability under changed climatic conditions through genetic adaptations is unknown. The main question is to what extent there is temporal synchronisation between rapid climate changes and possible genetic adaptations that extend over a period of several generations.

If applied to Austria, the findings to date would a decline in salmonids. Individual species, such as the grayling, are already so endangered as a result of other anthropogenic impacts that additional climate-related impacts could bring them to the brink of extinction. However, it is not just individual species that are affected by climate change, but the entire biotic community. This can be seen, for example, in a shift in the fish region index (Schmutz et al., 2004; Mattula et al., 2007). In spatial terms, this corresponds to a shift of fish regions upstream (phenomenon of potamalisation, see above).

5 Behavioural changes due to climate change

As mentioned above, temperature changes can significantly influence the behaviour of fish. Fish migrations are used here as an example to illustrate possible influences of climate change. Fish undertake short to long migrations between different habitats every year. Spawning migrations are the most pronounced. Long-term observations (44 years) of the spawning habits of fish in Estonia showed, for example, that spawning dates were shifted by 6 days for pike and 8 days for bream due to climate change (Ahas, 1999).

In addition to water temperature, day length is particularly important for gonad development. It is obvious that the success of spawning migration and reproduction depends on the syn-

chronity between day length and water temperature. Climate-related changes in the timing of migration behaviour can lead to drastic changes and threats for affected species.

Species with particularly specific habitat requirements, isolated habitats and low population densities are considered to be particularly endangered (Rabinowitz et al., 1986). Due to other pressures, around three quarters of Austrian fish species are already endangered (Spindler et al.). Additional pressures due to climate change could significantly worsen this situation.

5.1 Case study Mur

Schmutz et al. (2004) and Mattula et al. (2007) used a modelling chain to develop a forecasting system that local predictions of climate impacts on fish. Based on large-scale IPCC emission scenarios, local climate and water temperature changes were predicted for the Mur, an inner-Alpine river in Austria, using a downscaling procedure and the resulting changes in fish species composition were assessed using the fish region index. Methodologically, space-for-time substitution was used, a method in which the current biogeographical and longitudinal zonal boundaries are used to predict the future distribution of fish. This allows future scenarios of different temperature changes to be created (Flebbe et al., 2006). The models on the Mur were created using the monthly summer averages, as the highest and therefore most limiting temperatures occur in summer. Temperature increases of 0.5 °C (lower trout region) to 0.8 °C (barbel and bream region) were predicted for the summer months. The increase in temperature resulted in an increase in the fish region index of 0.22 units. This corresponds to a shift of the fish region by 70 metres above sea level or a longitudinal zonal shift of the fish regions upstream by approx. 30 km. According to this scenario, salmonids would be pushed back upstream and the habitat for cyprinids would be enlarged (potamalisation). Grayling are likely to be most affected by the temperature increases, as their shift is limited by their preference for larger waters upstream.

6 Influence of changing hydrology on the fish fauna

Fish distinct adaptations to the hydrological conditions of watercourses, particularly with regard to discharge levels, seasonal and spatial distribution and extreme events (low water, floods). Watercourses that were previously intermittent could dry up completely due to a lack of water. Watercourses with low discharge could become intermittent and thus unattractive for fish. Glacial streams, on the other hand, could even carry more water in the summer months, as was the case in Austria in the extremely warm summer of 2003, at least as long as the glaciers are not completely melted. Changes in the vegetation cover in the catchment area influence the erosion processes and the input of sediments into the watercourse. This can also affect fish, as their viability is heavily on the sediment conditions in the watercourse. For example, the increased input of fine sediments from the catchment area as a result of drought and/or an increase in heavy rainfall events can lead to the displacement of salmonids, as these species are particularly dependent on clear water and intact gravel gaps.

Most climate models predict an increase in extreme events. In addition to an increase in heat waves, an increase in heavy rainfall events and the resulting floods is . Catastrophic floods represent a massive stress for aquatic organisms and usually lead to a massive thinning of fish populations. However, recolonisation of depleted areas is not possible or only possible over very long periods of time due to the continuum interruptions that frequently occur today.

One of the few scientific studies on the topic of climate-induced changes in hydrology and their consequences was conducted by Xenopoulos & Lodge (2006). This study investigated how water scarcity could affect fish species diversity in (semi-)arid catchment areas. The methodological approach is based on the well-known species-area concept: The larger a catchment area is, the higher the number of species. Similarly, for this study they used discharge as a parameter for the dimension of the habitat. They predicted that an expected decrease in discharge of 20-90 % could lead to a loss of 2-38 % of fish species (Fig. 3).

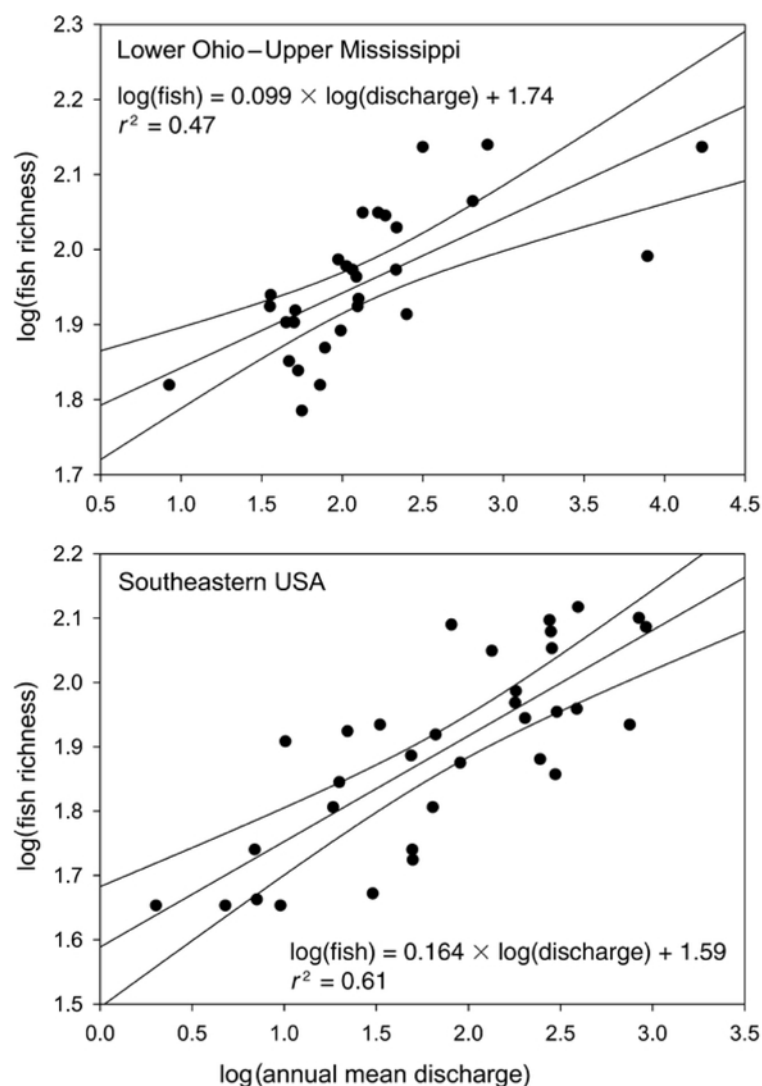


Figure 3: Relationship between mean annual discharge and number of fish species in catchment areas in the USA (Xenopoulos & Lodge, 2006)

7 Climate change as just one factor of global change

The fact that aquatic ecosystems, especially watercourses, are already heavily impaired by other human interventions and therefore often run the risk of climate-related effects being superimposed by other factors and thus not easily recognisable, complicates research into possible influences of climate change. In Austria, for example, the hydromorphological characteristics of around 80 % of larger watercourses are heavily influenced by hydropower plants and river regulation (Muhar et al., 1998). For research, this means that climate-related influences cannot be considered in isolation from other anthropogenic influences.

8 Socio-economic consequences

A decline in the habitat quality of these waters or the loss of individual stretches of water would have significant socio-economic as well as ecological consequences. Salmon waters are among the most attractive fishing waters in Austria. The number of sport fishermen in Austria has already risen to 400,000. Sport fishing is an important economic factor, especially when secondary markets such as leisure, tourism and the fishing tackle trade are included. Fishing grounds differ significantly in terms of their fishing value. For example, the annual lease price for 1 km of salmonid waters (fly-fishing waters) can up to 7,000 euros and more, whereas barbel and bream waters are priced at a much lower level. A change in species composition as a result of climate change, in particular a loss of highly attractive salmonids such as brown trout, grayling and huchen, can lead to a significant reduction in the value of a water body.

9 Outlook

The results presented here clearly demonstrate the many potential threats to fish fauna from climate change. While the fundamental influences of climate change are largely known, at least for North American conditions, there is a lack of corresponding large-scale studies in Austria and Europe. In particular, the specific situation in alpine landscapes is practically unexplored. Questions such as

- 1) How do climate-induced changes affect temperature-regulated mechanisms in behaviour and distribution in our Alpine rivers?
- 2) What interactions are there between climate-related and other human influences in our heavily utilised watercourses?
- 3) How do climate-related changes in fish fauna affect fishing ?

should be the subject of future research.

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Address of the author

Univ.-Prof. DI Dr Stefan Schmutz University of
Natural Resources and Life Sciences, Vienna
Institute of Hydrobiology and Water Management Max-
Emanuel-Straße 17, 1180 Vienna
Tel. +43-1-47654-5202
e-mailstefan.schmutz@boku.ac.at

Possible effects of climate change on forestry

Klaus Katzensteiner

1 Introduction

Society's demands on the forest are manifold. In Austria, modern, multifunctional forestry is legally obliged to simultaneously guarantee the utilisation, protection, welfare and recreational effects of the forest. On the one hand, forested catchment areas make a significant contribution to the supply of drinking water, and on the other, forest sites protect underlying landscape areas from runoff peaks, which could otherwise lead to flooding or mudslides, through their water consumption and the intermediate storage of water. The predicted increase in temperature in combination with a less favourable distribution of precipitation could lead to increased drought stress in forests in the future. Unstable forest stands, which are more susceptible to additional biotic or abiotic damaging factors, would be the result of such a scenario. Large-scale damage in mountain forests, for example due to windthrow and/or bark beetle calamities, as can currently be observed in Canada (Winkler et al., 2008), entails massive hydrological problems.

From this perspective, the following article will analyse possible medium-term effects of climate change on the condition of our forests and highlight adaptation strategies in forest management.

2 The current state of the Austrian forest

According to the Austrian Forest Inventory (ÖWI), 47 % of Austria's area is currently forested (Russ, 2004) - with an upward trend. The standing timber stock has increased from 780 million cubic metres in the 1961/70 inventory period to 1,095 million cubic metres in the 2000/02 period (Schadauer and Büchsenmeister, 2004). In addition to a decrease in utilisation in small forests, positive growth trends are cited as the cause. This trend of increasing timber growth has now lasted for several decades. There are a number of possible explanations for these improved growth rates in European forests (Kahle et al., 2008). Increased atmospheric nitrogen deposition is categorised as the most important factor. In addition, increased CO₂ concentrations and more favourable temperature and precipitation conditions are considered to be of great importance. Other influences - recovery of the soil after overexploitation in the past (Schadauer and Büchsenmeister, 2004) and particularly favourable growth conditions on afforested agricultural marginal soils - are also likely to contribute to increasing growth in Austrian forests.

3 Possible effects of climate change on the forest

The potential occurrence and competitiveness of species and vegetation communities are determined, among other things, by climate, location and soil conditions, which in turn control the availability of resources (*Figure 1*). Furthermore, disturbances ("stochastic effects") are a driving force for dynamic processes in ecosystems. This totality of environmental conditions acting on plants at a growth site, insofar as they are not controlled by the

Competition between plants is subsumed under the term "site". Only those environmental conditions that remain constant or recur regularly over a reasonable period of time are considered to characterise a site (Arbeitskreis Standortskartierung, 1996). The concept of constancy of environmental conditions was developed for forest site mapping as a basis for silvicultural planning.

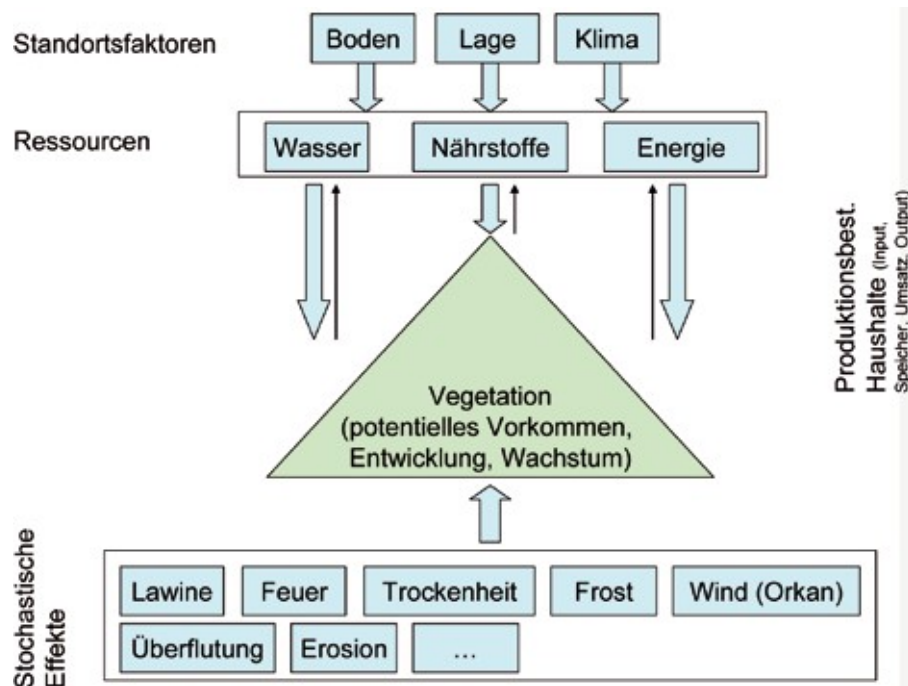


Figure 1: Location as the sum of environmental conditions. Climate change has an impact on the availability of resources and on stochastic effects

For long-lived species such as forest trees, rapid climate change means that the site conditions change dramatically from the time the seedlings are established over the potential lifespan of centuries.

Climate is the most important factor for vegetation development and temperature is a central controlling factor for numerous processes in the ecosystem. In addition to the mean annual temperature, whose change can be easily modelled and predicted within certain limits, numerous other climate parameters such as precipitation, seasonality, radiation, snow cover and thus the energy and water balance of plant populations also change. The type and frequency of disturbances are also changing.

3.1 Effects of an increase in temperature

Under conditions where temperature is a limiting factor, the current and predicted temperature increase will definitely have a positive effect on plant growth. This can be observed both in northern latitudes and in higher mountainous regions. A change in the phenological seasons (Menzel and Fabian, 1999) causes a shift in the vegetation zones. Due to Austria's geographical location on the border of different climatic provinces and its pronounced altitudinal range, it is obvious that the current

The distribution of vegetation communities can be used to predict the shifting of range boundaries.

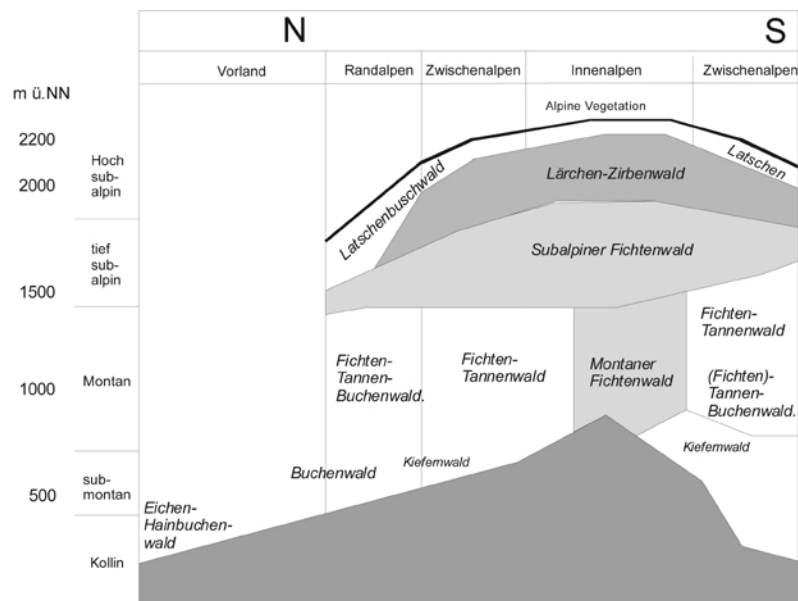


Figure 2: Geobotanical north-south profile through the Eastern Alps (greatly simplified after Ozenda, 1988)

Figure 2 shows the distribution of natural forest communities in a north-south transect across the Eastern Alps. According to the distribution of temperature, precipitation and continentality, there is a clear horizontal and vertical structure. The altitudinal gradient of the mean annual temperature is approx. -0.5°C per 100 m above sea level (Ozenda, 1988).

An increase in the mean annual temperature of 2.3°C by the middle of the century (Gobiet, 2008) thus corresponds in simplified terms to a vertical shift of the altitudinal zones by around 450 m upwards. A shift in the vegetation zones and a rise in the tree line are to be expected in any case and can already be observed in some mountains (Parmesan, 2006). Nicolussi and Patzelt (2006) document an increase in the distribution limit of Swiss stone pine in the Tyrolean Central Alps over the last 150 years, whereas the early 19th century was characterised by a decline in the tree line. Their dendrochronological investigations over the last

However, 9,000 years also show that between 7000 and 4000 BC the tree line was above the current potential level.

In addition to the positive effects of rising temperatures on growth, negative effects are also to be expected. One of the expected effects of global change for Central Europe is an increase in the frequency and intensity of dry periods (IPCC, 2007). Even if precipitation distribution remains the same, the increase in temperature and the associated decrease in relative humidity, especially in eastern Austria (Auer et al., 2001), will lead to an increase in evaporation requirements for vegetation. The water consumption of forests is generally significantly higher than in other cultivated species. Due to long development periods, forest ecosystems have only developed limited mechanisms for adapting to short-term changes in climatic conditions. Short-term changes in climatic conditions and periods of drought stress during the vegetation period can therefore have serious direct and indirect effects on forest ecosystems.

impact on the vitality and growth of forest trees. Leaf loss and decline in production up to the death of individual trees and stands are directly observable effects, whereby the symptoms sometimes occur with a time delay (review in Bre- da et al., 2006, Utschig et al., 2004). An indirect effect of drought stress is an increased vulnerability to biotic damaging factors such as infestation with phytophagous insects and pathogens. As the development of important insect pests is also controlled by temperature, calamities are expected to increase.

Lexer et al. (2002) modelled the effects of climate change on Austrian productive forests using a dynamic forest ecosystem model. They used stand and site data from the Austrian Forest Inventory (ÖWI) and regionalised climate scenarios as model input variables. A climate scenario with an increase in mean annual temperature by 2 °C by 2050 compared to the period 1961 to 1995 and a decrease in summer precipitation by 15 % showed clear effects on ecosystem dynamics. A change in the potential natural forest communities to more than three quarters of the inventory points is to be expected. While the proportion of deciduous forest communities increases at higher altitudes under this scenario, spruce loses its competitiveness and would no longer be sustainable at lower altitudes. Lexer and Seidl (2007) conclude that many of today's spruce thickets and poles will stand on mixed oak forest sites and dry beech sites up to medium altitudes and will therefore be more susceptible to disturbance, while at higher altitudes the silvicultural freedom in terms of tree species selection and regeneration methods will improve.

However, simplified analogies from vegetation composition under current climate constellations to vegetation distribution under future climate conditions are only permissible to a limited extent. As other important climate parameters change in addition to temperature, new competitive relationships arise and the utilisation of ecological niches can change. Using pollen analyses, Williams et al. (2007) were able to show that in the late glacial period around 14,000 years ago, vegetation communities with species that still exist today occurred in North America that would no longer be conceivable in this composition. The ability to regenerate and spread, on the one hand, and the availability of suitable habitats that can serve as retreat areas in the event of changing environmental conditions, on the other, are a prerequisite for the survival of species. Evolutionary adaptations of species with gene changes of more than one year are rather unlikely by the end of this century. Species that are geographically isolated, e.g. in the high mountains, are endangered by rapid climate change (Parmesan, 2006). The rise in the tree line can lead to loss of isolated alpine habitats, which jeopardises endemic species in particular. Shifts in the species composition of the vegetation in the alpine and nival levels of the Alps within just 10 years have already been demonstrated in the GLORIA project (www.gloria.ac.at; Pauli et al., 2007).

3.2 Effects of an increase in the CO₂ concentration

The increase in CO₂, the most important plant nutrient in the atmosphere, has a direct impact on growth and feedback effects on the water balance of plant populations. The photosynthetic uptake of CO₂ is not yet saturated. A positive "fertilisation effect" from CO₂ can therefore still be expected, although this is dependent on the

other site factors - soil, nutrient supply and water availability (Körner, 2003). C3 plants in particular, which include our forest trees, react to increased CO₂ concentrations with reduced stomatal conductivity, resulting in reduced water consumption and increased water utilisation efficiency (Leutzinger et al., 2005; Leutzinger and Körner, 2007). This factor thus partly counteracts the increased evaporation demand caused by the rise in temperature.

3.3 Effects of climate change on disturbance patterns

Disturbances such as storm and snow breakage damage or insect calamities, even if they have serious economic consequences, should always be seen as part of forest dynamics (Brang et al., 2004; Splechtna and Gratzner, 2005). Forests are not static and are subject to processes of development and decomposition.

From an economic perspective, however, a structured risk assessment is necessary. The following concept is used for this, based on Kobler (2006) and Brang et al. (2004):

A **hazard** a phenomenon that has a negative impact on an object of investigation when it occurs (storm, drought, insect calamity); a **risk** is then the probability of a potentially dangerous event occurring. How high the **risk**, the predicted damage is when the hazardous event occurs, depends in turn on the **vulnerability**, the degree of vulnerability of an object under investigation. The reaction of the forest to disturbances depends on the **resistance to disturbance**, i.e. the ability to withstand stress from disturbance factors without change (e.g. susceptible to windthrow or not), and the **resilience to disturbance**, the speed with which the system returns to its original state (e.g. quickly rejuvenates).

Even if the forecasts with regard to climate change are fraught with uncertainty, the risk, i.e. the probability of extreme events occurring, is likely to increase regionally. How the forests react to such disturbances can be seen from the long-term observations of the quantities of damaged timber due to storms and snow breakage as well as bark beetle infestation (*Figure 3*, Tomiczek and Steyrer, 2008).

The vulnerability of forest stands to hurricanes is modified regionally by the location. The susceptibility to windthrow depends very much on the stand structure, which in turn can be influenced by management. Foresters therefore avoid creating sharp stand edges that are exposed to the main wind direction. Trees that are planted all year round, such as spruce, also offer large areas exposed to the wind outside the growing season. Due to a rather shallow root system, spruces are less stable than other tree species in some locations. This results in low resistance, especially for spruce forests. If large quantities of damaged wood are present as a result of a windthrow event, conditions are favourable for consequential damage caused by insect pests. In spruce forests, the spruce bark beetle (*Ips typographus*) is a major damaging factor (Christiansen and Bakke, 1988). Under favourable temperature conditions, the spruce bark beetle can produce several generations per year. Long, warm periods of vegetation with little precipitation favour mass propagation of bark beetles. Model calculations by Seidl et al. (2007, 2008) therefore show a particularly high vulnerability of spruce stands

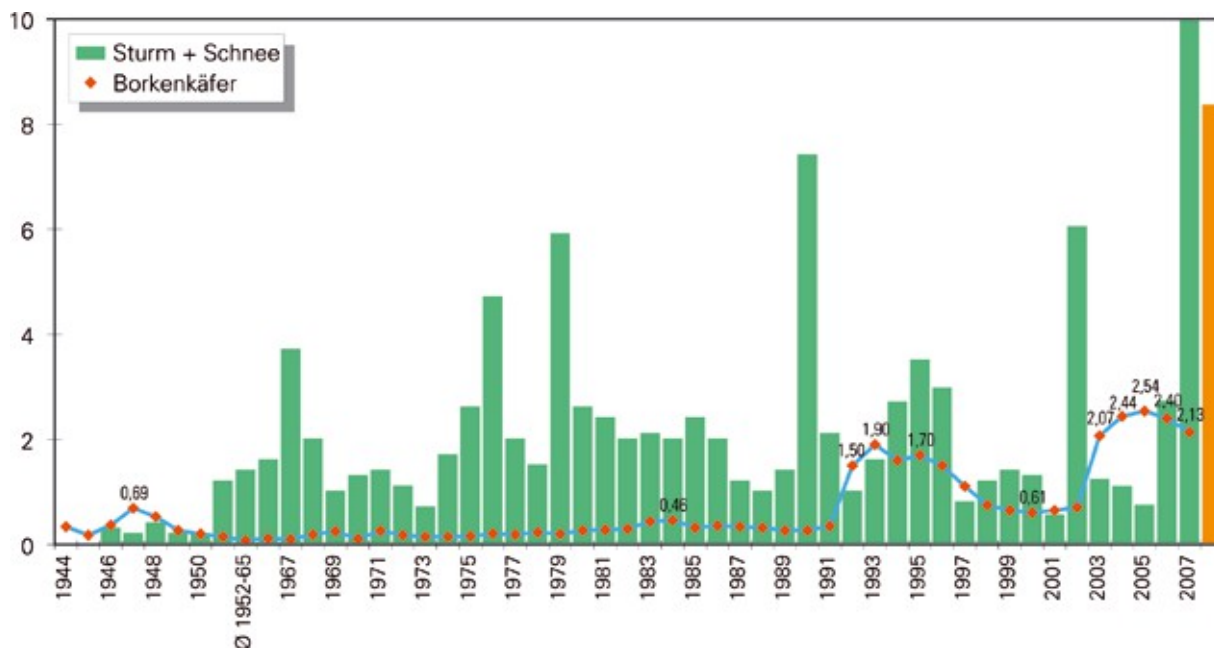


Figure 3: Time series of quantities of damaged timber due to bark beetle infestation, storm and snow breakage (Tomiczek and Steyrer, 2003, courtesy of BWF Vienna)

The results of these analyses are based on lowland locations and predict a sharp increase in bark beetle damage under the expected climate changes. According to Grabher (1998), 22 per cent of Austria's forests can be classified as heavily modified by humans. These forests have an altered tree species composition and altered structures compared to the potentially natural forest community. This includes, in particular, secondary spruce monocultures at low altitudes. However, the resilience to disturbance at low altitude sites can be classified as low, as reforestation or natural regeneration usually works relatively easily.

The threat posed by bark beetles to near-natural, subalpine spruce forests is generally lower than at low altitudes due to unfavourable temperature conditions in most years. However, the extremely dry year of 2002 showed that the bark beetle is also capable of mass reproduction at high altitudes (Schopf, 2007). Hurricane "Uschi" in 2002 and the subsequent warm, dry year of 2003 led to bark beetle infestations in Styria and Salzburg altitudes of over 1,800 metres above sea level (Wiener, 2008; Lick, 2008). This development has created new conditions for the management of protection forests. Poor access to these regions leads to more difficult conditions for the processing of damaged wood. In some mountain valleys, the bark beetle infestation has resulted in large clear-cut areas. Large sums of public funds are required to the protective effect of forests against avalanches, peak runoff and erosion (Lick, 2008). Although subalpine spruce forests are more resistant to disturbance than forests at lower altitudes, their resilience to disturbance is low for periods of 30 to 80 years until secure regeneration (Brang et al., 2004). The area of the Austrian protection forest is 776,000 ha, of which 303,000 ha are in yield. Rejuvenation is necessary on 2/3 of the protection forest in yield, but 80 % of this no regeneration. With regard to the multifunctional effects of the forest, these forests are therefore at particularly high risk.

How disturbance patterns will develop in the future can only be predicted to a limited extent. One possible effect of global warming is the shifting or disappearance of thermal barriers to the spread of plant pests. For example, the pine processionary moth (*Thaumetopoea pityocampa*) has expanded its range in France by 87 kilometres to the north over the last three decades. In the Venosta Valley, its altitudinal range has increased by 110 to 230 metres to an altitude of 1,400 metres above sea level in the same period (Battisti et al., 2005). The larvae of this butterfly feed on pine needles in winter. The rise in winter temperatures favours survival at higher altitudes and more northerly latitudes.

4 Mitigation strategies in forestry

The aim of forestry must be to increase resistance to disturbance and resilience. In view of rotation periods of 100 years and more, the changed climatic conditions must be taken into account when selecting tree species. The risk can be reduced in low-lying areas by avoiding spruce monocultures. A shift away from age-class forests towards permanent forests with natural regeneration and a larger, site-appropriate range of tree species is seen as one way of reducing the vulnerability of forests (Spiecker, 2003; Lexer et al., 2006). This trend can already be observed in forestry practice. In the forest inventory, a clear decline in pure conifer stands from 70 per cent in the 1971/80 inventory period to 62 % in the 2000/02 period and an increase in mixed stands and pure hardwood stands can be observed (Russ, 2004).

In younger secondary spruce stands, timely thinning measures can lower water consumption and reduce water stress for the individual tree (Katzensteiner et al., 2007 a, b).

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Author's address:

Univ.-Prof. DI Dr Klaus Katzensteiner
University of Natural Resources and Life
Sciences, Vienna Institute of Forest
Ecology
Peter-Jordan-Straße 82, 1190 Vienna
Tel. +43-1-47654-4103
e-mailklaus.katzensteiner@boku.ac.at

Climate change and its foreseeable consequences for agriculture

Josef Eitzinger, Gerhard Kubu and Sabina Thaler

The assessment of the effects of regional climate change (i.e. a change in the regional climate of the past decades and its natural variability) with regard to regional agricultural production risk or possible adaptation measures to reduce these risks is increasingly becoming the focus of research interest and is gaining in value for decision-makers from politics and practice or for farmers. Compared to the diverse influences of climate change on agricultural ecosystems, the analysis of potential adaptation measures due to human influence and numerous options is even more complex. Potentially realistic adaptation measures must in particular take into account regional conditions, be it with regard to climate scenarios, existing natural production resources, production systems and socio-economic scenarios. In addition, adaptation strategies should also guarantee and ensure sustainable agricultural production. All of these factors lead to relatively high uncertainties and ranges with regard to regional future scenarios, which makes it difficult for decision-makers to take concrete measures. Despite the great need for further research, a number of conclusions can be drawn and trends recognised with the help of current knowledge, which outline the future risk potential and adaptation strategies.

1 Regional climate change and effects on agricultural production resources

In order to be able to estimate the regional effects of climate change, it must first be known which climatic factors will change a) to what extent, b) within what period of time, c) likely to change at which location. This is achieved through regionalisation methods of global climate scenarios. The scenarios for temperature are generally less uncertain than those for precipitation and other climate parameters. The mean changes are also much more reliable than a possible change in climate variability, i.e. the frequency, intensity, duration, timing and spatial extent of extreme weather events (droughts, heavy precipitation, storms, hail, frost, heat waves, etc.). It must also be taken into account that the damage potential of these extreme events for agriculture depends largely on the time of their occurrence in connection with the variable sensitivity of the receptors over the course of the year (such as crops that in a certain phenological stage). Thus, research first assesses the consequences of an average temperature increase on the production potential of crops and gradually incorporates other parameters and climate variability.

The current climate scenarios show that temperatures in the main production areas of Upper Austria, Lower Austria and Styria will rise between approx. 0.8 °C and 2 °C (comparison period 1961-1990) by the 2050s (corresponds to the medium from the 30-year average), depending on the climate model and emission scenario. A further increase of between 2.5 °C and 4.5 °C is indicated by the 2080s. All climate models show a higher temperature increase during the winter and summer than in the transitional periods. There are also signs of greater warming with increasing altitude.

A key factor for agriculture - especially where water is a limiting factor for production - is how precipitation will develop in the climate scenarios. Here, Austria is basically in a transitional area, very unpleasant prospects in the Mediterranean region with up to 50 % less summer precipitation by the 2080s, while an increase in precipitation can be expected in the higher latitudes. From a regional perspective, the annual precipitation could decrease slightly in the east and south and increase in the west according to the present scenarios, albeit with a relatively high degree of uncertainty. Furthermore, the scenarios show a decrease in summer precipitation in particular and a significant increase in precipitation in the winter half-year, i.e. a change in the seasonal distribution of precipitation. Therefore, an increase in the frequency and intensity of agrometeorological dry periods and a decrease in precipitation in summer can also be expected in the northern Alpine region. In addition to the distribution of precipitation, a particular factor for the water supply in plant production is a possible increase in heavy precipitation, which can be observed regionally from measurements. Increasingly dry conditions are therefore to be expected, especially during the vegetation period. As a result, regional differences in water supply are also increasing, with soil water storage capacity playing a particularly important role for plant production (e.g. generally increasing water shortages in the north-east, east and south of Austria, but more so on the lighter soils).

Previous research results on the effects on agriculture in Austria show the following general picture:

- A number of studies for the Central European region have come to the conclusion that with a moderate average temperature increase of up to approx. 2 °C (as expected by the 2040s), the positive effects on yield potential in agriculture could outweigh the negative effects. However, the "more extreme" climatic conditions would also result in greater regional differences, which have hardly been analysed to date.
- There is a shift in temperature zones with corresponding effects on the phenology and growth of crops. The vegetation period is extended by approx. 7-10 days per decade, starts earlier and the development rates of the plants are accelerated. The production potential in cultivation regions previously limited by temperature would improve, particularly as a result of an extended vegetation period, such as fodder production in many high-precipitation grassland regions. The risk of frost damage due to late frosts, particularly in fruit crops, could also increase due to the earlier vegetation period at various locations.
- The rising temperatures increase the evaporation potential disproportionately, which means an increase in the utilisation of soil water resources by the vegetation. In Austria's low-precipitation growing regions, summer crops in particular would be exposed to increasing heat and drought stress. Water-saving crops, cultivation techniques and crop rotations will therefore become more important. An increase in the need for agricultural irrigation would also be a logical consequence. On the other hand, an increase in the number of weather-related working days available in the fields due to drier conditions combined with a longer vegetation period would increase the flexibility of the agricultural sector.

in production technology or allow certain machine capacities (as a cost factor) to be reduced.

- The fertilisation effect of the increasing carbon dioxide content of the air would increase the yield potential of common crops. Simulation studies have shown a slightly positive average yield trend for cereals up to the 2050s, assuming constant climate variability. However, the influence of increasing weather extremes is generally expected to result in lower yield stability and higher yield risk between years (with corresponding implications for farm management and risk hedging). Recent studies have shown that increasing carbon dioxide concentrations lead to changes in crop quality, such as a deterioration in the gluten quality of wheat.
- With climate variability remaining the same, rising temperatures increase the number of hot days and dry periods compared to the previous climate period, which means an increase in stress situations caused by drought and heat for crops grown to date, especially in warmer and drier locations. Other yield-impacting stress factors that occur in combination with heat, such as yield-reducing ozone damage, could also increase.
- An increase in temperature changes ecological niches for weeds, diseases and pests. These can spread further, appear or disappear, or develop more quickly (e.g. additional generations). Insects, which are very sensitive to temperature, are a particular threat here.
- An increase in extreme weather events generally harbours a very high potential for damage to agriculture and could negate the positive effects of a longer growing season, for example. In particular, drought and heat (drought and heat damage), heavy precipitation (soil erosion damage, wintering damage), an increase in the frequency of thunderstorms (hail damage) and storms (damage in the forestry sector) should be mentioned. The extent to which the frequency of these extremes will change under the climate scenarios is still subject to great uncertainty, although changes - which vary from region to region - are emerging from measurement series (e.g. increase in heat waves).
- Regional differences in yield potential are generally increasing due to differences in water supply (e.g. soil water storage capacity). Locations with poor soils lose out to locations with good soils. Similarly, in locations where the water supply acts as a limiting production factor, there is increasing inter-annual yield variability.

2 Effects on different production systems and possible adaptation measures

Adaptation measures in Austrian agriculture to climate change have so far only been examined to a limited extent (e.g. in the ongoing EU project ADAGIO), which means that there is still a great need for research into regional analyses. By simulating various scenarios, e.g. in production technology, possible adaptation measures can be assessed in terms of their impact on long-term production potential. Of course, impacts and possible adaptation measures can only be definitively assessed by taking into account the expected socio-economic boundary conditions, which will be influenced by the ongoing process of climate change.

structural change in agriculture is likely to have a greater impact in the medium term than climate change alone. However, by changing the production risk or yield potential, climate change can contribute to strengthening or weakening corresponding trends. The respective precipitation and temperature conditions and their possible change must always be seen in connection with a specific production system, as different critical climatic limits apply here. For example, different precipitation and temperature conditions are optimal for the production potential of grassland production than for arable farming. Different limits can also be drawn within arable farming, including crop-specific climatic limits. Overall, a clear shift in the agroclimatic zones can be expected (*Fig. 1 a)-c*). Production systems that are close to or at the climatic limits under the given site conditions would be the first to feel - positive or negative - changes in climate-related production potential or risk as a result of climate change.

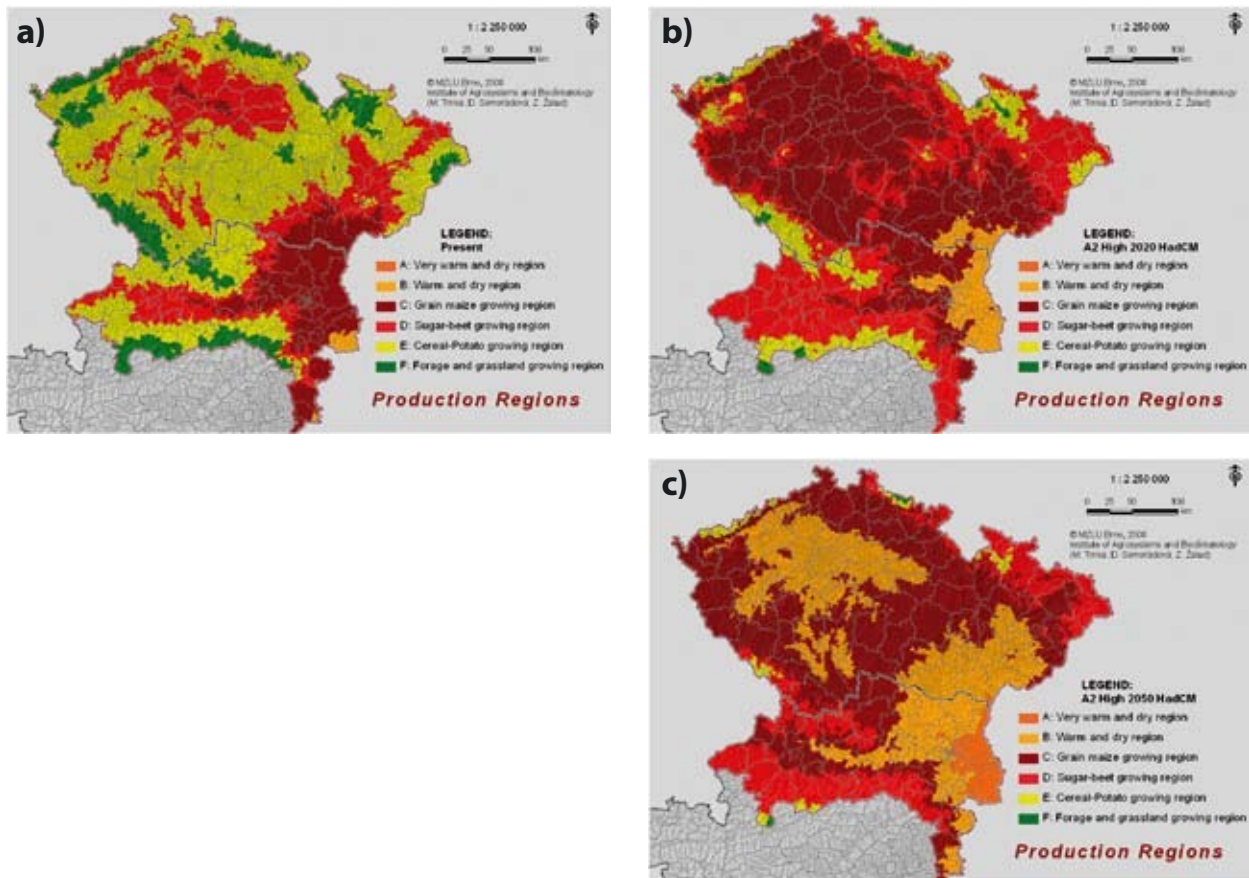


Figure 1: Shift of agroclimatic zones based on 1961-1990 conditions (a) under the HadCM AG4 climate scenario for the 2020s (b) and 2050s (c) (Trnka et al., 2008)

The following section outlines possible adaptation measures for Austria's most important agricultural production sectors to climate change:

2.1 Arable farming

A largely effective measure is the adaptation of cultivation measures to an extended vegetation period (earlier cultivation times in spring, selection of later maturing varieties) for many annual crops. In the case of summer cereals and maize this measure maintains or extends the growing season - a factor that has a significant influence on yield. However, droughts can have a particular impact on crops that grow well into the summer, although the negative effects are on average lower for early-harvested winter crops (e.g. winter cereals) due to better utilisation of winter moisture during critical phenological phases (e.g. flowering, grain filling). In the regions of Austria with low precipitation, for example, there was a clear dependence on soil water storage capacity (Fig. 2).

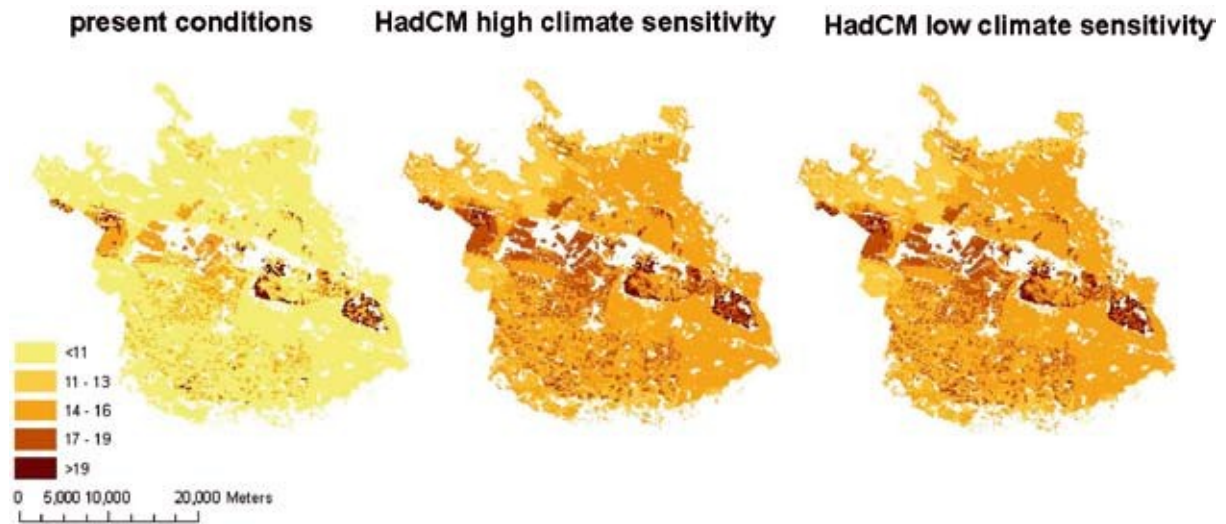


Figure 2: Spatial change of a relative water stress factor for spring barley in the Marchfeld for the 2050s (dark areas, mainly on the lighter soils, indicate a stronger increase). Simulation with the CERES-Barley growth model

In general, this means that any adaptation measures that increase the efficiency of water utilisation in crop production are important. These measures include appropriate landscaping (wind and evaporation protection through hedges), crop rotation that conserves soil water (e.g. favouring winter crops over summer crops), ground cover as protection against evaporation and erosion (mulch covers), irrigation systems that respond to soil variations and/or are efficient, selection of drought- and heat-resistant varieties or crops with low water consumption, improvement of soil water storage capacity by promoting a good soil structure and by building up humus, avoidance of soil damage, etc.

Due to their high water requirements, alternatives such as biomass production will have to be strongly orientated towards regional precipitation conditions and available soil water reserves. The increased or new occurrence of climate-sensitive harmful factors such as weeds, pests (especially insects) and diseases will require production technology to be highly flexible and able to react quickly. demands placed on plant breeding and

The methods of plant protection will increase as a result. A wider range of robust, stress- and disease-resistant varieties will be required, as well as an increase in cultivated variety and species diversity (also from a regional perspective) in order to minimise the production risk of individual farms and regions due to the various biogenic damaging factors and increasing weather extremes.

2.2 Grassland

In the case of grassland production, the threatened border areas in Austria are likely to be located at around 600-700 mm annual precipitation, whereby in addition to the topography, the temperature conditions and the soil properties (nutrient and water storage capacity) also have a significant influence on the production potential via the climatic water balance.

The constant or slightly decreasing precipitation on the eastern edge of the Alps and also north and south of the Alps indicated in the climate scenarios and the increasing unproductive evaporation due to the increase in temperature affect precisely these regions, as initial estimates using agroclimatic water balance models show (Figure 3).

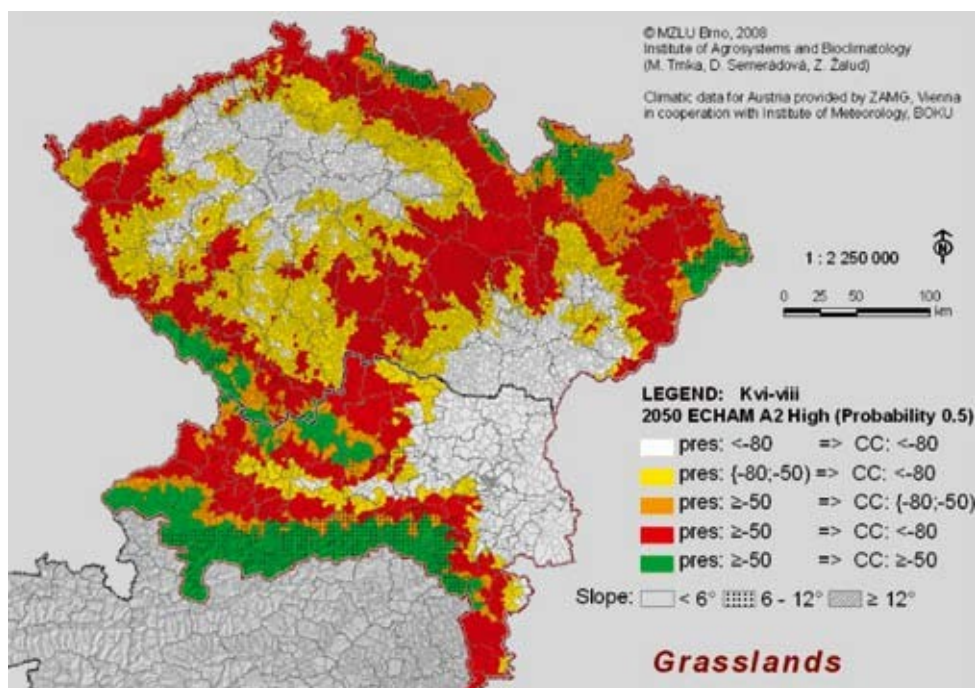


Figure 3: The negatively affected grassland regions north of the Alps under the ECHAM climate scenario for 2050 (in red). This is due to an increasingly poorer water balance and thus declining production potential and a significantly higher yield risk

Due to poor soil conditions, the Mühlviertel and Waldviertel in particular are likely to have to contend with declining production potential and increasing production risk (frequent extreme yield losses due to more frequent droughts) in the grassland sector. On the eastern edge of the Alps, it is primarily the transition regions between grassland and arable land that could be affected by a more significant decrease in precipitation and warmer temperatures in the east. In locations with significantly higher precipitation (approx. over 800 mm per year)

precipitation), the production potential would increase due to the higher temperatures. At higher altitudes, where temperature and the length of the growing season are the main limiting factors, this effect is likely to be even stronger. However, changes in the species composition of permanent grassland are also to be expected, which will influence forage quality. Climate-sensitive pests that are important for grassland, such as white grubs, could also occur more frequently and cause widespread damage.

As there are hardly any regional adaptation studies available to date, it is currently only possible to rely on expert judgements. Adaptation options are certainly much more limited in the grassland sector than in arable farming, as it is often difficult to switch to another form of production if boundary conditions (such as soil properties, topography or land use restrictions) do not permit arable farming, for example. Otherwise, a switch to biomass production and fodder crop production - even if only on partial areas - would be an option that would allow much greater flexibility. Another adaptation option would be to irrigate grassland if the water is available and accessible locally at low cost. Due to the ongoing structural change (trend towards increasing farm sizes), the pressure to maximise yields per hectare could also be partially offset by a larger available production area (although this would also result in higher production costs and is significantly influenced by basic costs).

2.3 Animal husbandry

In the livestock sector, the increase in hot days can be expected to lead directly to increased stress due to heat (i.e. poorer growth rate or milk yield, losses), and the requirements for corresponding housing systems or ventilation systems will increase in this respect. Indirectly, a deterioration in feed quality (e.g. from grassland due to a change in species composition or from maize production due to increased secondary *Fusarium* infestation) can also increase production costs in the medium term. In principle, the demands on the (often relatively cost-intensive) capacities of land maintenance will also increase in order to be better equipped for a possible increase in the frequency of major yield losses due to extreme events. The occurrence of new climate-related diseases in livestock farming is also a largely unexplored area.

2.4 Permanent crops

For fruit growing and viticulture, there are currently only isolated estimates from experts on adaptation measures. In the case of wine, for example, a change in regional wine qualities and a shift in varieties is expected, which is already beginning to take place. In the long term, the development of new wine and fruit-growing areas in regions that were previously too cool is also conceivable. One major unknown is the new or altered occurrence of climate-sensitive pests or diseases such as the vine cicada, which will require new methods of plant protection or rapid reactions in plant breeding. In the case of fruit crops, an increase in thunderstorm activity, as is expected in a warmer climate, could further increase the risk of hail. Appropriate protection through hail protection measures or hail insurance will become particularly important here. The increase in starling precipitation has negative effects on permanent crops in particular due to the often uncovered soil.

effects on soil water erosion. Here, more protective measures such as mulch covers or landscaping measures must be taken, which are particularly important on slopes in order to avoid enormous and irreversible damage to soil structure and fertility in the long term.

Of course, a number of adaptation measures are also essential at a general and supra-regional level, in the area of policy and management measures. In addition to the provision of emergency funds for compensation, these include an insurance system that covers damage from extreme weather events as comprehensively as possible. Other areas relate to regulations regarding the effective use of resources such as the use of water for irrigation purposes, which will certainly become increasingly important in the coming decades, especially in regions with low precipitation. The expansion of monitoring systems for the rapid spatial assessment of crop-specific damage that has occurred, for the early detection of damage or for warnings, e.g. in the case of diseases or pests, or the assessment of long-term risks will also become increasingly important. These and other similar measures should increasingly find their way into operational applications and be developed for practical use.

Address of the car:

Univ.-Prof. DI Dr Josef Eitzinger DI
Gerhard Kubu
Sabina Thaler
University of Natural Resources and
Life Sciences, Vienna Institute of
Meteorology
Peter-Jordan-Straße 82, 1190 Vienna
Tel. +43-1-47654-5610
E-Mail josef.eitzinger@boku.ac.at